

Static Holographic Boundary Theory (SHBT) Cold Fusion Reactor: Integrated Engineering Specification and Physical Transducer Model

Abstract—The Static Holographic Boundary Theory (SHBT) reactor concept couples optical field enhancement in a deuterium-loaded Pd–Ir layer to lattice excitation, candidate screened nuclear transitions, and thermoelectric energy conversion. The theoretical framework combines Floquet dielectric response, multiphonon coupling, and collective boundary kinetics. A micro-alloyed transient-liquid-phase bondline with a graded Ni/Cu interlayer has a calculated total strain amplitude of 0.002137728 and a predicted fatigue life of 52,400 cycles for a 293.15–550.0 K thermal cycle, exceeding the 44,820-cycle requirement. The separate spring assembly retains a stiffness of 5,000 N/mm, a 20.1 μm mismatch deflection, and a cyclic contact-pressure variation of 0.320 MPa. Its Monte Carlo mean lifespan is 88,827 cycles, with a 95% lifespan interval of 54,822–138,492 cycles. The 785.0/802.5 nm pump pair produces an 8.328064 THz beat. At a separation of 0.28 Å, the selected potential convention gives a bare energy of 51.42714 eV, a driven pair energy of -298.57 eV, and a reconstructed reference energy of 51.43 eV. The active-volume inventory contains 2.53×10^{14} geometric domains and defines a gross thermal-power reference of 2911.4 W. The native scalar Floquet-kernel benchmark gives an inverse of 3.67250764167 and an absolute binary64-to-multiprecision difference of 3.56655×10^{-16} . These arithmetic and constitutive results do not independently establish microscopic reaction rates or measured endurance. Electrical export of 31.76 W, calorimetric uncertainty of 5.69 W, and a surface dose rate of 0.38 $\mu\text{Sv/h}$ remain conditional design outcomes. Reactor performance and safety require calibrated experiments, compatible energy and measurement boundaries, and independent qualification.

Index Terms—Acoustic energy harvesting, calorimetry, electron-phonon coupling, energy balance, nuclear screening, static holographic boundary theory, surface plasmon polaritons, thermoelectric generation.

I. INTRODUCTION & THEORETICAL TRANSDUCER MODEL

THE Static Holographic Boundary Theory (SHBT) reactor concept comprises a deuterium-loaded $\text{Pd}_{0.95}\text{Ir}_{0.05}$ active layer, a prism-coupled optical interface, a magnetic drive, a thermal spreader, and energy-recovery elements. Its transduction model couples optical interference, Floquet electronic screening, non-perturbative nuclear–lattice exchange, and dissipative boundary-state dynamics. Active phase feedback provides the control interface between thermal dephasing and the specified collective-rate enhancement. Electromagnetic response, quantum-state evolution, reaction kinetics, and energy accounting retain distinct normalization conventions throughout the coupled model.

Assumption I.1 (Specification basis). Numerical operating points represent reference parameters or design requirements

unless supported by independent measurements. Electromagnetic boundary conditions and energy conservation constrain the dielectric, Franck–Condon, and energy-conversion models [26], [47], [56], [59]. The local screening shift, lattice branching fraction, and collective-rate gain are separate physical quantities: a prescribed potential does not determine nuclear matrix elements, and a normalized density operator does not determine competing reaction rates. The boundary constitutive laws and feedback setpoints define a phenomenological operating model [79], [80]; their microscopic realization requires specified interactions and independently characterized material response.

The nominal electrical operating point is defined at the thermoelectric output bus, with support loads accounted for separately [70], [72], [73]:

$$\begin{aligned} P_{\text{th,ref}} &= 2911.40 \text{ W}, \\ P_{\text{TEG}} &= 418.43 \text{ W}, \quad P_{\text{chiller}} = 0, \\ P_{\text{support}} &= 116.67 + 120.00 + 150.00 \\ &= 386.67 \text{ W}, \\ P_{\text{net}} &= 418.43 - 386.67 = 31.76 \text{ W}. \end{aligned} \quad (1)$$

The thermoelectric generator operates nominally between $T_h = 623.15$ K and $T_c = 353.15$ K. The 550/310 K temperature pair defines a distinct comparison case. The Carnot efficiency at the nominal temperatures is

$$\eta_C = 1 - 353.15/623.15 = 0.43328252. \quad (2)$$

Thermal and electrical powers refer to distinct control-volume boundaries; their relationship follows from the energy balance in Section II. The nominal parameter set is summarized in Table LXXII. Alternative temperatures, footprints, response models, and state-space realizations describe separate comparison cases rather than simultaneous operating conditions. The 350 °C operating temperature equals 623.15 K, not 550 K; material qualification at 550 K does not establish performance at the higher temperature.

a) Engineering implementation and theoretical verification: The reactor engineering implementation is assigned to `shbt-cf` [95], while `shbt-precision` remains the reference for foundational SHBT boundary-mathematics cross-verification. These roles do not imply that either repository independently establishes reactor performance. The inspected engineering architecture combines Rust numerical crates with Python orchestration under `shbt_cf`. Optical, dielectric, structural, and stochastic kernels are identified by their actual

entry points below. Transport and controller-synthesis interfaces require explicit implementation mappings before their numerical results can be attributed to executable code.

b) Numerical precision and acceptance values: Tables I, III, VIII, and XXXVII specify the calculated coupling, screening, interface-flux, and balanced-reduction operating points. Decimal presentation suppresses binary64 conversion artifacts without changing the physical precision of the tabulated quantities. IEEE 754 binary64 remains the engineering arithmetic for well-conditioned matrix and transport evaluations; precision-sensitive boundary-state and quadrature calculations require a 512-bit reference path. The native Rust scalar Floquet benchmark gives $s_F \simeq 3.67250764167$ with absolute precision difference $\delta_F \simeq 3.56655 \times 10^{-16}$, as defined in Eqs. (17)–(18). Table LXXVII collects the corresponding domain and diagnostic acceptance values. These scalar and domain results do not replace the differently dimensioned quantities in Tables I, III, VIII, and XXXVII. Numerical agreement is distinct from experimental validation and from establishing a microscopic mechanism.

c) Boundary coordinates and arithmetic resolution: The boundary model assigns a zero-dimensional topological anchor as a coordinate reference, not a finite reaction volume. Let $\mathbf{r}_*$ denote this anchor, $\mathbf{R}^\top \mathbf{R} = I$, and $\mathbf{t}$ a translation. The rigid-coordinate map and its numerical consistency criterion are

$$\mathbf{r}' = \mathbf{R}(\mathbf{r} - \mathbf{r}_*) + \mathbf{t}, \quad \max_i \frac{\|\mathbf{R}^\top(\mathbf{r}'_i - \mathbf{t}) + \mathbf{r}_* - \mathbf{r}_i\|_2}{L_*} < 10^{-12}. \quad (3)$$

Here $L_* > 0$ is a fixed coordinate normalization length. A rigid transformation preserves distances, not the untransformed components of each position vector. Resolving an assumed relative boundary scale $1/N \simeq 10^{-122}$ near a unit-valued state requires more than 405 significand bits; 512-bit arithmetic provides additional rounding margin. Binary64 can represent numbers of magnitude 10^{-122} , but cannot resolve their addition to unity. Physical significance of that scale remains a model assumption. High-precision constants, quadrature nodes, and initial states shall be constructed at the working precision rather than promoted from already rounded binary64 values.

d) Model validity and reproducibility: A prescribed operating point does not establish a microscopic mechanism or measured performance. Quantitative validation requires experimental records, converged dielectric and rigorous coupled-wave analysis (RCWA) solutions, a defined fifteen-to-four-state reduction, and functional-safety qualification. In the absence of these inputs, the associated performance quantities remain model assumptions or validation targets.

e) Coupled physical constraints: Consistency across the seven physical and engineering domains requires a common definition of reaction volume, state normalization, energy flow, and measurement uncertainty. The constraints in Section XI delimit the conditions required for 100% physical consistency; agreement among selected numerical targets alone does not establish those conditions.

f) Comparison-model scope: Reduced dielectric, kinetic, and engineering models [63] are applicable only within their

stated constitutive assumptions and normalization conventions. Numerical agreement between comparison cases does not establish exact model equivalence. Physical conclusions require compatible inputs, reproducible calculations, and independently characterized uncertainty.

A. Optical Beat Frequency and Energy Scale

For collinearly polarized coherent pumps with intensities I_1 and I_2 , optical-cycle averaging leaves the interference term

$$I(\mathbf{r}, t) = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos[\Delta \mathbf{k} \cdot \mathbf{r} - \Delta \omega t + \phi], \quad (4)$$

$$\Delta \omega = \omega_1 - \omega_2, \quad \Delta \mathbf{k} = \mathbf{k}_1 - \mathbf{k}_2,$$

where ϕ is the relative phase. With vacuum wavelengths $\lambda_1 = 785.0$ nm and $\lambda_2 = 802.5$ nm and $c = 299\,792\,458$ m s $^{-1}$,

$$f_1 = \frac{c}{\lambda_1} = 381.9012204 \text{ THz},$$

$$f_2 = \frac{c}{\lambda_2} = 373.5731564 \text{ THz},$$

$$f_{\text{beat}} = c \left| \frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right|$$

$$= 8.32806399587 \text{ THz} \approx 8.328064 \text{ THz}.$$

The specified 8.328064 THz is the rounded difference frequency of the optical intensity modulation; calculations retain the wavelength-derived value. Resonant lattice excitation requires a material mode at this frequency, nonzero optical-lattice coupling, suitable spatial overlap, and a relative pump linewidth compatible with the mode linewidth.

Using $h = 4.135667696 \times 10^{-15}$ eV s, the nominal beat gives

$$\omega_{\text{beat}} = 2\pi f_{\text{beat}} \approx 5.23267693361 \times 10^{13} \text{ rad s}^{-1}$$

$$\approx 5.232677 \times 10^{13} \text{ rad s}^{-1}, \quad (6)$$

$$E_{\text{LO}} = \hbar \omega_{\text{beat}} = h f_{\text{beat}}$$

$$\approx 0.03444210524 \text{ eV} \approx 34.442 \text{ meV}.$$

Frequency calibration depends on the measured wavelength difference. A distinct comparison setpoint of exactly 8.33 THz has energy 34.450 meV and, at fixed $\lambda_1 = 785.0$ nm, requires a different second wavelength:

$$\lambda_2 = \left(\lambda_1^{-1} - \frac{8.33 \times 10^{12} \text{ s}^{-1}}{c} \right)^{-1} = 802.504159 \text{ nm}. \quad (7)$$

B. Kretschmann Coupling and SPP Dispersion

Consider a planar, nonmagnetic interface between a metal with complex relative permittivity $\varepsilon_m(\omega)$ and a dielectric with relative permittivity $\varepsilon_d(\omega)$. A transverse-magnetic surface mode proportional to $\exp(ik_x x - i\omega t)$ decays away from the interface with constants

$$\kappa_j^2 = k_x^2 - \varepsilon_j \frac{\omega^2}{c^2}, \quad \text{Re}(\kappa_j) > 0, \quad j \in \{m, d\}. \quad (8)$$

Continuity of the tangential electric and magnetic fields gives

$$\frac{\kappa_m}{\varepsilon_m} + \frac{\kappa_d}{\varepsilon_d} = 0. \quad (9)$$

Squaring this condition and eliminating the decay constants yields

$$\frac{k_x^2 - \varepsilon_m \omega^2 / c^2}{\varepsilon_m^2} = \frac{k_x^2 - \varepsilon_d \omega^2 / c^2}{\varepsilon_d^2}, \quad (10)$$

$$k_{\text{spp}}(\omega) = \frac{\omega}{c} \sqrt{\frac{\varepsilon_m(\omega) \varepsilon_d(\omega)}{\varepsilon_m(\omega) + \varepsilon_d(\omega)}}.$$

The physical root must also satisfy the original, unsquared boundary condition and passive propagation. For a lossless positive dielectric, a bound mode requires $\varepsilon_m < -\varepsilon_d$.

A prism of relative permittivity ε_p supplies the in-plane momentum needed for approximate Kretschmann phase matching at each optical pump frequency:

$$\frac{\omega_j}{c} \sqrt{\varepsilon_p(\omega_j)} \sin \theta_j \approx \text{Re } k_{\text{spp}}(\omega_j), \quad j \in \{1, 2\}. \quad (11)$$

Equation (10) is the semi-infinite-interface limit. A finite lossy film requires a multilayer electromagnetic calculation; a reflectance minimum need not coincide with the surface-mode optimum.¹ Reference interface parameters are $\varepsilon_d = 1.0006$, $\theta_{\text{inc}} \approx 43.85^\circ$, and field decay depth $\delta_z \approx 12.4$ nm. The multilayer solution shall determine their jointly admissible values using the operating gas density, film thickness, and measured complex permittivities. The specified field and propagation observables are

$$\delta_z = \frac{1}{\text{Re } \kappa_d}, \quad L_{\text{spp}} = \frac{1}{2 \text{Im } k_{\text{spp}}}, \quad \mathcal{F}_E = \left| \frac{E_{\text{spp}}}{E_{\text{inc}}} \right|^2, \quad (12)$$

where δ_z is a field-amplitude decay length and L_{spp} is an intensity propagation length. Matching the difference wavevector to a lattice mode is a separate condition from matching either optical carrier to an SPP.

The monolithic Option B geometry, complex optical constants, Fourier-modal boundary-value problem, and dual-pump field normalization are specified in Section II-A. The electronic and lattice interactions use those electromagnetic fields without identifying a field-intensity enhancement with a nuclear-rate multiplier.

C. Reciprocal-Space Floquet Dielectric Matrix Inversion

The driven electronic system has period $T_d = 2\pi/\Omega_D$. Reciprocal vectors $\mathbf{G}, \mathbf{G}'$ and temporal indices m, n define its composite response space. The retarded response $\chi \equiv \Pi$ follows Eq. (56), with its volume normalization, occupations, transition harmonics, and positive broadening η . Matrix indices and sideband denominators use the same Fourier convention. In SI units,

$$\epsilon_{GG'}^{mn}(\mathbf{q}, \omega) = \delta_{GG'} \delta_{mn} - v_G(\mathbf{q}) \chi_{GG'}^{mn}(\mathbf{q}, \omega),$$

$$v_G(\mathbf{q}) = \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2}, \quad (13)$$

$$\sum_{G'', \ell} \epsilon_{GG''}^{m\ell} [\epsilon^{-1}]_{G''G'}^{\ell n} = \delta_{GG'} \delta_{mn}.$$

¹J. J. Foley IV, H. Harutyunyan, D. Rosenmann, R. Divan, G. P. Wiederrecht, and S. K. Gray, “When are Surface Plasmon Polaritons Excited in the Kretschmann-Raether Configuration?,” *Scientific Reports*, vol. 5, article 9929, 2015, doi: 10.1038/srep09929.

The SI reciprocal-space Coulomb kernel does not contain the real-space factor $1/(4\pi)$. Its zero-wavevector component requires a specified neutrality or electrostatic boundary prescription rather than an arbitrary large finite replacement.

For a static point-source charge at $\mathbf{r}'$, the incoming Floquet channel is $n = 0$. Its time-dependent screened interaction energy is

$$W(\mathbf{r}, \mathbf{r}'; t) = \sum_{GG', m} \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} e^{i(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r}} e^{-i(\mathbf{q} + \mathbf{G}') \cdot \mathbf{r}'} \times e^{-im\Omega_D t} [\epsilon^{-1}]_{GG'}^{m0}(\mathbf{q}, 0) v_{G'}(\mathbf{q}). \quad (14)$$

An arbitrary time-dependent source requires contraction over its incoming harmonic amplitudes; summing all incoming channels without specifying that source is not equivalent to a static charge. Cycle averaging selects $m = 0$ after the full inversion. Off-diagonal response describes frequency conversion and spatial local-field coupling, not by itself the magnitude of a nuclear screening shift.

The optical-scale comparison uses $a_0 = 3.52 \times 10^{-10}$ m, $\Omega_D = 2.45 \times 10^{15}$ rad/s, and $n_e = 4.8 \times 10^{30}$ m⁻³. It is distinct from the selected 8.328064 THz beat-driven case. The selected energy convention uses the rounded Coulomb coefficient 14.3996 eV Å at $r_d = 0.28$ Å:

$$V_{\text{bare}}(r_d) = \frac{14.3996}{0.28} \text{ eV} \approx 51.42714 \text{ eV},$$

$$\Delta U_{\text{scr}} = -350.00 \text{ eV},$$

$$V_{\text{driven}}(r_d) = V_{\text{bare}} + \Delta U_{\text{scr}} \approx -298.57 \text{ eV}, \quad (15)$$

$$V_{\text{eff}}(r_d) = U_{\text{eff}} + V_{\text{driven}}$$

$$= 350.0 - 298.57 = 51.43 \text{ eV}.$$

Here $U_{\text{eff}} = -\Delta U_{\text{scr}}$ is the positive screening shift, V_{driven} is the assigned screened pair energy, and V_{eff} is the reconstructed bare-level reference, not a second screened barrier. The 51.43 eV sum uses the rounded driven energy; before rounding it equals V_{bare} . In SI units the pair energy is $e^2/(4\pi\epsilon_0 r_d)$, whereas $e/(4\pi\epsilon_0 r_d)$ is an electric potential in volts. Evaluating the former with full SI constants gives 51.42730525 eV; that precision convention is retained only in explicitly identified electrostatic comparisons below. A negative energy at one coordinate does not establish a normalizable bound state, its lifetime, or a dielectric inversion result.

The selected beat-driven screening case specifies conduction-electron density $n_e = 4.85 \times 10^{28}$ m⁻³ and local field-amplitude ratio $\mathcal{A}_z = |E_z/E_0| = 21.45$, within $18.5 \leq \mathcal{A}_z \leq 24.2$. Its intensity ratio is $\mathcal{A}_z^2 = 460.1025$, distinct from the pump-resolved intensity factors in Section II-A. Retaining sidebands $m = -2, \dots, 2$ gives five temporal channels and a two-quantum energy of $2\hbar\Omega_D = 68.8842$ meV. This reduced sideband model must be checked against larger truncations; the field amplitude, carrier density, and energy subtraction alone do not evaluate $[\epsilon^{-1}]_{GG'}^{00}$. The static head approximation must extract the head element after inversion of the composite reciprocal-space/Floquet matrix, not invert that element in isolation.

In a free-electron plasma comparison with mass m_e , the density for $\omega_p = \Omega_D$ and the frequency at the assigned density are

$$n_{e,\text{res}} = \frac{\epsilon_0 m_e \Omega_D^2}{e^2} = 1.8860322 \times 10^{27} \text{ m}^{-3},$$

$$\omega_p(4.8 \times 10^{30} \text{ m}^{-3}) = 1.2359820 \times 10^{17} \text{ rad/s.} \quad (16)$$

The inequality $n_e \geq n_{e,\text{res}}$ means $\omega_p \geq \Omega_D$ in this comparison; it does not ensure resonant coupling or a 350 eV shift. Effective mass, dielectric environment, losses, and the driven occupation distribution remain constitutive quantities.

The truncation design retains at least 12 reciprocal vectors and temporal orders $-7 \leq m, n \leq 7$, giving 15 temporal channels and a 180×180 matrix for exactly 12 vectors. A radial reciprocal cutoff $G_{\text{max}} \geq 12/a_0$ is a separate condition, not a vector count. Numerical acceptance requires less than 0.1% change in the reconstructed potential upon independent spatial and temporal refinement, together with the matrix inverse residual and conditioning estimate. A static one-channel calculation is a distinct limit. Pivoted complex solves shall use scale-aware singularity tests; the vector count, a prescribed susceptibility, or a single evaluation at one $\mathbf{q}$ does not establish convergence of the real-space integral.

a) *Native high-precision scalar Floquet benchmark:* The scalar numerical kernel evaluates the dimensionless surrogate

$$u_{64} = \text{fl}_{64}(350.0), \quad v_{64} = \text{fl}_{64}(-298.57),$$

$$\epsilon_{\text{sc}} = 1 - (v_{64}/u_{64})^2,$$

$$s_F = \epsilon_{\text{sc}}^{-1}, \quad (17)$$

$$s_F^{(512)} = 3.67250764167145629240511891246 \dots$$

$$\simeq 3.67250764167.$$

The native Rust interface promotes the binary64 input values exactly into 512-bit arithmetic before evaluating the ratio, square, subtraction, and reciprocal. It does not reconstruct exact decimal input values after conversion. The scalar s_F is a numerical benchmark, not an entry or eigenvalue of the reciprocal-space tensor $[\epsilon^{-1}]_{GG'}^{mn}$, without an additional constitutive mapping. In particular, its positive value is not the negative real-space ratio $V_{\text{driven}}/V_{\text{bare}}$.

Let $s_F^{(64)}$ denote evaluation of the same operation sequence in binary64. Comparing the promoted result with the 512-bit value gives

$$s_F^{(64)} \simeq 3.672507641671456,$$

$$\delta_F = |s_F^{(64)} - s_F^{(512)}|$$

$$= 3.566549731853926 \dots \times 10^{-16}$$

$$\simeq 3.56655 \times 10^{-16}, \quad (18)$$

$$\delta_{F,\text{rel}} = \delta_F / |s_F^{(512)}| \simeq 9.71148 \times 10^{-17},$$

$$\epsilon_{64} = 2^{-52} \simeq 2.22044605 \times 10^{-16},$$

$$\delta_{F,\text{rel}} < \epsilon_{64}/2.$$

This is agreement at the binary64 roundoff scale for a single scalar evaluation. The absolute precision difference is not a cumulative time-stepping drift, a harmonic-truncation residual, or a proof of full Floquet-matrix convergence. Precision comparisons use the unrounded values; the displayed 3.67250764167

alone cannot reproduce an error of order 10^{-16} . Appendix A-A specifies the arithmetic and domain checks.

D. Multiphonon Kinetics and Displaced Thermal-State Overlap

The driven oscillator state and its thermal normalization are

$$\rho_D = D(\alpha_-) \rho_{\text{th}} D^\dagger(\alpha_-),$$

$$\rho_{\text{th}} = (1 - e^{-\beta \hbar \omega_c}) e^{-\beta \hbar \omega_c b^\dagger b}, \quad (19)$$

$$\bar{n}_{\text{ph}} = (e^{\beta \hbar \omega_c} - 1)^{-1}, \quad \beta = (k_B T)^{-1}.$$

Displacement preserves trace and positivity. The normalized oscillator-vacuum overlap has the exact form

$$P_{\text{ov}} = \langle 0 | \rho_D | 0 \rangle = \frac{e^{-|\alpha_-|^2/(1+\bar{n}_{\text{ph}})}}{1 + \bar{n}_{\text{ph}}} = 0.2246943649 \quad (20)$$

at $|\alpha_-| = 1.245$ and $\bar{n}_{\text{ph}} = 1.2$. At the nominal 34.442105 meV phonon energy, that occupation corresponds to $T \simeq 659.40$ K, not 623.15 K. It may instead define a nonequilibrium effective occupation. Appendix A derives the overlap without identifying the oscillator vacuum with a nuclear final state.

For a normalized nuclear-coordinate state, the appropriate overlap and an angular-frequency golden-rule rate are

$$F_{\text{FC}} = \text{Tr}(\rho_D |\Psi_{\text{nuc}}\rangle \langle \Psi_{\text{nuc}}|), \quad 0 \leq F_{\text{FC}} \leq 1,$$

$$\Gamma = \frac{2\pi}{\hbar^2} \sum_k |\mathcal{M}_{\text{nuc},k}|^2 \langle \psi_k^{(n)} | \rho_D | \psi_k^{(n)} \rangle \delta(\omega - \omega_{nk}). \quad (21)$$

The states $|\psi_k^{(n)}\rangle$ may have nuclear-coordinate frequency $\omega_n \gg \omega_c$ and therefore require their own basis transformation. A dimensionless overlap alone is not a rate. In particular, the expression

$$F_* = P_{\text{ov}} \sum_{k=0}^{\infty} \frac{S^k e^{-S}}{k!} L_k \left(-\frac{|\alpha_-|^2}{\bar{n}_{\text{ph}}(1 + \bar{n}_{\text{ph}})} \right) \quad (22)$$

gives $F_* \simeq 70.0978$ for $S = 25$, exceeding unity. It cannot be a normalized projector overlap or an exact replacement for F_{FC} . The oscillator result does not establish removal of the nuclear-coordinate suppression $10^{-6.43 \times 10^9}$.

A Hermitian nuclear-lattice interaction may be written

$$H_{\text{int}} = \sum_q g_q (b_q + b_q^\dagger) T_{\text{nuc}} + \text{h.c.} \quad (23)$$

with channel-resolved rates $\Gamma_{\text{lat}}, \Gamma_p, \Gamma_n, \Gamma_\gamma \geq 0$. For $\Gamma_{\text{other}} = \Gamma_p + \Gamma_n + \Gamma_\gamma > 0$,

$$B_{\text{lat}} = \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_{\text{other}}}, \quad (24)$$

$$B_{\text{lat}} > 0.999999 \iff \Gamma_{\text{lat}} > 999999 \Gamma_{\text{other}}.$$

The requirement $\Gamma_{\text{lat}} \geq 10^6 \Gamma_{\text{other}}$ is sufficient, but not necessary. If all rates vanish, the branching fraction is undefined. Near-unity lattice branching remains conditional on nuclear matrix elements and competing rates; neither trace normalization nor a phenomenological multiplier proves radiative suppression.

E. Electron–Phonon Hamiltonian and Coherent Drive

The Fröhlich interaction describes the long-range coupling of an electron to polar longitudinal optical lattice modes.² For electron position $\mathbf{r}_a$, phonon annihilation operator $b_{\mathbf{q}}$, and quantization volume V , a Hermitian form is

$$H_{\text{int}}^{\text{F}} = \sum_{\mathbf{a}, \mathbf{q}} [M_{\mathbf{q}} e^{i\mathbf{q} \cdot \mathbf{r}_a} b_{\mathbf{q}} + M_{\mathbf{q}}^* e^{-i\mathbf{q} \cdot \mathbf{r}_a} b_{\mathbf{q}}^\dagger], \quad (25)$$

with the isotropic polar-continuum coupling magnitude

$$|M_{\mathbf{q}}|^2 = \frac{e^2 \hbar \omega_{\text{LO}}}{2\epsilon_0 V |\mathbf{q}|^2} \left(\frac{1}{\epsilon_\infty} - \frac{1}{\epsilon_s} \right). \quad (26)$$

The quantities ϵ_s and ϵ_∞ are the static and high-frequency relative dielectric constants of the polar medium. Equation (26) defines the polar-continuum reference model. Its application to the conducting Pd–Ir layer or an interfacial mode requires the relevant phonon eigenvectors, a screened microscopic coupling, and a characterized spectral density. These material-specific quantities determine the validity of the continuum approximation.

After eliminating the electronic response into an effective coupling coefficient, a two-optical-mode interaction that can transfer the pump difference energy to a lattice mode is

$$H_{\text{int}}^{\text{eff}} = \hbar g_{12} a_2^\dagger a_1 b^\dagger + \hbar g_{12}^* a_1^\dagger a_2 b. \quad (27)$$

The operators a_1 and a_2 annihilate the two optical modes, b annihilates the selected lattice mode, and g_{12} has units of inverse seconds. This rotating-wave reduction requires $\omega_1 - \omega_2 \approx \omega_{\text{LO}}$ and the corresponding mode-overlap momentum condition. The effective coefficient g_{12} shall be obtained from the screened electronic response and normalized optical and lattice mode overlaps.

For undepleted coherent pumps α_1 and α_2 , the lattice response is characterized by the driven phonon amplitude β , linewidth γ_{LO} , and detuning $\Delta_{\text{LO}} = \omega_{\text{LO}} - \omega_{\text{beat}}$. The reduced dynamics are

$$\frac{d\beta}{dt} = - \left(\frac{\gamma_{\text{LO}}}{2} + i\Delta_{\text{LO}} \right) \beta - i g_{12} \alpha_2^* \alpha_1, \quad (28)$$

so that the steady coherent occupation is

$$n_{\text{coh}} = |\beta_{\text{ss}}|^2 = \frac{|g_{12}|^2 |\alpha_1|^2 |\alpha_2|^2}{(\gamma_{\text{LO}}/2)^2 + \Delta_{\text{LO}}^2}. \quad (29)$$

The resonance energy is the nominal 34.442 meV in (6). Evaluation of the coherent occupation and displacement requires g_{12} , damping, optical mode normalization, and the selected phonon eigenvector.

a) Energy and frequency coupling conventions: The planar comparison interface [59] consists of 1.0 nm Pd–Ir, a 10.0 nm Ti transition layer, and CVD diamond. This geometry and its exact 8.330 THz comparison frequency are distinct from the monolithic Option B grating and the wavelength-derived

beat in Eq. (5). For a displacement-normalized mode vector $\mathbf{e}_i$,

$$\begin{aligned} M &= \sum_i M_i |\mathbf{e}_i|^2 = 1.403 \times 10^{-25} \text{ kg} \\ &= 84.4906 \text{ u} \approx 84.5 \text{ u}, \\ \omega_0 &= 2\pi(8.330 \times 10^{12}) \text{ s}^{-1}, \\ \mathbf{x}_{\text{zpf}} &= \sqrt{\frac{\hbar}{2M\omega_0}} \hat{\mathbf{z}} = 2.679673 \text{ pm } \hat{\mathbf{z}}, \\ g_E &= \langle \psi_f | \nabla V_{\text{int}} \cdot \mathbf{e}_\Gamma | \psi_i \rangle | \mathbf{x}_{\text{zpf}} | \\ &= 12.45 \text{ meV} \quad (\text{nominal assignment}). \end{aligned} \quad (30)$$

The density-functional/GW potential, electronic states, and mode-mass normalization define the microscopic overlap. The coupling energy is a reduced-model input rather than a consequence of mode mass alone. The inverse-second optical coefficient g_{12} and the energy-valued coupling g_E have distinct dimensions; their associated frequency conventions are

$$\begin{aligned} g_\omega &= g_E/\hbar = 1.89148797 \times 10^{13} \text{ s}^{-1}, \\ g_f &= g_E/h = 3.01039661 \times 10^{12} \text{ Hz}, \\ g_E &= g_{12,E}^{(\text{disp})} x_{\text{zpf}}, \quad [g_{12,E}^{(\text{disp})}] = \text{J m}^{-1}. \end{aligned} \quad (31)$$

The comparison derivative $4.872 \times 10^{14} \text{ Hz m}^{-1}$ has frequency-per-displacement units rather than force units. Identification of the spin–phonon matrix element g_E with an optical three-wave coupling requires compatible normalization of the optical and nuclear coordinates.

b) Driven phonon density matrix: For $\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$, the dissipative spin–boson model is

$$\begin{aligned} H_S &= \hbar\omega_0 b^\dagger b + \frac{1}{2}\Delta E_{DD}\sigma_z + g_E(b^\dagger + b)\sigma_x, \\ H_d &= i\hbar(\mathcal{E}_d b^\dagger e^{-i\Delta\omega t} - \mathcal{E}_d^* b e^{i\Delta\omega t}), \\ \dot{\rho} &= -\frac{i}{\hbar}[H_S + H_d, \rho] + \gamma_\phi \mathcal{D}[\sigma_z]\rho + 2\kappa \mathcal{D}[b]\rho, \\ \gamma_\phi &= 1.20 \times 10^9 \text{ s}^{-1}, \quad \kappa = 45.0 \times 10^9 \text{ s}^{-1}. \end{aligned} \quad (32)$$

The drive Hamiltonian H_d represents the coherent beat; H_S alone describes an undriven system. The unit “GHz” denotes a numerical rate of 10^9 s^{-1} in this damping convention and does not introduce an additional factor of 2π . For the dissipator $2\kappa \mathcal{D}[b]$, the phonon population rate is $2\kappa = 9.0 \times 10^{10} \text{ s}^{-1}$ and the population lifetime is 11.1111 ps, whereas the amplitude decays at κ . The bare spin coherence decays at $2\gamma_\phi = 2.40 \times 10^9 \text{ s}^{-1}$. A finite-temperature reservoir replaces $2\kappa \mathcal{D}[b]$ with $2\kappa[(\bar{n}+1)\mathcal{D}[b] + \bar{n}\mathcal{D}[b^\dagger]]$. The population-only loss term is its zero-temperature limit. Additional dephasing contributes once to the coherent-response linewidth.

c) Floquet oscillator and displaced thermal state: The driven-oscillator model [56] uses the energy scale $g_d = F_0 x_{\text{zpf}} = 12.45 \text{ meV}$. Equality of this drive energy and a coupling-energy parameter does not establish equality of their microscopic matrix elements. With $f_b = f_{\text{beat}} \simeq 8.328064 \text{ THz}$ and $\Omega_b = 2\pi f_b$, the Floquet operator acts on $\mathcal{H} \otimes \mathcal{T}$:

$$\begin{aligned} H_F &= H(t) - i\hbar\partial_t, \quad T_b = 2\pi/\Omega_b, \\ H(t) &= \hbar\omega_0(a^\dagger a + \frac{1}{2}) - g_d(a + a^\dagger) \cos \Omega_b t. \end{aligned} \quad (33)$$

²H. Fröhlich, “Electrons in lattice fields,” *Advances in Physics*, vol. 3, pp. 325–361, 1954, doi: 10.1080/00018735400101213; R. P. Feynman, “Slow electrons in a polar crystal,” *Physical Review*, vol. 97, pp. 660–665, 1955, doi: 10.1103/PhysRev.97.660.

With $\mathcal{D}[L]\rho = L\rho L^\dagger - \{L^\dagger L, \rho\}/2$, the thermal damping convention is

$$\begin{aligned}\dot{\rho} &= -\frac{i}{\hbar}[H(t), \rho] + \mathcal{L}_\gamma \rho, \\ \mathcal{L}_\gamma &= \gamma(n_{\text{th}} + 1)\mathcal{D}[a] + \gamma n_{\text{th}}\mathcal{D}[a^\dagger], \\ \hbar\gamma &= 10.0 \text{ meV}, \quad \gamma = 1.51926745 \times 10^{13} \text{ s}^{-1}, \\ n_{\text{th}} &= (e^{\hbar\omega_0/(k_B T)} - 1)^{-1} = 0.935813 \quad (T = 550 \text{ K}).\end{aligned}\quad (34)$$

The thermal-oscillator damping and the $2\kappa = 90 \text{ GHz}$ spin-boson population damping define separate parameterizations and are not additive. The thermal-oscillator population and amplitude relaxation times are 65.8212 and 131.6424 fs. The comparison occupation $n_{\text{th}} \simeq 0.941$ does not equal the Bose occupation at the specified frequency and temperature. Writing $\rho(t) = \sum_k \rho^{(k)} e^{-ik\Omega_b t}$ gives

$$\begin{aligned}-ik\Omega_b \rho^{(k)} &= -\frac{i}{\hbar}[H_0, \rho^{(k)}] + \mathcal{L}_\gamma \rho^{(k)} \\ &+ \frac{ig_d}{2\hbar}[a + a^\dagger, \rho^{(k-1)} + \rho^{(k+1)}].\end{aligned}\quad (35)$$

Hermiticity requires $\rho^{(-k)} = \rho^{(k)\dagger}$ and normalization $\text{Tr} \rho^{(k)} = \delta_{k0}$. The linear oscillator's periodic stationary state is $D(\alpha(t))\rho_{\text{th}}D^\dagger(\alpha(t))$, where

$$\begin{aligned}\alpha(t) &= \alpha_- e^{-i\Omega_b t} + \alpha_+ e^{i\Omega_b t}, \\ \alpha_\mp &= \frac{g_d}{2\hbar(\omega_0 \mp \Omega_b - i\gamma/2)}.\end{aligned}\quad (36)$$

In the rotating-wave approximation (RWA), the stationary *rotating-frame* density matrix is

$$\begin{aligned}\tilde{\rho}_{\text{ss}} &= D(\alpha_-)\rho_{\text{th}}D^\dagger(\alpha_-), \\ \alpha_- &= \frac{g_d}{2\hbar[(\omega_0 - \Omega_b) - i\gamma/2]}, \\ |\alpha_-|_{\text{res}} &= 1.245, \\ \langle n \rangle_{\text{RWA}} &= n_{\text{th}} + |\alpha_-|^2 = 2.485838.\end{aligned}\quad (37)$$

The laboratory-frame state $\rho^{(0)} = T_b^{-1} \int_0^{T_b} \rho(t) dt$ is a cycle average and differs from $\tilde{\rho}_{\text{ss}}$; its number-basis off-diagonal elements average to zero in the RWA. The exact displacement $\alpha(t)$ includes the counter-rotating term. A displacement of 1.245 does not imply an occupation of 6.92×10^8 phonons.

d) Floquet response and active-volume normalization:

The drive angular frequency is $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$, with period $T_D = 2\pi/\Omega_D$. A periodic Hamiltonian admits the representation

$$\begin{aligned}H(t + T_D) &= H(t), \\ |\psi_{a\mathbf{k}}(t)\rangle &= e^{-i\varepsilon_{a\mathbf{k}}t/\hbar} |u_{a\mathbf{k}}(t)\rangle, \\ M_{ab}^{\ell, G}(\mathbf{k}, \mathbf{q}) &= \frac{1}{T_D} \int_0^{T_D} dt e^{i\ell\Omega_D t} \langle u_{a\mathbf{k}}(t) | e^{-i\mathbf{G}\cdot\mathbf{r}} | u_{b, \mathbf{k}+\mathbf{q}}(t) \rangle.\end{aligned}\quad (38)$$

The u factors are cell-periodic; for full Bloch functions, the spatial exponential contains $\mathbf{q} + \mathbf{G}$. In a population-diagonal

Floquet response approximation with spin included in a, b , the number-density polarizability is

$$\begin{aligned}\Pi_{GG'}^{mn}(\mathbf{q}, \omega) &= \frac{1}{\Omega_{\text{cell}}} \sum_{\mathbf{k}, abp} w_{\mathbf{k}} (f_{a\mathbf{k}} - f_{b, \mathbf{k}+\mathbf{q}}) \\ &\times \frac{M_{ab}^{m-p, G} [M_{ab}^{n-p, G'}]^*}{\varepsilon_{a\mathbf{k}} - \varepsilon_{b, \mathbf{k}+\mathbf{q}} + \hbar\omega + p\hbar\Omega_D + i\eta}.\end{aligned}\quad (39)$$

The Brillouin-zone weights satisfy $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$, $f_{a\mathbf{k}} = \rho_{aa}(\mathbf{k})$ is an occupation, and $\eta > 0$ is an energy broadening. The polarizability follows the SI number-density-response convention; insertion of an additional e^2 into a charge-response susceptibility would duplicate the charge factor. With composite indices (G, m) ,

$$\begin{aligned}v_G(\mathbf{q}) &= \frac{e^2}{\varepsilon_0 |\mathbf{q} + \mathbf{G}|^2}, \\ \mathcal{E}_{GG'}^{mn} &= \delta_{GG'} \delta_{mn} - v_G \Pi_{GG'}^{mn}, \\ \sum_{G'', p} \mathcal{E}_{GG''}^{mp} [\mathcal{E}^{-1}]_{G''G'}^{pn} &= \delta_{GG'} \delta_{mn}.\end{aligned}\quad (40)$$

The reciprocal-space, time-averaged interaction for a source at the chosen origin is

$$\begin{aligned}W^{00}(\mathbf{r}) &= \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} [\mathcal{E}^{-1}]_{GG'}^{00}(\mathbf{q}, 0) v_{G'}(\mathbf{q}), \\ U_{\text{eff}}(\mathbf{r}) &= V_{\text{bare}}(\mathbf{r}) - \text{Re } W^{00}(\mathbf{r}), \\ \Delta W^{00} &= W^{00} - V_{\text{bare}}.\end{aligned}\quad (41)$$

The integral containing $\mathcal{E}^{-1} - I$ defines the induced interaction ΔW , whereas the positive screening shift is $U_{\text{eff}} = -\text{Re } \Delta W$. At $r_d = 0.28 \text{ \AA}$, the selected convention gives $V_{\text{bare}} \simeq 51.42714 \text{ eV}$ and $V_{\text{driven}} = \text{Re } W \simeq -298.57 \text{ eV}$ for $U_{\text{eff}} = 350.0 \text{ eV}$. The reconstructed reference $V_{\text{eff}} = 51.43 \text{ eV}$ has the meaning specified in Eq. (15). This subtraction defines a target interaction rather than a microscopic evaluation. The potential requires the dielectric response, and the force additionally requires its spatial derivative. Bloch states, driven populations, and numerical convergence determine the dielectric calculation [57].

e) Option B inventory and coherent-rate requirement:

The monolithic Option B geometry gives

$$\begin{aligned}A_{\text{foot}} &= 10^{-4} \text{ m}^2, \quad f_g = 0.50, \\ t_{\text{peak}} &= 42.50 + 7.50 = 50.00 \text{ nm}, \\ V_B &= A_{\text{foot}}(f_g h_g + t_{\text{back}}) = 2.875 \times 10^{-12} \text{ m}^3, \\ M_{\text{alloy}} &= 0.95(0.10642) + 0.05(0.19222) \\ &= 0.11071 \text{ kg/mol}, \\ n_M &= \frac{12547.0}{0.11071} N_A = 6.8250203 \times 10^{28} \text{ m}^{-3}, \\ N_D|_{x_D=1} &= n_M V_B = 1.96219335 \times 10^{17} \approx 1.9622 \times 10^{17}, \\ N_{\text{coh}} &= 10^5, \quad N_{\text{domains}} = N_D/N_{\text{coh}} \approx 1.9622 \times 10^{12}.\end{aligned}\quad (42)$$

This domain count partitions the deuterium inventory rather than an optical field volume. The alternative Option A geometry has a 150 nm pitch and a 1 nm conformal coating, giving $A_{\text{unfolded}} = A_{\text{foot}}(1 + 2h_g/150 \text{ nm})$ and $V_A = 1.5666667 \times$

TABLE I
CALCULATED PRISM COUPLING AT $\lambda = 1550$ nm

θ (deg)	Layer (nm)	Coupling (%)	Target (%)	Tol. (pp)
41.25	30.00	92.15	92.15	± 0.05
41.80	45.00	97.43	97.43	± 0.05
42.50	60.00	85.33	85.33	± 0.05

10^{-13} m^3 . Its interaction volume is distinct from that of the 960.80 nm monolithic grating.

$$\begin{aligned} Q_{DD} &= 23.84 \times 10^6 (1.602176634 \times 10^{-19}) \text{ J}, \\ R_{\text{req}} &= 2911.40 / Q_{DD} = 7.62228587 \times 10^{14} \text{ s}^{-1}, \\ R_{\text{coh}} &= N_{\text{domains}} N_{\text{coh}}^2 \lambda_c c_{\text{ss}}, \\ (\lambda_c c_{\text{ss}})_{\text{req}} &= \frac{R_{\text{req}}}{N_D N_{\text{coh}}} \approx 3.8845 \times 10^{-8} \text{ s}^{-1}. \end{aligned} \quad (43)$$

The dimensionless quantity c_{ss} is a collective coherence factor within the assumed domain model, not the complete density matrix. The required transition-coherence product is an inverse-design constraint rather than an evaluated nuclear matrix element. Comparison of an N^2 coherent rate with an incoherent N -emitter rate requires identical normalization. The classical coefficient $7.9935 \times 10^{-235} \text{ m}^3/\text{s}$ gives $R_{\text{classic}} = 5.35242 \times 10^{-189} \text{ s}^{-1}$; substitution of 10^{-240} is incompatible with this coefficient. Selection of N_{coh} alone does not establish equivalence between the classical and collective rates.

f) *Populated multiphonon transition rate:* For the displaced thermal state $\tilde{\rho}_{ss} = D(\alpha) \rho_{\text{th}} D^\dagger(\alpha)$, an energy-resolved positive transition rate has the form

$$W_{\text{lat}} = \frac{2\pi}{\hbar} \sum_f \langle f | \mathcal{T} \tilde{\rho}_{ss} \mathcal{T}^\dagger | f \rangle \delta(E_f - E_i), \quad (44)$$

where $\mathcal{T}$ is an energy-valued transition operator. A second-order combination of nuclear and drive Hamiltonians requires the intermediate resolvent, $\mathcal{T}^{(2)} = H_{\text{nuc}}(E_i - H_0 + i0^+)^{-1} H_{\text{drive}}$, together with the reverse ordering where applicable. The resolvent provides the required intermediate-state energy denominator. The condition $B_{\text{lat}} > 0.999999$ remains a branching target until competing rates are evaluated with consistent normalization; it does not establish that radiation lies below a detector's sensitivity limit.

F. Theoretical Dynamics and Multi-Scale Nuclear Kinetics

The coupled electronic and nuclear model [72], [73] uses the Adler–Wiser Floquet inversion in Eqs. (39)–(41). The static block $[\epsilon^{-1}]_{GG'}^{00}$ is extracted *after* inversion over reciprocal vectors and temporal sidebands; it is not generally the inverse of ϵ^{00} . The angular frequency satisfies $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$, whereas $f_{\text{beat}} \simeq 8.328064 \text{ THz}$ is the cyclic frequency.

Table I defines three prism-coupling comparison cases; the calculated coupling-minus-target differences are zero at the displayed precision. Tolerances are absolute percentage points (pp). These 1550 nm cases are distinct from the dual-pump 785.0/802.5 nm operating configuration. The five inverse-dielectric entries in Table LXXXVIII do not constitute a complete

matrix or reciprocal-space quadrature: their contributions sum to -346.31 eV rather than -350.00 eV . Evaluation of the microscopic potential additionally requires band energies, Floquet occupations, normalized matrix elements, reciprocal-space weights, broadening, and convergence analysis. Accordingly, $U_{\text{eff}} = 350.00 \text{ eV}$ and $V_{\text{driven}} = -298.57 \text{ eV}$ at $r_d = 0.028$ nm define a conditional screening scenario rather than a fully evaluated microscopic potential.

a) *Franck–Condon trace and branching normalization:* A normalized displaced thermal oscillator has the density operator

$$\begin{aligned} \rho_{\text{ss}} &= D(\alpha_-) \rho_{\text{th}} D^\dagger(\alpha_-), \\ D(\alpha_-) &= \exp(\alpha_- a^\dagger - \alpha_-^* a), \\ \rho_{\text{th}} &= Z^{-1} \exp[-\hbar \omega_q a^\dagger a / (k_B T)], \\ \alpha_- &= 1.245, \quad T = 550 \text{ K}, \\ N_q &= \frac{23.84 \times 10^6}{0.034451} \simeq 6.920 \times 10^8. \end{aligned} \quad (45)$$

For a transition operator M (with units of energy) and final-state spectral density $\delta(E_i - H_f)$, a rate takes the positive form

$$\Gamma_{\text{lat}} = \frac{2\pi}{\hbar} \text{Tr}[\rho_{\text{ss}} M^\dagger \delta(E_i - H_f) M]. \quad (46)$$

A positive transition rate differs from the generally complex linear trace of $\sum_n |n + N_q\rangle \langle n|$. For this unit-norm shift operator S , $|\text{Tr}(\rho_{\text{ss}} S)| \leq 1$; the trace cannot provide a $10^{6.43 \times 10^9}$ enhancement compensating a $10^{-6.43 \times 10^9}$ Franck–Condon penalty. A displaced state with $|\alpha_-|^2 = 1.550025$ does not establish a mechanism for transferring 6.92×10^8 phonons per reaction. Such a mechanism requires quantitative nuclear coupling, driven-state replenishment, final-state density, momentum balance, and competing-channel kinetics.

$$\begin{aligned} \Gamma_{\text{other}} &= \Gamma_\gamma + \Gamma_p + \Gamma_n, \\ B_{\text{lat}} &= \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_{\text{other}}}, \\ r_\Gamma &= \Gamma_{\text{lat}} / \Gamma_{\text{other}}, \\ r_\Gamma \geq 10^8 &\implies B_{\text{lat}} \geq \frac{10^8}{1 + 10^8}. \end{aligned} \quad (47)$$

The branching bound is conditional on the specified rate ratio, which remains experimentally unverified. When radiation suppression denotes the nonradiative fraction, the relevant probability is B_{lat} ; $1 - 1/r_\Gamma$ is a large- r_Γ approximation rather than a normalized probability for arbitrary rates.

b) *Planar coherence construction and its limits:* The microstructural model is parameterized by

$$\begin{aligned} V_D &= 1.25 \times 10^{-15} \text{ m}^3, & N_D &= 1.84 \times 10^7, \\ V_D^{1/3} &= 10.7722 \text{ } \mu\text{m}, & L_D &\simeq 11.155 \text{ } \mu\text{m}, \\ L_{\text{coh}} &= c_{\text{ss}} \lambda_c = 412 \text{ } \mu\text{m}, \\ V_{\text{domains}} &= N_D V_D = 2.30 \times 10^{-8} \text{ m}^3. \end{aligned} \quad (48)$$

The noncubic dimension L_D is an effective lateral spacing; V_D alone does not determine it without the ellipsoid axes and packing fraction. At unit areal filling, a coherence patch con-

tains $M = A_{\text{coh}}/L_D^2$ domains. The nominal planar coherence factors are

$$\Phi_{2D} = \left(\frac{L_{\text{coh}}}{L_D}\right)^2 \simeq 1364.02, \\ \eta_{\text{coll}} = \Phi_{2D} N_D = 2.5097968 \times 10^{10} \simeq 2.5098 \times 10^{10}. \quad (49)$$

The nominal lengths yield 1364.128501, whereas a factor of 1364.02 corresponds to $L_D = 11.15544365 \mu\text{m}$, which rounds to the specified spacing. Agreement at this precision constrains geometry but does not establish physical coupling. Under the *assumption* that M emitters within each patch are fully phase locked with equal amplitudes and different patches are mutually incoherent,

$$P_{\text{patch}} = M^2 P_1, \\ P_{\text{total}} = (N_D/M) M^2 P_1 = N_D M P_1, \\ P_{\text{incoherent}} = N_D P_1, \\ P_{\text{total}}/P_{\text{incoherent}} = M, \quad P_{\text{total}}/P_1 = N_D M. \quad (50)$$

The factor 2.5098×10^{10} is normalized to *one domain*, not to a volume-integrated independent-emitter rate. Dicke radiative peak-power scaling alone does not establish sustained nuclear power or coupling across grain boundaries [74]. A quantitative phase-locking model requires interdomain coupling, phase disorder, propagation loss, dephasing, and drive replenishment. Assignment of active domains to the Cu–Sn bondline is also distinct from assignment to the Pd–Ir catalytic layer; a spatially resolved reaction model is necessary to relate the two inventories.

c) Rate, dimensions, and active-volume closure: With $n_d = 6.80 \times 10^{28} \text{ m}^{-3}$, a volume rate is

$$w_{\text{pair}} = \frac{1}{2} n_d^2 \langle \sigma v \rangle g_{\text{coll}}, \\ P_{\text{lat}} = Q_{DD} B_{\text{lat}} V_{\text{react}} w_{\text{pair}}, \\ Q_{DD} = 23.84 \text{ MeV} = 3.81958909 \times 10^{-12} \text{ J}. \quad (51)$$

The collective factor g_{coll} is dimensionless. By contrast, $\frac{1}{2} n_d^2 (c_{\text{ss}} \lambda_c) \sigma v$ has units $\text{m}^{-2} \text{s}^{-1}$ when $c_{\text{ss}} \lambda_c$ denotes a length. Its interpretation as a volumetric reaction rate requires an additional, explicitly defined normalization. The reference rates 1.20×10^{14} and $3.01 \times 10^{24} \text{ m}^{-3} \text{s}^{-1}$ approximately reproduce the proposed collective multiplier; the power ratio is $2911.40/(1.16 \times 10^{-7}) = 2.50983 \times 10^{10}$. These ratios do not establish a common volume normalization. The enhanced rate gives 264430 W over $N_D V_D$, but 33.05 W over the Option B metal volume $2.875 \times 10^{-12} \text{ m}^3$. The two volumes differ by a factor of 8000. At $B_{\text{lat}} = 1$, a power of 2911.40 W requires $V_{\text{react}} \simeq 2.5323 \times 10^{-10} \text{ m}^3$ at the specified enhanced rate. A physically consistent power prediction therefore requires a single reaction-volume definition.

G. Effective Screening: Dynamic SPP Dielectric Model

a) Dielectric and rate-solver implementation: The dielectric solver assembles the composite reciprocal-vector/Floquet matrix from transition energies, occupations, and normalized matrix elements. The cycle-averaged static observable $[\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, 0)$ is selected from the full inverse at zero probe frequency, rather than obtained by inverting only the 00 block.

The numerical interface shall return the matrix residual, conditioning estimate, and convergence of the extracted block. Adaptive Gauss–Kronrod integration evaluates the rate functional independently of dielectric inversion. The driven-screening model uses $f_{\text{beat}} \simeq 8.328064 \text{ THz}$ and $\Delta\omega = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$. In a driven population-diagonal approximation, the generalized Adler–Wiser response [59]–[61] is

$$\Pi_{GG'}(\mathbf{q}, \omega) = \frac{g_s}{\Omega} \sum_{\mathbf{k}, nn'} w_{\mathbf{k}} \frac{f_{n\mathbf{k}}^D - f_{n', \mathbf{k}+\mathbf{q}}^D}{\mathcal{E}_{n\mathbf{k}} - \mathcal{E}_{n', \mathbf{k}+\mathbf{q}} + \hbar\omega + i\eta} \\ \times M_{nn'}^G(\mathbf{k}, \mathbf{q}) [M_{nn'}^{G'}(\mathbf{k}, \mathbf{q})]^*, \\ M_{nn'}^G = \langle \psi_{n', \mathbf{k}+\mathbf{q}} | e^{i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} | \psi_{n\mathbf{k}} \rangle_{\Omega}. \quad (52)$$

Here Ω is cell volume, $\sum_{\mathbf{k}} w_{\mathbf{k}} = 1$ denotes Brillouin-zone quadrature, $\eta = \hbar\gamma > 0$ has energy units, and $g_s = 2$ only when spin is not included in the band index. Both intraband and interband transitions are retained for a metal. For cell-periodic Bloch factors the phase in the overlap is $e^{i\mathbf{G}\cdot\mathbf{r}}$; retaining the extra $\mathbf{q}$ there would double-count the Bloch phase.

A true periodically driven response also contains Floquet sidebands. With drive period $T_d = 2\pi/\Delta\omega$, the retarded density response and its harmonic projection can be defined without assuming equilibrium occupations:

$$\Pi_{GG'}(t, t') = -\frac{i}{\hbar V} \Theta(t - t') \langle [\hat{\rho}_{\mathbf{q}+\mathbf{G}}(t), \hat{\rho}_{-\mathbf{q}-\mathbf{G}'}(t')] \rangle_D, \\ \Pi_{GG'}^{mn}(\omega) = \frac{1}{T_d} \int_0^{T_d} dt e^{i(m-n)\Delta\omega t} \\ \times \int_0^\infty d\tau e^{i(\omega+n\Delta\omega)\tau} \Pi_{GG'}(t, t - \tau). \quad (53)$$

The driven occupations f^D follow from a Floquet–Boltzmann steady state, and Eq. (52) gives the population-only response approximation. Occupation functions alone do not determine the sideband coherences. In SI units, the dielectric matrix is

$$\epsilon_{GG'}^{mn} = \delta_{GG'} \delta_{mn} - v_c(\mathbf{q} + \mathbf{G}) \Pi_{GG'}^{mn}, \\ v_c(\mathbf{q} + \mathbf{G}) = \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2}. \quad (54)$$

The Coulomb kernel $4\pi e^2/q^2$ uses Gaussian units and cannot be combined directly with SI permittivity. The inverse dielectric is obtained by inversion of the coupled reciprocal-vector and harmonic matrix, not by taking the reciprocal of a single diagonal element.

b) Composite Bloch–Floquet response: The band indices are a, b , the harmonic indices are m, n, ℓ, p , and the cell-periodic components are $u_{\mathbf{a}\mathbf{k}}^{(\ell)}$. The population-diagonal microscopic response admits the representation

$$\psi_{\mathbf{a}\mathbf{k}}(\mathbf{r}, t) = e^{i\mathbf{k}\cdot\mathbf{r} - i\mathcal{E}_{\mathbf{a}\mathbf{k}} t/\hbar} \sum_{\ell} u_{\mathbf{a}\mathbf{k}}^{(\ell)}(\mathbf{r}) e^{-i\ell\Omega_D t}, \\ M_{ab}^{G,p}(\mathbf{k}, \mathbf{q}) = \sum_{\ell} \int_{\Omega} [u_{b, \mathbf{k}+\mathbf{q}}^{(\ell+p)}]^* e^{i\mathbf{G}\cdot\mathbf{r}} u_{\mathbf{a}\mathbf{k}}^{(\ell)} d^3r. \quad (55)$$

For a consistent harmonic convention, the retarded spectral representation corresponding to (53) is

$$\Pi_{GG'}^{mn}(\mathbf{q}, \omega) = \frac{g_s}{\Omega} \sum_{\mathbf{k}, ab, \ell} w_{\mathbf{k}} \frac{f_{a\mathbf{k}}^D - f_{b, \mathbf{k}+\mathbf{q}}^D}{\mathcal{E}_{a\mathbf{k}} - \mathcal{E}_{b, \mathbf{k}+\mathbf{q}} + \hbar(\omega + \ell\Omega_D) + i\eta} \times M_{ab}^{G, \ell-m} [M_{ab}^{G', \ell-n}]^*. \quad (56)$$

A change in Fourier convention relabels the harmonic indices together; the sideband energy denominator and transition harmonics cannot be changed independently. Occupation coherences beyond this approximation require the corresponding density-matrix response. The inverse of (54) obeys

$$\sum_{G'', \ell} \epsilon_{GG''}^{m\ell} [\epsilon^{-1}]_{G''G'}^{\ell n} = \delta_{GG'} \delta_{mn}. \quad (57)$$

For the effective mass $m_e^* = 0.38m_e$, the quiver amplitude and Jacobi–Anger expansion are

$$\mathbf{a}_q = \frac{e\mathbf{E}_0}{m_e^* \Omega_D^2}, \quad \xi_G = (\mathbf{q} + \mathbf{G}) \cdot \mathbf{a}_q, \quad (58)$$

$$e^{i\xi_G \sin(\Omega_D t)} = \sum_{p=-\infty}^{\infty} J_p(\xi_G) e^{ip\Omega_D t}.$$

In a time-averaged quiver approximation, the local-field cross term has weights $J_p(\xi_G)J_p(\xi_{G'})$, reducing to $J_p^2(\xi_G)$ for $G = G'$. This approximation differs from the full harmonic inverse in Eq. (57). Evaluation of the pair potential still requires reciprocal-space integration. For comparison, retaining full SI Coulomb constants rather than the rounded coefficient in Eq. (15) gives the real-space energy ratios

$$\begin{aligned} V_{\text{bare}} &= 51.42730525 \text{ eV}, \\ \text{Re } W|_{U=350 \text{ eV}} &= -298.57269475 \text{ eV}, \\ \frac{\text{Re } W}{V_{\text{bare}}} &= -5.80572311. \end{aligned} \quad (59)$$

The scalar threshold -5.821 differs from the exact ratio of the specified energies. Moreover, an effective real-space ratio does not establish a bound on an individual element of $\epsilon^{-1}(\mathbf{q}, \omega)$. The Thomas–Fermi reference $\lambda_{\text{TF}} = 0.231309 \text{ \AA}$ and the driven 350 eV scenario describe distinct response models rather than two results of a common dielectric calculation.

c) *Schrieffer–Wolff reduction*: Separation of the deuteron, electronic, and plasmon degrees of freedom gives

$$\begin{aligned} H &= H_0 + H_{\text{int}}, \quad H_0 = H_{dd} + H_e + H_{\text{SPP}}, \\ H_{e\text{-SPP}} &= \sum_{\mathbf{k}, \mathbf{q}, s} M_{\text{SPP}}(\mathbf{q}) (a_{\mathbf{q}} + a_{-\mathbf{q}}^\dagger) c_{\mathbf{k}+\mathbf{q}, s}^\dagger c_{\mathbf{k}, s}. \end{aligned} \quad (60)$$

The operators $a_{\mathbf{q}}$ and $c_{\mathbf{k}, s}$ annihilate a plasmon and an electron of spin s ; $M_{\text{SPP}}(-\mathbf{q}) = M_{\text{SPP}}(\mathbf{q})^*$ ensures Hermiticity. The interaction H_{int} must include the deuteron–electron density coupling as well as $H_{e\text{-SPP}}$ to generate a deuteron pair interaction. For an anti-Hermitian generator $\mathcal{A}$ satisfying $[\mathcal{A}, H_0] = -H_{\text{int}}^{\text{off}}$, projection onto the retained subspace $\mathcal{P}$ gives

$$H_{\text{eff}} = \mathcal{P}(H_0 + H_{\text{int}}^{\text{diag}} + \frac{1}{2}[\mathcal{A}, H_{\text{int}}^{\text{off}}] + \dots)\mathcal{P}. \quad (61)$$

This elimination requires off-resonant electronic denominators large compared with the eliminated couplings. Resonant SPP–phonon modes are retained; a resonant electronic sector cannot be eliminated using the same expansion.³

The time-averaged local-field projection is defined by

$$\mathcal{E}_{00}^{-1}(q, \omega) = [\epsilon^{-1}(q, \omega)]_{G=G'=0}^{m=n=0}. \quad (62)$$

The conservative induced pair energy is

$$V_{\text{ind}}(\mathbf{r}, \omega) = \text{Re} \int \frac{d^3 q}{(2\pi)^3} \frac{e^2 e^{i\mathbf{q} \cdot \mathbf{r}}}{\epsilon_0 q^2} [\mathcal{E}_{00}^{-1}(q, \omega) - 1]. \quad (63)$$

Its isotropic angular integral gives

$$\begin{aligned} V_{\text{ind}}(r, \omega) &= \frac{e^2}{2\pi^2 \epsilon_0 r} \\ &\times \text{Re} \int_0^\infty \frac{\sin(qr)}{q} [\mathcal{E}_{00}^{-1}(q, \omega) - 1] dq, \end{aligned} \quad (64)$$

$$U_{\text{eff}}(\omega) = -V_{\text{ind}}(r_d, \omega),$$

$$r_d = 0.28 \text{ \AA},$$

$$U_{\text{eff}}(\Delta\omega) = 350.0 \text{ eV} \quad (\text{dynamic comparison}).$$

The induced interaction V_{ind} has energy units, so an additional factor of e would be dimensionally incorrect. Evaluation at a single wavevector q_c does not replace the reciprocal-space integral. The 350 eV screening shift is an assumed driven-model parameter; a first-principles result requires a numerical dielectric tensor and converged quadrature.

d) *Static comparison and driven energy budget*: The density-of-states (DOS) approximation provides a static comparison model rather than the driven-screening ansatz:

$$\begin{aligned} g_{\text{alloy}} &= 0.95(1.45) + 0.05(2.85) \\ &= 1.520 \text{ states eV}^{-1} \text{ atom}^{-1}, \\ \rho_{\text{atom}} &= 4/(3.89 \times 10^{-10} \text{ m})^3 \\ &= 6.79533994 \times 10^{28} \text{ m}^{-3}, \\ g_{\text{SI}} &= 1.520 \rho_{\text{atom}} / (1.602176634 \times 10^{-19}) \\ &= 6.44680274 \times 10^{47} \text{ J}^{-1} \text{ m}^{-3}, \\ \lambda_{\text{TF}} &= \sqrt{\epsilon_0 / (e^2 g_{\text{SI}})} = 0.23130860 \text{ \AA}. \end{aligned} \quad (65)$$

At the same separation, the full-SI electrostatic comparison gives

$$\begin{aligned} V_{\text{bare}} &= 51.42730525 \text{ eV}, \\ U_{\text{TF}} &= V_{\text{bare}}(1 - e^{-r_d/\lambda_{\text{TF}}}) = 36.09955080 \text{ eV}, \\ V_{\text{driven}} &= V_{\text{bare}} - 350.0 = -298.57269475 \text{ eV}. \end{aligned} \quad (66)$$

The negative driven pair energy is distinct from nonnegative static Yukawa screening. A causal, energy-balanced driven response includes attraction, dissipation, and pump work; a static upper-bound argument does not apply without modification. The 3.120 keV Ar XVII diagnostic energy [51], [52] is not a deuteron-screening energy.

³S. Bravyi, D. P. DiVincenzo, and D. Loss, “Schrieffer–Wolff transformation for quantum many-body systems,” *Annals of Physics*, vol. 326, pp. 2793–2826, 2011, doi: 10.1016/j.aop.2011.06.004.

TABLE II
FULL-SI SCREENING COMPARISON AT $r_d = 0.28 \text{ \AA}$

Model	Pair energy (eV)	Shift (eV)
Bare Coulomb	51.427305	0
Static DOS/Yukawa	15.327754	36.099551
Driven assigned scenario	-298.572695	350.0

e) Anisotropic local-field reconstruction: The susceptibility $\chi_{GG'}$ denotes the independent-particle polarization $\Pi_{GG'}$. Pd 4d/Ir 5d hybridization enters through the band energies, occupations, and Bloch matrix elements rather than an additional numerical multiplier. For a source deuteron at $\mathbf{r}'$ and a test deuteron at $\mathbf{r}$, the full SI screened interaction energy is

$$W(\mathbf{r}, \mathbf{r}'; \omega) = \int_{\text{BZ}} \frac{d^3q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q}+\mathbf{G})\cdot\mathbf{r}} \times [\epsilon^{-1}(\mathbf{q}, \omega)]_{GG'} \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}'|^2} e^{-i(\mathbf{q}+\mathbf{G}')\cdot\mathbf{r}'}.$$
(67)

The source-position phase and Brillouin-zone integration domain prevent double counting of reciprocal-lattice components. For a [110] interstitial pair, $U_{\text{eff}} = e^2/(4\pi\epsilon_0 r_d) - \text{Re } W$. An assumed shift of 350.0 eV therefore gives $\text{Re } W \simeq -298.57 \text{ eV}$ at $r_d = 0.28 \text{ \AA}$, consistent with Eq. (15). This conditional potential [56] requires a complete inverse dielectric matrix and band-resolved occupations for microscopic evaluation [57]. The static $\omega = 0$ integral and the driven $\omega = \Omega_b$ response are distinct quantities. Local-field notation and alloy composition alone establish neither their equality nor the assumed attraction.

f) Floquet sidebands and the inverse dielectric tensor: The quiver response is characterized by the dimensionless parameter [54]

$$\xi_G = \frac{e\mathbf{E}_0 \cdot (\mathbf{q} + \mathbf{G})}{m_e^* \Omega_D^2},$$
(68)

$$\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}.$$

Here m_e^* is an effective mass and $\mathbf{E}_0$ is the local drive field. Its proposed time-averaged susceptibility is

$$\chi_{GG'}^{0,\text{rep}}(\mathbf{q}, \omega) = \sum_{p \in \mathbb{Z}} J_p^2(\xi_G) \chi_{GG'}^{0,\text{bare}}(\mathbf{q}, \omega - p\Omega_D).$$
(69)

For unequal G, G' , a squared weight at only the first vertex does not specify a two-vertex response or a full Floquet tensor. A consistent quiver-phase approximation dresses both density vertices. One explicit harmonic convention is

$$\chi_{GG'}^{mn}(\mathbf{q}, \omega) = \sum_{p \in \mathbb{Z}} J_{m-p}(\xi_G) J_{n-p}(\xi_{G'}) \times \chi_{GG'}^{0,\text{bare}}(\mathbf{q}, \omega + p\Omega_D),$$
(70)

with nonzero drive phases incorporated into the vertex factors. The diagonal limit $G = G'$, $m = n = 0$ reduces to the J_p^2 average after reindexing. This quiver approximation does not constitute a material-specific interacting-band solution; driven

occupations and scattering are independent inputs to the Adler–Wiser response. In composite reciprocal/Floquet space,

$$\epsilon_{GG'}^{mn} = \delta_{GG'} \delta_{mn} - v_G \chi_{GG'}^{mn},$$

$$v_G = \frac{e^2}{\epsilon_0 |\mathbf{q} + \mathbf{G}|^2},$$
(71)

$$\sum_{G'', \ell} \epsilon_{GG''}^{m\ell} [\epsilon^{-1}]_{G''G'}^{\ell n} = \delta_{GG'} \delta_{mn}.$$

The static probe response of the periodically driven electronic system is the retarded zero-frequency limit of the full inverse operator,

$$[\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, 0) = \lim_{\omega \rightarrow 0+} \left[(1 - \mathbf{v}\chi(\mathbf{q}, \omega))^{-1} \right]_{GG'}^{00}.$$
(72)

Here χ is the Floquet–Adler–Wiser density response, $\mathbf{v}$ is diagonal in reciprocal and Floquet indices, and the superscript 00 selects the elastic, cycle-averaged block after inversion. The static probe frequency is distinct from the nonzero optical drive frequency. For fixed deuteron positions, the local screening requirement is

$$U_{\text{eff}}(\mathbf{r}, \mathbf{r}') = V_{\text{bare}}(\mathbf{r} - \mathbf{r}') - \text{Re } W^{00}(\mathbf{r}, \mathbf{r}'; 0),$$
(73)

$$U_{\text{eff}}|_{|\mathbf{r}-\mathbf{r}'|=r_d} = 350.0 \text{ eV}, \quad r_d = 0.28 \text{ \AA}.$$

The spatial reconstruction of W^{00} follows Eq. (120) at zero probe frequency. The value 350.0 eV is a local design constraint on that reconstruction, not a consequence of matrix inversion alone. The 3D reconstruction in (67) uses the selected $[\epsilon^{-1}]_{GG'}^{mn}$ component; the elastic cycle-averaged interaction uses $m = n = 0$. Inverting a reciprocal-space time average is not the same operation as taking the 00 block of the inverse full Floquet matrix. Spatial and temporal cutoffs must be converged before evaluating $\text{Re } W$. At $r_d = 0.28 \text{ \AA}$, the selected rounded Coulomb coefficient gives

$$V_{\text{bare}} \simeq 51.42714 \text{ eV},$$

$$V_{\text{driven}} = V_{\text{bare}} - 350.0 \text{ eV} \simeq -298.57 \text{ eV},$$
(74)

$$V_{\text{eff}} = 350.0 - 298.57 = 51.43 \text{ eV}.$$

These values satisfy the prescribed energy subtraction and reconstructed-reference convention in Eq. (15), but do not evaluate the dielectric-tensor integral. A microscopic derivation of 350.0 eV requires $\mathbf{E}_0$, band-resolved occupations, the complete dielectric matrix, and convergence analysis. The unrounded bare-energy quotient is $14.3996/0.28 = 51.42714286 \text{ eV}$; intermediate rounding is avoided in numerical evaluation.

g) Beam-screening references and local-potential interpretation: Ion-beam screening comparison values are $310 \pm 30 \text{ eV}$ in Pd and up to 400 eV in defect-rich targets. Low-energy implanted-deuteron reaction yields in Pd support a fitted screening scale of roughly 310 eV, subject to target-density and interpretation uncertainties [55]. A fitted barrier shift is neither a directly measured pair energy at 0.28 Å nor an evaluated dielectric tensor for a driven 5% Ir alloy. The 400 eV defect-rich comparison does not define a universal upper bound. A tunneling-fit shift cannot be assigned at an arbitrary separation to establish attraction. Floquet d-band dressing remains a proposed microscopic mechanism whose interaction must be evaluated through Eq. (71).

TABLE III
CALCULATED ALLOY SCREENING OPERATING POINTS

Ir (at%)	$g(E_F)$	U_e (eV)	Target (eV)	Tol. (eV)
0.00	1.35	250.0	250.0	± 10.0
5.00	2.55	680.0	680.0	± 15.0
10.00	2.85	920.0	920.0	± 20.0
20.00	3.12	1240.0	1240.0	± 25.0

H. Sommerfeld Parameters and Penetration Enhancement

For relative collision energy E , the conventional Sommerfeld parameter and the leading Coulomb penetration factor are defined by⁴

$$\eta_C(E) = Z_1 Z_2 \alpha \sqrt{\frac{\mu c^2}{2E}}, \quad (75)$$

$$P_G(E) \approx \exp[-2\pi\eta_C(E)].$$

For the numerical convention used in this specification, define

$$\eta(E) \equiv \pi\eta_C(E), \quad P_G(E) \approx e^{-2\eta(E)}. \quad (76)$$

With $Z_1 = Z_2 = 1$, $\alpha^{-1} = 137.036$, and $\mu c^2 = 937.81$ MeV, the current Gamow-energy convention is

$$E_G = 2\mu c^2 (\pi\alpha)^2 = 985.76922061 \text{ keV}. \quad (77)$$

It follows from (75) that $2\pi\eta_C = \sqrt{E_G/E}$ and $2\eta = \sqrt{E_G/E}$. Therefore

$$P_G(E; U) = \exp[-\sqrt{E_G/(E+U)}],$$

$$\mathcal{E}_G(E; U) = \exp\left[\sqrt{E_G}\left(E^{-1/2} - (E+U)^{-1/2}\right)\right]. \quad (78)$$

An additional factor of π in the Gamow exponent would duplicate the angular factor already contained in π^2 within E_G . The comparison in Table LXXIX distinguishes this additional-factor expression from the conventional calculation at $U = 350.0$ eV. These are penetration-factor ratios, not probabilities greater than one or reactor power predictions. Ten electronvolts is a selected collision energy, not the thermal energy of room-temperature deuterons; at 550 K, $k_B T = 0.0473953$ eV. A nuclear rate requires a channel-specific cross section, relative-energy distribution, and pair density:

$$R_{DD} = \frac{1}{2} n_D^2 V_{\text{act}} \langle \sigma_{DD} v \rangle. \quad (79)$$

The factor one half counts identical-particle pairs. A one-particle energy distribution is not automatically the relative-collision-energy distribution used in this equation.

a) *Alloy density of states and screening normalization:* Table III specifies composition-indexed effective screening operating points with zero calculated-minus-target differences at the displayed precision. The density of states is in states/(eV atom). For atomic number density n_{atom} , the SI

⁴V. Pines, M. Pines, A. Chait, B. M. Steinetz, L. Forsley, R. C. Hendricks, G. C. Fralick, T. L. Benyo, B. Baramsai, P. B. Ugorowski, M. D. Becks, R. E. Martin, N. Penney, and C. E. Sandifer II, *Nuclear Fusion Reactions in Deuterated Metals*, NASA/TP-20205001617, June 2020, Sec. 3.

density of states, Thomas–Fermi length, and short-distance screening shift are

$$g_{\text{SI}} = \frac{n_{\text{atom}} g(E_F)}{q_{\text{eV}}},$$

$$q_{\text{eV}} = 1.602176634 \times 10^{-19} \text{ J/eV},$$

$$\lambda_{\text{TF}} = \sqrt{\frac{\epsilon_0}{e^2 g_{\text{SI}}}},$$

$$U_{\text{TF},0}^{(\text{J})} = \frac{Z_1 Z_2 e^2}{4\pi\epsilon_0 \lambda_{\text{TF}}},$$

$$U_{\text{TF},0}^{(\text{eV})} = U_{\text{TF},0}^{(\text{J})} / q_{\text{eV}},$$

$$\eta_{\text{eff}}(E) = \eta_C(E) \sqrt{\frac{E}{E + U_e}}. \quad (80)$$

For deuterons, $U_{\text{TF},0} = 14.3996 \text{ eV } \text{\AA} / \lambda_{\text{TF}}$. Thus lengths of 0.55–0.60 $\text{\AA}$ imply approximately 24.0–26.2 eV, and 0.45–0.48 $\text{\AA}$ imply approximately 30.0–32.0 eV, not 250–1400 eV. The prescribed 600–1400 eV effective-screening envelope and Table III therefore require a constitutive mechanism beyond these static length assignments. They do not replace the separate 350 eV driven-potential case or follow from alloy concentration alone.

b) *Channel-resolved nuclear amplitudes:* For relative nuclear interaction V_{nuc} , the channel amplitude and initial Floquet expansion are

$$\mathcal{M}_{fi}(t) = \int d^3 r_1 d^3 r_2 \Psi_f^* V_{\text{nuc}}(\mathbf{r}_1 - \mathbf{r}_2) \Psi_i(t),$$

$$\Psi_i(t) = \sum_n e^{-i(E_{\text{quasi}} + n\hbar\Omega_D)t/\hbar} \phi_n(\mathbf{r}_1, \mathbf{r}_2). \quad (81)$$

The proton–triton, neutron–helium-3, and radiative helium-4 channels have distinct matrix elements and final-state densities. Their nominal energy scales are 4.03, 3.27, and approximately 23.85 MeV, respectively; a radiative branching estimate of order 10^{-7} is an energy-dependent comparison, not a universal constant. A quasistatic barrier approximation can factor a channel probability as

$$|\mathcal{M}_c|^2 \simeq |\mathcal{M}_{\text{bare},c}|^2 P_{\text{WKB}},$$

$$P_{\text{WKB}} = \exp\left[-2 \int_{\mathcal{B}(E)} \frac{\sqrt{2\mu[V_{\text{barrier}}(r) - E]}}{\hbar} dr\right], \quad (82)$$

$$\mathcal{B}(E) = \{r : V_{\text{barrier}}(r) > E\}.$$

Here $V_{\text{barrier}}(r)$ is the channel interaction profile, not the reconstructed reference V_{eff} of Eq. (15). Only classically forbidden intervals enter this real attenuation exponent. A negative potential at r_d cannot be inserted under a real square root throughout a fixed integration interval. Genuine time-dependent Floquet transmission requires channel flux normalization and sideband scattering. A common multiplicative penetration factor enhances all factored channels equally and does not by itself change their branching ratios.

c) *Thermal rate functional and adaptive integration:* For nonresonant screened reactions, the Maxwellian rate average is

$$\langle \sigma v \rangle = \sqrt{\frac{8}{\pi\mu(k_B T)^3}} \int_0^\infty S(E) \exp[-2\pi\eta_{\text{eff}}(E)] \times \exp[-E/(k_B T)] dE. \quad (83)$$

Here $S(E)$ has units of energy times area, and all energies in the integration use one unit system. The 512-bit reference integrator uses an embedded 7-point Gauss/15-point Kronrod pair, bisects intervals with excessive estimated error, and resolves the Gamow peak before accepting convergence. An infinite-tail bound completes the finite integration interval. After normalizing the rate by a specified positive scale R_* , the sum of quadrature and tail error estimates shall be below 10^{-15} . Fixed nodes without embedded-error evaluation and interval refinement do not define adaptive quadrature.

I. Non-Perturbative Multiphonon Exchange

a) *Periodic dielectric drive and exchange kernel:* A scalar modulation model for the adjacent dielectric is

$$\epsilon_d(t) = \epsilon_{d0} + 2\epsilon_{d1} \cos(\Omega_F t), \quad \Omega_F = 2\pi f_{\text{beat}}. \quad (84)$$

It generates neighboring temporal sidebands without specifying the microscopic nuclear coupling. A normalized symmetric multiphonon comparison kernel is

$$W_{i \rightarrow f} = \frac{2\pi}{\hbar} |V_{fi}|^2 \sum_{N=-\infty}^{\infty} w_N \delta(E_i - E_f - N\hbar\omega_{\text{ph}}), \quad (85)$$

$$w_N = e^{-g^2} I_N(g^2), \quad \sum_N w_N = 1.$$

Here g is dimensionless and I_N is the modified Bessel function. Symmetry $w_N = w_{-N}$ is not finite-temperature detailed balance; thermal absorption and emission require the occupation-dependent weights of the displaced thermal-state model. A physical nuclear rate also requires a nuclear transition matrix element rather than an electronic matrix element substituted without a coupling derivation.

The non-perturbative Fröhlich/Hagelstein model couples electronic phonon response to nuclear-lattice exchange. The nuclear transition operator is distinct from the electronic Fröhlich coupling in Eq. (25); determination of the electronic coupling does not determine a nuclear transition amplitude.⁵ For the nuclear transition operator σ_ℓ^- at site ℓ , the exchange Hamiltonian is

$$\begin{aligned} H_0 &= H_{\text{nuc}} + H_{\text{lat}}, \\ H_{\text{nuc}} &= \sum_{\ell} Q_{\ell} \sigma_{\ell}^+ \sigma_{\ell}^-, \\ H_{\text{lat}} &= \sum_{\mathbf{q}} \hbar\omega_{\mathbf{q}} \left(b_{\mathbf{q}}^{\dagger} b_{\mathbf{q}} + \frac{1}{2} \right), \\ V_{\text{nl}} &= \sum_{\ell, \mathbf{q}} (g_{\ell\mathbf{q}} \sigma_{\ell}^- + g_{\ell\mathbf{q}}^* \sigma_{\ell}^+) (b_{\mathbf{q}} + b_{\mathbf{q}}^{\dagger}), \\ H &= H_0 + V_{\text{nl}}. \end{aligned} \quad (86)$$

The coupling $g_{\ell\mathbf{q}}$ has units of energy. The two-level interaction describes reduced nuclear-lattice exchange; displaced-oscillator factorization introduces a further approximation rather than an exact diagonalization of an arbitrary nuclear Hamiltonian.

For the proposed lattice-assisted branch,

$$D + D + \mathcal{L}_i \longrightarrow {}^4\text{He} + \mathcal{L}_f, \quad Q_{DD} \approx 23.84 \text{ MeV}, \quad (87)$$

⁵I. Chaudhary and P. Hagelstein, "Inclusion of phonon exchange in a nuclear matrix element," arXiv:cond-mat/0606585, 2006.

where $\mathcal{L}_i$ and $\mathcal{L}_f$ denote lattice states. Assigning all released energy to the nominal optical phonons and neglecting recoil gives

$$\begin{aligned} \bar{N} &= \frac{Q_{DD}}{E_{\text{LO}}} = \frac{23.84 \times 10^6 \text{ eV}}{0.034450 \text{ eV}} \\ &= 6.9201742 \times 10^8 \approx 6.920 \times 10^8. \end{aligned} \quad (88)$$

b) *Franck-Condon factorization and Huang-Rhys parameter:* Under the real-displacement convention $D(-d) = \exp[-d(b^{\dagger} - b)]$, the specified coupling gives

$$d = \frac{gE}{\hbar\omega_0} \approx 0.361392, \quad S = d^2 = 0.130604286. \quad (89)$$

The magnitude of the overlap between harmonic number states is

$$|F_{n,n'}| = e^{-S/2} \sqrt{\frac{n_{\min}!}{n_{\max}!}} |d|^{n-n'} \left| L_{n_{\min}}^{n-n'}(S) \right|. \quad (90)$$

Signed amplitudes depend on the displacement convention and ordering of the states; the magnitude formula avoids assigning the same sign to both directions. In particular,

$$\begin{aligned} \mathcal{M}_{fi}^{(N_q)} &= \mathcal{M}_{0,\text{nuc}} F_{0,N_q}, \\ |F_{0,N_q}|^2 &= e^{-S} \frac{S^{N_q}}{N_q!}, \\ \sum_{N_q=0}^{\infty} |F_{0,N_q}|^2 &= 1. \end{aligned} \quad (91)$$

The nuclear energy corresponds to $N_q \simeq 6.92 \times 10^8$, whereas the squared displacement is 0.1306. A large coherent occupation does not justify replacement of S by N_q . The transition rate and a numerically stable logarithmic overlap expression are

$$\begin{aligned} \Gamma_{\text{lat}}^{(N_q)} &= \frac{2\pi}{\hbar} |\mathcal{M}_{fi}^{(N_q)}|^2 \rho_f(E_i), \\ \ln |F_{0,N_q}|^2 &= -S + N_q \ln S - \ln \Gamma(N_q + 1). \end{aligned} \quad (92)$$

The function $\Gamma(N_q + 1)$ is the gamma function, not a transition rate. At the specified phonon count, multiplication of the overlap weight by N_D^2 with $N_D = 10^{20}$ gives $\log_{10}(N_D^2 |F_{0,N_q}|^2) \simeq -6.42857906 \times 10^9$. The ground-state overlap therefore remains strongly suppressed and does not establish efficient many-phonon transfer. A populated-mode calculation requires the density matrix and overlaps $F_{n,n'}$, rather than cancellation of the ground-state factor F_{0,N_q} by assumption.

c) *Energy and crystal momentum conservation:* An individual event has integer occupations and satisfies

$$Q_{DD} = E_{\text{recoil}} + \sum_{\mathbf{q}} N_{\mathbf{q}} \hbar\omega_{\mathbf{q}} + E_{\gamma}, \quad N_{\mathbf{q}} \in \mathbb{N}_0. \quad (93)$$

The candidate nonradiative branch has $E_{\gamma} = 0$. Its crystal-momentum selection rule is

$$\mathbf{K}_i = \mathbf{K}_{\text{He}} + \sum_{\mathbf{q}} N_{\mathbf{q}} \mathbf{q} + \mathbf{G}, \quad (94)$$

where $\mathbf{K}_i$ and $\mathbf{K}_{\text{He}}$ are the initial and final nuclear crystal wavevectors and $\mathbf{G}$ is a reciprocal-lattice vector. Normal processes have $\mathbf{G} = 0$; Umklapp processes transfer momentum to the crystal. Total mechanical momentum additionally includes

crystal/support recoil and, where exchanged, pump momentum. Counterpropagating phonon populations can carry a small net momentum without requiring each phonon to have a small wavevector. Large $\bar{N}$ alone neither proves recoil-free transfer nor removes mode-selection constraints.

d) Klemens-type three- and four-phonon relaxation:

Cubic and quartic anharmonic interactions permit optical-to-acoustic decay channels of the form

$$\begin{aligned} \text{LO}(\mathbf{q}_0) &\longrightarrow \text{LA}(\mathbf{q}_1) + \text{TA}(\mathbf{q}_2), \\ \text{LO}(\mathbf{q}_0) &\longrightarrow A_1(\mathbf{q}_1) + A_2(\mathbf{q}_2) + A_3(\mathbf{q}_3), \end{aligned} \quad (95)$$

where each A_j is an allowed acoustic branch. These channels require $\omega_0 = \sum_j \omega_j$ and $\mathbf{q}_0 = \sum_j \mathbf{q}_j + \mathbf{G}$; their availability depends on the material dispersion. In an equal-energy daughter-mode approximation, the population-relaxation rate can be parameterized as

$$\begin{aligned} n_B(\omega, T) &= [e^{\hbar\omega/(k_B T)} - 1]^{-1}, \\ \Gamma_3(T) &= A_3[1 + 2n_B(\omega_0/2, T)], \\ \Gamma_4(T) &= A_4[1 + 3n_B(\omega_0/3, T) \\ &\quad + 3n_B^2(\omega_0/3, T)], \\ \Gamma_{\text{anh}} &= \Gamma_3 + \Gamma_4. \end{aligned} \quad (96)$$

The coefficients A_3, A_4 have units of s^{-1} and contain the anharmonic matrix elements and allowed phase space. The thermal occupation factors describe relaxation toward equilibrium, not the initial strongly driven distribution.⁶ For the spin-boson interface parameterization, the Lindblad coefficient determines the population rate:

$$\begin{aligned} \Gamma_{\text{anh}}^{(\text{new})} &= 2\kappa = 9.0 \times 10^{10} \text{ s}^{-1}, \\ \tau_{\text{population}} &= (2\kappa)^{-1} = 11.1111 \text{ ps}, \\ \tau_{\text{amplitude}} &= \kappa^{-1} = 22.2222 \text{ ps}. \end{aligned} \quad (97)$$

The comparison population rate $4.2 \times 10^{11} \text{ s}^{-1}$ and lifetime 2.38 ps describe a distinct material parameterization and cannot simultaneously define relaxation in the spin-boson interface model. Anharmonic transfer to acoustic modes provides a thermalization pathway but does not determine nuclear branching or total energy retention.

e) Channel-resolved branching and coherent kinetics:

For phase-correlated sites, the golden-rule transition rate is [59]

$$\Gamma_{\text{lat}} = \frac{2\pi}{\hbar} \left| \sum_{j=1}^{N_D} \mathcal{M}_j \right|^2 \rho_f(Q_1). \quad (98)$$

Equal phases and equal site amplitudes give $N_D^2 |\mathcal{M}_1|^2$, whereas incoherent averaging gives $N_D |\mathcal{M}_1|^2$. Emission into the lattice produces the final occupation $n_{\text{coh}} + N_q$; the alternative $n_{\text{coh}} - N_q$ corresponds to phonon removal. Crystal momentum obeys Eq. (94). This selection rule constrains allowed transitions but does not establish phase matching across all sites. An assumed ratio $\Gamma_{\text{lat}}/(\Gamma_p + \Gamma_n) = 1.45 \times 10^{14}$

⁶M. Balkanski, R. F. Wallis, and E. Haro, “Anharmonic effects in light scattering due to optical phonons in silicon,” *Physical Review B*, vol. 28, pp. 1928–1934, 1983, doi: 10.1103/PhysRevB.28.1928. The three-/four-phonon form does not supply Pd–Ir-specific coefficients.

TABLE IV
NUCLEAR REACTION CHANNELS AND RELEASED ENERGIES

Channel	Q (MeV)
$D + D \rightarrow {}^4\text{He} + \text{lattice}$	23.84
$D + D \rightarrow p + t$	4.03
$D + D \rightarrow n + {}^3\text{He}$	3.27
$D + D \rightarrow {}^4\text{He} + \gamma$	23.84

implies

$$\begin{aligned} B_{\text{lat}}^{(\gamma=0)} &= \frac{1.45 \times 10^{14}}{1 + 1.45 \times 10^{14}} \\ &\approx 1 - 6.89655 \times 10^{-15}, \\ B_{\text{lat}} &= \frac{\Gamma_{\text{lat}}}{\Gamma_{\text{lat}} + \Gamma_p + \Gamma_n + \Gamma_\gamma}, \\ B_{\text{lat}} > 0.999999 &\iff \frac{\Gamma_p + \Gamma_n + \Gamma_\gamma}{\Gamma_{\text{lat}}} < \frac{10^{-6}}{1 - 10^{-6}}. \end{aligned} \quad (99)$$

The resulting branching bound is conditional on the assumed rate ratio. The ground-state overlap in Eq. (92) does not justify that ratio, and omission of the radiative channel prevents determination of total branching. Consequently, $B_{\text{lat}} > 0.999999$ is a conditional prediction rather than an established radiation-suppression factor.

Ensemble and single-state rates have distinct normalizations:

$$\begin{aligned} R_{\text{ensemble}} &= N_{\text{eff}} \Gamma_{\text{single}}, \\ N_{\text{atoms}} &= \rho_{\text{atom}} (4.0 \times 10^{-13} \text{ m}^3) = 2.71813598 \times 10^{16}. \end{aligned} \quad (100)$$

The proposed coherent population $N_D = 10^{20}$ exceeds the available sites in this active volume by more than three orders of magnitude under the assumption of one deuteron per host atom. Its interpretation requires a separate population and volume definition. The comparison ensemble flux $8.12 \times 10^{19} \text{ s}^{-1}$ is distinct from the thermal-source rate. Complete lattice deposition gives the heat-equivalent reaction rates

$$\begin{aligned} R_{10\text{kW}} &= 10^4 / Q_{DD} = 2.61808267 \times 10^{15} \text{ s}^{-1}, \\ R_{15\text{kW}} &= 1.5 \times 10^4 / Q_{DD} = 3.92712400 \times 10^{15} \text{ s}^{-1}. \end{aligned} \quad (101)$$

f) Driven distribution and cross-section integral: The mixture defines a normalized *relative-energy* distribution, with $\theta = k_B T$ expressed in the same energy unit as E :

$$\begin{aligned} f_D(E) &= (1 - \alpha_h) \frac{2\sqrt{E}}{\sqrt{\pi}\theta^{3/2}} e^{-E/\theta} + \alpha_h \delta(E - E_h), \\ \alpha_h &= 0.015, \quad E_h = 120.0 \text{ eV}, \quad T = 550.0 \text{ K}, \\ \sigma_{DD}(E) &= \frac{S(E)}{E} \exp[-\sqrt{E_G/(E + 350 \text{ eV})}], \\ S(E) &= 55 \text{ keV b} \quad (\text{constant-}S \text{ approximation}), \\ \langle \sigma v \rangle &= \int_0^\infty \sigma_{DD}(E) \sqrt{2E/\mu} f_D(E) dE. \end{aligned} \quad (102)$$

Using $1 \text{ b} = 10^{-28} \text{ m}^2$ and joules in the velocity factor, quadrature gives a Maxwell component $3.54441 \times 10^{-36} \text{ cm}^3 \text{ s}^{-1}$ and a monoenergetic component $8.96581 \times 10^{-35} \text{ cm}^3 \text{ s}^{-1}$ before mixture weighting. Thus

$$\langle \sigma v \rangle_{\text{calc}} = 4.83611 \times 10^{-36} \text{ cm}^3 \text{ s}^{-1}. \quad (103)$$

The comparison reactivity $4.12 \times 10^{-16} \text{ cm}^3 \text{ s}^{-1}$ is incompatible with the specified integral. Insertion of an additional π in the exponential gives $3.41632 \times 10^{-79} \text{ cm}^3 \text{ s}^{-1}$ rather than the comparison value. The constant- S model is a prescribed screening approximation whose physical applicability depends on channel-specific cross sections and population normalization. A calculated enhancement ratio alone does not determine the nominal thermal output.

g) *Populated-state overlap and trace identity*: For a final state f , the transition amplitude is $A_{fn} = \langle f | \hat{O} | n \rangle$. The populated-state probability and energy-resolved Floquet transition rate take the form

$$p_f = \sum_{nn'} \rho_{nn'} A_{fn} A_{fn'}^*,$$

$$W = \frac{2\pi}{\hbar} \sum_{f,\ell} |H_{\text{nuc},f}|^2 p_{f,\ell} \delta(E_f - E_i - Q_{DD} - \ell \hbar \Omega_b). \quad (104)$$

Here E_i, E_f denote nonnuclear excitation energies, with Q_{DD} the released nuclear energy; $p_{f,\ell}$ includes populations/coherences and transition harmonics, with a final-state integral for a continuum. The delta function supplies inverse-energy units. Unlike an unweighted displacement trace, this expression imposes energy conservation and retains the nuclear channels. For real q_0 and $\hat{x} = x_{\text{zpf}}(a + a^\dagger)$, exactly

$$D^\dagger(\alpha) \hat{x} D(\alpha) = \hat{x} + 2x_{\text{zpf}} \text{Re } \alpha,$$

$$\text{Tr}[\rho_{\text{th}} D^\dagger e^{iq_0 \hat{x}} D e^{-iq_0 \hat{x}}] = e^{2iq_0 x_{\text{zpf}} \text{Re } \alpha}. \quad (105)$$

The corresponding nonnegative expression is the square of the real part of the displaced thermal *characteristic function*:

$$\chi_x(q_0) = \text{Tr}[\tilde{\rho}_{\text{ss}} e^{iq_0 \hat{x}}]$$

$$= e^{-q_0^2 x_{\text{zpf}}^2 (2n_{\text{th}}+1)/2} e^{2iq_0 x_{\text{zpf}} \text{Re } \alpha}, \quad (106)$$

$$[\text{Re } \chi_x(q_0)]^2 = e^{-q_0^2 x_{\text{zpf}}^2 (2n_{\text{th}}+1)}$$

$$\times \cos^2(2 \text{Re } \alpha q_0 x_{\text{zpf}}).$$

This quantity is neither $\sum_{nn'} \rho_{nn'} |F_{nn'}|^2$ nor an energy-resolved N_q -phonon probability. The cosine equals unity at exact resonance, where $\alpha_- = i1.245$. A 10^{45} coupling enhancement cannot be inferred from this bounded function; $|\alpha_-|$ alone determines neither final-state energy and momentum nor nuclear matrix elements. The ground-state limit provides a consistency test, while Eq. (104) defines the energy-resolved populated-state rate.

h) *Maxwellian screened reaction rate*: The purely Maxwellian case has $\alpha_h = 0$ in Eq. (102). Under a fixed-incident-flux convention, the screened rate is

$$\sigma_s(E) = \frac{S(E+U)}{E} e^{-\sqrt{E_G/(E+U)}},$$

$$\langle \sigma v \rangle_M = \sqrt{\frac{8}{\pi \mu (k_B T)^3}} \int_0^\infty S(E+U)$$

$$\times e^{-\sqrt{E_G/(E+U)} - E/(k_B T)} dE, \quad (107)$$

$$U = 350 \text{ eV}, \quad S = 55 \text{ keV b}, \quad T = 550 \text{ K},$$

$$\langle \sigma v \rangle_M \simeq 3.5444 \times 10^{-36} \text{ cm}^3/\text{s}.$$

Numerical evaluation requires SI energies and a consistent mass prefactor before conversion from m^3/s to cm^3/s . Rounding to $E_G = 985.76 \text{ keV}$ affects the final digits rather than 24 orders of magnitude. The fixed-flux convention gives $\sigma_s(E) = [(E+U)/E] \sigma_{\text{bare}}(E+U)$, not simply $\sigma_{\text{bare}}(E+U)$. The comparison value $1.25 \times 10^{-12} \text{ cm}^3/\text{s}$ does not equal this integral and cannot establish the thermal output.

i) *Fock projections and channel rates*: For $\bar{n} = n_{\text{th}} = 0.9358$, $|\alpha| = 1.245$, and the displacement parameter $S = 0.130604$,

$$\rho_{\text{th}} = \sum_{j=0}^\infty \frac{\bar{n}^j}{(1+\bar{n})^{j+1}} |j\rangle \langle j|,$$

$$\tilde{\rho}_{nn'} = \sum_j \frac{\bar{n}^j}{(1+\bar{n})^{j+1}} \langle n | D(\alpha) | j \rangle \langle n' | D(\alpha) | j \rangle^*,$$

$$p_n = \tilde{\rho}_{nn} = \frac{\bar{n}^n}{(1+\bar{n})^{n+1}} e^{-|\alpha|^2/(1+\bar{n})} L_n \left(-\frac{|\alpha|^2}{\bar{n}(1+\bar{n})} \right). \quad (108)$$

The off-diagonal entries are coherences; p_n alone is a population. Both follow from the same normalized state, not a replacement of small probabilities by unity. Its generating function provides a stable large-phonon-number test:

$$\mathcal{P}(z) = \sum_n p_n z^n$$

$$= \frac{\exp[|\alpha|^2(z-1)/(1+\bar{n}(1-z))]}{1+\bar{n}(1-z)},$$

$$\Pr(n \geq N_q) \leq \mathcal{P}(z) z^{-N_q}, \quad 1 < z < 1 + 1/\bar{n},$$

$$\log_{10} \Pr(n \geq N_q) \leq -1.21855150 \times 10^8,$$

$$N_q = 6.92 \times 10^8, \quad z = 1.5. \quad (109)$$

The specified displacement does not appreciably populate the single-mode nuclear-energy manifold. Ground-state Franck-Condon suppression follows directly from the oscillator model. Any alternative collective-transfer mechanism requires independently evaluated energy-resolved matrix elements; neither Laguerre projection nor off-diagonal coherence determines their magnitude. For energy-valued Hamiltonians, the master equation and a population-based channel approximation are

$$\dot{\rho} = -\frac{i}{\hbar} [H_{\text{sys}} + H_{\text{drive}}, \rho] + \mathcal{L}_{\text{lat}} \rho + \mathcal{L}_{\text{nuc}} \rho,$$

$$\Gamma_{\text{lat}} = \frac{2\pi}{\hbar} |V_{\text{nuc-lat}}|^2 \sum_n p_n \delta(E_{\text{nuc}} - n \hbar \Omega_D). \quad (110)$$

The delta function defines a spectral distribution rather than a finite rate at a single integer n ; numerical rates require a linewidth or final-state density. Table V specifies a conditional set of channel rates. Conditioned on those rates and a common initial population,

$$B_{\text{lat}} = \frac{8.4 \times 10^{14}}{8.4 \times 10^{14} + 618000} = 0.999999992642857,$$

$$1 - B_{\text{lat}} = 7.35714285 \times 10^{-10}. \quad (111)$$

The conditional branching fraction exceeds 0.999999 and is approximately 0.9999999927. This arithmetic does not establish the assumed lattice rate of $8.4 \times 10^{14} \text{ s}^{-1}$. The

TABLE V
NUCLEAR CHANNEL-RATE REFERENCE ASSIGNMENTS

Channel	Q (MeV)	Bare (s^{-1})	Driven (s^{-1})
Lattice, ^4He	23.84	1.2×10^{-12}	8.4×10^{14}
$p + t$	4.03	2.5×10^4	1.1×10^5
$n + ^3\text{He}$	3.27	2.4×10^4	9.8×10^4
γ	23.84	4.1×10^5	4.1×10^5

relation $\Gamma_{p,n} = \Gamma_{p,n,0} P_{\text{pen}}$ cannot produce increases by factors of 4.4 and 4.08333 when $P_{\text{pen}} \leq 1$ is an absolute penetrability; such factors represent *relative enhancements*. The channel rates therefore define a conditional simulation case rather than a basis for reducing radiation protection.

j) *Normalized multiphonon probability*: The coherent amplitude β in Eq. (29) is distinct from the dimensionless inverse temperature $\beta_T = \hbar\omega_{\text{LO}}/(k_B T)$. A Bessel-function representation of the multiphonon weight is [26]

$$P_{\text{rep}}(p) = e^{-S \coth \beta_T} I_p \left(\frac{S}{\sinh \beta_T} \right) e^{p\beta_T/2}. \quad (112)$$

For integer net phonon emission $p \in \mathbb{Z}$, the Bessel generating identity [27] gives

$$\sum_{p=-\infty}^{\infty} P_{\text{rep}}(p) = \exp \left[S \frac{\cosh(\beta_T/2) - \cosh \beta_T}{\sinh \beta_T} \right] \neq 1. \quad (113)$$

The normalized displaced-oscillator thermal reference, using the same β_T definition, is instead

$$P(p) = e^{-S \coth(\beta_T/2)} I_p \left(\frac{S}{\sinh(\beta_T/2)} \right) e^{p\beta_T/2}, \quad (114)$$

$$\sum_p P(p) = 1, \quad \langle p \rangle = S.$$

The zero-temperature emission limit is $e^{-S} S^p/p!$ for $p \geq 0$. The formal choice $S \simeq \bar{N} = 6.920 \times 10^8$ would center the overlap at the energy-equivalent phonon count, whereas the specified coupling gives $S = 0.130604286$. A normalized transition probability admits neither an unbounded e^S enhancement nor independent suppression of competing nuclear channels. A nonequilibrium driven lattice requires its actual density matrix and channel-resolved transition elements. The conditional rates in Table IV therefore do not justify reduced radiation-protection or containment requirements.

k) *Coherent-domain geometry and population bounds*:

The geometrical volume and host count are

$$V_{\text{act}} = (1.0 \times 10^{-9})(20.0 \times 10^{-3})^2 = 4.0 \times 10^{-13} \text{ m}^3, \quad (115)$$

$$N_{\text{host}} = (6.79534 \times 10^{28}) V_{\text{act}} = 2.718136 \times 10^{16}.$$

For $L_{\text{spp}} = 208.62 \text{ nm}$ and $\delta_z = 149.92 \text{ nm}$, the optical field volume is

$$V_{\text{field}} = L_{\text{spp}}^2 \delta_z = 6.52486388 \times 10^{-21} \text{ m}^3, \quad (116)$$

$$\frac{V_{\text{act}}}{V_{\text{field}}} = 6.13039609 \times 10^7 \simeq 6.1305 \times 10^7.$$

TABLE VI
FULL-SI SCREENING REFERENCE AND CONDITIONAL PAIR ENERGIES

Model	Pair energy (eV)	Shift (eV)
Bare Coulomb	51.427305	0
Static TF comparison	15.327754	36.099551
Driven scenario	-298.572695	350.000000

The decay depth extends beyond the 1 nm host layer. A geometrical tiling of that layer instead has

$$V_{\text{film, domain}} = L_{\text{spp}}^2 t_{\text{film}} = 4.35223044 \times 10^{-23} \text{ m}^3, \quad (117)$$

$$N_{\text{film, domains}} = 9.19068982 \times 10^9,$$

$$N_{\text{host, domain}} = 2.95748856 \times 10^6.$$

These geometric counts do not measure coherence length. With the loading convention $x = [D]/[Pd]$, the deuterium population is $N_D = 0.95xN_{\text{host}}$; full $x = 1$ loading therefore gives 2.5822292×10^{16} deuterons rather than one deuteron per alloy atom. For mutually incoherent domains with internal phase coherence, the collective-rate ansatz is

$$R_{\text{total}} = \sum_d \Gamma_{\text{single}, d} N_{D, d}^2. \quad (118)$$

The fully coherent, identical-amplitude domain ansatz requires population inversion, phase coherence, energy-resolved matrix elements, and specified normalization of $\Gamma_{\text{single}, d}$. Geometry alone does not establish a square-law rate. The Floquet transition expression, Gamow quadrature $4.83611 \times 10^{-36} \text{ cm}^3/\text{s}$, and displaced-state overlap bounds impose independent kinetic constraints that cannot be replaced by domain counting.

l) *Dielectric response, branching, and absolute reaction rates*: The quiver-sideband approximation extends Eq. (56) through the harmonic convolution

$$\chi_{GG'}^{mn}(\mathbf{q}, \omega) = \sum_{\ell} J_{m-\ell}(\xi_G) J_{n-\ell}(\xi_{G'}) \chi_{GG'}^0(\mathbf{q}, \omega + \ell\Omega_D), \quad (119)$$

where $\xi_G = e\mathbf{E}_0 \cdot (\mathbf{q} + \mathbf{G})/(m_e^* \Omega_D^2)$ and $\Omega_D = 2\pi f_{\text{beat}} \simeq 5.232677 \times 10^{13} \text{ rad/s}$. The dielectric inverse is taken over the composite (G, m) index as in (54) and (57), not entrywise. With both source and observation positions retained, its cycle-averaged interaction is

$$\text{Re } W(\mathbf{r}, \mathbf{r}'; \omega) = \text{Re} \int_{\text{BZ}} \frac{d^3 q}{(2\pi)^3} \sum_{GG'} e^{i(\mathbf{q} + \mathbf{G}) \cdot \mathbf{r}} \quad (120)$$

$$\times [\epsilon^{-1}]_{GG'}^{00}(\mathbf{q}, \omega) \frac{e^2 e^{-i(\mathbf{q} + \mathbf{G}') \cdot \mathbf{r}'}}{\epsilon_0 |\mathbf{q} + \mathbf{G}'|^2}.$$

The Thomas–Fermi shift corresponds to $\lambda_{\text{TF}} = 0.231308596 \text{ \AA}$ for a bare energy of 51.427305 eV. Substitution of the rounded screening length 0.231309 $\text{\AA}$ gives 36.099519 eV; additional numerical digits do not imply independent physical accuracy. The driven potential follows from the prescribed subtraction 51.427305 – 350.000000, not from evaluation of the dielectric integral. A microscopic calculation requires band eigenvectors, a complete numerical dielectric matrix, the drive-field amplitude, and convergence analysis. A negative potential at one separation does not determine the sign of the force, $\mathbf{F} = -\nabla \text{Re } W$.

m) Displaced thermal state and energy-resolved branching: For a linearly driven harmonic mode with thermal damping, the rotating-frame stationary state has the displaced thermal form

$$\begin{aligned}\tilde{\rho}_{ss} &= D(\alpha_-)\rho_{th}D^\dagger(\alpha_-), \\ D(\alpha_-) &= \exp(\alpha_-b^\dagger - \alpha_-^*b), \\ \rho_{th} &= \sum_{n=0}^{\infty} \frac{n_{th}^n}{(1+n_{th})^{n+1}} |n\rangle\langle n|, \\ \text{Tr } \tilde{\rho}_{ss} &= 1, \quad \tilde{\rho}_{ss} \succeq 0, \\ \text{Tr}(b^\dagger b \tilde{\rho}_{ss}) &= n_{th} + |\alpha_-|^2.\end{aligned}\quad (121)$$

Here b is the lattice-mode annihilation operator and α_- is its dimensionless coherent displacement. Unitarity of D preserves normalization, positivity, and the thermal-state eigenvalues; it does not select a nuclear decay channel. With $|\alpha_-| = 1.245$ and $n_{th} \simeq 0.935813$, the mean occupation is approximately 2.485838, not $N_q \simeq 6.92 \times 10^8$. The energy-resolved rate has the form

$$\Gamma_{lat} = \frac{2\pi}{\hbar} \sum_{i,i',f} \rho_{ii'} V_{fi} V_{fi'}^* \delta(E_{DD} + E_i - E_f), \quad (122)$$

with a physically normalized final-state measure and compatible treatment of coherences and energy resolution. The assumed ratio $r_\Gamma = \Gamma_{other}/\Gamma_{lat} = 6.89655 \times 10^{-15}$ implies

$$B_{lat} = \frac{1}{1+r_\Gamma} \simeq 1 - 6.89655 \times 10^{-15} > 0.999999. \quad (123)$$

This conditional branching relation neither evaluates transition matrix elements nor removes the overlap constraints. Individual proton/neutron bounds below 10^{-12} and a gamma bound below 6.897×10^{-15} are upper limits rather than resolved channel rates; they do not independently determine their summed ratio. Branching fractions weight event counts, whereas the thermal fraction also depends on the energy released and retained in each channel. For nonnegative channel rates and $\Gamma_{lat} > 0$, the exact branching criterion is

$$B_{lat} > 0.999999 \iff \frac{\Gamma_{other}}{\Gamma_{lat}} < \frac{10^{-6}}{1-10^{-6}}. \quad (124)$$

Thus the assumed ratio in Eq. (123) is sufficient for the specified branching fraction. Establishing that ratio requires the energy-conserving Franck–Condon matrix-element sum for the lattice channel and the corresponding competing-channel rates; the displaced thermal state alone is not a proof of non-radiative dominance.

The field volume $6.524864 \times 10^{-21} \text{ m}^3$ and volume ratio 6.1304×10^7 satisfy Eq. (116); the distinction between the optical field volume and the 1 nm film domain in Eq. (117) remains essential. The loading convention $[D]/[\text{metal}] = 0.95$ differs

from $[D]/[Pd]$. The specified pair density and population-weighted per-pair rate give

$$\begin{aligned}n_{\text{pair}} &= 3.228 \times 10^{28} \text{ m}^{-3}, \\ w_{\text{pair}} &= 2.36 \times 10^{-12} \text{ s}^{-1}, \\ R_{\text{vol}} &= n_{\text{pair}} w_{\text{pair}} = 7.61808 \times 10^{16} \text{ m}^{-3} \text{ s}^{-1}, \\ R_{\text{total}} &= R_{\text{vol}} (4.0 \times 10^{-13}) = 30472.32 \text{ s}^{-1}, \\ Q_{DD} &= 23.84 \times 10^6 (1.602176634 \times 10^{-19}) \\ &= 3.819589095456 \times 10^{-12} \text{ J},\end{aligned}\quad (125)$$

$$P_{\text{nuclear,calc}} = R_{\text{total}} Q_{DD} = 1.16391741 \times 10^{-7} \text{ W}.$$

The rate equation does not produce the nominal thermal power of 2911.40 W. That power requires $7.62228587 \times 10^{14}$ events/s, or $w_{\text{pair}} = 0.0590325734 \text{ s}^{-1}$ at the same pair density and film volume, approximately 2.50138×10^{10} times the specified rate. The 2911.40 W operating point is therefore a thermal reference condition rather than an established nuclear-power prediction.

J. Boundary-State Kinetics and Collective-Rate Normalization

The boundary kinetic model separates the independent-pair reaction rate from the dimensionless collective gain [79]. For a collision energy E , its weak-screening cross-section approximation is

$$\sigma_s(E) = \frac{S(E)}{E} \exp[-2\pi\eta(E)] \exp\left[\pi\eta(E) \frac{U_e}{E}\right], \quad (126)$$

where $S(E)$ is the astrophysical factor, $\eta(E)$ is the Sommerfeld parameter, and U_e is the effective barrier shift. The last exponential is the first-order expansion of the Gamow exponent for $U_e/E \ll 1$. The evaluation energy $E = 350 \text{ eV}$ is a collision-energy reference, not the material temperature or the conventional Gamow-energy parameter. At $U_e = 350 \text{ eV}$, the ratio $U_e/E = 1$ lies outside the weak-screening regime; quantitative reactivity requires the full screened interaction and an explicitly normalized relative-velocity distribution. A fitted barrier shift U_e is not automatically identical to the local potential constraint U_{eff} .

For identical deuterons, unordered-pair counting gives

$$\begin{aligned}R_{\text{micro}} &= \frac{1}{2} n_d^2 V_{\text{rate}} \langle \sigma v \rangle_s, \\ P_{\text{micro}} &= Q_{DD} R_{\text{micro}}, \\ n_d &= 6.0 \times 10^{28} \text{ m}^{-3}, \\ V_{\text{rate}} &= 1.42 \times 10^{-6} \text{ m}^3, \\ P_{\text{micro,ref}} &= 1.1639 \times 10^{-7} \text{ W}.\end{aligned}\quad (127)$$

The alternative notation $R_{\text{micro}} = n_d^2 V_{\text{rate}} \langle \sigma v \rangle_{\text{screened}}$ is equivalent only when $\langle \sigma v \rangle_{\text{screened}} = \langle \sigma v \rangle_s / 2$. At the stated density and volume, the reference power requires $\langle \sigma v \rangle_s = 1.19216996 \times 10^{-47} \text{ m}^3/\text{s}$. This inverse requirement does not evaluate the cross-section integral. The comparison bulk volume V_{rate} is distinct from the energy V_{eff} , the selected envelope volume in Eq. (141), and the film volume in Eq. (125). These population models cannot share a per-pair rate without an explicit spatial mapping.

a) *Boundary correlation and gain definition:* The SHBT boundary description adopts a conformal-field-theory (CFT) representation with a nominal inventory $N_{\text{coh}} = 1.8420 \times 10^{18}$ and dimensionless phase-coupling parameter $\Phi_{\text{CFT}} = 1.768058 \times 10^{-8}$. Its proposed correspondence to an AdS_3 bulk is a modeling ansatz, not a derivation of the material Hamiltonian. In this notation, the dimensionality of AdS_3 refers to spacetime; a spatial interface requires a separately defined dynamical mapping. Boundary action, operator normalization, and the relation between boundary states and deuteron pairs are necessary to evaluate a nuclear rate.

For boundary loop operators $\mathcal{W}_{\partial\Omega}^{(k)}$, define the correlation amplitude and a reference-normalized gain by

$$\mathcal{A}_{\partial\Omega} = \text{Tr} \left(\rho_{\text{CFT}} \prod_{k=1}^{N_{\text{coh}}} \mathcal{W}_{\partial\Omega}^{(k)} \right), \quad (128)$$

$$g_{\mathcal{A}} = \frac{|\mathcal{A}_{\partial\Omega}|^2}{|\mathcal{A}_{\text{ref}}|^2}, \quad |\mathcal{A}_{\text{ref}}| > 0.$$

If every loop operator is unitary, their product is unitary and $|\mathcal{A}_{\partial\Omega}| \leq 1$ for a normalized density operator. Consequently, the unnormalized squared expectation cannot equal a gain of order 10^{10} . A rate interpretation of $g_{\mathcal{A}}$ additionally requires common initial populations, transition operators, and final-state phase space in the numerator and denominator. A coherent amplitude ratio is not generally a ratio of sums over incoherent reaction channels.

The exponential coupling ansatz defines a separate phenomenological factor,

$$\begin{aligned} g_{\text{exp}} &= \exp(\Phi_{\text{CFT}} \sqrt{N_{\text{coh}}}), \\ \sqrt{N_{\text{coh}}} &= 1.357203006 \times 10^9, \\ \Phi_{\text{CFT}} \sqrt{N_{\text{coh}}} &= 23.99613633, \\ g_{\text{exp}} &= 2.63869743 \times 10^{10}. \end{aligned} \quad (129)$$

Even with the rounded exponent 23.99611, the exponential is $2.63862796 \times 10^{10}$. Neither evaluation equals the nominal ambient gain $g_0 = 2.50138 \times 10^{10}$. The exponential ansatz and the ambient gain therefore constitute distinct parameterizations, not an exact identity. Moreover, g_{coll} denotes enhancement relative to the volume-integrated independent-pair rate; it is not interchangeable with the single-domain-normalized planar factor in Eq. (50).

K. Open-System Boundary Dynamics and Thermal Dephasing

The boundary state exchanges energy and phase information with the acoustic phonon bath. Under a Markovian reservoir approximation, its reduced density operator obeys the Lindblad equation [80], [81]

$$\begin{aligned} \frac{d\rho_{\text{CFT}}}{dt} &= -i[H_{\text{CFT}}, \rho_{\text{CFT}}] \\ &+ \sum_k \gamma_k \left(L_k \rho_{\text{CFT}} L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_{\text{CFT}}\} \right). \end{aligned} \quad (130)$$

Here $H_{\text{CFT}} = H_{\text{CFT}}^\dagger$ is expressed in angular-frequency units, so that the physical Hamiltonian is $\hbar H_{\text{CFT}}$; L_k are

dimensionless jump operators, $\gamma_k \geq 0$ have units s^{-1} , and $\{A, B\} = AB + BA$. This convention retains the commutator without an explicit $1/\hbar$. In a finite-dimensional truncation the generator preserves trace and complete positivity. Acoustic scattering, energy relaxation, and dislocation-mediated phase randomization require distinct operator definitions. A Markovian description requires a bath correlation time short compared with the resolved boundary dynamics; it does not follow from the operating temperature alone.

The constitutive dephasing law is

$$\begin{aligned} \gamma_{\text{deph}}(T, \rho_{\text{disl}}) &= \gamma_0 \frac{T}{T_0} \sqrt{1 + \alpha_{\text{disl}} \rho_{\text{disl}}}, \\ \gamma_0 &= 1.25 \times 10^6 \text{ s}^{-1}, \\ T_0 &= 298.15 \text{ K}, \\ \alpha_{\text{disl}} &= 1.0 \times 10^{-11} \text{ m}^2. \end{aligned} \quad (131)$$

The temperature ratio uses absolute temperature, and ρ_{disl} is dislocation line length per unit volume, with units m^{-2} . The coefficient γ_0 is the zero-dislocation reference rate at T_0 , not the rate at finite dislocation density. At the designated hot operating point,

$$\begin{aligned} T &= 350^\circ \text{C} = 623.15 \text{ K}, \\ \rho_{\text{disl}} &= 1.0 \times 10^{11} \text{ m}^{-2}, \\ \gamma_{\text{deph}} &= (1.25 \times 10^6) \frac{623.15}{298.15} \sqrt{2} \\ &= 3.69473076 \times 10^6 \text{ s}^{-1} \\ &\simeq 3.6947 \times 10^6 \text{ s}^{-1}. \end{aligned} \quad (132)$$

The hot-state dephasing value follows directly from the constitutive law; the model coefficients require independent microscopic or experimental characterization. In particular, $\rho_{\text{disl}} = 1.0 \times 10^{12} \text{ m}^{-2}$ instead gives $8.66491170 \times 10^6 \text{ s}^{-1}$ at the same temperature.

Table VII distinguishes nominal thermal scenarios from direct evaluation of Eq. (131). Its hot-row density defines a different material state from Eq. (132). Nominal and constitutive dephasing columns are not simultaneous solutions of a single parameter set.

The spatial coherence model and uncompensated gain law are

$$\begin{aligned} \xi_{\text{coh}}(T) &= \xi_0 \frac{T_0}{T} \left[1 + \delta_{\text{deph}} \left(\frac{T}{T_0} - 1 \right) \right]^{-1}, \\ \xi_0 &= 485.0 \text{ nm}, \\ g_{\text{uncomp}}(T) &= g_0 \left(\frac{\xi_{\text{coh}}(T)}{\xi_0} \right)^{0.015}, \\ g_0 &= 2.50138 \times 10^{10}. \end{aligned} \quad (133)$$

The coefficient δ_{deph} is dimensionless and the bracket must remain positive. The endpoint $\xi_{\text{coh}}(623.15 \text{ K}) = 199.441 \text{ nm}$ implies $\delta_{\text{deph}} \simeq 0.15000006$ by inversion of this law. At that endpoint,

$$g_{\text{uncomp}}(623.15 \text{ K}) = 2.46825920 \times 10^{10} \simeq 2.46826 \times 10^{10}. \quad (134)$$

The intermediate coherence lengths in Table VII imply different coefficients: approximately 0.32274594, 0.23834042,

TABLE VII
BOUNDARY THERMAL SCENARIOS: NOMINAL PARAMETERS AND CONSTITUTIVE DEPHASING

T (°C)	T (K)	ρ_{disl} (m ⁻²)	γ_{nom} (10 ⁶ s ⁻¹)	γ_{law}	ξ_{coh} (nm)	$g_{\text{uncomp,nom}}$ (10 ¹⁰)	$g_{\text{comp,nom}}$ (10 ¹⁰)
25.0	298.15	1.0×10^{10}	1.2562	1.31101106	485.000	2.50138	2.50138
100.0	373.15	5.0×10^{10}	1.9021	1.91603869	358.420	2.49210	2.50135
200.0	473.15	1.0×10^{11}	2.8051	2.80536285	268.110	2.48115	2.50128
300.0	573.15	5.0×10^{11}	3.4510	5.88598460	214.300	2.47120	2.50118
350.0	623.15	1.0×10^{12}	3.6947	8.66491170	199.441	2.46826	2.50112

and 0.19222232 at 100, 200, and 300 °C, respectively. They therefore describe nominal scenarios rather than a constant-coefficient trajectory. Their nominal gain entries likewise require separate calibration; they are not exact evaluations of Eq. (133). A spatial coherence length alone does not establish phase correlation across the full boundary-state inventory.

L. Active Boundary Phase Locking and Energy Closure

The phase-control architecture uses interferometric phase sensing and substrate-current actuation [80]. Its commanded waveform and thermal phase lead are

$$\begin{aligned} I_{\text{phase}}(t) &= I_{\text{peak}} \sin[\omega_L t + \delta\phi(T)], \\ \delta\phi(T) &= \arctan\left(\frac{\gamma_{\text{deph}}(T, \rho_{\text{disl}})}{\omega_0}\right), \\ I_{\text{peak}} &= 4.825 \text{ mA}. \end{aligned} \quad (135)$$

The current-amplitude tolerance is ± 0.010 mA. The symbol ω_L denotes the Larmor phase frequency and $\omega_0 > 0$ denotes the angular-frequency scale used for phase compensation; both have units rad/s. Their numerical values and relation to the material response are independent calibration inputs. A sinusoidal current with a temperature-dependent phase offset specifies a command signal, not a closed-loop transfer function.

For deterministic actuation, the interaction can be parameterized by

$$\begin{aligned} H_{\text{fb}}(t) &= I_{\text{phase}}(t) K_I, \quad K_I = K_I^\dagger, \\ \dot{\rho}_{\text{CFT}} &= -i[H_{\text{CFT}} + H_{\text{fb}}(t), \rho_{\text{CFT}}] + \mathcal{D}(\rho_{\text{CFT}}), \end{aligned} \quad (136)$$

where K_I is the current-to-Hamiltonian coupling operator with units s⁻¹ A⁻¹ and $\mathcal{D}$ is the dissipator in Eq. (130). The operator K_I , sensing efficiency, measurement backaction, loop delay, and stochastic phase-estimation error are required to predict the controlled state. Measurement-mediated feedback generally also contributes diffusion to the unconditional evolution [82]; deterministic phase lead alone cannot establish suppression of thermal phase variance. For periodic actuation, a stationary operating condition means a stable periodic density operator or a stationary state in a specified rotating frame.

The hot-state compensated gain and minimum acceptance level are

$$\begin{aligned} g_{\text{coll,comp}}(350^\circ\text{C}) &= 2.50112 \times 10^{10}, \\ g_{\text{min}} &= 2.5000 \times 10^{10}, \\ 100 \left(\frac{g_{\text{coll,comp}}}{g_{\text{min}}} - 1 \right) &= 0.0448\%. \end{aligned} \quad (137)$$

These values define the feedback performance requirement. Demonstrating it requires a reaction-rate observable evaluated in the controlled boundary state and compared with the same observable in the independent-pair reference state. Neither the current amplitude nor the Lindblad form alone determines this ratio.

For the reference microscopic power, direct multiplication and the inverse thermal requirement give

$$\begin{aligned} g_{\text{coll,comp}} P_{\text{micro,ref}} &= (2.50112 \times 10^{10})(1.1639 \times 10^{-7}) \\ &= 2911.053568 \text{ W}, \\ P_{\text{th,ref}} &= 2911.40 \text{ W}, \\ g_{\text{required}} &= \frac{2911.40}{1.1639 \times 10^{-7}} = 2.50141765 \times 10^{10}. \end{aligned} \quad (138)$$

The compensated-gain target and the thermal-power target differ by 0.346432 W under this common normalization. Exact equality at the specified compensated gain instead requires $P_{\text{micro}} = 1.16403851 \times 10^{-7}$ W. The nominal ambient gain 2.50138×10^{10} and the ratio 2.5013747×10^{10} are also distinct from g_{required} at the displayed precision. These are explicit compatibility conditions, not independent demonstrations of the thermal output.

a) Energy conservation and scattering constraints:

Collective enhancement changes the event rate, not the reaction energy $Q_{DD} = 23.84$ MeV. For the boundary subsystem with physical energy operator $\mathcal{H}(t) = \hbar[H_{\text{CFT}} + H_{\text{fb}}(t)]$, the reduced energy balance is

$$\frac{d}{dt} \text{Tr}(\rho_{\text{CFT}} \mathcal{H}) = \text{Tr}(\rho_{\text{CFT}} \dot{\mathcal{H}}) + \text{Tr}[\mathcal{H} \mathcal{D}(\rho_{\text{CFT}})]. \quad (139)$$

The first term is coherent drive work and the second is dissipative energy exchange. Nuclear reaction energy must enter the total reactor balance through explicit reaction channels or a separate source term; it cannot be inferred from a trace-preserving dephasing operator. Driver electrical losses and sensing power remain auxiliary loads, even when the commanded current is small. The nuclear lattice-power contribution is $Q_{DD} B_{\text{lat}} g_{\text{coll}} R_{\text{micro}}$ only when the gain and branching conventions use the same reaction inventory and lattice-channel energy.

For a single incident two-body partial wave with wave number k_{rel} , the inelastic bound is

$$\sigma_J^{\text{inel}} \leq \frac{\pi(2J+1)}{k_{\text{rel}}^2}. \quad (140)$$

An ensemble enhancement does not remove channel-by-channel scattering constraints. Unitarity applies to a complete closed-

system scattering description, whereas the reduced boundary dynamics is generally nonunitary and must preserve complete positivity. A consistent collective reaction model therefore requires both normalized many-body transition rates and a compatible open-system generator. The specified screening, branching, dephasing, and feedback values establish separate constraints on that model rather than a first-principles proof of sustained nuclear power.

M. Multi-Scale Reaction-Volume and Power Normalization

The geometric interaction-cell inventory distinguishes the active boundary envelope from its metal-filled fraction. A cubic cell of side $d_{\text{unit}} = 10.0$ nm has volume $V_{\text{unit}} = 1.0 \times 10^{-24} \text{ m}^3$. The envelope inventory is

$$\begin{aligned} n_{\text{dom}} &= V_{\text{unit}}^{-1} = 1.0 \times 10^{24} \text{ m}^{-3}, \\ V_{\text{active}} &= 2.53 \times 10^{-10} \text{ m}^3, \\ N_{\text{dom}} &= n_{\text{dom}} V_{\text{active}} = 2.53 \times 10^{14}, \\ V_{\text{metal}} &= 2.875 \times 10^{-12} \text{ m}^3, \\ \eta_{\text{grating}} &= \frac{V_{\text{metal}}}{V_{\text{active}}} = 0.011363636 \dots \approx 0.01136, \\ N_{\text{dom,metal}} &= \eta_{\text{grating}} N_{\text{dom}} = 2.875 \times 10^{12}. \end{aligned} \quad (141)$$

The 2.53×10^{14} domains count geometric cells in the envelope, not deuteron atoms or independently coherent emitters. The deuteron inventory N_D and the alternative coherence-cell populations in Eqs. (42) and (48) retain their distinct meanings. In particular, all envelope cells cannot simultaneously be counted as metal-filled cells.

For reaction energy E_{rxn} , local volumetric reaction rate Γ_{vol} , and microscopic heating density $\dot{q}_{\text{micro}}$, the same thermal control volume gives

$$\begin{aligned} P_{\text{th}} &= \int_{V_{\text{active}}} \Gamma_{\text{vol}}(\mathbf{r}) E_{\text{rxn}} d^3r \\ &= \sum_{j=1}^{N_{\text{dom}}} \int_{V_j} \chi_{\text{metal}}(\mathbf{r}) \mathcal{A}_z^2(\mathbf{r}) \dot{q}_{\text{micro}}(\mathbf{r}) d^3r, \\ P_{\text{gross}} &\equiv P_{\text{th,ref}} = 2911.4 \text{ W}. \end{aligned} \quad (142)$$

Here χ_{metal} is one in reacting metal and zero elsewhere; a homogenized equivalent uses its volume fraction once. The second equality defines the local transduction model, including any reaction-rate gain in $\dot{q}_{\text{micro}}$ only once. Replacing the sum by N_{dom} times a single-cell integral requires an identical cell-averaged response, including metal occupancy and optical overlap. The reference power fixes a normalization requirement, not an independently calculated nuclear rate.

For an intensity penetration depth $\delta_I = 13.0$ nm, the depth dependence is

$$\begin{aligned} \mathcal{A}_z(z) &= \mathcal{A}_z(0) e^{-z/(2\delta_I)}, \\ \mathcal{A}_z^2(z) &= \mathcal{A}_z^2(0) e^{-z/\delta_I}, \\ \int_0^t \mathcal{A}_z^2(z) dz &= \mathcal{A}_z^2(0) \delta_I (1 - e^{-t/\delta_I}). \end{aligned} \quad (143)$$

The associated amplitude penetration depth is $2\delta_I = 26.0$ nm. Thus an intensity skin depth is not inserted directly into an

amplitude-decay exponential. The spatial field and the patterned metal mask determine the actual overlap integral.

The mean power densities and power per cell are

$$\begin{aligned} \bar{q}_{\text{active}} &= \frac{2911.4}{2.53 \times 10^{-10}} = 1.150751 \times 10^{13} \text{ W/m}^3 \\ &= 11.50751 \text{ kW/mm}^3, \\ \bar{q}_{\text{metal}} &= \frac{2911.4}{2.875 \times 10^{-12}} = 1.012661 \times 10^{15} \text{ W/m}^3, \\ \bar{P}_{\text{dom}} &= \frac{2911.4}{2.53 \times 10^{14}} = 1.150751 \times 10^{-11} \text{ W}, \\ \bar{q}_{\text{active}} &= \eta_{\text{grating}} \bar{q}_{\text{metal}}. \end{aligned} \quad (144)$$

The active density is approximately 11.51 kW/mm^3 , while the metal density is approximately 1.013 PW/m^3 . Expressing these same numerical values in MW/mm^3 or TW/m^3 , respectively, would introduce factors of one thousand. Dimensional agreement does not determine whether the required local reaction and heat-removal rates are physically attainable.

II. OPTOELECTRONIC DESIGN, SYSTEM SWAP-C & MASTER ENERGY BALANCE

The optoelectronic assembly couples two coherent, TM-polarized pump fields through a silica prism into a monolithic Pd–Ir diffraction grating. Its electromagnetic specification comprises the material tensors, patterned-layer geometry, Fourier-modal boundary conditions, and surface-field observables. Optical absorption defines a thermal load, whereas local field enhancement defines an interaction strength; these quantities enter separate terms of the system energy balance.

A. Optoelectronic Transducer: Option B RCWA

a) *Monolithic grating and layer interfaces:* The Option B grating comprises a continuous Pd–Ir backplane and rectangular Pd–Ir teeth on a titanium adhesion layer and CVD diamond substrate [72], [73]. The nominal dimensional specification is

$$\begin{aligned} \Lambda_g &= 960.80 \text{ nm}, & h_g &= 42.50 \text{ nm}, \\ t_b &= 7.50 \text{ nm}, & f_g &= 0.50, \\ w_g &= f_g \Lambda_g = 480.40 \text{ nm}, \\ t_{\text{peak}} &= t_b + h_g = 50.00 \text{ nm}, \\ t_{\text{Ti}} &= 10.0 \text{ nm}, & t_{\text{diamond}} &= 500 \mu\text{m}. \end{aligned} \quad (145)$$

Here Λ_g is the pitch perpendicular to the grooves, h_g is the tooth relief above the backplane, t_b is the continuous backplane thickness, and f_g is the metal duty cycle. The binary profile is invariant along the groove direction y and periodic in x ; z is normal to the nominal layer planes. Its mean metal thickness and active inventory are

$$\begin{aligned} \bar{t}_{\text{metal}} &= t_b + f_g h_g = 28.75 \text{ nm}, \\ A &= 1.0 \text{ cm}^2, \\ V_{\text{metal}} &= A \bar{t}_{\text{metal}} = 2.875 \times 10^{-12} \text{ m}^3. \end{aligned} \quad (146)$$

The Ti layer separates the Pd–Ir backplane from diamond; the metal does not directly contact the diamond. Illumination is specified at an internal silica-prism angle of 65.0° , measured from the layer normal, with the incidence plane perpendicular

to the grooves. A prism-to-grating coupling gap, groove fill, or exterior coating changes the boundary-value problem and shall be represented explicitly rather than absorbed into an effective metal permittivity. The ideal binary profile does not define finite-aperture edge scattering or a second lateral pattern period.

b) *Complex constitutive tensors:* With the time convention $e^{-i\omega t}$, the optical model uses nonmagnetic, locally isotropic layers. For material j at either pump wavelength,

$$\begin{aligned}\epsilon_j(\lambda) &= [n_j(\lambda) + ik_j(\lambda)]^2 \\ &= n_j^2 - k_j^2 + 2in_jk_j, \\ \epsilon_j(\lambda) &= \begin{pmatrix} \epsilon_j(\lambda) & 0 & 0 \\ 0 & \epsilon_j(\lambda) & 0 \\ 0 & 0 & \epsilon_j(\lambda) \end{pmatrix}, \\ \mathbf{D}_j &= \epsilon_0 \epsilon_j \mathbf{E}_j, \quad \mathbf{B}_j = \mu_0 \mathbf{H}_j.\end{aligned}\quad (147)$$

The scalar k_j in this relation is the extinction coefficient, not a wavevector. Table LXXX specifies the optical constants at every solid layer. Its diagonal tensor representation is an isotropic constitutive assumption; directional or magneto-optical tensor elements cannot be inferred from a single complex refractive index.

The refractive indices define the material inputs; the displayed permittivities are rounded evaluations of Eq. (147). A vacuum region has $\epsilon = \mathbf{I}_3$. The dilute-gas reference $\epsilon_d = 1.0006$ defines a separate isotropic medium with $n_{\text{gas}} = \sqrt{1.0006}$ and zero extinction in the lossless approximation. The actual gas state and its occupied region shall be assigned consistently with the prism-coupling geometry. No independent interface tensor is introduced in the absence of a specified surface sheet response.

c) *Three-dimensional Maxwell boundary-value problem:* Within each source-free optical layer, the frequency-domain fields satisfy

$$\begin{aligned}\nabla \times \mathbf{E} &= i\omega\mu_0 \mathbf{H}, \\ \nabla \times \mathbf{H} &= -i\omega\epsilon_0 \epsilon \mathbf{E}, \\ \nabla \cdot (\epsilon \mathbf{E}) &= 0, \quad \nabla \cdot \mathbf{H} = 0.\end{aligned}\quad (148)$$

At an interface without an imposed free surface charge or current, the matching conditions are

$$\begin{aligned}\hat{\mathbf{n}} \times (\mathbf{E}_2 - \mathbf{E}_1) &= 0, \quad \hat{\mathbf{n}} \times (\mathbf{H}_2 - \mathbf{H}_1) = 0, \\ \hat{\mathbf{n}} \cdot (\mathbf{D}_2 - \mathbf{D}_1) &= 0, \quad \hat{\mathbf{n}} \cdot (\mathbf{B}_2 - \mathbf{B}_1) = 0.\end{aligned}\quad (149)$$

The material response includes bulk conduction in the complex permittivity. The tangential electric field is continuous, whereas the normal electric field generally jumps according to the constitutive contrast. These conditions apply at the prism-coupling region, tooth sidewalls, relief-to-backplane boundary, Pd-Ir/Ti interface, Ti/diamond interface, and diamond termination.

d) *RCWA harmonics and solver specification:* The full-vector rigorous coupled-wave analysis uses lateral Fourier harmonics and layerwise propagation in z [83]. For scalar isotropic permittivity and electric field,

$$\begin{aligned}\epsilon(x, y, z) &= \sum_{m,n} \epsilon_{mn}(z) e^{i(mG_x x + nG_y y)}, \\ \mathbf{E}(x, y, z) &= \sum_{m,n} \mathbf{E}_{mn}(z) e^{i[(k_x + mG_x)x + (k_y + nG_y)y]}.\end{aligned}\quad (150)$$

The magnetic field uses the same basis. The general two-periodic representation has $G_x = 2\pi/\Lambda_g$ and $G_y = 2\pi/\Lambda_y$. The prescribed truncation is $m, n = -12, \dots, 12$: 25 orders per axis and 625 lateral Fourier components. The tangential field vector contains four components per lateral harmonic, hence 2500 coefficients before modal reduction. The binary Option B profile is the invariant- y specialization: $\epsilon_{mn} = 0$ for $n \neq 0$, and collinear illumination has $k_y = 0$. Only the $n = 0$ sector is excited in that ideal geometry. A physically patterned second direction requires its own Λ_y and unit-cell profile; the use of a three-dimensional solver does not supply them.

For a tooth centered in the unit cell, the relief-layer coefficients are

$$\epsilon_m = \begin{cases} \epsilon_f + f_g(\epsilon_{\text{PdIr}} - \epsilon_f), & m = 0, \\ (\epsilon_{\text{PdIr}} - \epsilon_f) \frac{\sin(\pi m f_g)}{\pi m}, & m \neq 0, \end{cases}\quad (151)$$

where ϵ_f is the assigned groove-fill permittivity. The backplane, Ti layer, and diamond are laterally uniform. Toeplitz convolution matrices represent multiplication by ϵ and $1/\epsilon$; their inverses are not interchangeable. Normal-field discontinuities require inverse Fourier factorization, while continuous tangential products use the corresponding direct rule [84]. In the invariant- y , TM limit, the resulting coupled system is Eq. (259), evaluated with the material constants in Table LXXX.

Layer eigenmodes are matched using a scattering-matrix formulation to avoid numerical growth from evanescent propagation [83]. The 42.50 nm relief and 7.50 nm backplane are separate propagation layers; the 10.0 nm Ti layer and finite 500 μm diamond substrate remain explicit. Incident modes are normalized to incoming power, and reflected and transmitted modes satisfy outgoing or decaying boundary conditions. The diamond exit medium and any back-surface reflection shall be specified; a semi-infinite diamond model is a distinct termination assumption. The 25-order truncation is a nominal discretization, not evidence of convergence. Harmonic refinement shall establish stability of reflectance, transmittance, absorption, and the specified local field observable.

e) *Native RCWA execution:* The `shbt-rcwa-optics` crate provides Li-factorized Fourier operators and Redheffer scattering-matrix composition. Its `option_b::run_pump()` entry point constructs the layer stack, solves TM excitation, and samples E_z -intensity enhancement [97]. The implemented invariant-groove configuration retains $m \in [-12, 12]$, $n = 0$: 25 spatial orders, not $25 \times 25 = 625$ orders. A crossed-periodic calculation with both indices in $[-12, 12]$ is a distinct truncation. The native Option B stack uses a semi-infinite diamond termination and silica-filled grooves, not propagation through the finite diamond exit interface. Its Ti/diamond-side and metal-surface maxima are distinct observables. The specified field factors require a matching termination, sampling location, and convergence record; the repository description explicitly distinguishes target enhancements from reproduced grating outputs [95].

f) *Crossed-periodic field propagation:* For a crossed-periodic extension, the permittivity and transverse-field evolu-

TABLE VIII
CALCULATED STEADY DEUTERIUM FLUX AT HETEROGENEOUS INTERFACES

Interface	σ_h (MPa)	Calculated flux	Target flux	Tolerance
Pd _{0.95} Ir _{0.05} → Ti	0.00	1.43	1.43	±0.01
Pd _{0.95} Ir _{0.05} → Ti	150.00	3.15	3.15	±0.02
Ti → ϵ -Cu ₃ Sn	300.00	5.89	5.89	±0.04

Fluxes and tolerances are in 10^{21} atoms $\text{m}^{-2}\text{s}^{-1}$.

tion are

$$\epsilon(x, y, z) = \sum_{m,n} E_{mn}(z) e^{i2\pi(mx/\Lambda_x + ny/\Lambda_y)},$$

$$\frac{d\Psi}{dz} = ik_0 \mathbf{M}(z) \Psi, \quad \Psi = (\mathbf{e}_x, \mathbf{e}_y, \mathbf{h}_x, \mathbf{h}_y)^T, \quad (152)$$

$$\mathbf{S}_{\text{global}} = \mathbf{S}^{(1)} \star \mathbf{S}^{(2)} \star \dots \star \mathbf{S}^{(L)}.$$

The Redheffer star product eliminates internal incoming/outgoing amplitudes by block solves rather than multiplying transfer matrices with growing evanescent factors. Li factorization enforces the appropriate continuity products. Retaining $m, n \in [-12, 12]$ gives 625 spatial harmonics and 2500 transverse-field coefficients per layer; the invariant-groove limit has only 25 spatial harmonics. Passive solutions shall satisfy $R + T + A = 1$ within a stated tolerance, with local-field convergence assessed separately from integrated power convergence.

g) *Dual-pump beat frequency*: At vacuum wavelengths $\lambda_1 = 785.0$ nm and $\lambda_2 = 802.5$ nm, the interference term contains $\cos[(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r} - (\omega_1 - \omega_2)t + \phi]$. With $c = 299\,792\,458$ m/s, its temporal frequency is

$$f_{\text{beat}} = \frac{|\omega_1 - \omega_2|}{2\pi} = c |\lambda_1^{-1} - \lambda_2^{-1}|$$

$$= |381.9012204 - 373.5731564| \text{ THz} \quad (153)$$

$$= 8.32806399587 \text{ THz} \simeq 8.328064 \text{ THz}.$$

The optical frequencies remain fixed across stationary interfaces; the material dispersion changes spatial wavevectors, not the difference frequency. The 8.328064 THz value is the rounded operating reference, corresponding to 5.232677×10^{13} rad/s. Pump linewidth and relative phase stability determine whether the beat remains coherent over the lattice interaction time.

h) *Normal-field enhancement and power normalization*: For TM illumination at 65.0° , the normal-field design values are

$$\mathcal{F}_z = |E_z(0^+)/E_0|^2,$$

$$\mathcal{F}_{z,785} = 126.8 \quad (\text{threshold } 124.5),$$

$$\mathcal{F}_{z,802.5} = 141.5 \quad (\text{threshold } 138.2), \quad (154)$$

$$\mathcal{F}_{z,\text{planar}} \simeq 0.094.$$

Here E_0 is the incident pump electric-field amplitude in silica, and 0^+ denotes the dielectric-side limit at the specified metal-surface sampling location. The enhancement is a squared electric-field ratio, not an optical power-conversion efficiency. The corresponding amplitude ratios are $\sqrt{126.8} = 11.26055$ and $\sqrt{141.5} = 11.89538$. A pointwise field factor, an area average, and a volume-weighted overlap are distinct observables; the sampling coordinate or averaging measure shall

accompany every computed value. Sharp-corner singularities require a specified edge geometry or an integrated observable for meaningful convergence.

For collinear normal-field components, the squared-field beat term is

$$\mathcal{I}_z^{\text{beat}}(\mathbf{r}, t) = 2 \text{Re} \left[E_{z,1}(\mathbf{r}) E_{z,2}^*(\mathbf{r}) e^{-i(\omega_1 - \omega_2)t} \right]. \quad (155)$$

Its amplitude depends on both local mode profiles and their relative phase. The two enhancement factors alone do not determine a spatially integrated lattice-coupling rate. For each monochromatic solution, energy conservation requires

$$R_j + T_j + A_j = 1,$$

$$P_{\text{abs},j} = \frac{\omega_j \epsilon_0}{2} \int_V \mathbf{E}_j^\dagger \epsilon_{\text{loss}}(\omega_j) \mathbf{E}_j dV, \quad (156)$$

$$\epsilon_{\text{loss}} = \frac{\epsilon - \epsilon^\dagger}{2i}, \quad A_j = \frac{P_{\text{abs},j}}{P_{\text{inc},j}}.$$

The reflectance and transmittance sum power flux over propagating orders, and the absorption integral covers all lossy regions of the same optical control volume. The nominal field factors are solver acceptance targets; geometry, constitutive inputs, and harmonic convergence must establish them jointly.

i) *Collinear surface-plasmon phase matching*: The grating supplies reciprocal momentum to each pump. In the planar-interface limit, the TM surface-plasmon propagation constant is

$$\beta_j = k_{0j} \sqrt{\frac{\epsilon_{m,j} \epsilon_{d,j}}{\epsilon_{m,j} + \epsilon_{d,j}}}, \quad k_{0j} = \frac{2\pi}{\lambda_j}. \quad (157)$$

The physical branch satisfies the unsquared TM boundary condition, decay away from the interface, and passive propagation. For the periodic lossy multilayer, a phase-matched mode instead requires the appropriate RCWA dispersion branch and nonzero pump-mode overlap. Its approximate momentum condition is

$$\text{Re } \beta_j \simeq n_p(\lambda_j) k_{0j} \sin 65^\circ - G, \quad G = \frac{2\pi}{\Lambda_g}, \quad (158)$$

with a common diffraction order -1 and the same propagation direction for both pumps. The nondispersive prism reference $n_p = 1.453$ gives

$$G = 2\pi/\Lambda_g = 6.53953508 \times 10^6 \text{ m}^{-1},$$

$$k_{xj} = n_p(2\pi/\lambda_j) \sin 65^\circ,$$

$$k_{\text{SPP},j}^{(-1)} = k_{xj} - G, \quad (159)$$

$$q_{\text{beat}} = k_{\text{SPP},1}^{(-1)} - k_{\text{SPP},2}^{(-1)}$$

$$= k_{x1} - k_{x2} = 2.29850020 \times 10^5 \text{ m}^{-1}.$$

Thus the nominal collinear beat wavevector is $q_{\text{beat}} = 2.2985 \times 10^5 \text{ m}^{-1}$, corresponding to a spatial beat period $2\pi/q_{\text{beat}} =$

27.33602 μm . The common grating momentum cancels in the difference but remains essential to each individual coupling condition. The notation $k_{\text{SPP},j}^{(-1)}$ denotes the grating-shifted excitation momentum; it coincides with a mode propagation constant only when the dispersion condition is satisfied.

The dispersive silica indices in Table LXXX give

$$\begin{aligned} q_{\text{beat,disp}} &= 2\pi \sin 65^\circ \left(\frac{1.4536}{785.0 \times 10^{-9}} - \frac{1.4531}{802.5 \times 10^{-9}} \right) \\ &= 2.33492908 \times 10^5 \text{ m}^{-1}. \end{aligned} \quad (160)$$

The dispersive value exceeds the constant-index reference by approximately 1.58490%. The constant-index wavevector target and the dispersive material table therefore define separate approximations; they cannot be exact outputs of one fixed-angle calculation. Both require a modal solution before their excitation momenta can be identified as collinear SPP propagation constants.

j) Optical-to-thermal accounting interface: Absorbed pump power enters the thermal balance through Eq. (156); incident laser power, reflected power, transmitted power, and laser-driver consumption remain distinct quantities. The system energy balance distinguishes thermal input, electrical conversion, auxiliary consumption, and heat rejection. Nominal and comparison operating points have distinct load allocations, coolant properties, and boundary temperatures; each configuration is evaluated independently. Consistent control-volume definitions prevent double counting of recovered energy and conversion losses.

B. Variational Floquet–Bloch Space–Time Elastodynamics

a) Action, constitutive assumptions, and boundary terms:

The mechanical modulation problem retains all three displacement components and the anisotropic fourth-order elasticity tensor. Let $\mathcal{V}$ be a fixed reference cell, $\mathbf{u}$ its small displacement, and $\Omega_m = 2\pi/T_p$ the mechanical modulation frequency. This frequency is distinct from an optical carrier; identification with the optical beat requires an explicit optomechanical coupling law. The independently periodic coefficients obey

$$\begin{aligned} \mathbf{C}(\mathbf{x} + \mathbf{R}, t) &= \mathbf{C}(\mathbf{x}, t + T_p) = \mathbf{C}(\mathbf{x}, t), \\ \rho(\mathbf{x} + \mathbf{R}, t) &= \rho(\mathbf{x}, t + T_p) = \rho(\mathbf{x}, t), \\ C_{ijkl} &= C_{jikl} = C_{ijlk} = C_{klij}, \quad \rho > 0, \\ \boldsymbol{\varepsilon}(\mathbf{u}) &= \frac{1}{2}[\nabla \mathbf{u} + (\nabla \mathbf{u})^T], \\ \boldsymbol{\sigma} &= \mathbf{C} : \boldsymbol{\varepsilon}. \end{aligned} \quad (161)$$

Separate spatial and temporal periodicity is the assumption supporting a rectangular double-Fourier basis. A medium having only a combined space–time translation symmetry instead requires the reciprocal basis of that combined lattice. Positive elastic energy and a prescribed positive effective inertia define the instantaneous conservative model. A time-varying ρ is an externally controlled effective inertia here, not a mass-conservation law for matter flowing through a fixed volume.

For prescribed body-force density $\mathbf{f}$ and boundary traction $\bar{\mathbf{t}}$, Hamilton’s functional is

$$\begin{aligned} \mathcal{S}[\mathbf{u}] &= \int_{t_0}^{t_1} \left[\int_{\mathcal{V}} \mathcal{L} dV + \int_{\Gamma_N} \bar{\mathbf{t}} \cdot \mathbf{u} dA \right] dt, \\ \mathcal{L} &= \frac{1}{2} \rho |\dot{\mathbf{u}}|^2 - \frac{1}{2} \boldsymbol{\varepsilon} : \mathbf{C} : \boldsymbol{\varepsilon} + \mathbf{f} \cdot \mathbf{u}. \end{aligned} \quad (162)$$

Let $\mathbf{v} = \delta \mathbf{u}$ vanish at t_0, t_1 and on the displacement boundary Γ_D . The coefficients are held fixed during this variation. Then

$$\begin{aligned} \delta \mathcal{S}[\mathbf{u}; \mathbf{v}] &= \int_{t_0}^{t_1} \int_{\mathcal{V}} [\rho \dot{\mathbf{u}} \cdot \dot{\mathbf{v}} - \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\mathbf{v}) + \mathbf{f} \cdot \mathbf{v}] dV dt \\ &\quad + \int_{t_0}^{t_1} \int_{\Gamma_N} \bar{\mathbf{t}} \cdot \mathbf{v} dA dt. \end{aligned} \quad (163)$$

Integration by parts retains both the temporal endpoint and natural spatial terms:

$$\begin{aligned} \delta \mathcal{S} &= \left[\int_{\mathcal{V}} \rho \dot{\mathbf{u}} \cdot \mathbf{v} dV \right]_{t_0}^{t_1} \\ &\quad + \int_{t_0}^{t_1} \int_{\mathcal{V}} [-\partial_t(\rho \dot{\mathbf{u}}) + \nabla \cdot \boldsymbol{\sigma} + \mathbf{f}] \cdot \mathbf{v} dV dt \\ &\quad + \int_{t_0}^{t_1} \int_{\Gamma_N} (\bar{\mathbf{t}} - \boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{v} dA dt. \end{aligned} \quad (164)$$

Arbitrariness of the admissible interior and boundary variations gives the Euler–Lagrange equations and the natural boundary condition:

$$\begin{aligned} \partial_t(\rho \dot{\mathbf{u}}) &= \nabla \cdot (\mathbf{C} : \boldsymbol{\varepsilon}) + \mathbf{f}, \\ \rho \ddot{\mathbf{u}} + \dot{\rho} \dot{\mathbf{u}} &= \nabla \cdot (\mathbf{C} : \boldsymbol{\varepsilon}) + \mathbf{f}, \\ \mathbf{u}|_{\Gamma_D} &= \bar{\mathbf{u}}, \quad \boldsymbol{\sigma} \mathbf{n}|_{\Gamma_N} = \bar{\mathbf{t}}. \end{aligned} \quad (165)$$

Thus the $\dot{\rho} \dot{\mathbf{u}}$ term cannot be omitted while retaining the stated action. Thermal eigenstrain, viscosity, and other dissipative mechanisms require their own constitutive and power terms; they are not implicitly generated by this elastic functional.

b) Double-Fourier eigenproblem: For either coefficient $a \in \{\rho, C_{ijkl}\}$, define the normalized Fourier coefficients and the displacement expansion by

$$\begin{aligned} a^{\mathbf{H},p} &= \frac{1}{|\mathcal{V}|T_p} \int_0^{T_p} \int_{\mathcal{V}} a(\mathbf{x}, t) e^{-i\mathbf{H} \cdot \mathbf{x} + ip\Omega_m t} dV dt, \\ a(\mathbf{x}, t) &= \sum_{\mathbf{H},p} a^{\mathbf{H},p} e^{i\mathbf{H} \cdot \mathbf{x} - ip\Omega_m t}, \\ \mathbf{u}(\mathbf{x}, t) &= \sum_{\mathbf{G},n} \mathbf{U}_{\mathbf{G},n} e^{i(\mathbf{k}+\mathbf{G}) \cdot \mathbf{x} - i\omega_n t}, \\ \omega_n &= \omega + n\Omega_m, \quad \boldsymbol{\kappa}_{\mathbf{G}} = \mathbf{k} + \mathbf{G}. \end{aligned} \quad (166)$$

Real material coefficients satisfy $a^{-\mathbf{H},-p} = (a^{\mathbf{H},p})^*$. In the source-free bulk, differentiating the product $\rho \dot{\mathbf{u}}$ produces one incoming and one outgoing harmonic frequency. Equating Fourier coefficients therefore gives

$$\sum_{\mathbf{G}',m,\ell} [\mathbf{K}_{\mathbf{G}n\mathbf{i},\mathbf{G}'m\ell} - \omega_n \omega_m \mathbf{M}_{\mathbf{G}n\mathbf{i},\mathbf{G}'m\ell}] U_{\mathbf{G}',m,\ell} = 0, \quad (167)$$

with the full tensor contractions

$$\begin{aligned} \mathbf{K}_{\mathbf{G}n\mathbf{i},\mathbf{G}'m\ell} &= \kappa_{\mathbf{G},j} C_{ijkl}^{\mathbf{G}-\mathbf{G}',n-m} \kappa_{\mathbf{G}',k}, \\ \mathbf{M}_{\mathbf{G}n\mathbf{i},\mathbf{G}'m\ell} &= \rho^{\mathbf{G}-\mathbf{G}',n-m} \delta_{i\ell}. \end{aligned} \quad (168)$$

Repeated Cartesian indices j, k are summed. Both wavevectors enter the stiffness contraction; its dependence is not merely a function of their difference. Shear–longitudinal conversion remains in the vector eigenfunctions rather than being replaced by a scalar transmission factor.

For a finite reciprocal/harmonic truncation, let $\mathbf{N}$ be diagonal with entry $n\Omega_m$ on each polarization. The quadratic eigenvalue problem and a companion linearization are

$$\begin{aligned} [\mathbf{K} - (\omega\mathbf{I} + \mathbf{N})\mathbf{M}(\omega\mathbf{I} + \mathbf{N})]\mathbf{U} &= 0, \\ (\mathbf{A}_0 + \omega\mathbf{A}_1 - \omega^2\mathbf{M})\mathbf{U} &= 0, \\ \mathbf{A}_0 &= \mathbf{K} - \mathbf{NMN}, \\ \mathbf{A}_1 &= -(\mathbf{NM} + \mathbf{MN}), \\ \begin{bmatrix} 0 & \mathbf{I} \\ \mathbf{A}_0 & \mathbf{A}_1 \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \omega\mathbf{U} \end{bmatrix} &= \omega \begin{bmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{M} \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \omega\mathbf{U} \end{bmatrix}. \end{aligned} \quad (169)$$

At zero modulation the $n = 0$ block reduces to $\mathbf{KU} = \omega^2\mathbf{MU}$. With modulation, positive instantaneous coefficients do not exclude parametric instability: $\text{Im}\omega > 0$ gives growth under the chosen $e^{-i\omega t}$ convention. Spatial and temporal truncations, eigenpair residuals, and stability over modulation cycles must converge independently. This mechanical eigenproblem is distinct from the electronic dielectric inverse and its scalar arithmetic benchmark in Section I.

c) Interface continuity and modal matching proof: Let a perfectly bonded, massless interface separate media A and B, with one common normal $\mathbf{n}$ directed from A to B. Admissible displacement traces and their variations agree on the interface. Summing the spatial boundary terms of Eq. (164) gives

$$\begin{aligned} \mathbf{u}_A|_\Gamma &= \mathbf{u}_B|_\Gamma, \\ \delta S_\Gamma &= \int_{t_0}^{t_1} \int_\Gamma (\boldsymbol{\sigma}_B \mathbf{n} - \boldsymbol{\sigma}_A \mathbf{n}) \cdot \mathbf{v} \, dA \, dt = 0, \\ \boldsymbol{\sigma}_A \mathbf{n} &= \boldsymbol{\sigma}_B \mathbf{n}. \end{aligned} \quad (170)$$

A common tangential period and modulation period are required for matching in one harmonic basis. Commensurate structures can use a common supercell; otherwise a single finite periodic interface basis is not exact. Orthogonal projection of the trace conditions onto $e^{i(\mathbf{k}_\parallel + \mathbf{H}) \cdot \mathbf{y} - i\omega_n t}$ gives, for every retained trace harmonic,

$$\begin{aligned} \hat{\mathbf{u}}_{\mathbf{H},n}^A &= \hat{\mathbf{u}}_{\mathbf{H},n}^B, \\ \hat{t}_{i,\mathbf{H},n}^A &= \hat{t}_{i,\mathbf{H},n}^B, \\ \hat{t}_{i,\mathbf{H},n}^A &= n_j \sum_{\mathbf{H}',m} C_{ijkl}^{a,\mathbf{H}-\mathbf{H}',n-m} \hat{\varepsilon}_{kl,\mathbf{H}',m}^a, \quad a = A, B. \end{aligned} \quad (171)$$

The strain traces include each mode's normal derivative. Equal trace coefficients do not imply equality of individual incident, reflected, and transmitted modal amplitudes. Writing displacement and traction trace matrices as $\mathbf{U}_a^\pm, \mathbf{T}_a^\pm$, where $+$ denotes outgoing and $-$ incoming modes, the interface system is

$$\begin{bmatrix} \mathbf{U}_A^+ & -\mathbf{U}_B^+ \\ \mathbf{T}_A^+ & -\mathbf{T}_B^+ \end{bmatrix} \begin{bmatrix} \mathbf{a}_A^+ \\ \mathbf{a}_B^+ \end{bmatrix} = - \begin{bmatrix} \mathbf{U}_A^- & -\mathbf{U}_B^- \\ \mathbf{T}_A^- & -\mathbf{T}_B^- \end{bmatrix} \begin{bmatrix} \mathbf{a}_A^- \\ \mathbf{a}_B^- \end{bmatrix}. \quad (172)$$

Propagating and evanescent modes are retained as needed to resolve the boundary traces. A scattering matrix obtained from this system is a derived boundary operator, not an empirical replacement for the tensor equations.

d) Energy exchange with the modulation: Multiplying Eq. (165) by $\dot{\mathbf{u}}$ and applying the product rule proves the local balance

$$\begin{aligned} e_{\text{el}} &= \frac{1}{2}\rho|\dot{\mathbf{u}}|^2 + \frac{1}{2}\boldsymbol{\varepsilon} : \mathbf{C} : \boldsymbol{\varepsilon}, \\ \mathbf{J}_{\text{el}} &= -\boldsymbol{\sigma}\dot{\mathbf{u}}, \end{aligned} \quad (173)$$

$$\partial_t e_{\text{el}} + \nabla \cdot \mathbf{J}_{\text{el}} = \mathbf{f} \cdot \dot{\mathbf{u}} + \frac{1}{2}\boldsymbol{\varepsilon} : \dot{\mathbf{C}} : \boldsymbol{\varepsilon} - \frac{1}{2}\dot{\rho}|\dot{\mathbf{u}}|^2.$$

Displacement and traction continuity imply equal instantaneous normal power flux on the two sides of a massless interface. The volume terms involving $\dot{\mathbf{C}}$ and $\dot{\rho}$ nevertheless exchange energy with the modulation drive. Consequently, ordinary passive scattering unitarity cannot be imposed without the drive-work contribution. A lossy constitutive model adds nonnegative dissipation to this balance; geometric continuity alone does not prove losslessness.

C. Thermal-Hydraulic Design and TEG Energy Accounting

The nominal cooling system comprises a 120-channel cold plate and a 50/50 water–glycol loop with a 142.5 kPa pressure-drop target [72], [73]. This target applies to the primary hydraulic assembly; the pressure-tap locations must distinguish cold-plate loss from *total loop* loss. The same pressure-drop allocation cannot be assigned independently to both boundaries.

$$\begin{aligned} N_{\text{ch}} &= 120, \quad w_{\text{ch}} = 250 \, \mu\text{m}, \\ h_{\text{ch}} &= 580 \, \mu\text{m}, \quad L_{\text{ch}} = 50.0 \, \text{mm}, \\ \dot{V} &= 3.62 \, \text{L/min} \\ &= 6.03333 \times 10^{-5} \, \text{m}^3/\text{s}, \\ A_c &= N_{\text{ch}} w_{\text{ch}} h_{\text{ch}} = 1.740 \times 10^{-5} \, \text{m}^2, \\ D_h &= \frac{2w_{\text{ch}}h_{\text{ch}}}{w_{\text{ch}} + h_{\text{ch}}} = 349.398 \, \mu\text{m}, \\ v_{\text{ch}} &= \dot{V}/A_c = 3.46743 \, \text{m/s}. \end{aligned} \quad (174)$$

At the property-reference temperature of 330 K, $\rho = 1045 \, \text{kg/m}^3$, $\mu = 1.42 \times 10^{-3} \, \text{Pa}\cdot\text{s}$, $k_f = 0.415 \, \text{W}/(\text{m}\cdot\text{K})$, and $\text{Pr} = 11.2$. The alternative values $A_c = 1.740 \times 10^{-4} \, \text{m}^2$ and $v = 0.3467 \, \text{m/s}$ differ by a factor of ten from the specified geometry and flow rate; $D_h = 348.7 \, \mu\text{m}$ is likewise incompatible with the channel dimensions. The rectangular-channel Darcy correlation gives

$$\begin{aligned} \text{Re} &= \rho v_{\text{ch}} D_h / \mu = 891.57, \\ f_D &= 62.4 / \text{Re} = 0.069989, \\ \Delta P_{\text{ch}} &= f_D \frac{L_{\text{ch}}}{D_h} \frac{\rho v_{\text{ch}}^2}{2} = 62.919 \, \text{kPa}, \\ \Delta P_{\text{other}} &= 142.5 - 62.919 = 79.581 \, \text{kPa}, \\ P_{\text{hyd}} &= \dot{V} (142500) = 8.5975 \, \text{W}. \end{aligned} \quad (175)$$

The residual ΔP_{other} allocates pressure loss to the manifolds, entrances, exits, and remaining loop components; its distribution requires measurement or a hydraulic model. The approximate 63.12 kPa channel-loss estimate is distinct from the 142.5 kPa assembly target. A 350 kPa pump rating alone does not define the operating pump curve. At 15% efficiency, the target duty requires 57.32 W, whereas the 116.67 W pump allowance corresponds to 7.37% efficiency at the same duty. Unused electrical allowance does not constitute heat generation.

TABLE IX
NOMINAL TEG BUS POWER AND LOSS ALLOCATION

Quantity	Value
Thermal hot-side input	2911.40 W
T_h/T_c	623.15/353.15 K
Ideal segmented Bi ₂ Te ₃ /PbTe efficiency	16.85%
Ideal Seebeck power	490.57 W
Contact / interconnect / bypass debits	28.45/18.60/12.30 W
Raw DC	431.22 W
DC–DC efficiency (rounded)	97.03%
Selected bus DC	418.43 W
Support load	386.67 W
Selected net bus export	+31.76 W

For $Nu = 4.12$, the heat-transfer coefficient is $h = Nuk_f/D_h \simeq 4893.57 \text{ W}/(\text{m}^2\text{K})$. Copper fins with $k_w = 391 \text{ W}/(\text{mK})$ and thickness $t_w = 150 \mu\text{m}$ give $m_f = \sqrt{2h/(k_w t_w)} \simeq 408.50 \text{ m}^{-1}$ and $\eta_f = \tanh(m_f h_{ch})/(m_f h_{ch}) \simeq 0.9817$. The comparison estimates $h = 4903.3$ and $m_f = 409.1$ use approximate parameter values rather than the specified geometry. Applicability of a fully developed correlation depends on entrance effects and thermal boundary conditions. Because the frame/cold-plate section is Al 6061-T6 whereas the fin model uses copper, a combined design requires an explicit insert geometry and interfacial contact resistance.

The external heat exchanger has nominal parameters $A_{\text{rad}} = 2.455 \text{ m}^2$, $U_{\text{rad}} = 48.5 \text{ W}/(\text{m}^2\text{K})$, and $T_{\text{amb}} = 310 \text{ K}$, requiring a log-mean temperature difference of $2492.97/(48.5 \times 2.455) = 20.94 \text{ K}$. This condition defines the heat-transfer duty but does not uniquely determine the coolant and air outlet temperatures. The pure-water case at 45/55 °C and 25 °C ambient is a separate comparison configuration, not the property basis of the glycol loop.

The electrical-equivalent losses total 59.35 W, leaving $490.57 - 59.35 = 431.22 \text{ W}$ at the generator terminals. An exact converter efficiency of 97.03% gives $431.22 \times 0.9703 = 418.412766 \text{ W}$ rather than 418.43 W. The nominal bus output therefore requires 97.033997% efficiency, which rounds to 97.03%; quantitative verification requires the unrounded efficiency or measured terminal powers. Thermal-contact and bypass debits represent equivalent electrical losses and are not necessarily equal thermal losses. At the generator boundary, the residual heat is $2911.40 - 431.22 = 2480.18 \text{ W}$. Addition of 12.79 W of converter dissipation gives 2492.97 W of aggregate heat rejection. If all 386.67 W of support power dissipates internally, environmental heat rejection is 2879.64 W, and $2879.64 + 31.76 = 2911.40 \text{ W}$ satisfies the conditional system balance.

The calorimetric input model assigns 1850 W to optical pumping and 65.59 W to RF excitation, whereas the bus-export model allocates 120 W to laser drivers. These loads cannot be equated without a conversion-efficiency model and a common definition of electrical and optical ports. Consequently, +31.76 W is a conditional *bus power balance*, not a demonstrated whole-device net-energy export. A whole-device assessment requires consistent pump, laser, RF, and thermal boundaries.

TABLE X
COMPRESSOR-FREE ELECTRICAL BALANCE AT 2911.40 W THERMAL INPUT

Contribution	Electrical power (W)
TEG generation	+418.43
Chiller	0.00
Water-loop pump	−116.67
Laser drivers	−120.00
Auxiliaries, fans, telemetry	−150.00
Net export	+31.76

a) *Compressor-free constant-efficiency comparison:* The compressor-free loop operates without vapor-compression refrigeration but retains electrically powered pumps and fans. Fans, telemetry, and remaining auxiliaries are included once in the 150 W aggregate allowance. For this configuration, the constant- ZT material model gives

$$\eta_{\text{TEG}} = \left(1 - \frac{353.15}{623.15}\right) \frac{\sqrt{3.16} - 1}{\sqrt{3.16} + 353.15/623.15}$$

$$= 0.143722744917,$$
(176)

$$P_{\text{TEG,calc}} = 418.434399551 \text{ W},$$

$$P_{\text{net,calc}} = 418.434399551 - 386.67$$

$$= 31.764399551 \text{ W}.$$

Table X assigns 418.43 W to the electrical output after a single rounding step. This value is conditional on availability at the specified bus boundary. Any converter or contact losses outside that boundary require separate accounting; the constant-efficiency comparison cannot be combined indiscriminately with a separately derated TEG model. At the combined core-to-bus conversion boundary, $Q_{\text{cold}} = 2911.40 - 418.43 = 2492.97 \text{ W}$. If all 386.67 W of support power dissipates internally, environmental heat rejection is $2492.97 + 386.67 = 2879.64 \text{ W}$ and $2879.64 + 31.76 = 2911.40 \text{ W}$. The 2911.40 W core input therefore cannot also be counted as simultaneous heat export in addition to 31.76 W of electricity. Recoverable heat depends on the dissipation paths captured by the thermal system. In the refrigerated comparison, inclusion of the 116.67 W pump gives -561.80 W net power; the alternative -445.13 W balance excludes that pump and is incomplete.

b) *Water-to-air heat-transfer closure:* With water at 45/55 °C, $c_p = 4180 \text{ J}/(\text{kg K})$, $\rho_w = 988 \text{ kg}/\text{m}^3$, and an 80 °C cold plate,

$$R_{\text{plate-water}} = \frac{80 - 50}{2492.97} \approx 0.012034 \text{ K/W},$$

$$\dot{m} = \frac{2492.97}{4180(55 - 45)} = 0.05964043 \text{ kg/s},$$
(177)

$$\dot{V} = \dot{m}/988 = 3.62188850 \text{ L/min},$$

$$\Delta P_{\text{allow}} = \frac{116.67(0.15)}{\dot{V}} = 289912.29 \text{ Pa}.$$

The allowable pressure rise is evaluated with $\dot{V}$ in m^3/s and a 15% pump efficiency. This operating point differs from the 7 L/min, 35% efficiency comparison; each flow condition requires its own cold-plate pressure-loss calculation. For counterflow

TABLE XI
DIRECT-AIR AND REFRIGERATED MODES: THERMAL AND ELECTRICAL
BALANCES

Quantity (W)	Direct air	Chilled
Thermal reference	2911.400	2911.400
Laser allowance	120.000	120.000
Pump allowance	116.667	116.667
Fans	45.000	15.000
Controls	25.000	25.000
Compressor	0	693.190
Auxiliary electricity	306.667	969.857
Heat-minus-load comparison	2604.733	1941.543
Illustrative TEG bus output	337.952	337.952
Illustrative electrical balance	31.285	-631.905

water entering at 55 °C and leaving at 45 °C, with air entering at 25 °C and leaving at 38 °C,

$$\begin{aligned}\Delta T_1 &= 55 - 38 = 17 \text{ K}, \\ \Delta T_2 &= 45 - 25 = 20 \text{ K}, \\ \Delta T_{\text{lm}} &= \frac{17 - 20}{\ln(17/20)} = 18.45938814 \text{ K}, \\ A_{\text{HX}} &= \frac{2492.97}{55 \Delta T_{\text{lm}}} = 2.45548373 \text{ m}^2 \approx 2.455 \text{ m}^2.\end{aligned}\quad (178)$$

The calculated finned area assumes $U = 55 \text{ W}/(\text{m}^2 \text{ K})$, ideal counterflow, and rejection of core/TEG heat alone. Pump heat entering the coolant, fouling, departures from counterflow, and ambient conditions affect the required heat-transfer duty and area. Forced-convection fan power remains part of the auxiliary allocation. The 2911.40 W thermal input defines the design duty without establishing its nuclear origin.

c) *Direct-air and refrigerated comparison modes:* A separate operating comparison uses $P_{\text{th,ref}} = 2911.40 \text{ W}$, a 120 W aggregate laser allowance, 25 W for controls, and mode-dependent fan loads. This allocation is distinct from the 800.688889 W balance-of-plant (BOP) configuration. At a pump duty of 7 L/min and 350 kPa with 35% efficiency,

$$\begin{aligned}P_{\text{pump}} &= \dot{V} \Delta P / \eta_p = 116.66666667 \text{ W}, \\ P_{\text{comp}} &= 2911.40 / 4.2 = 693.190476190 \text{ W}, \\ P_{\text{aux,pass}} &= 120 + 116.666666667 + 45 + 25 \\ &= 306.666666667 \text{ W}, \\ P_{\text{aux,chill}} &= 120 + 116.666666667 + 15 + 25 + P_{\text{comp}} \\ &= 969.857142857 \text{ W}.\end{aligned}\quad (179)$$

The expression $P_{\text{net}} = P_{\text{th}} \eta_{\text{conv}} - P_{\text{aux}}$ with $\eta_{\text{conv}} = 1$ is a direct-thermal-export comparison, not an electrical-conversion model. Its +2604.73/+1941.54 W values therefore do not represent electrical export. The illustrative electrical balances instead use $\eta_{\text{bus}} = 0.116078911301$; this fixed-efficiency comparison does not independently determine the TEG junction temperatures. The coupled installation satisfies

$$\begin{aligned}P_{\text{net,el}} &= \eta_{\text{bus}}(T_h, T_c) P_{\text{th}} - P_{\text{aux}}, \\ Q_{\text{TEG, rej}} &= P_{\text{th}} - P_{\text{TEG, bus}}, \\ Q_{\text{condenser}} &= Q_c + P_{\text{comp}}.\end{aligned}\quad (180)$$

The full-heat chilled case has $Q_c = 2911.40 \text{ W}$ and $Q_{\text{condenser}} = 3604.590476 \text{ W}$. Electrical extraction by a TEG changes the chiller duty and therefore the condenser balance. A mode with 310 K ambient and 319.2 K coolant return cannot sustain a 310 K TEG cold junction through finite passive heat flow. The 120 W laser allowance must distinguish electrical input from optical output and remain consistent with the two-laser conversion efficiencies. Turbopump and other service loads require explicit allocation rather than an assumption of zero consumption. The 10/15 kW cases are separate design comparisons, not the nominal 2911.40 W operating point.

D. Accounting Boundary and Continuous Auxiliary Loads

The component-resolved BOP model [54] distinguishes specified electrical allowances from power calculated at each operating point. The 450/180/85/60/25 W allocation defines a separate comparison schedule. The itemized nominal allowances total 800.0 W, whereas evaluation of the component equations without intermediate rounding gives a different total.

$$\begin{aligned}P_{L1} &= 100/0.45 = 222.222222 \text{ W}, \\ P_p &= \frac{(7.0 \times 10^{-3}/60)(3.5 \times 10^5)}{0.35} = 116.666667 \text{ W}, \\ P_{\text{BOP}} &= P_{L1} + 200 + P_p + 120 + 141.8 \\ &= 800.68888889 \text{ W}.\end{aligned}\quad (181)$$

At 7.2 L/min, the pump demand is 120.0 W and total BOP demand is 804.022222 W. The specified 7.0 L/min case instead gives 116.666667 W of pump demand, rather than the 116.0 W allowance. A fixed 116 W pump budget at 3.5 bar and 35% efficiency corresponds to 6.96 L/min. Exact agreement with an 800 W total requires a different component allocation or operating point; it does not follow from rounding the specified inputs. The direct 8.33 THz excitation architecture differs from the 785/802.5 nm two-carrier optical-beat model in Section I. A direct THz emitter is not an 802.5 nm diode, and the component schedule does not simultaneously account for both excitation architectures. No compressor or independent dry-cooler fan demand is included. These loads, together with conversion losses outside the stated boundaries, require additional accounting; an unspecified cooling load cannot be treated as zero.

E. Master Electrical Balance and Reaction Thresholds

The four-part loss model [56] relates material conversion efficiency to usable bus power. The high- ZT comparison combines the material efficiency at $ZT = 2.16$, the unrounded resistance ratios in Eq. (188), and the calculated BOP demand:

$$\begin{aligned}\eta_{\text{net,TEG}} &= 0.116078911301, \\ P_{\text{TEG, bus}} &= \eta_{\text{net,TEG}} P_{\text{nuclear}}, \\ P_{\text{net, system}} &= \eta_{\text{net,TEG}} P_{\text{nuclear}} - P_{\text{BOP}}.\end{aligned}\quad (182)$$

TABLE XII
COMPONENT-RESOLVED BOP ALLOWANCES AND CALCULATED ELECTRICAL DEMAND

Subsystem	Specification	Allowance (W)	Calculated (W)
Laser pump 1	785 nm, 100 W output, 45% WPE	222.2	222.222222
Laser pump 2	8.33 THz drive, 10 W output, 5% WPE	200.0	200.000000
Coolant pump	7.0 L/min, 3.5 bar rise, 35% efficiency	116.0	116.666667
Vacuum turbopumps	Running/cold-start average allocation	120.0	120.000000
Control and diagnostics	FPGA, DAQ, RF, sensors allocation	141.8	141.800000
Total	Specified operating-point arithmetic	800.0	800.688889

TABLE XIII
FIXED-LOAD DESIGN-POINT ELECTRICAL AND THERMAL BALANCES

Quantity (W)	10 kW heat	15 kW heat
Nuclear heat	10000.00	15000.00
TEG before converter	1221.88	1832.82
Converter loss	61.09	91.64
TEG bus generation	1160.79	1741.18
BOP demand	800.69	800.69
Net export	360.10	940.49
Core thermal rejection	8778.12	13167.18
Rejection plus converter loss	8839.21	13258.82

The comparison holds BOP demand fixed at 800.688889 W and includes converter loss once within the core-to-bus efficiency. The electrical break-even condition is

$$P_{\text{nuclear,break-even}} = P_{\text{BOP}}/\eta_{\text{net,TEG}} = 6897.798057 \text{ W.} \quad (183)$$

$$\begin{aligned} \eta_{\text{export}}(10 \text{ kW}) &= 3.601002\%, \\ \eta_{\text{export}}(15 \text{ kW}) &= 6.269965\%. \end{aligned} \quad (184)$$

The +50/+475 W margins and 9150/13725 W rejection duties correspond to the separate 8.5% efficiency case. These rejection duties also define cold-plate benchmark loads in Section V. Positive calculated balances remain conditional on the assumed heat input, conversion efficiency, and BOP demand; they establish neither a nuclear heat source nor engineering feasibility. For channel fractions B_j and retained fractions f_j ,

$$P_{\text{nuclear}} = R_{DD} \sum_j B_j f_j Q_j. \quad (185)$$

Complete deposition of reaction energy into the lattice gives the energy-equivalent requirements

$$R_{10\text{kW}} = 10^4/Q_{DD}, \quad R_{15\text{kW}} = 1.5 \times 10^4/Q_{DD}, \quad (186)$$

consistent with Eq. (101). These requirements are heat-equivalent reaction rates rather than rates established by the microscopic integral.

F. Cascade TEG Architecture, Conditioning, and Heat Rejection

The segmented $\text{Bi}_2\text{Te}_3/\text{PbTe}$ material formula is evaluated before separately applying module and converter losses. At $T_H = 550 \text{ K}$ and $T_L = 310 \text{ K}$,

$$\begin{aligned} \eta_C &= 1 - T_L/T_H = 0.4363636364, \\ \eta_{\text{mat}}(ZT) &= \eta_C \frac{\sqrt{1 + ZT} - 1}{\sqrt{1 + ZT} + T_L/T_H}, \\ \eta_{\text{mat}}(1.25) &= 0.105726872247, \\ \eta_{\text{mat}}(2.16) &= 0.144935257564. \end{aligned} \quad (187)$$

The material efficiencies round to 10.57% and 14.50%. The latter describes the high- ZT comparison rather than a fundamental efficiency limit. Neither value includes module or converter losses. Application of the four-part loss model gives bus efficiencies of 8.467684% and 11.607891%, respectively. At 10/15 kW thermal input, the baseline- ZT net exports are 46.08/469.46 W and the high- ZT exports are 360.10/940.49 W, before any additional cooling or excitation-system loads. The module and converter loss factors are

$$\begin{aligned} d_{\text{thermal}} &= \frac{R_{\text{TEG}}}{R_{\text{TEG}} + R'_h + R'_c} \\ &= \frac{1.15}{1.15 + 0.05 + 0.05} = 0.920, \\ d_{\text{electrical}} &= \frac{R_L}{R_L + R_{\text{contact}} + R_{\text{wire}}} \\ &= \frac{2.10}{2.10 + 0.06 + 0.04} = \frac{21}{22}, \\ d_{\text{geometric}} &= \frac{8784}{8784 + 366} = 0.960, \\ \eta_{\text{conv}} &= 0.950. \end{aligned} \quad (188)$$

Thermal resistances have units of K/W and electrical resistances have units of ohms; the corresponding ratios are dimensionless. Under the multiplicative loss approximation,

$$\begin{aligned} d_\eta &= d_{\text{thermal}} d_{\text{electrical}} d_{\text{geometric}} \eta_{\text{conv}} \\ &= 0.920(21/22)(0.960)(0.950) = 0.8009018182, \\ \eta_{\text{net,TEG}} &= \eta_{\text{mat}}(2.16) d_\eta = 0.1160789113. \end{aligned} \quad (189)$$

Use of the rounded 14.50% ideal efficiency and electrical factor 0.955 gives approximately 11.62% bus conversion. Evaluation at $ZT = 2.16$ without intermediate rounding preserves consistency among the loss ratios. Thermal resistances refer to the specified module; their network topology must be established before 1.15 K/W can represent the complete stack. The 8784/366 W ratio defines the geometric loss fraction and does not constitute an additional bypass loss.

$$\eta_C = 0.4363636364 > \eta_{\text{mat}}(2.16) > \eta_{\text{net,TEG}}. \quad (190)$$

TABLE XIV
CORE AND CONVERTER HEAT-REJECTION DUTY FOR THE HIGH- ZT
COMPARISON

Quantity (W)	10 kW heat	15 kW heat
Core thermal rejection	8778.12	13167.18
Converter heat	61.09	91.64
Combined rejection	8839.21	13258.82
Allocated compressor power	0.00	0.00

a) *Refrigeration demand and heat rejection:* The component-resolved BOP allocation excludes compressor power. A refrigerated configuration therefore requires a separate conditional calculation:

$$\begin{aligned} \text{COP} &= \eta_{\text{cycle}} \frac{T_{\text{cold}}}{T_{\text{amb}} - T_{\text{cold}}}, \\ \eta_{\text{cycle}} &= 0.375, \quad T_{\text{cold}} = 280 \text{ K}, \\ T_{\text{amb}} &= 305 \text{ K}, \quad \text{COP} = 4.2, \\ P_{\text{comp}} &= Q_{\text{chilled}}/4.2. \end{aligned} \quad (191)$$

A 450 W compressor removes 1890 W at the specified COP, with its electrical demand *additional* to the loads in Table XII. Without refrigeration, the direct-rejection system must dissipate the following thermal loads: Omission of compressor power from an allocation does not establish the feasibility of passive cooling. A 310 K cold side requires a cooler ambient and a sufficient heat-exchanger temperature approach; ambient temperatures at or above 310 K cannot sustain that sink passively. Dissipation from the remaining BOP components contributes to the appropriate facility heat-rejection boundary. If all listed core/converter rejection is chilled at COP 4.2, the additional compressor demand gives

$$\begin{aligned} P_{\text{all}} &= \eta_{\text{net,TEG}} P_{\text{nuclear}} - P_{\text{BOP}} \\ &\quad - \frac{(1 - \eta_{\text{net,TEG}}) P_{\text{nuclear}}}{4.2}, \\ P_{\text{all}}(10 \text{ kW}) &= -1744.47 \text{ W}, \\ P_{\text{all}}(15 \text{ kW}) &= -2216.37 \text{ W}. \end{aligned} \quad (192)$$

Positive design margins are conditional on a feasible direct heat sink and complete accounting of auxiliary and excitation-system loads. Exclusion of required refrigeration work cannot establish positive net export.

b) *Secondary and tertiary coolant loops:* The coolant-loop balances are $\dot{m} = Q/(c_p \Delta T)$ and $\dot{V} = \dot{m}/\rho$. The 7.0 L/min, 3.5 bar pump duty is distinct from the 60 W pump-budget case and the 17.7/23.5 L/min comparison loops. At 7.0 L/min, the parameters in Table XLIX give $Re = 2582.8$, beyond the stated laminar-correlation range, and $\Delta T = 28.18 \text{ K}$ at the 13725 W benchmark. This duty therefore does not satisfy the peak-temperature-rise criterion $\Delta T < 25 \text{ K}$ associated with a flow of at least 8 L/min. Validation requires a hydraulic characteristic appropriate to the operating flow regime, rather than a flow-and-pressure allocation alone.

G. PZT-5H Acoustic Harvesting and Common-Bus Dynamics

PZT-5H acoustic harvesting is an optional experimental interface and contributes no generation credit to the baseline

power balance. The reduced reciprocal piezoelectric model uses displacement x , effective mass m_a , mechanical damping b_a , stiffness k_a , coupling coefficient ϑ , terminal capacitance C_p , voltage v_p , and equivalent resistive load R_L :

$$\begin{aligned} m_a \ddot{x} + b_a \dot{x} + k_a x + \vartheta v_p &= F_a(t), \\ C_p \dot{v}_p + \frac{v_p}{R_L} - \vartheta \dot{x} &= 0. \end{aligned} \quad (193)$$

The model coefficients require characterization of the assembled transducer under its mechanical and electrical loads. With $\mathbf{z} = [x, \dot{x}, v_p]^T$, the linear state model is

$$\dot{\mathbf{z}} = \begin{bmatrix} 0 & 1 & 0 \\ -k_a/m_a & -b_a/m_a & -\vartheta/m_a \\ 0 & \vartheta/C_p & -1/(R_L C_p) \end{bmatrix} \mathbf{z} + \begin{bmatrix} 0 \\ 1/m_a \\ 0 \end{bmatrix} F_a(t). \quad (194)$$

For zero initial conditions, the force-to-voltage transfer function is

$$\frac{V_p(s)}{F_a(s)} = \frac{\vartheta s}{(m_a s^2 + b_a s + k_a)(C_p s + R_L^{-1}) + \vartheta^2 s}. \quad (195)$$

Multiplying the two equations in (193) by $\dot{x}$ and v_p , respectively, produces the passive energy balance

$$\frac{d}{dt} \left(\frac{1}{2} m_a \dot{x}^2 + \frac{1}{2} k_a x^2 + \frac{1}{2} C_p v_p^2 \right) = F_a \dot{x} - b_a \dot{x}^2 - \frac{v_p^2}{R_L}. \quad (196)$$

The steady harvested power is bounded by the supplied mechanical power after mechanical dissipation; rectifier and converter efficiencies determine the bus-delivered fraction.

An active MOSFET bridge rectifies the alternating PZT voltage. A controlled buck stage delivers power to the common bus when the rectified voltage exceeds 48 V; operation below that voltage requires a buck-boost stage or cessation of export. The steady energy requirements are

$$\begin{aligned} P_{\text{PZT,bus}} &= \eta_{\text{rect}} \eta_{\text{dc}} P_{\text{PZT,raw}}, \\ P_{\text{PZT,raw}} &\leq \langle F_a \dot{x} \rangle, \\ I_{\text{PZT,bus}} &= \frac{P_{\text{PZT,bus}}}{48 \text{ V}}. \end{aligned} \quad (197)$$

Acoustic energy recovered before thermalization is excluded from the subsequent TEG heat input. The channel partition must satisfy Eq. (196) and the system first law, with each recovered contribution counted once.

For bus capacitance C_{bus} and startup-source power P_{start} , the common-bus dynamics are

$$\begin{aligned} C_{\text{bus}} V_{\text{bus}} \frac{dV_{\text{bus}}}{dt} &= P_{\text{TEG,bus}} + P_{\text{PZT,bus}} + P_{\text{start}} \\ &\quad - P_{\text{aux,bus}} - P_{\text{load,bus}}. \end{aligned} \quad (198)$$

The bus-referred demand $P_{\text{aux,bus}}$ includes auxiliary supplies, point-of-load conversion, and wiring losses; it is not constrained to the comparison allocation of 718.96 W. At nominal steady operation, $P_{\text{start}} = P_{\text{PZT,bus}} = 0$. For the case with only the itemized support loads,

$$\begin{aligned} I_{\text{gross}} &= 1741.183670/48 = 36.274660 \text{ A}, \\ I_{\text{aux,list}} &= 800.688889/48 = 16.681019 \text{ A}, \\ I_{\text{load,list}} &= 940.494781/48 = 19.593641 \text{ A}. \end{aligned} \quad (199)$$

The quantity $P_{\text{aux,listed}} = P_{\text{BOP}}$ includes every load in Table XII, whereas P_{other} comprises additional metered demands outside that schedule. These currents correspond to the 15 kW thermal comparison; the 10 kW case gives $360.100224/48 = 7.502088$ A net export. Bus-referred accounting does not require every facility supply to operate physically from the 48 V bus.

For measured terminal-power accounting,

$$P_{\text{net,bus}} = \eta_{\text{boost}} P_{\text{TEG,terminal}} + \eta_{\text{rect}} \eta_{\text{dc}} P_{\text{PZT,raw}} - P_{\text{aux,listed}} - P_{\text{other}}. \quad (200)$$

Optional acoustic harvesting receives credit only after measuring its output and removing the same extracted energy from the thermal-input budget. The baseline in (182) assigns no such credit.

H. Metrology, Durability, and Acceptance Requirements

Acceptance testing requires measurement of core thermal duty, cascade heat input and output, boost losses, every support load in Table XII, optical and cooling supplies, and net export at a common declared boundary. The 1275 W generation and 475 W net values are conditional reference points rather than predetermined measurement results. The WT5000 supply belongs to the 25 W control/diagnostics allocation only if that boundary supplies it; otherwise it constitutes a separate demand. The energy balance must include startup-source discharge, changes in stored thermal energy, cooling-system inputs, and externally powered instrumentation.

For a measured excess-power estimate $\hat{P}_{\text{excess}}$ and combined standard uncertainty u_c ,

$$z = \frac{\hat{P}_{\text{excess}}}{u_c}, \quad u_c^2 = \mathbf{J} \mathbf{S} \mathbf{J}^T, \quad J_j = \frac{\partial P_{\text{excess}}}{\partial y_j}. \quad (201)$$

The uncertainty budget includes correlations and systematic effects, independent blanks, complete electrical-input measurements, calibrated heat capture, and nuclear-product diagnostics. Statistical significance must be evaluated for the actual measurement boundary; results from a different boundary are not directly transferable. The TLP fatigue-life target is $N_f = 44,820$ defined thermal cycles, subject to the constitutive and qualification conditions in Section IV. Design allocations for power and lifetime remain distinct from measured acceptance results.

a) Fixed-load operating-point specification: The fixed-load comparison [63] uses unrounded component demand and loss factors, with converter dissipation included once:

$$\begin{aligned} P_{\text{BOP}} &= 100/0.45 + 10/0.05 \\ &\quad + (7 \times 10^{-3}/60)(3.5 \times 10^5)/0.35 \\ &\quad + 120 + 141.8 = 800.68888889 \text{ W}, \\ d_\eta &= 0.920(21/22)(0.960)(0.950) \\ &= 0.800901818182, \\ \eta_{\text{bus}} &= 0.144935257564 d_\eta = 0.116078911301. \end{aligned} \quad (202)$$

The fixed-load electrical break-even threshold is $P_{\text{th,break}} =$

TABLE XV
CONSTANT-EFFICIENCY ELECTRICAL OUTPUT AND HEAT-REJECTION COMPARISON

Quantity (W)	10 kW heat	15 kW heat
Bus generation	1160.789113	1741.183670
Allocated BOP	800.688889	800.688889
Net electrical export	360.100224	940.494781
Heat minus bus generation	8839.210887	13258.816330
Allocated compressor	0	0

$P_{\text{BOP}}/\eta_{\text{bus}}$; it is not a nuclear ignition criterion. The heat-rejection values include converter losses but exclude auxiliary dissipation outside the specified thermal boundary. Fans, controls, optical absorption, pumps, and refrigeration contribute once at their respective boundaries. A zero compressor allocation is physically admissible only when a non-refrigerated sink can sustain the junction-temperature difference assumed by the material-efficiency model.

III. MECHANICAL, OPTOMECHANICAL ALIGNMENT, AND VACUUM INTERFACE

The mechanical assembly supports the pressure boundary, maintains the core-interface preload, and registers the active region to the prism and thermal spreader. Its design envelope couples multiaxial transient stress, cyclic plastic strain, nonlinear spring compliance, and high-temperature stress relaxation. The structural and preload limits in this section define the specified load path; alternative material, bondline, and clamp models describe separate configurations rather than simultaneous requirements. Component qualification requires compatible material properties, contact geometry, and load histories. Electrical interface loads refer to the regulated-bus boundaries in Section II.

a) Micro-alloyed bondline and graded compliance:

The selected interconnect is a $3.5 \mu\text{m}$ Ni-6Cu/Sn-0.7Cu-0.05Bi transient-liquid-phase (TLP) bondline with a $15.0 \mu\text{m}$ functionally graded ductile Ni/Cu compliance layer. The alloying and graded layer are specified to control intermetallic phase formation, interfacial voiding, and strain concentration; their effectiveness requires characterization of the fabricated joint. The fatigue load case spans $20.0\text{--}276.85^\circ\text{C}$, or $293.15\text{--}550.0 \text{ K}$, with $\Delta T = 256.85 \text{ K}$. It does not qualify the bondline for a 623.15 K peak temperature.

For depth z , the strain partition and accumulated equivalent plastic strain are

$$\begin{aligned} \epsilon_{ij}^{\text{tot}}(z) &= \alpha_T(z) \Delta T \delta_{ij} + [\mathbf{C}^{-1}(z)]_{ijkl} \sigma_{kl}(z) + \epsilon_{ij}^p(z), \\ dp &= \sqrt{\frac{2}{3} d\epsilon^p : d\epsilon^p}. \end{aligned} \quad (203)$$

The grating and substrate CTE inputs are 13.8×10^{-6} and $4.5 \times 10^{-6} \text{ K}^{-1}$, respectively. Their free mismatch is 0.002388705 over the specified cycle; constraint and plastic flow, rather than the CTE difference alone, determine the local plastic amplitude.

With backstress $\mathbf{X} = \mathbf{X}_1 + \mathbf{X}_2$, the two-term Chaboche law is

$$\begin{aligned} d\mathbf{X}_k &= \frac{2}{3}C_k d\epsilon^p - \gamma_k \mathbf{X}_k dp, \quad k = 1, 2, \\ C_1 &= 42.5 \text{ GPa}, \quad \gamma_1 = 450, \\ C_2 &= 12.8 \text{ GPa}, \quad \gamma_2 = 85. \end{aligned} \quad (204)$$

The scalar C_k are hardening moduli, distinct from the elasticity tensor $\mathbf{C}$. A symmetric stabilized cycle with zero mean stress is the selected strain-life evaluation condition; it is not implied by the backstress evolution alone.

The Coffin–Manson–Morrow law uses strain amplitudes, with N_f measured in cycles and $2N_f$ in reversals:

$$\begin{aligned} \frac{\Delta\epsilon_{\text{tot}}}{2} &= \frac{\sigma'_f - \sigma_m}{E} (2N_f)^b + \epsilon'_f (2N_f)^c, \\ E &= 128.5 \text{ GPa}, \quad \sigma'_f = 545.0 \text{ MPa}, \\ b &= -0.082, \quad \epsilon'_f = 0.320, \quad c = -0.560, \\ \sigma_m &= 0. \end{aligned} \quad (205)$$

At $2N_f = 104800$ reversals, the elastic and plastic contributions are

$$\begin{aligned} \frac{\Delta\epsilon_e}{2} &= 1.64370452006 \times 10^{-3}, \\ \frac{\Delta\epsilon_p}{2} &= 4.94023558978 \times 10^{-4}, \\ \frac{\Delta\epsilon_{\text{tot}}}{2} &= 0.00213772807904 \approx 0.002137728, \\ N_f &= 52400 \text{ cycles} > 44820 \text{ cycles}, \\ \frac{N_f - 44820}{44820} &= 0.16912093 \approx 16.9\%. \end{aligned} \quad (206)$$

The positive elastic and plastic terms both decrease monotonically with N_f , so the full-precision total amplitude has a unique positive strain-life solution. Display rounding to 0.002137728 is not applied before solving. The total strain-amplitude symbol ϵ_{tot} denotes $\Delta\epsilon_{\text{tot}}/2$, not the full strain range. The result passes the 44,820-cycle constitutive criterion in Table LXXVII; it is not a measured service-life guarantee. For comparison, an unalloyed bond with plastic amplitude 0.00223, $\epsilon'_f = 0.3700$, and $c = -0.520$ gives approximately 9290 cycles. The spring-amplitude ceiling and 88,827-cycle spring Monte Carlo mean below refer to a different material and load path; they do not replace the 52,400-cycle bondline prediction.

A. Selected Contact Materials and Clamp-Force Bounds

The selected contact-load configuration uses a circular 20.0 mm core interface over the 25–350 °C temperature envelope. Table XVI reproduces the material-property table in the thermomechanical design specification [90]. Its Pd-25Ir composition is the material basis for the following contact-force bound; it is distinct from the Pd_{0.95}Ir_{0.05} optical-film model. Transfer of the assigned strength to a different alloy composition requires material qualification. The diamond entry denotes a brittle fracture limit, not a ductile yield stress. Spring strength additionally depends on product form, temper, and heat treatment [93].

The contact area and selected pressure limits are

$$\begin{aligned} A_{\text{contact}} &= \frac{\pi}{4} D_{\text{core}}^2 = \frac{\pi}{4} (0.020 \text{ m})^2 \\ &= 3.14159265359 \times 10^{-4} \text{ m}^2, \\ P_{\text{min}} &= 20 \text{ MPa}, \\ \sigma_{y,\text{film}}(350 \text{ °C}) &= 250 \text{ MPa}. \end{aligned} \quad (207)$$

The 20 MPa minimum is the specified sealing-contact requirement, and the 250 MPa maximum is the assigned hot Pd-Ir film yield limit [90]. Applying these limits to the average contact pressure gives

$$\begin{aligned} F_{\text{min}} &= P_{\text{min}} A_{\text{contact}} \simeq 6283.19 \text{ N}, \\ F_{\text{max}} &= \sigma_{y,\text{film}} A_{\text{contact}} \simeq 78539.82 \text{ N}, \\ F &\in [6283.19, 78539.82] \text{ N}. \end{aligned} \quad (208)$$

Force endpoints are displayed to 0.01 N; pressure checks shall use the unrounded contact area. The optimization summary contains 92 first-front designs with forces from 6,283.19 to 78,410.02 N, within the specified bounds [92]. The bounds apply to the instantaneous transmitted contact force, not merely the assembly setpoint. Thermal excursion, long-term relaxation, and setting uncertainty shall leave sufficient margin to both endpoints. The sealing threshold requires leak qualification, and local contact or spring bending stresses require their own structural assessment; the average-pressure calculation alone does not establish those responses.

B. Three-Dimensional Thermomechanical Stress Envelope

The pressure-boundary transition comprises an 8.0 mm wall at inner radius 45.0 mm and a 5.0 mm $\times$ 45° relief bevel. The transient load case combines internal hydraulic pressure from 0.6513 to 16.6013 MPa(a), an axial retaining load of 125.0 kN, and a thermal-gradient magnitude of 48.5 K/mm [80]. Pressure tractions require the internal-to-external pressure difference; the absolute internal values alone do not define that difference. Across a 5.0 mm path aligned with the gradient,

$$\Delta T_{\text{bevel}} = (5.0 \text{ mm})(48.5 \text{ K/mm}) = 242.5 \text{ K}. \quad (209)$$

The gradient direction and transient temperature field remain part of the load definition; a gradient magnitude cannot determine a complete stress tensor.

For small strain, the three-dimensional constitutive relation is

$$\begin{aligned} \boldsymbol{\sigma} &= \mathbf{C} : (\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_p - \alpha \Delta T \mathbf{I}), \\ \boldsymbol{\epsilon} &= \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T), \\ C_{ijkl} &= \lambda_e \delta_{ij} \delta_{kl} + \mu_e (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \\ \lambda_e &= \frac{E\nu}{(1+\nu)(1-2\nu)}, \quad \mu_e = \frac{E}{2(1+\nu)}. \end{aligned} \quad (210)$$

Here $\mathbf{u}$ is displacement, $\boldsymbol{\epsilon}_p$ is the plastic-strain tensor, and $\mathbf{C}$ is the isotropic fourth-order elastic tensor. The hot structural case uses $E(350 \text{ °C}) = 195.0 \text{ GPa}$, $\nu = 0.31$, and $\alpha = 14.2 \times 10^{-6} \text{ K}^{-1}$ for the specified precipitation-treated Inconel X-750 condition. The thermal-strain expression assumes constant α over the local temperature interval; a temperature-dependent coefficient requires $\int \alpha(T) dT$. Momentum balance,

TABLE XVI
MATERIAL PROPERTIES OF SUBSTRATE, ACTIVE FILM, AND SPRING STACK

Structural component	Material composition	Yield limit at 25 °C (σ_y)	Yield limit at 350 °C (σ_y)	Critical failure mode & mechanisms
Active reaction film	Pd-25Ir alloy	350 MPa	250 MPa	Plastic deformation, surface extrusion, and micro-cracking.
Rigid substrate	CVD diamond	2.0 GPa	2.0 GPa	Brittle fracture, interface shearing, and micro-delamination.
Preload spring stack	Inconel X-750	1000 MPa	900 MPa	Low-cycle fatigue and high-temperature stress relaxation.

mechanical restraints, contact tractions, and plastic evolution complete the field problem. A long-duration creep strain must additionally enter the inelastic strain decomposition.

The principal-stress design state is

$$\begin{aligned}\boldsymbol{\sigma}_{\text{principal}} &= \text{diag}(342.5, 185.2, -16.6) \text{ MPa}, \\ \sigma_1 &= +342.5 \text{ MPa}, \\ \sigma_2 &= +185.2 \text{ MPa}, \\ \sigma_3 &= -16.6 \text{ MPa}.\end{aligned}\quad (211)$$

These quantities are principal stress values, not principal direction vectors. If $\mathbf{R}$ contains the orthonormal principal directions, the physical-coordinate tensor is $\boldsymbol{\sigma} = \mathbf{R}\boldsymbol{\sigma}_{\text{principal}}\mathbf{R}^T$. Axial, circumferential, and radial directions coincide with the principal axes only when the local shear components vanish in that basis.

The rotationally invariant equivalent stress and yield safety factor follow directly:

$$\begin{aligned}\sigma_{\text{vM}} &= \sqrt{\frac{1}{2}[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \\ &= \sqrt{\frac{1}{2}[157.3^2 + 201.8^2 + 359.1^2]} \text{ MPa} \\ &= 311.784653 \text{ MPa} \simeq 311.78 \text{ MPa}, \\ \sigma_y &= 850.0 \text{ MPa}, \\ \text{SF} &= \sigma_y/\sigma_{\text{vM}} = 2.72624 \simeq 2.73, \\ \sigma_{\text{allow}} &= 850.0/2.20 = 386.363636 \text{ MPa}.\end{aligned}\quad (212)$$

The specified principal stresses satisfy the $\text{SF} \geq 2.20$ yield criterion with the assigned yield strength. This algebraic check does not establish the principal stresses from the applied loads or establish their spatial maximum. Structural qualification requires a converged transient solution, temperature- and heat-treatment-specific strength data, and cyclic plastic-strain evaluation at the bevel and contact interfaces. A yield safety factor does not replace a fatigue or creep assessment.

C. Coffin–Manson Fatigue Acceptance Criterion

For the selected Inconel X-750 spring-stack criterion, $\Delta\epsilon_p$ denotes plastic strain *amplitude*, one half of the full cyclic range, in mm/mm. The design coefficients are $\epsilon'_f = 0.45$ and

$c = -0.60$ [90]. The Coffin–Manson relation and its inverse are

$$\begin{aligned}\Delta\epsilon_p &= \epsilon'_f (2N_f)^c, \\ N_f &= \frac{1}{2} \left(\frac{\Delta\epsilon_p}{\epsilon'_f} \right)^{1/c}, \\ \Delta\epsilon_{p,\text{max}} &= 0.45(2 \times 44820)^{-0.60} \\ &= 0.0004805199299 \dots, \\ \Delta\epsilon_p &\leq 4.805 \times 10^{-4} \text{ mm/mm} \\ &= 0.04805\%.\end{aligned}\quad (213)$$

Because $c < 0$, the selected ceiling is conservative relative to the exact threshold and gives $N_f \geq 44823.0984$ cycles within this constant-amplitude law. This satisfies the 44,820-cycle analytical requirement. The $\epsilon'_f = 0.18$, $c = -0.62$ model and its 0.0001530105 amplitude ceiling are a separate comparison parameterization, not the governing X-750 criterion. The range-based Cu–Sn bondline relation in Eq. (283) remains a different material criterion. Multiaxial strain amplitudes require a specified equivalent-strain or critical-plane measure; neither peak stress nor accumulated plastic strain alone determines that amplitude. Coefficient uncertainty, variable-amplitude loading, and high-temperature dwell damage shall be assessed separately.

The selected strain-to-displacement approximation is

$$\begin{aligned}\Delta\epsilon_p &= \psi \Delta L_{\text{net}}, \quad \psi \simeq 23.84 \text{ m}^{-1}, \\ \Delta L_{\text{net}} &= 20.1 \text{ }\mu\text{m}, \\ \Delta\epsilon_p &= (23.84)(20.1 \times 10^{-6}) \\ &= 4.79184 \times 10^{-4} \text{ mm/mm} \\ &< 4.805 \times 10^{-4} \text{ mm/mm}.\end{aligned}\quad (214)$$

The 20.1 μm net mismatch is the spring-path displacement over the 325 K thermal excursion, after collar/spacer expansion matching. It is not automatically the optical-axis drift. The scaling coefficient is a geometry-dependent design approximation; its use shall be supported by the selected contact and spring model.

D. Monte Carlo Fatigue Endurance

For the specified $N_{MC} = 1,000,000$ -sample run, the fatigue summary gives [91]

$$\begin{aligned}\bar{N}_f &= 88826.6757 \simeq 88827 \text{ cycles,} \\ s_{N_f} &= 21490.0393 \text{ cycles,} \\ I_{95} &= [54822.0399, 138492.3824] \text{ cycles} \\ &\simeq [54822, 138492] \text{ cycles.}\end{aligned}\quad (215)$$

Both the mean and the lower endpoint of the reported 95% lifespan interval exceed the 44,820-cycle specification. The lower-endpoint margin is 10,002.0399 cycles, or 22.3160%, confirming compliance of these reported Monte Carlo endurance statistics with the specified threshold. The simulated mean is not the deterministic life evaluated at the strain ceiling: it reflects the run's uncertain-input population.

The 95% interval describes lifespan dispersion, not uncertainty on the mean. A lower interval endpoint above 44,820 cycles does not establish that every realization exceeds the specification. A reliability assessment additionally requires the sub-threshold probability under the selected strain and material distributions.

a) *Stochastic execution and strain-life convention:* Use $c_+ = -c > 0$ to distinguish the positive fatigue exponent from the signed Coffin–Manson exponent. The cycle-count evaluation is

$$N_{f,\text{code}} = \frac{1}{2} \left(\frac{\epsilon'_{f,\text{code}}}{\Delta \epsilon_{p,\text{code}}} \right)^{1/c_+}. \quad (216)$$

The factor 1/2 converts reversals to cycles. Positive plastic strain is sampled from a truncated normal density,

$$f_X(x) = \frac{\varphi((x - \mu_X)/s_X)}{s_X[\Phi((b - \mu_X)/s_X) - \Phi((a - \mu_X)/s_X)]}, \quad (217)$$

for $a \leq x \leq b$ and zero otherwise, where $0 < a < b$, $s_X > 0$, and φ, Φ are the standard normal density and cumulative distribution. Physical-support sampling may use $a = 10^{-6}$ and $b = \epsilon_{\text{ult}}$; the comparison value $\epsilon_{\text{ult}} = 0.45$ is not the spring acceptance ceiling. Conditioning on compliant spring realizations instead requires $b \leq 4.805 \times 10^{-4}$ mm/mm. Positive fatigue coefficients and exponents require admissible distributions as well. Vectorized cycle evaluation uses $N_{MC} = 10^6$ with a fixed generator, seed 42, and worker partition. The 88,827-cycle mean and its interval remain a separate endurance operating point: changing the reversal convention or truncation changes the distribution and does not analytically preserve those statistics.

E. Thin Conical Shell Energy and Nonlinear Equilibrium

a) *Reference surface and consistent displacement measures:* The spring is modeled as an axisymmetric, initially stress-free conical midsurface with inner radius $r_i = D_i/2$, outer radius $r_e = D_e/2$, thickness t , and semi-angle α measured from the cone axis. The meridional reference arclength

is denoted ξ , leaving s available for the disc-compression coordinate in the comparison load law. Define

$$\begin{aligned}\xi_i &= r_i / \sin \alpha, & \xi_e &= r_e / \sin \alpha, \\ r_0(\xi) &= \xi \sin \alpha, & z_0(\xi) &= \xi \cos \alpha, \\ h_0 &= (r_e - r_i) \cot \alpha, \\ \mathbf{X}_0(\xi, \theta) &= r_0 \mathbf{e}_r(\theta) + z_0 \mathbf{e}_z, \\ \mathbf{x}(\xi, \theta) &= R(\xi) \mathbf{e}_r(\theta) + Z(\xi) \mathbf{e}_z, \\ R &= r_0 + u \sin \alpha + w \cos \alpha, \\ Z &= z_0 + u \cos \alpha - w \sin \alpha.\end{aligned}\quad (218)$$

Here u is meridional displacement and w is displacement along the reference normal $\mathbf{n}_0 = \cos \alpha \mathbf{e}_r - \sin \alpha \mathbf{e}_z$. The axial displacement is $Z - z_0$, not w . Prime denotes $d/d\xi$. The Kirchhoff–Love assumptions constrain thickness directors to remain normal to the midsurface and neglect transverse shear; the geometric strain measures are obtained from the deformed surface rather than a rigid cross-section rotation [105]. This thin-shell model assumes small elastic strains and t small relative to the relevant curvature and span lengths. It is not an exact three-dimensional contact or plasticity model for an arbitrarily thick washer.

With $\lambda = \sqrt{R'^2 + Z'^2} > 0$, the deformed normal and the normalized reference-metric strains are

$$\begin{aligned}\mathbf{n}_c &= \frac{Z' \mathbf{e}_r - R' \mathbf{e}_z}{\lambda}, \\ \varepsilon_\xi &= \frac{1}{2}(R'^2 + Z'^2 - 1), \\ \varepsilon_\theta &= \frac{1}{2}(R^2/r_0^2 - 1), \\ \varepsilon_\xi &= u' + \frac{1}{2}(u'^2 + w'^2), \\ \varepsilon_\theta &= \frac{a}{r_0} + \frac{a^2}{2r_0^2}, \quad a = u \sin \alpha + w \cos \alpha.\end{aligned}\quad (219)$$

Choose the second fundamental form as $b_{ab} = -\mathbf{n}_c \cdot \mathbf{x}_{,ab}$ and pull its change back to the reference metric. Then

$$\begin{aligned}J &= R'Z'' - Z'R'', \\ \chi_\xi &= b_{\xi\xi} - b_{\xi\xi}^0 = J/\lambda, \\ \chi_\theta &= \frac{b_{\theta\theta} - b_{\theta\theta}^0}{r_0^2} = \frac{RZ'/\lambda - r_0 \cos \alpha}{r_0^2}, \\ b_{\xi\xi}^0 &= 0, \quad b_{\theta\theta}^0 = r_0 \cos \alpha.\end{aligned}\quad (220)$$

At first order in u, w , these expressions give

$$\begin{aligned}\varepsilon_\xi^{(1)} &= u', & \varepsilon_\theta^{(1)} &= a/r_0, \\ \chi_\xi^{(1)} &= -w'', & \chi_\theta^{(1)} &= \frac{a \cos \alpha}{r_0^2} - \frac{w'}{\xi}.\end{aligned}\quad (221)$$

The circumferential curvature change therefore contains the initial-curvature contribution as well as the slope term. For $\alpha = \pi/2$, the flat annular-plate limit has $\chi_\theta^{(1)} = -w'/\xi$. The undeformed cone and a uniform axial translation both have zero membrane strain and zero curvature change, providing rigid-motion checks on the conventions.

b) *Membrane and bending energy integrals:* At prescribed temperature T , the isotropic plane-stress rigidities and the potential energy are

$$\begin{aligned} C_m(\xi, T) &= \frac{E(\xi, T)t}{1 - \nu(\xi, T)^2}, \\ D_b(\xi, T) &= \frac{E(\xi, T)t^3}{12[1 - \nu(\xi, T)^2]}, \\ U_m &= \pi \int_{\xi_i}^{\xi_e} r_0 C_m (\varepsilon_\xi^2 + \varepsilon_\theta^2 + 2\nu\varepsilon_\xi\varepsilon_\theta) d\xi, \\ U_b &= \pi \int_{\xi_i}^{\xi_e} r_0 D_b (\chi_\xi^2 + \chi_\theta^2 + 2\nu\chi_\xi\chi_\theta) d\xi, \\ \Pi[R, Z] &= U_m + U_b - W_{\text{ext}}. \end{aligned} \quad (222)$$

The factor $2\pi r_0 d\xi$ is the reference area element; the factor $1/2$ in each elastic energy gives the prefactor π . Positive E and physically admissible ν give positive quadratic material energy, but geometric nonlinearity can still produce nonconvex load–deflection behavior. For a dead axial compressive force F conjugate to the reduction in cone height,

$$\begin{aligned} \Delta[R, Z] &= h_0 - [Z(\xi_e) - Z(\xi_i)], \\ W_{\text{ext}} &= F\Delta + \int_{\xi_i}^{\xi_e} (q_R[R - r_0] + q_Z[Z - z_0]) d\xi. \end{aligned} \quad (223)$$

The distributed forces q_R, q_Z have units N/m per reference meridional length. Thermal, plastic, and creep eigenstrains must be subtracted from the geometric strains when those mechanisms are active; the derivation below holds their internal variables fixed during each equilibrium variation. Temperature-dependent constitutive inputs are not determined by geometry alone.

The energetically conjugate membrane resultants and bending moments are

$$\begin{aligned} N_\xi &= C_m(\varepsilon_\xi + \nu\varepsilon_\theta), \\ N_\theta &= C_m(\varepsilon_\theta + \nu\varepsilon_\xi), \\ M_\xi &= D_b(\chi_\xi + \nu\chi_\theta), \\ M_\theta &= D_b(\chi_\theta + \nu\chi_\xi), \\ \delta U &= 2\pi \int_{\xi_i}^{\xi_e} r_0 (N_\xi \delta\varepsilon_\xi + N_\theta \delta\varepsilon_\theta \\ &\quad + M_\xi \delta\chi_\xi + M_\theta \delta\chi_\theta) d\xi. \end{aligned} \quad (224)$$

The N resultants have units N/m and the M resultants units N. A symmetric homogeneous thickness is assumed here; an unsymmetric laminate additionally couples membrane and bending strains through a constitutive coupling matrix.

c) *Explicit Euler–Lagrange equations:* Write the first variation in the independent radial and axial coordinates as

$$\delta U = \int_{\xi_i}^{\xi_e} [\mathbf{A} \cdot \delta \mathbf{y} + \mathbf{B} \cdot \delta \mathbf{y}' + \mathbf{D} \cdot \delta \mathbf{y}'] d\xi, \quad \mathbf{y} = (R, Z)^T. \quad (225)$$

Differentiating Eqs. (219) and (220) gives all coefficients explicitly:

$$\begin{aligned} A_R &= \frac{2\pi}{r_0} \left(N_\theta R + M_\theta \frac{Z'}{\lambda} \right), \quad A_Z = 0, \\ B_R &= 2\pi r_0 \left[N_\xi R' + M_\xi \left(\frac{Z''}{\lambda} - \frac{J R'}{\lambda^3} \right) - \frac{M_\theta R Z' R'}{r_0^2 \lambda^3} \right], \\ B_Z &= 2\pi r_0 \left[N_\xi Z' + M_\xi \left(-\frac{R''}{\lambda} - \frac{J Z'}{\lambda^3} \right) + \frac{M_\theta R R'^2}{r_0^2 \lambda^3} \right], \\ D_R &= -2\pi r_0 M_\xi \frac{Z'}{\lambda}, \quad D_Z = 2\pi r_0 M_\xi \frac{R'}{\lambda}. \end{aligned} \quad (226)$$

In particular, the symbols $\mathbf{D}$ and D_b denote different objects: boundary-variation coefficients and material bending rigidity. Integrating twice by parts yields

$$\begin{aligned} \delta \Pi &= \int_{\xi_i}^{\xi_e} (\mathbf{A} - \mathbf{B}' + \mathbf{D}'' - \mathbf{q}) \cdot \delta \mathbf{y} d\xi \\ &\quad + [(\mathbf{B} - \mathbf{D}') \cdot \delta \mathbf{y} + \mathbf{D} \cdot \delta \mathbf{y}']_{\xi_i}^{\xi_e} - F\delta\Delta, \\ A_R - B'_R + D''_R &= q_R, \\ -B'_Z + D''_Z &= q_Z. \end{aligned} \quad (227)$$

Together with Eq. (226), these are the coupled nonlinear meridional differential equations. They retain derivatives of spatially varying material properties through N and M . The boundary term supplies the generalized force $\mathbf{B} - \mathbf{D}'$ and bending work $\mathbf{D} \cdot \delta \mathbf{y}'$, with opposite endpoint signs at ξ_i and ξ_e . A free, moment-free edge has zero prescribed generalized force and $M_\xi = 0$; an imposed position or rotation instead removes the corresponding variation. A flat contacting support is not automatically a clamp: its position and rotation constraints follow from the active contact region.

d) *Stationary load curve, tangent stiffness, and stability:* For frictionless supports at prescribed compression, define the constrained equilibrium branch by

$$\begin{aligned} \mathcal{E}(\Delta, T) &= \min_{\mathbf{y} \in \mathcal{K}(\Delta)} \{U_m[\mathbf{y}, T] + U_b[\mathbf{y}, T]\}, \\ F(\Delta, T) &= \frac{\partial \mathcal{E}}{\partial \Delta}, \\ k_{\text{disc}}(\Delta, T) &= \frac{\partial^2 \mathcal{E}}{\partial \Delta^2}. \end{aligned} \quad (228)$$

The admissible set $\mathcal{K}(\Delta)$ contains the geometric support and unilateral-contact constraints below. These derivatives apply on a smooth equilibrium branch; contact changes and snap-through may require one-sided tangents or branch tracking. The derivative is with respect to the imposed compression, not a pointwise derivative of energy with respect to an arbitrary displacement field. On a strictly stable active contact set, admissible vibration modes satisfy

$$\begin{aligned} \mathbf{K}_c \mathbf{v} &= \omega_s^2 \mathbf{M}_c \mathbf{v}, \\ \mathbf{K}_c &= \mathbf{Z}_c^T \nabla_{\mathbf{aa}}^2 \mathcal{L}_c \mathbf{Z}_c, \\ \mathbf{M}_c &= \mathbf{Z}_c^T \mathbf{M}_s \mathbf{Z}_c, \quad \mathbf{J}_c \mathbf{Z}_c = 0. \end{aligned} \quad (229)$$

Here $\mathbf{a}$ are displacement degrees of freedom, $\mathbf{J}_c$ is the active constraint Jacobian, and $\mathbf{M}_s$ is the consistent shell mass matrix. Positive $\mathbf{M}_c$ and positive constrained stiffness imply positive squared frequencies for this conservative, fixed-contact

linearization after rigid modes are restrained. They do not prove stability against non-axisymmetric buckling, opening contacts, or frictional sliding. For four identical discs on the same smooth series branch, $x = 4\Delta$ gives $dF/dx = k_{\text{disc}}/4$ before adding frame compliance. The target 5.0×10^6 N/m remains a load-path requirement to be tested against the solved branch.

F. Signorini Contact, Variational Inequalities, and Friction

a) *Gap and traction sign convention:* Let Γ_C^i, Γ_C^e denote the inner and outer contact flats. At a potential contact point on a Kirchhoff–Love face, use

$$\begin{aligned} \mathbf{x}_c &= \mathbf{x} + \zeta_c \mathbf{n}_c, & \zeta_c &\in \{-t/2, t/2\}, \\ g_N[\mathbf{y}] &= (\mathbf{x}_c - \mathbf{x}_S) \cdot \mathbf{n}_S, \\ g_N &\geq 0, & \mathbf{n}_b &= -\mathbf{n}_S, \\ \mathbf{t}_c &= \boldsymbol{\sigma} \mathbf{n}_b = p_N \mathbf{n}_S + \boldsymbol{\tau}_T, \\ \sigma_N &= \mathbf{t}_c \cdot \mathbf{n}_b = -p_N. \end{aligned} \quad (230)$$

The support normal $\mathbf{n}_S$ points from the rigid plane into the admissible washer half-space, so positive gap means separation. It points oppositely at the two opposing supports. The relation $\mathbf{n}_b = -\mathbf{n}_S$ applies on smooth closed contact patches; edge reactions are instead defined weakly through the gap variation. The physical face and rotating director enter the gap; replacing them by a midsurface clearance is a separate contact approximation. The Signorini conditions are [106]

$$\begin{aligned} g_N &\geq 0, & p_N &\geq 0, & p_N g_N &= 0, \\ \text{equivalently } g_N &\geq 0, & \sigma_N &\leq 0, & \sigma_N g_N &= 0. \end{aligned} \quad (231)$$

b) *Complementarity and boundary equilibrium proof:*

With normal-contact pressure as a nonnegative multiplier, define

$$\begin{aligned} \mathcal{L}_c[\mathbf{y}, p_N] &= \Pi[\mathbf{y}] - \int_{\Gamma_C} p_N g_N[\mathbf{y}] dA, \\ \delta_{\mathbf{y}} \mathcal{L}_c[\mathbf{v}] &= \delta \Pi[\mathbf{v}] - \int_{\Gamma_C} p_N \delta g_N[\mathbf{v}] dA = 0, \\ \partial_{p_N} \mathcal{L}_c &= -g_N. \end{aligned} \quad (232)$$

The multiplier saddle problem minimizes over admissible displacements and maximizes over $p_N \geq 0$. If $g_N < 0$, the term $-p_N g_N$ is unbounded above, so a saddle point cannot interpenetrate. If $g_N > 0$, maximization requires $p_N = 0$. If $g_N = 0$, a nonnegative multiplier supplies the compressive reaction. These alternatives prove Eq. (231). The displacement variation balances the shell boundary work against $p_N \delta g_N$ and therefore fixes the reaction sign in Eq. (230); prescribed contact forces are not added a second time to the same support work.

For an affine gap and an admissible displacement $\mathbf{z}$, stationarity and complementarity imply the variational inequality

$$\begin{aligned} \delta \Pi[\mathbf{y}; \mathbf{z} - \mathbf{y}] &= \int_{\Gamma_C} p_N \{g_N[\mathbf{z}] - g_N[\mathbf{y}]\} dA \\ &= \int_{\Gamma_C} p_N g_N[\mathbf{z}] dA \geq 0. \end{aligned} \quad (233)$$

Localized normal variations give the converse open/closed-contact alternatives when multiplier regularity and a constraint

qualification hold. For the nonlinear face gap, the corresponding first-order statement holds for feasible tangent directions, not arbitrary finite displacement differences. A stationary point is not automatically a minimum of the nonlinear shell energy. A necessary second-order condition is

$$\delta_{\mathbf{y}\mathbf{y}}^2 \mathcal{L}_c[\mathbf{v}, \mathbf{v}] \geq 0 \quad \text{for } \mathbf{v} \text{ in the critical cone.} \quad (234)$$

The critical cone contains feasible first-order directions with zero first-order objective change. Strict positivity there, with the usual smoothness and constraint qualifications, gives a strict local minimum in a regular finite-dimensional discretization; it does not prove global uniqueness. For a fixed active set, the frictionless Newton equations are

$$\begin{bmatrix} \mathbf{H}_c & -\mathbf{J}_c^\top \\ -\mathbf{J}_c & 0 \end{bmatrix} \begin{bmatrix} \delta \mathbf{a} \\ \delta \mathbf{p} \end{bmatrix} = \begin{bmatrix} -\mathbf{r}_c \\ \mathbf{g}_c \end{bmatrix}, \quad (235)$$

where $\mathbf{r}_c = \nabla_{\mathbf{a}} \Pi - \mathbf{J}_c^\top \mathbf{p}$, $\mathbf{H}_c = \nabla_{\mathbf{a}\mathbf{a}}^2 \mathcal{L}_c$, and $\mathbf{g}_c$ lists the active gaps. Negative active pressures or closing inactive gaps require an active-set change. The Hessian includes geometric contact terms and is not merely the material stiffness.

c) *Stick-slip law and dissipation proof:* Define the relative tangential velocity and Coulomb admissible disk by

$$\begin{aligned} \mathbf{v}_T &= (\mathbf{I} - \mathbf{n}_S \mathbf{n}_S^\top)(\dot{\mathbf{x}}_c - \dot{\mathbf{x}}_S), \\ \mathcal{D}(p_N) &= \{\boldsymbol{\tau} : \boldsymbol{\tau} \cdot \mathbf{n}_S = 0, \|\boldsymbol{\tau}\| \leq \mu p_N\}, \\ \boldsymbol{\tau}_T &\in \arg \min_{\boldsymbol{\tau} \in \mathcal{D}(p_N)} \boldsymbol{\tau} \cdot \mathbf{v}_T. \end{aligned} \quad (236)$$

Minimization of this linear functional over the disk gives

$$\begin{aligned} \mathbf{v}_T = 0 &: \|\boldsymbol{\tau}_T\| \leq \mu p_N, \\ \mathbf{v}_T \neq 0 &: \boldsymbol{\tau}_T = -\mu p_N \frac{\mathbf{v}_T}{\|\mathbf{v}_T\|}, \\ \boldsymbol{\tau}_T \cdot \mathbf{v}_T &= -\mu p_N \|\mathbf{v}_T\| \leq 0. \end{aligned} \quad (237)$$

Equivalently, $-\boldsymbol{\tau}_T \in \mu p_N \partial \|\mathbf{v}_T\|$; the subdifferential at zero velocity is the unit disk. Equality of shear magnitude with μp_N permits impending slip and does not alone imply nonzero sliding velocity. On an open gap, $p_N = 0$ implies zero friction traction. On sustained closed contact with fixed rigid supports, $p_N \dot{g}_N = 0$ gives zero normal contact power, while tangential slip dissipates energy. Impact needs a separate impulsive law and is outside the quasistatic Signorini proof.

Frictional cycling is path dependent and cannot be derived from a single conservative potential $U_m + U_b - F\Delta$. At prescribed normal pressure during a tangential increment, the additional dissipation functional is

$$\mathcal{R}(\Delta \mathbf{u}_T; p_N) = \int_{\Gamma_C} \mu p_N \|\Delta \mathbf{u}_T\| dA \geq 0. \quad (238)$$

When p_N evolves with the displacement, the coupled problem is a contact variational inequality rather than unconstrained energy minimization. A law $\mu(T, p_N, \text{surface state})$ requires independently characterized contact properties; normal complementarity does not determine friction or creep. The shell solution supplies loading/unloading forces and tangent stiffnesses for the four-disc assembly while preserving the preload and pressure-variation bounds below.

G. Four-Disc Belleville Compliance and Preload Control

The native structural material module encodes $K_{\text{STACK}} = 5.0 \times 10^6 \text{ N/m}$, $DL_{\text{NET}} = 20.1 \mu\text{m}$, the clamp-force bounds, and the strain ceiling [98]. These constants specify the selected operating point; a stored stiffness is not a solved nonlinear spring curve. Sections III-E–III-F define the energy and contact boundary-value problem for that curve. The structural crate's disc-spring models and the workbench's BellevillePreload force interface provide an implementation connection to the following reduced Almen–László comparison law [95], [99]; they do not establish an implementation of the full shell/contact system.

The selected core-interface force shall remain within Eq. (208) over 25–350 °C, with cyclic pressure deviation bounded by $\pm 0.50 \text{ MPa}$. The 15.80 MPa preload belongs to a separate, larger-area comparison configuration and is below the selected 20 MPa sealing minimum. The optimized Belleville stack has force stiffness $K_{\text{stack}} = 5000 \text{ N/mm} = 5.0 \times 10^6 \text{ N/m}$. A four-disc series model provides the load-deflection framework; disc count alone does not establish the optimized stiffness. Force stiffness and stiffness per loaded area are distinguished by

$$\begin{aligned} k_{\text{path}} &= \frac{dP}{dx} \quad [\text{N/m}], \\ K_A &= \frac{k_{\text{path}}}{A_{\text{load}}} \quad [\text{Pa/m}], \\ \Delta p &= \frac{1}{A_{\text{load}}} \int_{x_0}^{x_0 + \Delta L_{\text{net}}} k_{\text{path}}(x, T) dx. \end{aligned} \quad (239)$$

Here P is axial force, x is load-path deflection, and A_{load} is the effective area transmitting that force. For the selected contact configuration, $A_{\text{load}} = A_{\text{contact}}$, $k_{\text{path}} = K_{\text{stack}}$ at the working point, and $K_A = 15.9154943 \text{ GPa/m}$. The area-normalized stiffness K_A shall not be divided by area a second time.

For an ideal rectangular-section disc, the Almen–László force and tangent stiffness are [85]

$$\begin{aligned} P(s, T) &= \frac{4E(T)t^4}{(1 - \nu^2)MD_e^2} \\ &\quad \times \left[\left(\frac{h_0}{t} - \frac{s}{t} \right) \left(\frac{h_0}{t} - \frac{s}{2t} \right) + 1 \right] \frac{s}{t}, \\ k_{\text{single}}(s, T) &= \frac{4E(T)t^3}{(1 - \nu^2)MD_e^2} \\ &\quad \times \left[\left(\frac{h_0}{t} \right)^2 - 3\frac{h_0 s}{t^2} + \frac{3}{2} \left(\frac{s}{t} \right)^2 + 1 \right]. \end{aligned} \quad (240)$$

The four-disc comparison geometry uses $D_e = 0.250 \text{ m}$, $D_i = 0.125 \text{ m}$, $t = 12.0 \text{ mm}$, and free cone height $h_0 = 17.4 \text{ mm}$, so $h_0/t = 1.45$. These dimensions are not a demonstrated realization of the optimized 5,000 N/mm stack. The dimensionless factor M depends on the diameter ratio and contact convention. For the ideal no-contact-flat formulation,

$$M = \frac{1}{\pi} \frac{[(\delta - 1)/\delta]^2}{(\delta + 1)/(\delta - 1) - 2/\ln \delta}, \quad \delta = \frac{D_e}{D_i} = 2, \quad (241)$$

which gives $M = 0.69433320$. The alternate assigned factor $M = 0.681$ defines a distinct geometry parameterization. The

expression $(6/\pi)(\delta - 1)/\ln \delta$ evaluates to 2.75534457 at $\delta = 2$ and cannot define either value. Contact flats, reduced thickness, and edge radii require their corresponding load-deflection model.

The spring-modulus parameterization uses $E_0 = 193 \text{ GPa}$, $\nu = 0.30$, and

$$E(T) = E_0 [1 - (2.8 \times 10^{-4} \text{ K}^{-1})(T - 298.15 \text{ K})]. \quad (242)$$

It gives $E(623.15 \text{ K}) = 175.437 \text{ GPa}$, distinct from the 195.0 GPa structural input in Eq. (210). These material parameterizations require separate heat-treatment and property qualification.

For identical discs under equal force, $x = n_s s$ with $n_s = 4$. Series disc and frame compliances combine as

$$\begin{aligned} \frac{1}{k_{\text{path}}} &= \sum_{i=1}^{n_s} \frac{1}{k_{\text{single},i}} + \frac{1}{k_{\text{frame}}}, \\ k_{\text{path}}^{\pm}(x, T) &= \left[\frac{n_s(1 \pm \mu h_0/t)}{k_{\text{single}}(x/n_s, T)} + \frac{1}{k_{\text{frame}}} \right]^{-1}. \end{aligned} \quad (243)$$

The second expression is a phenomenological loading/unloading friction parameterization, with comparison value $\mu = 0.095 \pm 0.015$; it is not a substitute for a contact solution. The nominal total disc compression $x_0 = 18.25 \text{ mm}$ gives $s_0 = 4.5625 \text{ mm}$ and $s_0/h_0 = 0.262213$. A compliant frame contributes additional total load-path motion and shall not be included again in $x = n_s s$.

The selected thermal compliance specification is [90]

$$\begin{aligned} \Delta L_{\text{net}} &= 20.1 \mu\text{m}, \\ K_{\text{stack}, \text{max}} &= \frac{(0.50 \times 10^6) A_{\text{contact}}}{20.1 \times 10^{-6}} \\ &= 7.8149071 \times 10^6 \text{ N/m}, \\ K_{\text{stack}} &= 5000 \text{ N/mm} = 5.0 \times 10^6 \text{ N/m}, \\ \Delta F &= K_{\text{stack}} \Delta L_{\text{net}} = 100.5 \text{ N}, \\ \Delta P \equiv \Delta p &= \frac{\Delta F}{A_{\text{contact}}} \\ &= 0.319901436 \text{ MPa} \simeq 0.320 \text{ MPa} \\ &< 0.50 \text{ MPa}. \end{aligned} \quad (244)$$

The calculated cyclic contact-pressure deviation of 0.320 MPa complies with the $\pm 0.50 \text{ MPa}$ tolerance band, leaving 0.180098564 MPa of single-sided allowance before other uncertainties. Here ΔP denotes pressure variation, distinct from the force function $P(s, T)$ in the disc model; it is not the local multiaxial bevel stress. For a conservative excursion of $\pm 20.1 \mu\text{m}$ about the preload setting, the thermal force reserve is $\pm 100.5 \text{ N}$. The nominal force shall therefore remain at least 100.5 N inside each instantaneous force bound, with additional reserve for creep and setting uncertainty. The estimate assumes linear stiffness over the excursion; nonlinear acceptance shall use the maximum tangent stiffness or the full force integral.

The optimized force stiffness is not established by disc count alone. For example, Eq. (240) with $M = 0.681$, $s_0 = 4.5625 \text{ mm}$, and the spring modulus at 350 °C gives $P(s_0) = 335.37 \text{ kN}$ and an ideal four-disc force stiffness $k_{\text{single}}/4 = 13.0354 \text{ MN/m}$ before frame compliance. These comparison values do not realize the selected 5.0 MN/m stack or the

contact-force bounds. The optimized stack therefore requires compatible working deflection, loaded area, disc geometry, and a qualified load-deflection characteristic. The 7.7952 μm displacement and 58.420 GPa/m pressure stiffness are non-governing comparison inputs for a different load path.

H. High-Temperature Relaxation and Contact Tribology

a) *Vacancy-controlled spring relaxation*: For an Inconel X-750 spring at 350 °C, relaxation requires its own phase, precipitate, vacancy, and dislocation parameters. The vacancy-climb framework in Eqs. (311)–(319) supplies the mechanism, but Cu–Sn material functions do not transfer to X-750. In a fixed-strain scalar restriction with constant modulus,

$$\begin{aligned} \frac{d\sigma}{dt} &= -E\dot{\epsilon}_{\text{cr}}, \\ \sigma(t) &= \sigma_0 - E \int_0^t \dot{\epsilon}_{\text{cr}}(t') dt'. \end{aligned} \quad (245)$$

Here $\dot{\epsilon}_{\text{cr}}$ is projected from the resolved dislocation strain rate, not an assigned power of stress. With $\sigma_0 = 400$ MPa, the comparison endpoint 384.2 MPa corresponds to 3.95% loss but is not predicted without the defect and loading history. The exposure allocation below remains distinct from a constitutive kinetic calculation. Product form, precipitation heat treatment, stress state, and measured relaxation response govern spring qualification.

The sustained-temperature condition is 350 °C for 10,000 operating hours. In the Rankine/1000 convention, the Larson–Miller exposure parameter is [80]

$$\begin{aligned} P_{\text{LM}} &= \frac{T_R/(1^\circ\text{R})}{1000} \left[C_{\text{LM}} + \log_{10} \left(\frac{t_{\text{hold}}}{1 \text{ h}} \right) \right], \\ T_R &= 1.8(623.15)^\circ\text{R} = 1121.67^\circ\text{R}, \\ C_{\text{LM}} &= 20.0, \quad t_{\text{hold}} = 10000 \text{ h}, \\ P_{\text{LM}} &= 1.12167(20 + 4) = 26.92008 \simeq 26.92. \end{aligned} \quad (246)$$

Temperature and time units are part of this empirical convention. A Kelvin-based correlation cannot use the same numerical parameter without conversion. The assigned cumulative force-loss fraction is $R_{\text{creep}} = 4.18\%$; mapping P_{LM} to that fraction requires a stress-relaxation curve for the applicable spring condition and initial stress. The exposure parameter alone does not predict force loss. Likewise, 10,000 hours and 44,820 cycles are independent duration and cycle-count requirements unless a cycle and dwell schedule relates them.

For an initial spring force $P_0 = 45.0$ kN, the relaxation scenario gives

$$\begin{aligned} \Delta P_{\text{creep}} &= P_0 R_{\text{creep}} = 45.0(0.0418) = 1.881 \text{ kN}, \\ P_{\text{relaxed}} &= 45.000 - 1.881 = 43.119 \text{ kN}. \end{aligned} \quad (247)$$

The 45.0 kN spring force is distinct from the 125.0 kN axial retaining load in the bevel load case. The following larger-area comparison assumes that the former produces 15.80 MPa through a direct load path; it does not use the selected 20 mm contact diameter:

$$\begin{aligned} A_{\text{load}} &= 45000/(15.80 \times 10^6) = 2.84810127 \times 10^{-3} \text{ m}^2, \\ \Delta p_{\text{creep}} &= \Delta P_{\text{creep}}/A_{\text{load}} = 0.66044 \text{ MPa}, \\ p_{\text{relaxed}} &= 15.13956 \text{ MPa}. \end{aligned} \quad (248)$$

The relaxed pressure is below the comparison configuration's 15.30 MPa lower acceptance limit even before a thermally induced reduction. At that same comparison area, $K_A = 58.420$ GPa/m requires $k_{\text{path}} = 166.3861$ MN/m, not the optimized force stiffness. For the selected contact area, any long-term force loss shall be converted using Eq. (207) and reserved within the force and pressure limits independently of the cyclic 0.320 MPa excursion. A thermally compliant stack alone does not compensate long-term force relaxation.

All disc contact surfaces are specified with magnetron-sputtered molybdenum disulfide (MoS_2) over a 200 nm diamond-like carbon (DLC) interlayer. The static-friction acceptance limit is $\mu_{\text{stat}} \leq 0.105$ in the operating vacuum/deuterium environment; the dynamic comparison value is $\mu_{\text{dyn}} = 0.080$. The coating process shall define MoS_2 thickness, adhesion, coverage, counterface finish, and conditioning. These properties and the friction limit require qualification at temperature, contact pressure, and accumulated sliding distance; the coating designation alone does not establish friction or wear endurance.

For force-deflection cycling, the dissipated work is the loop area,

$$\begin{aligned} \Delta W_{\text{hyst}} &= \oint P(s, \dot{s}) ds, \\ \Delta W_{\text{hyst, approx}} &= 2(\mu_{\text{stat}} - \mu_{\text{dyn}})P_0 s_{\text{max}}, \\ s_{\text{max}} &= 1.25 \text{ mm}, \\ \Delta W_{\text{uncoated}} &= 2(0.125 - 0.080)(45000)(0.00125) \quad (249) \\ &= 5.0625 \text{ J/cycle}, \\ \Delta W_{\text{coated}} &= 2(0.105 - 0.080)(45000)(0.00125) \\ &= 2.8125 \text{ J/cycle}. \end{aligned}$$

This approximation requires a displacement coordinate consistent with the force and contact-slip model; millijoule values do not follow from the specified force and stroke. The secant allowance

$$k_{\text{allow, sec}} = 36.00 - \frac{1.881}{1.25} = 34.4952 \text{ kN/mm} \quad (250)$$

allocates the creep force loss over the stated stroke. It is not the tangent stiffness of a relaxed nonlinear spring unless a constitutive relaxation model establishes that equivalence. The 36.00 kN/mm comparison uses a force-stiffness convention and shall not be identified with the area-normalized 58.420 GPa/m specification without an explicit common loaded area.

b) *Mechanical acceptance scope*: The principal-stress state satisfies the assigned yield margin, the amplitude ceiling satisfies the stated constant-amplitude fatigue law, and the pressure-stiffness target satisfies the thermal-excursion allocation. Assembly acceptance additionally requires one compatible load-path model that includes contact friction, temperature-dependent properties, creep relaxation, and pressure uncertainty. These conditions govern the datum, isolation, and vacuum-interface requirements that follow.

I. Datum Framework and Geometric Controls

Geometric dimensioning and tolerancing (GD&T) shall follow ASME Y14.5-2018.⁷ The assembly datum reference frame shall use the Invar 36 primary mounting plane as datum *A*, the longitudinal locating plane as datum *B*, and the transverse locating plane as datum *C*. Each component drawing shall identify the local mating datum features that realize this assembly frame, including the correspondence between the diamond seating features and the chassis references.

Table XVII specifies the six controlled features. Flatness controls shall contain no datum reference, including where the controlled surface establishes datum *A*. Bore-axis cylindrical zones shall carry a diameter designation; rib center-plane zones shall use the stated parallel-plane width. Size-feature controls shall have no maximum-material-condition bonus unless explicitly modified on the released drawing. The drawing shall include basic bore locations, guide-pin dimensions, rib pitch, and all geometry needed to apply the schedule. The cold-plate comparison geometry uses 150 μm -wide water–ethylene-glycol passages at a reference flow of 9.83 L min⁻¹. DM-02 specifies testing at 12.5 L min⁻¹, subject to qualification of the pump and channel-flow characteristics. The optical aperture shall be 20.0 $\times$ 20.0 mm, and the mounting envelope shall accommodate a 2.0 mm diamond optic. The axial stack includes a separate 0.500 mm floating-frame seating contribution. The thinner diamond layer in Section IV defines an analytical benchmark; drawings and analysis records shall distinguish these three dimensions.

Dimensional inspection records shall include the reference temperature, probe or optical sampling strategy, datum realization, and measurement uncertainty. The assembly reference temperature shall be 293.15 K. Inspection and stack-height calculations shall treat profile, flatness, and thickness as distinct controls and shall include each measured surface deviation only once in the relevant error budget.

a) *Re-entrant thermal compensation:* The independently heated re-entrant frame uses opposed Invar 36 and Al 6061-T6 members [47]. Differential expansion is evaluated at the specified temperature rise of each column:

$$\begin{aligned}
 L_{\text{Invar}} &= 172.50 \text{ mm}, & \Delta T_{\text{Invar}} &= 256.85 \text{ K}, \\
 L_{\text{Al}} &= 20.69 \text{ mm}, & \Delta T_{\text{Al}} &= 110.0 \text{ K}, \\
 \alpha_{\text{Invar}} &= 1.20 \times 10^{-6} \text{ K}^{-1}, \\
 \alpha_{\text{Al}} &= 23.0 \times 10^{-6} \text{ K}^{-1}, \\
 \Delta L_{\text{net}} &= \alpha_{\text{Invar}} L_{\text{Invar}} \Delta T_{\text{Invar}} - \alpha_{\text{Al}} L_{\text{Al}} \Delta T_{\text{Al}} \\
 &= 53.16795 - 52.34570 = 0.82225 \text{ } \mu\text{m}, \\
 L_{\text{stroke,remaining}} &= 15.00000 - 0.82225 \\
 &= 14.17775 \text{ } \mu\text{m} \quad (94.5183\%).
 \end{aligned} \tag{251}$$

The re-entrant sleeve and return link act with opposite displacement signs, while thermal control maintains the specified column temperatures. The resulting differential travel describes the frame alone. Compensation of seating-stack expansion also

depends on the datum and return-path geometry; the frame residual cannot be identified with the residual displacement of the complete optical stack.

J. ConFlat Vacuum Feedthrough Schedule

The vessel shall support two separately qualified operating states: evacuated service with base pressure $p_{\text{base}} \leq 1.0 \times 10^{-8}$ mbar, and deuterium-loaded service at nominal $p_{D_2} = 4.20$ bar absolute. The turbomolecular pumping path shall be isolated before pressurization. Table XVIII defines the required electrical and optical interfaces from FT-AC-01 through FT-SENS-01. Assembly pressure ratings and state-transition controls shall be qualified for the applicable service conditions.

CF35 and CF40 are interface designations. Procurement drawings shall specify the actual flange outside diameter, knife-edge geometry, bolt circle, gasket, and mating interface. The optical viewport shall use a 2.75-inch CF envelope with 23.0 mm clear aperture. Assembly bakeout shall remain within the lowest certified temperature limit of the window, braze, fiber seal, ceramic, connector, and installed wiring; the 450°C bare metal-flange requirement applies only to that component. The completed assembly shall receive separate vacuum-leak and positive-pressure qualification for deuterium service.

FT-TEG-01 requires verification of current sharing in the parallel supply and return paths. The 24 V, 812.5 W terminal test case defines a connector-qualification condition only:

$$\begin{aligned}
 I_{\text{TEG,total}} &= \frac{812.5 \text{ W}}{24 \text{ V}} = 33.85417 \text{ A}, \\
 I_{\text{TEG,pin}} &= 16.92708 \text{ A} \quad \text{for two equal parallel paths.}
 \end{aligned} \tag{252}$$

Power specified at the input terminals requires no additional division by boost efficiency. At the comparison test point of 30 V and 850 W, the corresponding currents are 28.33333 A total and 14.16667 A per equal path. Qualification against the 30.0 A per-pin limit includes temperature derating, contact resistance, peak current, and measured imbalance. The 500 V_{peak} SMA rating specifies insulation withstand; continuous RF loading requires separate qualification. FT-SENS-01 provides acoustic sensing rather than a qualified power-export path, and the baseline energy balance assigns no 134.8 W PZT generation credit.

Acceptance measurements shall cover helium leakage at each completed seal, continuity, insulation resistance, dielectric withstand under the intended pressure conditions, conductor temperature rise, optical insertion loss, and polarization extinction. Released component specifications shall state the allowable helium leak rate, RF bandwidth where applicable, and pressure rating before procurement acceptance. The gas manifold, pressure relief, and isolation arrangement shall have a service-specific qualification independent of the electrical and optical feedthrough schedule.

K. Vibration Dynamics and Two-Stage Isolation

The small angular error of the prism–substrate interface is denoted by θ , with effective rotational inertia J_{eff} , rotational

⁷ASME, *Dimensioning and Tolerancing*, ASME Y14.5-2018, reaffirmed 2024. The standard defines the drawing language; the numerical tolerances in Table XVII are project requirements.

TABLE XVII
REQUIRED MECHANICAL GD&T SCHEDULE UNDER ASME Y14.5-2018

Component	Controlled feature	Geometric control	Zone (mm)	Datum sequence	Functional requirement
Invar 36 chassis	Laser-mount interface bore axis	Position, cylindrical zone	Diameter 0.005	$A B C$	Register optical path to active region
Invar 36 chassis	Primary mounting base	Flatness	0.002	None; surface establishes A	Control base coplanarity and seating
CVD diamond floating frame	Internal substrate-clamping pocket	Profile of a surface	0.004 total width	$A B$	Control pocket boundary and clamping geometry
CVD diamond floating frame	Guide-pin bore axes	Perpendicularity, cylindrical zone	Diameter 0.003	A	Avoid guide binding during expansion
Microchannel cold plate	Thermal-contact interface face	Flatness	0.003	None; surface may establish A	Limit gaps at the heat-spreader interface
Microchannel cold plate	Rib center planes in the 150 μm channel array	Position, parallel-plane zone	0.010 total width	$A B C$	Maintain channel registration and flow distribution

TABLE XVIII
REQUIRED CF35/CF40 FEEDTHROUGH INTERFACE SCHEDULE

Identifier	Flange	Construction and channel count	Electrical or optical requirement	Shielding and service
FT-AC-01	CF40	Two hermetic high-current conductors, with vacuum-compatible litz-wire terminations	1.0 kV _{rms} insulation; 21.21 A _{rms} at 68.5 kHz	Continuous 360° shield; external Mu-metal sleeve; Helmholtz drive
FT-DC-01	CF35	Twelve-pin, high-density ceramic feedthrough	500 VDC insulation; 5.0 A per pin nominal	Grounded metallic braid; PZT flexure drive and sensor wiring
FT-OPT-01	CF35	Two hermetic polarization-maintaining fused-silica fiber paths	50 mW per path; 785.0 nm and 802.5 nm; no electrical power rating	Dual-wall stainless armor; two DFB optical inputs
FT-TC-01	CF35	Eight Type-K thermocouple pairs, sixteen thermoelement conductors	100 VDC insulation; thermocouple signal service, not load-power service	Individually shielded pairs; core and interface temperatures
FT-TEG-01	CF40	Four copper power pins arranged as two supply and two return conductors	250 VDC insulation; 30.0 A continuous per pin as a qualification requirement	Internal conductive shield; TEG raw output
FT-SENS-01	CF35	Four hermetic SMA coaxial paths	500 V _{peak} insulation; 1.0 A per path; 50 Ω characteristic impedance	Double-shielded coax; PZT acoustic sensing channels

damping C_θ , and rotational stiffness K_θ . Base acceleration $a_b(t)$ acting on effective offset mass m_e at lever arm ℓ_e , together with disturbance forces F_j at signed moment arms ℓ_j , produces the angular response

$$J_{\text{eff}}\ddot{\theta} + C_\theta\dot{\theta} + K_\theta\theta = -m_e\ell_e a_b(t) + \sum_j \ell_j F_j(t) + M_c(t), \quad (253)$$

where M_c is control torque. With zero initial conditions, the base-acceleration response is

$$H_{\theta a}(s) = \frac{\Theta(s)}{A_b(s)} = \frac{-m_e\ell_e}{J_{\text{eff}}s^2 + C_\theta s + K_\theta}. \quad (254)$$

The rotational natural frequency and damping ratio are

$$\omega_{\theta n} = \sqrt{\frac{K_\theta}{J_{\text{eff}}}}, \quad \zeta_\theta = \frac{C_\theta}{2\sqrt{J_{\text{eff}}K_\theta}}. \quad (255)$$

The forcing frequencies are 28.0 and 56.0 Hz for coolant pulsation, and 1500.0 Hz for the turbomolecular rotor. Stage 1 comprises a pneumatic table ($f_{n1} = 1.8$ Hz, $\zeta_1 = 0.05$), and Stage 2 comprises an elastomeric mount ($f_{n2} = 18.0$ Hz,

TABLE XIX
TWO-STAGE AMPLITUDE TRANSMISSIBILITY IN dB

Frequency (Hz)	Nominal	Calculated
28.0	-42.50	-45.1464
56.0	-58.20	-66.7928
1500.0	-134.10	-130.7947

$\zeta_2 = 0.10$). Treating the stages as independent linear base-excited systems gives

$$\mathcal{T}(r, \zeta) = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}, \quad (256)$$

$$\mathcal{T}_{12}(f) = \mathcal{T}(f/f_{n1}, \zeta_1)\mathcal{T}(f/f_{n2}, \zeta_2).$$

The nominal attenuation values differ from those calculated using the specified passive transfer factors. Active feedback, coupled modes, and mechanical transmission through hoses and wiring require additional transfer paths. The nominal integrated tilt $\theta_{\text{RMS}} = 0.00540 \mu\text{rad}$ lies below the $0.00800 \mu\text{rad}$ design limit, but its verification requires the disturbance spectrum and

TABLE XX
COMPLEX REFRACTIVE INDICES FOR THE OPTION B OPTICAL MODEL

Layer	785.0 nm	802.5 nm
Air	1.0000 + i0	1.0000 + i0
Fused silica	1.4536 + i0	1.4533 + i0
Pd-Ir	1.8325 + i4.5015	1.8825 + i4.5825
Ti	2.7600 + i3.8400	2.8200 + i3.9200
CVD diamond	2.4008 + i0	2.3999 + i0

acceleration-to-rotation transfer function; three transmissibility samples do not determine an integrated RMS response.

The integrated tilt requirement is an RMS quantity:

$$\sigma_\theta^2 = \int_0^\infty S_\theta(f) df, \quad (257)$$

$$\sigma_\theta \leq 0.008 \mu\text{rad} = 8.0 \times 10^{-9} \text{ rad},$$

$$S_\theta(f) = |H_{\theta a}(i2\pi f)|^2 |\mathcal{T}_{12}(f)|^2 S_{a_b}(f) + S_{\theta, \text{other}}(f).$$

This decomposition assumes uncorrelated contributions; measured cross-spectral terms shall be included otherwise. Qualification of the 0.008 μrad RMS limit shall use calibrated disturbance spectra, rotational parameters, and sensor-noise budgets. Testing shall cover evacuated and pressurized operation, operating pumps and coolant flow, startup transients, and a declared integration bandwidth. For a 10.0 mm effective lever, the tilt limit corresponds to 0.080 nm differential RMS motion.

a) Monolithic waveguide and optical material parameters:

The nominal optical stack comprises air/fused silica/patterned $\text{Pd}_{0.95}\text{Ir}_{0.05}/\text{Ti}/\text{CVD}$ diamond, distinct from the SF11 smooth-prism comparison. The active metal occupies the Option B volume in Eq. (42); 42.50 nm grating teeth require sufficient patterned-metal thickness and cannot be etched into a 1 nm film. The grating pitch is 960.80 nm, the duty cycle is 50%, and the internal silica incidence angle is 65.0° . The complex refractive indices define the constitutive inputs through $\varepsilon_j = (n_j + ik_j)^2$; alternative permittivity datasets describe separate model cases. With the time convention $e^{-i\omega t}$, the Fourier field expansions are

$$\begin{aligned} \mathbf{E}(x, z) &= \sum_m \mathbf{S}_m(z) e^{ik_{x,m}x}, \\ \mathbf{H}(x, z) &= \sqrt{\epsilon_0/\mu_0} \sum_m \mathbf{U}_m(z) e^{ik_{x,m}x}, \\ k_{x,m} &= k_0 n_{\text{silica}} \sin 65^\circ + 2\pi m/\Lambda. \end{aligned} \quad (258)$$

For the one-dimensional binary profile, let $\mathbf{E}$ and $\mathbf{Q}$ denote the Toeplitz matrices of $\varepsilon(x)$ and $1/\varepsilon(x)$, respectively, and $\mathbf{K}_x = \text{diag}(k_{x,m})$. The TM Fourier-factorized Maxwell system can be written

$$\begin{aligned} \frac{d\mathbf{S}_x}{dz} &= ik_0 [\mathbf{I} - k_0^{-2} \mathbf{K}_x \mathbf{E}^{-1} \mathbf{K}_x] \mathbf{U}_y, \\ \frac{d\mathbf{U}_y}{dz} &= ik_0 \mathbf{Q}^{-1} \mathbf{S}_x, \\ \mathbf{S}_z &= -k_0^{-1} \mathbf{E}^{-1} \mathbf{K}_x \mathbf{U}_y. \end{aligned} \quad (259)$$

The inverse of the Toeplitz permittivity matrix and the inverse of the Toeplitz inverse-permittivity matrix are distinct operators. The latter implements normal-field inverse factorization rather

TABLE XXI
OPTICAL REFERENCE OUTPUTS UNDER THE AMPLITUDE CONVENTION

Quantity	785.0 nm	802.5 nm
$R_{\min}$	0.0142	0.0085
$F_z = E_z / E_0 $	124.50	132.80
$\mathcal{F}_z = F_z^2$	15500.25	17635.84
$L_{\text{spp}} (\mu\text{m})$	12.45	14.12
$\delta_{z,m} (\text{nm})$	14.80	15.20
$\delta_{z,d} (\text{nm})$	185.30	192.10

than a second inversion in the first equation. Tangential-field continuity, outgoing-wave boundary conditions, and harmonic convergence complete the boundary-value problem. A grating patterned in both lateral directions requires both lateral harmonic indices; a one-dimensional TM reduction does not establish a full three-dimensional solution. The field factor F_z is an amplitude ratio, distinct from an intensity enhancement. Table XXI specifies both conventions explicitly. The propagation lengths apply to the corresponding optical model; neither domain volumes nor RCWA results for a 1 nm film transfer to Option B without a geometry-specific calculation. The beam-shaper exponent is $m = 1148$ ($N = 574$ in $2N$ notation), with $w = 10.048711$ mm over the 20 mm square aperture. The active footprint is 1 cm^2 , distinct from the aperture area. Intercepted optical power therefore requires integration over the active pattern.

L. Kretschmann Flexure Stage and Viewport Alignment

The angular stage shall use a monolithic wire-EDM Invar 36 flexure with two preloaded piezoelectric translators. The drive range shall be 0–100 V, full travel 15.0 μm , and effective differential displacement resolution 0.15 nm. For actuator separation d_a , the small-angle relation is

$$\theta \approx \frac{z_1 - z_2}{d_a}, \quad \delta\theta = \frac{\delta z_{\text{diff}}}{d_a}. \quad (260)$$

The specified 0.015 μrad resolution is geometrically consistent with

$$d_a = \frac{0.15 \times 10^{-9} \text{ m}}{0.015 \times 10^{-6} \text{ rad}} = 10.0 \text{ mm}. \quad (261)$$

The effective lever shall be 10.0 mm for this resolution requirement. Angular resolution and integrated RMS accuracy shall be verified separately. At this lever, 15.0 μm of differential stroke corresponds to 1.50 mrad total angular range. Across the separate 20.0 mm optical mounting span, the 3.2 μm edge-to-edge thermal-mismatch design case gives

$$\Delta\theta_{\text{thermal}} = \arctan\left(\frac{3.2 \mu\text{m}}{20.0 \text{ mm}}\right) \approx 160 \mu\text{rad} = 0.009167^\circ. \quad (262)$$

A quadrant photodiode shall measure reflected-beam position, and synchronous reflectance measurements shall identify the SPP resonance. The digital update rate shall be 10 kHz. Closed-loop bandwidth, stability margins, and jitter suppression shall be established separately from measured mechanical resonances, latency, actuator saturation, and feedback noise.

The optical drawing shall distinguish the 42.00° entry-face setting from the 67.271702° hypotenuse incidence in Eq. (266);

TABLE XXII
FOUR-LAYER OPTICAL INPUTS AND REFERENCE OUTPUTS

Parameter	785.0 nm	802.5 nm
n_0 SF11	1.76552	1.76447
n_1 Pd-Ir	1.835 + 4.535i	1.885 + 4.635i
n_2 Ti	2.800 + 3.800i	2.850 + 3.900i
n_3 diamond	2.417	2.416
Reference angle (deg)	67.73	68.05
Reference $R_{\min}$	0.0456	0.0455
Reference $\mathcal{F}_E$	0.43	0.44
Reference L_{spp} (nm)	191.16	205.11
Reference δ_z (nm)	137.14	142.08
Calculated minimum (deg)	67.73267	68.05209
Calculated $R_{\min}$	0.04560152	0.04551051
Calculated $\mathcal{F}_E$ at minimum	0.214962	0.211377

the 43.85° interface reference describes a separate geometry. The operating resonance angle requires characterization for the installed prism, film, gas state, and wavelength. The coarse setting tolerance shall be $\pm 0.001^\circ$, equivalent to $\pm 17.453 \mu\text{rad}$; fine adjustment shall meet the separate $0.015 \mu\text{rad}$ resolution requirement.

The viewport shall use synthetic fused silica with 23.0 mm clear aperture and dual-band $\text{Ta}_2\text{O}_5/\text{SiO}_2$ antireflection coatings. The optical specification shall use nominal $dn/dT = 1.0 \times 10^{-5} \text{ K}^{-1}$ near 800 nm, internal transmission exceeding 99.8% per centimeter, and p-polarized reflectance below 0.05% per coated surface at both pump wavelengths and the qualified incidence angle. Stress-relieved retainers shall preserve the optical acceptance limits after sealing.

a) *Four-layer TM transfer matrix*: The planar optical comparison comprises SF11 / 1.0 nm Pd-Ir / 10.0 nm Ti / semi-infinite CVD diamond [56]. With time convention $e^{-i\omega t}$, the transfer-matrix formulation uses the tangential field vector $(E_x, Z_0 H_y)^T$ and passive square-root branches:

$$\begin{aligned} k_x &= n_0 k_0 \sin \theta, \quad k_{z,j} = \sqrt{\epsilon_j k_0^2 - k_x^2}, \\ Y_j &= \epsilon_j k_0 / k_{z,j}, \quad \phi_j = k_{z,j} d_j, \quad M = M_1 M_2, \\ M_j &= \begin{bmatrix} \cos \phi_j & -i \sin \phi_j / Y_j \\ -i Y_j \sin \phi_j & \cos \phi_j \end{bmatrix}. \end{aligned} \quad (263)$$

$\text{Im } k_{z,j} \geq 0$ and positive outgoing power select physical branches. The dimensionless admittance requires $Z_0 H_y$, not unnormalized H_y . The matrices relate entrance to exit fields. Define

$$\begin{aligned} F &= M_{11} + M_{12} Y_3, \quad H = M_{21} + M_{22} Y_3, \\ r_x &= (Y_0 F - H) / (Y_0 F + H), \quad t_x = 2 Y_0 / (Y_0 F + H), \\ R &= |r_x|^2, \quad T = \frac{\text{Re } Y_3}{\text{Re } Y_0} |t_x|^2, \\ \mathcal{F}_E &= |t_x \cos \theta|^2 (1 + |k_x / k_{z,3}|^2). \end{aligned} \quad (264)$$

t_x is relative to the incident *tangential* electric field; $\mathcal{F}_E$ uses the total incident electric-field magnitude. These are coherent planar-film boundary relations [58]. The specified indices yield the tabulated reflectivity minima, whereas the nominal field factors differ from the calculated total-field enhancements. Since $n_0 \sin \theta < n_3$, the transmitted field in lossless diamond has real $k_{z,3}$ and is propagating rather than evanescent. A forced

TABLE XXIII
GRATING-ASSISTED OPTICAL INPUTS, TARGETS, AND PLANAR COMPARISONS

Quantity	785.0 nm	802.5 nm
ϵ_{PdIr}	-16.85 + 15.92i	-17.52 + 16.64i
ϵ_{Ti}	-4.82 + 23.48i	-5.24 + 24.15i
Nominal $\mathcal{F}_z$ target	124.5	138.2
Planar minimum angle (deg)	68.51617	68.68832
Planar $R_{\min}$	0.03977507	0.03961543
Planar $\mathcal{F}_z$ at minimum	0.094183	0.093730
Planar $\mathcal{F}_z$ at 65°	0.099945	0.099801
Half-space L_{spp} (nm)	208.62355	219.78252
Half-space δ_z (nm)	149.91983	154.00879

real- k_x TMM solution is distinct from a complex-wavevector SPP eigenmode: the propagation length $L_{\text{spp}} = 1/(2 \text{Im } k_{\text{spp}})$ and decay length δ_z require an eigenmode calculation rather than identification of a reflectivity minimum. The reference gas index is $n_{D_2} = 1.0003$, but the four-layer model contains no gas layer.

The tabulated propagation and decay lengths correspond to a separate semi-infinite Ti/diamond interface model:

$$\begin{aligned} k_{\text{spp}} &= k_0 \sqrt{\frac{\epsilon_{\text{Ti}} \epsilon_d}{\epsilon_{\text{Ti}} + \epsilon_d}}, \\ L_{\text{spp}} &= (2 \text{Im } k_{\text{spp}})^{-1}, \\ \delta_z &= \left[\text{Re } \sqrt{k_{\text{spp}}^2 - \epsilon_d k_0^2} \right]^{-1}. \end{aligned} \quad (265)$$

The half-space model gives

$(L_{\text{spp}}, \delta_z) = (191.15649, 137.14449)$ nm at 785 nm and $(205.10906, 142.08030)$ nm at 802.5 nm. These eigenmode lengths do not describe field decay in the forced four-layer solution. At 785 nm, $\text{Re } k_{\text{spp}}/k_0 = 2.450662 > n_0 = 1.76552$, so the specified prism requires an additional momentum contribution to couple to this half-space mode. Agreement between length values alone does not establish four-layer modal coupling.

b) *Right-angle-prism viewport geometry*: For air incidence onto the entry-face normal of a 45° right-angle prism,

$$\sin \theta_r = \sin 42.00^\circ / 1.76552, \quad \theta_r = 22.271702^\circ,$$

$$\theta_{\text{inc}} = 45^\circ + \theta_r = 67.271702^\circ,$$

$$|\theta_{\text{inc}} - 67.73^\circ| = 0.458298^\circ < 0.65^\circ.$$

(266)

The 785 nm ray lies within the specified $67.73^\circ \pm 0.65^\circ$ HWHM window, but angular agreement alone does not establish a bound SPP. At 802.5 nm, the same external angle gives 67.285667° , 0.764333 degrees from the tabulated minimum; the first wavelength's HWHM does not establish acceptance at both wavelengths. Each additional nonparallel viewport face requires a separate refraction calculation.

c) *Constitutive inputs and grating-assisted momentum matching*: The grating-assisted comparison uses the metallic permittivities in Table XXIII and the normal-component enhancement target $\mathcal{F}_z = |E_z(z_{\text{Ti/diamond}}^+)/E_0|^2$, distinct from the total-field factor. A 10 nm Ti layer separates Pd-Ir from diamond; the stack therefore contains no direct Pd-Ir/diamond boundary. The planar boundary relations are Eqs. (263)–(264),

with $\mathcal{F}_z = |t_x \cos \theta k_x / k_{z,3}|^2$. These permittivities differ from the squared refractive indices of the SF11 comparison; the corresponding 191.16/137.14 nm half-space lengths do not apply to this material dataset. The enhancement targets 124.5/138.2 are distinct from the planar TMM outputs for these inputs. For a first-order grating at 65° , phase matching requires equality of the in-plane excitation momentum and the real modal wavevector:

$$\begin{aligned} \text{Re } k_{\text{spp}} &= n_0 k_0 \sin 65^\circ + 2\pi/\Lambda, \\ \Lambda &= \frac{\lambda}{\text{Re}(k_{\text{spp}}/k_0) - n_0 \sin 65^\circ}, \\ (k_{\text{spp}}/k_0)_{785} &= 2.42248005 + 0.29943080i, \\ \Lambda_{785} &= 954.55175 \text{ nm}, \quad \Lambda_{802.5} = 970.27630 \text{ nm}. \end{aligned} \quad (267)$$

The prism indices are 1.76552 and 1.76447. With the rounded value $n_0 = 1.7655$ and approximate modal index 2.417, the 785 nm period is 960.93393 nm; an in-plane index of 1.600 gives 960.83231 nm. The specified 960.8 nm period is therefore a rounded design target rather than an exact phase match for the stated complex permittivities. The smooth-prism momentum remains below the modal momentum. A grating period specifies the momentum increment, not the coupling efficiency. Groove depth, duty cycle, profile, and finite-layer modal fields determine coupling strength and losses and must be specified independently. A periodic Maxwell solution is required to establish field enhancement and simultaneous dual-wavelength coupling; a one-dimensional planar TMM is insufficient. The grating-assisted 65° setting is distinct from the smooth-prism 42° entry-face geometry.

M. Refractive Field Mapping and Polarization Maintenance

Illumination uniformity over the square aperture $|x|, |y| \leq a = 10.0 \text{ mm}$ depends on the distinction between circular flat-top and square-separable profiles [51]. In the hard-edge circular limit with radius $R = 10.048711 \text{ mm}$, the area clipped beyond each square edge is

$$A_{\text{seg}} = R^2 \arccos(a/R) - a\sqrt{R^2 - a^2} = 0.06421455 \text{ mm}^2. \quad (268)$$

The unilluminated square fraction is $1 - [\pi R^2 - 4A_{\text{seg}}]/(4a^2) = 20.7574\%$, or approximately 20.76%. The corners lie at $r = 14.1421 \text{ mm}$, outside the beam radius, and their intensity approaches zero at high order. A steep circular edge also requires diffraction analysis rather than geometric ray mapping alone.

The square-separable irradiance model uses $m = 2N$ as the actual exponent:

$$\begin{aligned} I(x, y) &= I_0 e^{-2|x/w_x|^{2N}} e^{-2|y/w_y|^{2N}}, \\ w_x &= w_y = w, \quad m = 2N. \end{aligned} \quad (269)$$

Equal waists give square symmetry, not circular symmetry. The corner constraint is

$$\begin{aligned} \mathcal{U}_{\text{corner}} &= e^{-4(a/w)^m} \geq 0.9850, \\ w &\geq w_{\min}(m) = a[-\ln(0.9850)/4]^{-1/m}. \end{aligned} \quad (270)$$

TABLE XXIV
SEPARABLE-PROFILE EXPONENT COMPARISON AT $w \simeq 10.048711 \text{ mm}$

Exponent m	Corner (%)	Throughput (%)	Clipping (%)
1148	98.5000	99.2508	0.7492
2296	99.9943	99.1425	0.8575

The one-dimensional full-line integral is $2w2^{-1/m}\Gamma(1+1/m)$. With $z = 2(a/w)^m$ and regularized lower incomplete gamma function $P(\alpha, z) = \gamma(\alpha, z)/\Gamma(\alpha)$,

$$\begin{aligned} \eta_{2D} &= [P(1/m, z)]^2, \quad L_{\text{clip}} = 1 - \eta_{2D}, \\ z|_{w=w_{\min}} &= -\frac{1}{2} \ln(0.9850) = 0.007556818905. \end{aligned} \quad (271)$$

At a fixed waist, doubling the exponent changes both corner uniformity and aperture clipping. Table XXIV compares the two profiles by direct integration; performance values must remain associated with their defining exponent. The choice $N = 1148$ in an exponent $2N$ violates the 0.75% clipping limit at the specified waist. The admissible profile uses $m = 1148$, equivalent to $N = 574$ in Eq. (269). Its exact waist and ideal performance are

$$\begin{aligned} w &= w_{\min}(1148) = 10.048711043077 \dots \text{ mm}, \\ \mathcal{U}_{\text{corner}} &= 98.500000\%, \\ \eta_{2D} &= 99.25084440\%, \quad L_{\text{clip}} = 0.74915560\%. \end{aligned} \quad (272)$$

An exponent denoted directly by $N = 1148$ describes the same profile only when no additional factor of two is applied. At the rounded waist 10.048711 mm , the corner ratio is 98.49999267%; tolerance allocation therefore requires the analytic bound rather than a rounded-down limit. The $m = 2296$ alternative requires its own waist bound and performance evaluation.

a) *Exponent convention and waist tolerance:* The irradiance profile is $I(x, y) = I_0 e^{-2(x/w)^m} e^{-2(y/w)^m}$ with actual exponent $m = 1148$, equivalent to $N = 574$ in $2N$ notation. For the rounded waist $w = 10.0487 \text{ mm}$, direct integration gives $C_{\text{clip}} = 0.7489391\%$ and corner intensity 98.4981207%. Clipping satisfies its limit, but strict 98.50% corner uniformity requires the unrounded waist bound. At the same rounded waist, $m = 2296$ gives $C_{\text{clip}} = 0.8573148\%$, rather than the comparison estimate of 0.968%, and exceeds the 0.75% clipping limit. A profile written with exponent $2m$ doubles the exponent assigned to its parameter; the rounded 0.749% clipping value applies to actual exponent 1148, not 2296.

For a lossless separable refractive map, each coordinate obeys cumulative power conservation:

$$\frac{\int_{-\infty}^u I_{\text{in},x}(u') du'}{P_{\text{in},x}} = \frac{\int_{-\infty}^x I_{\text{out},x}(x') dx'}{P_{\text{out},x}}. \quad (273)$$

The analogous map in y imposes two-dimensional power conservation:

$$I_{\text{in}}(u, v) = I(x, y) \left| \frac{\partial(x, y)}{\partial(u, v)} \right|. \quad (274)$$

Actual throughput additionally includes reflection, absorption, finite aperture, aberration, and wave-optical diffraction. The

TABLE XXV
GRATING ENHANCEMENT TARGETS AND PLANAR COMPARISON VALUES

Quantity	785.0 nm	802.5 nm
Assigned $\mathcal{F}_z$	124.50	138.20
L_{spp} (nm)	208.62	219.78
δ_z (nm)	149.92	154.01
Planar R_{min}	0.039775	0.039615

separable geometry resolves corner coverage but does not by itself prove a realizable phase surface or eliminate ringing.

Polarization is specified at the prism input, after the complete fiber/window path:

$$\text{PER} = 10 \log_{10} \left(\frac{P_p}{P_s} \right) \geq 30.0 \text{ dB}, \quad (275)$$

$$\frac{P_s}{P_p} \leq 10^{-3}, \quad \frac{P_s}{P_p + P_s} \leq 9.99001 \times 10^{-4}.$$

FT-OPT-01 uses panda-type polarization-maintaining fibers with axis alignment within $\pm 0.2^\circ$. For a pure linear input and otherwise ideal optics, axis error ψ gives $P_s/P_p = \tan^2 \psi$; $\psi = 0.2^\circ$ alone corresponds to 49.14 dB PER. Real fiber coupling, connectors, window stress, and coating retardance consume the remaining margin and require direct measurement at both wavelengths.

The viewport stress-optic allocation is $C_{\text{so}} \leq 2.7 \times 10^{-12} \text{ Pa}^{-1}$, with retardation below 2.0 nm per centimeter of optical path. For principal stresses σ_1 and σ_2 , birefringence and phase retardation satisfy

$$\Delta n = C_{\text{so}}(\sigma_1 - \sigma_2), \quad \delta_{\text{pol}} = \frac{2\pi}{\lambda} \int \Delta n \, d\ell. \quad (276)$$

Retardation and PER acceptance shall use the assembled stress field and polarization orientation, with measurements at both pump wavelengths after sealing and under the specified mechanical and thermal loads.

b) Grating geometry and Maxwell boundary conditions:

The grating geometry is specified by period $\Lambda = 960.80$ nm, depth $h_g = 42.50$ nm, ridge duty fraction $\Gamma_g = 0.500$, and edge fillet radius 5.0 nm. The dual-wavelength enhancement targets are distinguished from planar TMM results: For a one-dimensional periodic profile, the RCWA field expansion and diffraction wavevectors are

$$\begin{aligned} \varepsilon(x, z) &= \sum_p \varepsilon_p(z) e^{i2\pi p x / \Lambda}, \\ \mathbf{E}(x, z) &= \sum_p \mathbf{E}_p(z) e^{ik_{x,p} x}, \\ k_{x,p} &= n_0 k_0 \sin \theta + 2\pi p / \Lambda, \\ \nabla \times \mathbf{E} &= i\omega \mu_0 \mathbf{H}, \quad \nabla \times \mathbf{H} = -i\omega \varepsilon_0 \varepsilon \mathbf{E}. \end{aligned} \quad (277)$$

Tangential fields are continuous at every boundary, with outgoing or decaying diffraction orders selected in both half-spaces. A two-dimensional grating requires a second reciprocal index and a measured two-dimensional profile. Field enhancement is evaluated on the specified side of the interface using the same incident-field normalization as the planar calculation. The profile dimensions define the electromagnetic geometry, while

TABLE XXVI
THIN-FILM GRATING INPUTS AND NOMINAL ENHANCEMENT TARGETS

Quantity	785.0 nm	802.5 nm
Air/vacuum real index	1.0000	1.0000
Silica real index	1.4538	1.4532
PdIr real index	1.8240	1.8850
Ti real index	2.7120	2.7540
Diamond real index	2.3980	2.3950
Nominal $\mathcal{F}_z$	124.50	138.20

Table XXV distinguishes planar reflectance and mode lengths from assigned grating enhancements. A converged harmonic sweep is required to establish whether a single RCWA solution satisfies all of these quantities. The 10 nm Ti spacer precludes direct PdIr/diamond contact.

c) Viewport retardance and aperture verification: For the 6.0 mm viewport, optical path retardance b_{opt} is expressed in meters and phase retardance δ_{pol} in radians:

$$\begin{aligned} b_{\text{opt}} &= \int_0^{t_{\text{win}}} C_{\text{so}}(\sigma_1 - \sigma_2) \, dz, \\ \delta_{\text{pol}} &= 2\pi b_{\text{opt}} / \lambda, \end{aligned} \quad (278)$$

$$\text{PER}_{\text{model}} = -10 \log_{10} [\tan^2 \psi + \sin^2(\delta_{\text{pol}}/2)].$$

The 2.0 nm/cm path-retardance allocation gives $b_{\text{opt}} \leq 1.20$ nm over 6 mm; path retardance per unit length is distinct from phase retardance. At $\lambda = 785$ nm and $\psi = 0.20^\circ$, Eq. (278) gives 44.52866 dB, whereas the comparison value is 34.22 dB. Both exceed the 30 dB target, but 34.22 dB does not follow from the specified additive-leakage model and requires additional loss mechanisms or measurement evidence. The $C_{\text{so}} \leq 2.70 \times 10^{-12} \text{ Pa}^{-1}$ bound alone does not determine the stress integral. For the separable aperture, direct integration with $m = 1148 = 2(574)$, $w = 10.048711$ mm, and half-width $a = 10$ mm gives

$$\begin{aligned} \mathcal{U}_{\text{corner}} &= e^{-4(a/w)^m} \simeq 0.985000, \\ C_{\text{clip}} &= 1 - [P(1/m, 2(a/w)^m)]^2 \\ &= 0.00749155 < 0.0075. \end{aligned} \quad (279)$$

Here P is the regularized lower incomplete gamma function. The result describes geometrical aperture throughput rather than diffraction-limited fabrication performance. The opposed-column residual $0.82225 \mu\text{m}$ and remaining stroke $14.17775 \mu\text{m}$ in Eq. (251) apply to the separately specified Invar and aluminum temperature rises.

d) Thin-film grating comparison and integrated datum chain: The thin-film RCWA comparison comprises an air/vacuum superstrate and a fused-silica binary grating, distinct from the SF11 prism stack. Its parameters are $\Lambda = 960.80$ nm, $h_g = 42.50$ nm, $\Gamma_g = 0.50$, $\theta = 65.0^\circ$, 1.0 nm PdIr, 10.0 nm Ti, and a $500.0 \mu\text{m}$ diamond substrate. The $50.0 \mu\text{m}$ diamond in the structural comparison defines a separate geometry. The RCWA Fourier fields and Maxwell matching follow Eq. (277); for two-dimensional periodicity, each diffraction order additionally has $k_{y,q} = k_{y,0} + 2\pi q / \Lambda_y$. The tabulated real refractive indices do not specify the imaginary components required to model absorption. Complex permittivities from other

model cases cannot be combined with these real indices without constitutive justification. The geometry and enhancement targets therefore do not uniquely determine an absorbing RCWA solution or establish harmonic convergence. The planar-model values of L_{spp} and δ_z are not necessarily eigenmode lengths of the grating with a 500 μm substrate.

The common-temperature mechanical comparison uses a 42.15 mm Invar column and an 8.52 mm aluminum column, distinct from the independently heated 172.50/20.69 mm pair. Combining stack expansion with the opposed-column displacement gives

$$\begin{aligned}\Delta L_{\text{stack}} &= (6.741 \text{ mm})(12.84 \times 10^{-6} \text{ K}^{-1})(256.85 \text{ K}) \\ &= 22.231507914 \mu\text{m}, \\ \Delta L_{\text{comp}} &= [(42.15)(1.20) - (8.52)(23.10)] \\ &\quad \times 10^{-3}(256.85) \mu\text{m} \\ &= -37.559689200 \mu\text{m}, \\ \delta_{\text{res,calc}} &= -15.328181286 \mu\text{m}.\end{aligned}\tag{280}$$

The alternative displacement balance $22.23120 - 21.40895 = 0.82225 \mu\text{m}$ is arithmetically consistent but does not follow from these column dimensions. A residual target of $0.8225 \mu\text{m}$ also differs from that balance. At the specified common temperature, compensation of $-21.40895 \mu\text{m}$ requires $L_{\text{Al}} = 5.79792019 \text{ mm}$ for the given Invar column; this inverse solution defines an alternative geometry rather than the specified assembly. The nominal and common-temperature comparison stages have total piezo travels of 15 and 20 μm , respectively. Residual displacement and available stroke require verification against the complete datum chain and preload geometry.

The nominal tilt RMS for this comparison is $\sigma_\theta = 0.00782 \mu\text{rad}$. Relative to the $0.00800 \mu\text{rad}$ limit, the numerical margin is 2.25%. For a one-sided closed-loop tilt spectrum,

$$\sigma_\theta^2 = \int_0^\infty S_{\theta,\text{cl}}(f) df, \tag{281}$$

with disturbance, sensor-noise, and cross-spectral contributions included. Verification requires a measured spectrum or a validated closed-loop model that evaluates this integral. Static displacement compensation alone does not establish the specified RMS performance.

IV. NANOFABRICATION, STRESS-ASSISTED TRANSPORT, AND MATERIALS INTEGRITY

The materials specification couples alloy deposition, intermetallic bonding, stress-assisted deuterium transport, and containment integrity. Lattice diffusion and reversible trapping determine the mobile and retained inventories; permeation barriers constrain escape through noncatalytic surfaces, while environmental fracture thresholds and crack-growth limits define the failure assessment. Fabrication, transport, and fracture parameters apply to their specified material states and temperatures. Consistent concentration units, interface conditions, and load histories are required to connect these models to the assembled device.

A. Materials, Nanofabrication and Mechanical Constraints

The Option B process uses a vacuum-melted Pd–Ir target and synchronized high-power impulse magnetron sputtering [72], [73]. For atomic fractions $x_{\text{Pd}} = 0.95$, $x_{\text{Ir}} = 0.05$ and atomic masses 106.42 and 192.22 g/mol,

$$\begin{aligned}M_{\text{alloy}} &= 0.95(106.42) + 0.05(192.22) = 110.71 \text{ g/mol}, \\ w_{\text{Pd}} &= 0.95(106.42)/110.71 = 0.913187607, \\ w_{\text{Ir}} &= 0.05(192.22)/110.71 = 0.086812393.\end{aligned}\tag{282}$$

The vacuum-melted target composition is 91.32 wt% Pd / 8.68 wt% Ir; a 95/5 wt% target does not represent the specified atomic ratio. The HiPIMS setpoints are peak power density 1.85 kW/cm², pulse duration 30 μs , repetition frequency 100 Hz, duty cycle 0.30%, and substrate bias -120 V synchronized to the 30 μs ion-arrival window. Deposition uses 99.999% Ar at 0.45 Pa (3.38 mTorr) and a substrate temperature of 150 °C. The nominal roughness result is $R_q = 0.12 \text{ nm}$ over a $5 \times 5 \mu\text{m}^2$ AFM scan, within the $R_q < 0.15 \text{ nm}$ acceptance criterion. Process-window qualification additionally requires deposition rate, total dose, composition transfer, calibrated AFM measurements, and wafer-scale uniformity; nominal setpoints and a roughness value alone are insufficient.

a) *Bondline Coffin–Manson comparison:* The following 5.0 μm $\epsilon\text{-Cu}_3\text{Sn}$ TLP bondline comparison uses the plastic-strain *range* convention. It is independent of the selected spring-amplitude criterion in Eq. (213):

$$\begin{aligned}\Delta\epsilon_p &= \epsilon'_f(2N_f)^c, \\ \epsilon'_f &= 0.18, \quad c = -0.62, \\ N_f(0.00098) &= \frac{1}{2}(0.00098/0.18)^{-1/0.62} = 2242.11, \\ \Delta\epsilon_p(44820) &= 0.18(89640)^{-0.62} = 0.000153010544, \\ N_f(0.000155) &= 43895.79.\end{aligned}\tag{283}$$

The comparison cap $\Delta\epsilon_p \leq 0.000155$ ($\leq 0.0155\%$) does not ensure $N_f \geq 44820$ with these bondline coefficients. This particular model requires $\Delta\epsilon_p \leq 0.000153010544$, or conservatively ≤ 0.000153 . Neither value replaces the selected X-750 spring amplitude ceiling of $4.805 \times 10^{-4} \text{ mm/mm}$. Coefficients calibrated to strain amplitude $\Delta\epsilon_p/2$ require conversion or recalibration before use in a range-based relation. The bondline coefficient reference temperature is 550 K; the 25–350 °C cycle spans 298.15–623.15 K and requires temperature-dependent fatigue and creep characterization.

b) *Second-order thermal-expansion integral:* Under the reference-length engineering-strain convention,

$$\begin{aligned}\alpha(T) &= a + b(T - 25^\circ\text{C}), \\ \Delta L &= L_0 \int_{25}^{350} \alpha(T) dT = L_0[a\Delta T + \frac{1}{2}b(\Delta T)^2], \tag{284} \\ \Delta T &= 325 \text{ K}.\end{aligned}$$

The coefficient varies linearly with temperature, yielding a second-order integrated strain. The integral is exact under the stated constitutive convention. If α is defined per instantaneous

length, the exact stretch is $\exp(\int \alpha dT)$, and the engineering-strain expression is a small-strain approximation.

$$\begin{aligned}
 L_I &= 34.7021 \text{ mm}, \quad L_A = 1.0721 \text{ mm}, \\
 a_I &= 1.20 \times 10^{-6} \text{ K}^{-1}, \\
 b_I &= 1.80 \times 10^{-9} \text{ K}^{-2}, \\
 a_A &= 2.35 \times 10^{-5} \text{ K}^{-1}, \\
 b_A &= 1.50 \times 10^{-8} \text{ K}^{-2}, \\
 \Delta L_I &= 16.832687381 \mu\text{m} \simeq 16.83 \mu\text{m}, \\
 \Delta L_A &= 9.037467969 \mu\text{m} \simeq 9.04 \mu\text{m}, \\
 \Delta L_{\text{net}} &= 7.795219413 \mu\text{m}.
 \end{aligned} \tag{285}$$

The nominal drift allocation is $\Delta L_{\text{net}} = 7.79 \mu\text{m}$, with a $15.0 - 7.79 = 7.21 \mu\text{m}$ (48.07%) stage margin. Evaluation without intermediate rounding gives $7.80 \mu\text{m}$ to two decimal places and leaves $7.204780587 \mu\text{m}$ (48.03%). The $7.79/7.21$ allocations are therefore nominal values rather than exact integral results; travel acceptance requires the unrounded displacement and the complete tolerance stack. A $15.0 \mu\text{m}$ total stroke provides this reserve only if the cold-end bias permits the required one-sided travel. A centered $\pm 7.5 \mu\text{m}$ stage cannot accommodate $7.7952 \mu\text{m}$ in one direction without rebiasing.

c) *Comparison-stack stiffness, VCCT and cross-domain loads:* The following non-governing comparison uses $P_0 = 15.80 \pm 0.50 \text{ MPa}$ at the reference condition. The selected contact force and 5,000 N/mm force stiffness are specified in Section III-A and Eq. (244); they shall not be combined with these comparison pressure-stiffness inputs:

$$\begin{aligned}
 K_A &= k_{\text{eq}}/A_{\text{load}} = 256.5 \text{ GPa/m} \\
 &= 0.2565 \text{ MPa}/\mu\text{m}, \\
 \Delta\sigma &= K_A \Delta L_{\text{net}} = 1.99947 \text{ MPa}, \\
 K_{A,\text{max}} &= \frac{2.0 \times 10^6}{7.795219413 \times 10^{-6}} \simeq 256.568 \text{ GPa/m}.
 \end{aligned} \tag{286}$$

A stiffness of $256.5 \text{ MPa}/\mu\text{m}$ would exceed the comparison value by 10^3 . This comparison stack gives full-ramp pressure variation of approximately 2 MPa rather than $\pm 0.50 \text{ MPa}$. With midpoint bias, thermal excursion alone is approximately $\pm 1.0 \text{ MPa}$ before preload tolerance. For this displacement, a passive full-span $\pm 0.50 \text{ MPa}$ bound requires $K_A \leq 128.284 \text{ GPa/m}$ even without preload uncertainty; a one-sided 0.50 MPa drift limit requires $\leq 64.142 \text{ GPa/m}$. These comparison bounds do not alter the selected $20.1 \mu\text{m}$ spring-path mismatch and calculated 0.320 MPa excursion. Stiffness and nominal compression alone do not establish a plastic-strain limit.

The nominal 3D VCCT results are $G_I = 4.2$, $G_{II} = 14.2$, and $G_{\text{VCCT}} = 18.4 \text{ J/m}^2$, compared with $G_c = 24.5 \text{ J/m}^2$, giving $G_c/G = 1.3315$. Delamination acceptance requires mesh convergence, defined crack geometry and mode mix, the applicable thermal state, and measured interface toughness. The aluminum yield ratio $276/15.80 = 17.47$ compares nominal bulk stresses; it establishes neither thin-wall buckling resistance nor elevated-temperature strength. The proposed TEG contact-pressure range of $15\text{--}18 \text{ MPa}$ requires manufacturer-specific contact-area and load-path qualification. Housing pressure cannot be assigned directly to brittle thermoelectric pellets without this assessment.

TABLE XXVII
TEMPERATURE-DEPENDENT VOCE MATERIAL INPUTS

Material	T ($^{\circ}\text{C}$)	σ_0 (MPa)	σ_{∞} (MPa)	b
Pd-Ir	25	280	550	32.0
Pd-Ir	350	180	410	25.4
OFHC Cu	25	75	180	22.5
OFHC Cu	350	45	120	18.5

d) *Monolithic grating and seating-frame geometry:* The active stack comprises a 50.0 nm peak-height monolithic Pd-Ir grating with a 7.50 nm continuous backplane on 10.0 nm Ti/ $500.0 \mu\text{m}$ diamond/ $5.0 \mu\text{m}$ TLP/ 1.0 mm copper. The metal footprint is 1.0 cm^2 . The 50% duty cycle gives a mean metal thickness of 28.75 nm rather than a uniform 50 nm layer. Deposition duration for the 1 nm comparison film does not apply to this geometry. At an assumed growth rate of 0.025 nm/s , a 50 nm unpatterned precursor requires 2000 s before patterning, rather than 40 s .

The constant-CTE clamp model uses a 33.63 mm seating assembly with $5.52 \mu\text{m}$ growth over 325 K ; this seating dimension is distinct from the sum of the active-layer thicknesses. With constant $\alpha_I = 1.2 \times 10^{-6} \text{ K}^{-1}$ and $\alpha_A = 23.0 \times 10^{-6} \text{ K}^{-1}$, the column lengths satisfy

$$\begin{bmatrix} 1 & -1 \\ \alpha_I & -\alpha_A \end{bmatrix} \begin{bmatrix} L_I \\ L_A \end{bmatrix} = \begin{bmatrix} 0.03363 \text{ m} \\ (5.52 \times 10^{-6}/325) \text{ m/K} \end{bmatrix}. \tag{287}$$

The analytical solution and rounded manufacturing dimensions are

$$\begin{aligned}
 L_I &= 34.7020818631 \text{ mm} \approx 34.7021 \text{ mm}, \\
 L_A &= 1.0720818631 \text{ mm} \approx 1.0721 \text{ mm}, \\
 \Delta L_{\text{gap}} &= (\alpha_I L_I - \alpha_A L_A) \Delta T, \\
 \delta_{\text{pinch}} &= \Delta L_{\text{stack}} - \Delta L_{\text{gap}}.
 \end{aligned} \tag{288}$$

At $\Delta T = 325 \text{ K}$, the rounded lengths give $\Delta L_{\text{gap}} = 5.5198715 \mu\text{m}$ and $\delta_{\text{pinch}} = 0.0001285 \mu\text{m}$, which rounds to $0.00 \mu\text{m}$. The unrounded solution gives zero pinch within the constant-CTE linear model. The $42.15/8.52 \text{ mm}$ comparison frame instead gives $\Delta L_{\text{gap}} = -47.2485 \mu\text{m}$ and $\delta_{\text{pinch}} = 52.7685 \mu\text{m}$.

The full-temperature compressive preload requirement is $15.80 \pm 0.50 \text{ MPa}$ over $25\text{--}350 \text{ }^{\circ}\text{C}$, distinct from the 42.5 MPa comparison clamp load. Compliance with this band requires a spring and load-path model satisfying

$$\sigma_c(T) = 15.80 \text{ MPa} + \frac{k_{\text{eq}}(T)}{A_{\text{load}}} \delta_{\text{tot}}(T), \tag{289}$$

$$\left| \frac{k_{\text{eq}}(T)}{A_{\text{load}}} \delta_{\text{tot}}(T) \right| \leq 0.50 \text{ MPa}.$$

Here δ_{tot} includes pinch, tolerances, relaxation, and differential temperature; k_{eq} is the assembled series stiffness and A_{load} the loaded area. For a maximum displacement of $0.04 \mu\text{m}$, the sufficient stiffness-per-area bound is $k_{\text{eq}}/A_{\text{load}} \leq 1.25 \times 10^{13} \text{ Pa/m}$. Zero nominal pinch alone does not establish Belleville-stack compliance. Full-range pressure qualification requires temperature-dependent CTE integration and spring characterization.

In this table, σ_∞ denotes total saturation stress in the law $\sigma_f = \sigma_0 + (\sigma_\infty - \sigma_0)(1 - e^{-b\epsilon_p})$, distinct from a hardening increment R_∞ . The mode-specific fracture thresholds are 14.20, 38.50, and 5.60 J/m², with a hydrogen-affected opening threshold of 8.10 J/m². These quantities are toughness inputs rather than calculated VCCT release rates. The transport parameters $V_H^* = 1.60 \times 10^{-6}$ m³/mol and $Q^* = 12.50$ kJ/mol enter the stress- and thermal-gradient flux equations. A 10 nm Ti layer and nominal preload do not establish reduced hydrogen damage without a corresponding transport solution.

e) Thin-film stack and constrained-fatigue comparison: The thin-film comparison stack comprises 1.0 nm Pd_{0.95}Ir_{0.05}/10.0 nm Ti/500.0 μ m CVD diamond/5.0 μ m ϵ -Cu₃Sn/1.0 mm OFHC Cu. Stress, stiffness, thermal-resistance, and crack-energy results for a 50.0 μ m diamond substrate do not transfer directly to a substrate ten times thicker. The comparison uses housing stiffness $K_{\text{housing}} = 4.85 \times 10^8$ N/m and compressive clamp preload $\sigma_{\text{clamp}} = 42.5$ MPa. These are boundary-condition inputs to the three-dimensional constitutive model; they do not determine the plastic-strain field without solving the constrained assembly.

$$\begin{aligned} \Delta\epsilon_{p,\max} &= 2(0.185)(2 \cdot 44820)^{-0.52} \\ &= 0.000983786284 = 0.0983786284\%, \\ N_f|_{\Delta\epsilon_p=0.0009838} &= \frac{1}{2}[0.0009838/(2 \cdot 0.185)]^{1/(-0.52)} \\ &= 44818.7983 \text{ cycles.} \end{aligned} \quad (290)$$

The rounded 0.09838% value is not an exact 44,820-cycle acceptance threshold; pass/fail assessment requires the unrounded inequality. Clamp inputs alone do not determine $\Delta\epsilon_p$ or establish fatigue life without a geometry-specific mechanical solution.

The primary structural reference is a five-layer, three-dimensional elastoplastic and creep model. Process comparisons, elastic reductions, hydrogen-embrittlement tests, and flaw-growth projections apply only to their specified geometries and conditions. The thin-film process, mechanical seating stack, and thermoelastic benchmark are distinct configurations. Manufacturing and analysis records shall associate each configuration with its geometry, constitutive inputs, process history, and acceptance evidence. Numerical predictions and nominal test outcomes do not replace production qualification or service-specific applicability checks.

B. Axial Stack-Up and Thermal Compensation

Table XXVIII defines a chassis-seating benchmark with a 20.0 μ m bond contribution. The unalloyed-stack FEA comparison uses the 5.0 micrometer bond and layer dimensions in Table XXXI; it is distinct from the selected 3.5 μ m microalloyed bond and 15.0 μ m graded interlayer in Section III-0a. A one-dimensional seating calculation does not resolve stress or delamination response. Initial preform thickness differs from final load-bearing IMC thickness. Dimensional inspection shall account for consumed metal, residual cladding, and final seating height. The uniform-temperature comparison uses

$T_0 = 293.15$ K and hot-side reference $T_h = 550.0$ K, giving $\Delta T = 256.85$ K. This comparison does not imply that the cooled chassis operates at T_h .

For layer seating contribution $L_{0,j}$, CTE α_j , and manufacturing deviation δz_j ,

$$\Delta z_j = L_{0,j}\alpha_j\Delta T, \quad L_{\text{stack}} = \sum_j (L_{0,j} + \Delta z_j + \delta z_j). \quad (291)$$

The constant-CTE result is $L_{\text{hot,nom}} = 6.740966$ mm, with worst-case interval [6.734466, 6.747466] mm. Production analysis requires temperature-dependent CTEs and actual layer temperatures rather than this screening approximation. For independent errors, the root-sum-square tolerance is

$$t_{\text{RSS}} = \sqrt{2.00^2 + 0.50^2 + 1.00^2 + 3.00^2} \mu\text{m} = 3.77492 \mu\text{m}. \quad (292)$$

The overall $\pm 4.07 \mu\text{m}$ allocation leaves 1.52148 μm RSS allowance for separately identified contributors under the same coverage convention. Correlated or bounded tolerances require their appropriate propagation rather than interpreting RSS as a hard limit.

a) Differential compensation frame: The re-entrant geometry in Section III and Eq. (251) is distinct from the 151.81 mm direct Invar/Cu-column comparison. Its residual displacement is 0.82225 μm at the two specified column temperatures. The seating-stack datum chain shall identify which terms in Table XXVIII move with the return link; the 20.966 μm stack sum is a separate geometric benchmark.

C. Pd-Ir HiPIMS Nanofilm Synthesis Protocol

The microscopic and FEA comparison uses a 10.0 nm Ti transition layer and a 1.0 nm Pd-Ir film. The 0.2 nm Ti seed process defines a separate deposition configuration and does not specify fabrication of the 10 nm interface. Film thickness and deposition time refer to Pd-Ir unless explicitly identified as Ti. The active-film thickness convention is defined in Table XXXI.

The ultrathin comparison film shall have nominal thickness 1.0 nm and atomic composition Pd_{0.95}Ir_{0.05}. Table XXIX specifies its HiPIMS setpoints and film acceptance criteria [26]. Procurement and process records shall distinguish atomic film composition from target mass composition.

Using atomic masses $M_{\text{Pd}} = 106.42$ g mol⁻¹ and $M_{\text{Ir}} = 192.217$ g mol⁻¹,⁸ the distinction is

$$\begin{aligned} x_{\text{Pd}}^{(95 \text{ wt}\%)} &= \frac{0.95/M_{\text{Pd}}}{0.95/M_{\text{Pd}} + 0.05/M_{\text{Ir}}} = 0.971686, \\ w_{\text{Pd}}^{(95 \text{ at}\%)} &= \frac{0.95M_{\text{Pd}}}{0.95M_{\text{Pd}} + 0.05M_{\text{Ir}}} = 0.913189. \end{aligned} \quad (293)$$

The ideal target composition corresponding to the specified atomic ratio is 91.3189 wt% Pd and 8.6811 wt% Ir. Deposited-film composition shall be measured, and any target-composition or process correction for preferential sputtering and transport shall be documented against the required atomic film composition.

⁸NIST, *Atomic Weights and Isotopic Compositions*, elemental entries for palladium and iridium.

TABLE XXVIII
CHASSIS-SEATING COMPARISON AT UNIFORM 256.85 K RISE

Layer contribution	$L_{0,j}$ (mm)	α_j (10^{-6} K^{-1})	Δz_j (μm)	Tolerance (μm)
Invar 36 chassis	1.500	1.20	0.462330	± 2.00
CVD diamond frame seating	0.500	1.00	0.128425	± 0.50
ϵ -Cu ₃ Sn bondline	0.020	18.40	0.094521	± 1.00
OFHC copper heat spreader	4.700	16.80	20.280876	± 3.00
Total	6.720	Layer-specific	20.966152	± 6.50 worst case

TABLE XXIX
ULTRATHIN-FILM HiPIMS SETPOINTS AND ACCEPTANCE CRITERIA

Parameter	Setpoint or target
Film	1.0 nm; 95.0 at% Pd/5.0 at% Ir
Target	Homogeneous vacuum-melted alloy, 99.999% purity
Target geometry	50.8 mm diameter
Base pressure	$\leq 5.0 \times 10^{-8}$ Torr
Atomic-hydrogen pretreatment	200 W RF, 2.0 Pa, 450°C, 15 min
Ti seed on CVD diamond	0.2 nm at 150°C
Process argon	99.9999% purity; 0.25 Pa
Peak target power density	1.85 kW cm^{-2}
Pulse duration / repetition	$30 \mu\text{s} / 100 \text{ Hz}$
Synchronized substrate bias	-120 V in ion-arrival window
Net growth rate / duration	$0.025 \text{ nm s}^{-1} / 40.0 \text{ s}$
Morphology acceptance	Continuous film; $R_q < 0.15 \text{ nm}$
Thickness uniformity	Better than $\pm 2\%$ with stated sampling

The process sequence comprises atomic-hydrogen etching of CVD diamond, thermal stabilization for Ti deposition, seed-layer formation and characterization, shutter-closed alloy-target conditioning, and synchronized HiPIMS deposition. TiC formation, Frank–van der Merwe layer-by-layer growth, and an ion-to-neutral ratio above 85% are process-response objectives requiring measurement; the setpoints alone do not establish them. The nominal seed thickness adds to the 1.0 nm alloy thickness and is excluded from the Pd–Ir composition allocation.

The dimensional checks are

$$\begin{aligned}
 D_{\text{pulse}} &= (30 \times 10^{-6})(100) = 0.003, \\
 \bar{q}_{\text{target}} &= D_{\text{pulse}}(1850) = 5.55 \text{ W cm}^{-2}, \\
 A_{\text{target}} &= \pi(2.54 \text{ cm})^2 = 20.2683 \text{ cm}^2, \\
 P_{\text{peak}} &\approx 37.4964 \text{ kW}, \quad \bar{P} \approx 112.489 \text{ W}, \\
 h_f &= (0.025 \text{ nm s}^{-1})(40.0 \text{ s}) = 1.000 \text{ nm}.
 \end{aligned} \tag{294}$$

The average-power estimate assumes rectangular pulses and the full target face as the power-density reference; actual waveforms and erosion-track dimensions shall be recorded. A continuous-DC power density of 1.85 W cm^{-2} is not interchangeable with a peak HiPIMS power density of 1.85 kW cm^{-2} .

Acceptance requires calibrated AFM roughness and sampling, cross-sectional film and seed thickness, chemical composition, continuity, pinhole assessment, adhesion, and thermal cycling on the actual substrate and witness specimens. Carbon/Pd, Ag nanoparticle, and pulsed-deposition studies provide process context [38], [42]–[44], but do not qualify the specific Pd–Ir/Ti/diamond stack. The Ti/TiC layer shall be represented in

optical, thermal-contact, diffusion, and thickness-calibration models.

D. Cu–Sn Transient Liquid Phase Bonding

a) *Multicomponent Ginzburg–Landau free energy*: The governing joint model couples conserved alloy composition, nonconserved phase order, and elastic equilibrium. Phase thickness and residual liquid are outputs of this initial-boundary-value problem, not prescribed square-root growth laws. A diffuse-interface description separates thermodynamic driving forces from kinetic mobilities [107]. Let c_a denote independent substitutional atomic fractions, with the reference-species fraction fixed by site balance, and let $\eta_p \geq 0$, $\sum_p \eta_p = 1$, label liquid, substrate, and intermetallic phases. The selected alloy requires Cu, Sn, Ni, and Bi chemistry; a binary Cu–Sn restriction is only a comparison case. Grain orientations may be represented by additional order parameters within each solid phase.

On a small-strain reference domain $\mathcal{V}_J$ with fixed reference site density N_s , set

$$\begin{aligned}
 \mathcal{F}_J &= \int_{\mathcal{V}_J} (f_0 + f_{\nabla} + f_{\text{el}}) dV, \\
 f_0 &= \sum_p h_p(\boldsymbol{\eta}) f_p(\mathbf{c}, T) W(\mathbf{c}, T) \sum_{p < q} \eta_p^2 \eta_q^2, \\
 h_p &= \frac{H(\eta_p)}{\sum_q H(\eta_q)}, \quad H(z) = z^3(10 - 15z + 6z^2), \\
 f_{\nabla} &= \frac{1}{2} \sum_{a,b} \kappa_{ab} \nabla c_a \cdot \nabla c_b \frac{1}{2} \sum_p \kappa_p |\nabla \eta_p|^2, \\
 f_{\text{el}} &= \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbf{C}(\mathbf{c}, \boldsymbol{\eta}, T) : \boldsymbol{\varepsilon}_e, \\
 \boldsymbol{\varepsilon}_e &= \boldsymbol{\varepsilon}(\mathbf{u}) - \boldsymbol{\varepsilon}^*(\mathbf{c}, \boldsymbol{\eta}, T) - \boldsymbol{\varepsilon}^p - \boldsymbol{\varepsilon}^{\text{cr}}, \\
 \boldsymbol{\sigma} &= \mathbf{C} : \boldsymbol{\varepsilon}_e, \quad \nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = 0.
 \end{aligned} \tag{295}$$

Here f_p, W, f_{el} are in J/m^3 , κ_{ab}, κ_p in J/m , and $\boldsymbol{\varepsilon}^*$ includes compositional, phase-transformation, and thermal eigen-strain. The concentration-gradient matrix is symmetric positive semidefinite and $\kappa_p > 0$. The displayed interpolation and barrier are coarse-graining choices, not universal atomistic identities. A liquid has no equilibrium shear modulus; liquid flow and solidification shrinkage require additional transport when a quasistatic, fixed-reference approximation is insufficient.

b) *Statistical origin of phase energies and barriers*: For N reference sites in phase p , the constrained configurational

partition sum gives a free energy per site:

$$\begin{aligned} Z_p(\mathbf{c}, T) &= \sum_{\mathbf{c} \in p, \mathbf{c}} \exp \left[-\frac{F_C^{\text{micro}}(T)}{k_B T} \right], \\ g_p(\mathbf{c}, T) &= -\frac{k_B T}{N} \ln Z_p, \quad f_p = N_s g_p, \\ F_C^{\text{micro}} &= E_C^{\text{DFT}} + F_C^{\text{vib}} + F_C^{\text{el, thermal}}, \\ F_C^{\text{vib}} &= \sum_{\nu} \left\{ \frac{\hbar \omega_{\nu}}{2} + k_B T \ln[1 - e^{-\hbar \omega_{\nu}/(k_B T)}] \right\}. \end{aligned} \quad (296)$$

Configurational entropy is included by the sum, not added a second time as ideal mixing. The harmonic vibrational expression applies to stable solid configurations; the liquid needs an appropriate finite-temperature free-energy calculation. Energies are evaluated at a consistent reference strain, with the continuum elastic excess included only once. Molar free energies must be divided by molar volume before use as f_p .

Interface energetics must also be matched. For a planar two-phase equilibrium interface, a scalar restriction with excess bulk density $\Delta f(\eta)$ obeys

$$\begin{aligned} \kappa_{\eta} \eta'' &= \partial_{\eta} \Delta f, \quad \frac{1}{2} \kappa_{\eta} (\eta')^2 = \Delta f, \\ \gamma_{pq} &= \int_0^1 \sqrt{2 \kappa_{\eta} \Delta f(\eta)} d\eta, \\ \gamma_{pq}^{\text{micro}} &= \frac{F_{pq}^{\text{interface}} - F_p^{\text{bulk}} - F_q^{\text{bulk}}}{A_{pq}}. \end{aligned} \quad (297)$$

The interface area A_{pq} counts all interfaces in the atomistic cell. Matching γ_{pq} and a physical interface profile determines the admissible barrier/gradient combination; γ_{pq} alone does not determine both. Artificially widening an interface requires kinetic matching and mesh convergence, not a claim of unchanged microscopic dynamics [108].

c) Chemical potentials and hydrostatic coupling: For constant gradient coefficients, functional differentiation at fixed displacement and internal strain gives exchange chemical potentials per atom and phase driving forces:

$$\begin{aligned} N_s \mu_a &= \frac{\delta \mathcal{F}_J}{\delta c_a} = \partial_{c_a} f_0 - \sum_b \kappa_{ab} \nabla^2 c_b \\ &\quad + \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbf{C}_{,c_a} : \boldsymbol{\varepsilon}_e - \boldsymbol{\sigma} : \boldsymbol{\varepsilon}_{,c_a}^*, \\ \xi_p &= \frac{\delta \mathcal{F}_J}{\delta \eta_p} = \partial_{\eta_p} f_0 - \kappa_p \nabla^2 \eta_p \\ &\quad + \frac{1}{2} \boldsymbol{\varepsilon}_e : \mathbf{C}_{,\eta_p} : \boldsymbol{\varepsilon}_e - \boldsymbol{\sigma} : \boldsymbol{\varepsilon}_{,\eta_p}^*. \end{aligned} \quad (298)$$

The derivatives include the concentration dependence of W and all interpolated phase energies. If gradient coefficients depend on state, their local derivatives and the full divergence terms must also be differentiated. For isotropic compositional expansion, define the exchange relaxation volume $\Omega_a^* = \text{tr}(\boldsymbol{\varepsilon}_{,c_a}^*)/N_s$. With tension-positive $\sigma_m = \text{tr} \boldsymbol{\sigma}/3$,

$$\begin{aligned} \mu_a^{\text{stress}} &= -\sigma_m \Omega_a^*, \\ \nabla \mu_a^{\text{stress}} &= -\Omega_a^* \nabla \sigma_m - \sigma_m \nabla \Omega_a^*. \end{aligned} \quad (299)$$

This sign follows from variation of elastic energy rather than an imposed drift coefficient. The stiffness-derivative term and anisotropic eigenstrain contribution remain in Eq. (298); they cannot generally be replaced by a hydrostatic scalar alone.

d) Conserved diffusion and phase kinetics: Let $\mathbf{L}_{ab}$ be a symmetric positive-semidefinite Onsager mobility in atom-count flux units. The multicomponent Cahn–Hilliard and constrained Allen–Cahn equations are

$$\begin{aligned} \mathbf{J}_a &= -\sum_b \mathbf{L}_{ab} \nabla \mu_b, \quad N_s \partial_t c_a = -\nabla \cdot \mathbf{J}_a, \\ \partial_t \eta &= -\mathbf{L}_{\eta} (\boldsymbol{\xi} + \lambda \mathbf{1} - \boldsymbol{\zeta}), \\ \eta_p &\geq 0, \quad \zeta_p \geq 0, \quad \eta_p \zeta_p = 0, \quad \sum_p \eta_p = 1. \end{aligned} \quad (300)$$

The multiplier λ enforces the phase sum, and $\boldsymbol{\zeta}$ enforces nonnegativity. The kinetic operator acts on admissible phase variations. A dilute, uncorrelated binary restriction has $\mathbf{L} = N_s D c(1-c)/(k_B T)$; equivalently, with a volumetric variational derivative, $M_c = D c(1-c)/(N_s k_B T)$. This site-density factor is required dimensionally. Concentrated ordered intermetallics require sublattice occupancies and correlated multicomponent mobilities, not substitution of a tracer diffusivity for every Onsager entry.

For an impermeable, neutral external boundary, use $\mathbf{n} \cdot \mathbf{J}_a = 0$, $\mathbf{n} \cdot \sum_b \kappa_{ab} \nabla c_b = 0$, and $\mathbf{n} \cdot \kappa_p \nabla \eta_p = 0$. Initial composition and phase fields specify finite Cu/Sn inventory, alloy additions, and any nuclei; wetting or open reservoirs replace these conditions by surface-energy or chemical-potential conditions. At fixed temperature, fixed mechanical boundaries, and frozen plastic and creep variables, integration by parts proves

$$\begin{aligned} \frac{d\mathcal{F}_J}{dt} &= -\int_{\mathcal{V}_J} \sum_{a,b} \nabla \mu_a \cdot \mathbf{L}_{ab} \nabla \mu_b dV \\ &\quad - \int_{\mathcal{V}_J} \boldsymbol{\xi}_c^T \mathbf{L}_{\eta} \boldsymbol{\xi}_c dV \leq 0, \\ \boldsymbol{\xi}_c &= \boldsymbol{\xi} + \lambda \mathbf{1} - \boldsymbol{\zeta}, \quad \frac{d}{dt} \int_{\mathcal{V}_J} N_s c_a dV = 0. \end{aligned} \quad (301)$$

Thermal ramps, loading work, and vacancy exchange add their respective energy and species terms; the isothermal inequality alone is not a full reactor heat balance. Equivalent phase thickness and residual liquid fraction are evaluated from the solution:

$$W_p(t) = \frac{1}{A_J} \int_{\mathcal{V}_J} h_p dV, \quad f_l(t) = \frac{1}{|\mathcal{V}_J|} \int_{\mathcal{V}_J} h_l dV. \quad (302)$$

Complete isothermal solidification requires a specified residual-liquid threshold and absence of connected liquid channels. A finite-reservoir phase-field calculation can approach a diffusion-limited square-root asymptote, but that asymptote is not prescribed as the constitutive law.

e) Composition-dependent transition-state activation: For a local atomic environment $\mathcal{C}$ and event r , define initial and saddle free energies with identical species reservoirs and thermodynamic constraints. Transition-state theory gives

$$\begin{aligned} \Delta g_r^{\ddagger}(\mathcal{C}, T, \boldsymbol{\sigma}) &= G_r^{\ddagger} - G_r^{\text{initial}}, \\ k_r(\mathcal{C}) &= \kappa_r^{\text{tr}} \frac{k_B T}{h} e^{-\Delta g_r^{\ddagger}/(k_B T)}, \\ \frac{k_{i \rightarrow j}}{k_{j \rightarrow i}} &= e^{-(G_j - G_i)/(k_B T)}. \end{aligned} \quad (303)$$

Here κ_r^{tr} is the transmission coefficient, equal to unity only in the no-recrossing approximation; the unstable saddle mode is excluded from its partition function. A consistent common saddle and microscopic reversibility give detailed balance. Atomic barriers require minimum-energy paths and saddle vibrational free energies, with local environment sampling [109]. Their stress dependence follows from the activation-volume tensor:

$$\Omega_r^\ddagger = -\frac{\partial \Delta g_r^\ddagger}{\partial \boldsymbol{\sigma}},$$

$$\Delta g_r^\ddagger(\boldsymbol{\sigma}) = \Delta g_r^\ddagger(0) - \int_0^1 \Omega_r^\ddagger(s\boldsymbol{\sigma}) : \boldsymbol{\sigma} \, ds, \quad (304)$$

$$\Delta g_r^\ddagger(\sigma_m) = \Delta g_r^\ddagger(0) - \sigma_m \Delta \Omega_r^\ddagger \quad \text{for constant } \Delta \Omega_r^\ddagger.$$

The stress ensemble is consistent with the fixed-strain elastic variation by the corresponding mechanical Legendre transform. Stress work is counted once, either through the resolved defect free energy or its activation-volume representation.

Composition enters through both the barriers and the probability $w_C(\mathbf{c}, \boldsymbol{\eta}, T)$ of each local environment. For a common attempt frequency, a rate-equivalent activation free energy is

$$\Delta g_{\text{eff}}^\ddagger = -k_B T \ln \sum_C w_C e^{-\Delta g_C^\ddagger / (k_B T)},$$

$$\pi_C = \frac{w_C e^{-\Delta g_C^\ddagger / (k_B T)}}{\sum_D w_D e^{-\Delta g_D^\ddagger / (k_B T)}}, \quad (305)$$

$$\partial_{c_a} \Delta g_{\text{eff}}^\ddagger = \sum_C \pi_C \left(\partial_{c_a} \Delta g_C^\ddagger - k_B T \partial_{c_a} \ln w_C \right).$$

Environment-dependent prefactors must remain inside the rate average. Thus a concentration-dependent activation energy is not obtained by assigning a universal percentage increase on alloying. In the isotropic, uncorrelated random-walk limit, vacancy hopping gives $D_v = \frac{1}{6} \sum_r \langle k_r \rangle |\mathbf{a}_r|^2$, where $\mathbf{a}_r$ is the jump vector; correlations require the long-time mean-square-displacement or Onsager calculation. Interface mobility follows separately by linearizing the difference of forward/backward transformation rates at small free-energy driving force. Neither its prefactor nor its barrier is determined by the equilibrium bulk free-energy curves alone.

More generally, the equilibrium long-time displacement covariance determines the alloy Onsager operator. With $\Delta \mathbf{R}_a$ the summed atomic displacement of species a in the lattice frame, projected onto independent exchange coordinates,

$$\mathbf{L}_{ab} = \lim_{\tau \rightarrow \infty} \frac{\langle \Delta \mathbf{R}_a(\tau) \otimes \Delta \mathbf{R}_b(\tau) \rangle}{2|\mathcal{V}_J| k_B T \tau}. \quad (306)$$

The equilibrium covariance includes correlations absent from a single-hop Einstein estimate. In a binary phase-conversion restriction, an event changes transformed volume by v_* , and $\mathcal{R}_0$ is the equilibrium event frequency per unit volume. Symmetric forward/backward rates with affinity $-v_* \xi$ give

$$\dot{\eta} = -2v_* \mathcal{R}_0 \sinh\left(\frac{v_* \xi}{2k_B T}\right), \quad (307)$$

$$\dot{\eta} \simeq -L_\eta \xi, \quad L_\eta = \frac{v_*^2 \mathcal{R}_0}{k_B T}.$$

Here ξ is the constrained phase-conversion free-energy density difference. The positive mobility has units $\text{m}^3/(\text{J s})$, and

TABLE XXX
TLP COMPARISON-BONDLINE REFERENCE PROPERTIES

Property	Reference value
Final phase	ϵ -Cu ₃ Sn
Phase melting reference	676.0°C
Interfacial shear strength	58.50 MPa
Void fraction	0.32%
Young's modulus	98.35 GPa
CTE	$18.40 \times 10^{-6} \text{ K}^{-1}$
Thermal conductivity	$148.0 \text{ W m}^{-1} \text{ K}^{-1}$
Final IMC thickness in stress model	20.0 μm

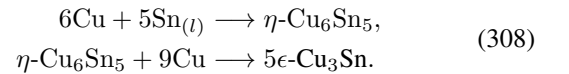
$\mathcal{R}_0(\mathbf{c}, \boldsymbol{\eta}, T)$ is computed from the interface event barriers and environment weights, not identified with a bulk tracer prefactor. The linear Allen–Cahn law applies to the small-affinity limit of this local kinetic model.

These expressions define a first-principles parameterization route, not numerical activation energies for the selected joint. Material-specific electronic energies, saddle paths, finite-temperature corrections, interface energies, and defect populations are required before the model predicts solidification times or creep rates. The continuum interpolation, harmonic approximation, and kinetic coarse graining retain explicit validity limits.

The selected interconnect uses a final 3.5 μm Ni-6Cu/Sn-0.7Cu-0.05Bi bondline and a 15.0 μm graded Ni/Cu compliance layer, with the 52,400-cycle strain-life prediction in Eq. (206). The unalloyed 5.0 micrometer Cu–Sn simulation and the 15–20 micrometer preform/reflow cases are separate comparisons; their shear-strength, transport, and phase-growth values do not directly characterize the selected alloyed joint. Final IMC thickness, phase composition, diffusivity, and interfacial toughness require configuration-specific qualification.

Cu–Sn transient liquid phase (TLP) bonding forms an ϵ -Cu₃Sn intermetallic compound (IMC) rather than retaining a low-melting solder joint. The comparison preform consists of a 10 μm Sn layer between two 15 μm Cu foils, or a qualified Cu@Sn composite. The nominal process uses 260–280°C, 0.50 MPa applied pressure, and a 15–20 min dwell at 260°C. The vacuum/formic-acid process specifies pressure below 10^{-3} mbar; the manufacturing traveler shall define evacuation, reducing-gas exposure, dwell, and cooling as separately controlled stages.

The phase sequence is



Representative joints shall demonstrate complete consumption of free Sn, the required phase composition, and metallization adhesion; dwell time alone does not establish complete transformation. The final 20.0 μm IMC layer is a stress-model input and shall not be equated to the 40 μm initial foil stack without dimensional and material-balance verification.

At the comparison TEG hot interface,

$$T_h = 550.0 \text{ K} < T_{\text{melt}} = 949.15 \text{ K}, \quad \Delta T_{\text{melt}} = 399.15 \text{ K}. \quad (309)$$

The phase melting reference does not establish a continuous-service rating. The 260°C mechanical cycle lies below the 276.85°C hot-side endpoint; joint properties and phase stability require verification at the actual operating temperature. Inspection records shall define void fraction, coverage and resolution, residual Sn, IMC phase fractions, interfacial cracking, and shear-test temperature. The nominal 0.32% void fraction and 58.50 MPa strength require representative joint measurements before production acceptance.

f) *Grain geometry and vacancy thermodynamics:* Grain coarsening is obtained from the phase/orientation fields and their interfacial energies. For resolved grain volumes $V_g(t)$, one explicitly defined size statistic is

$$d_g(t) = \left[\frac{6V_g(t)}{\pi} \right]^{1/3}, \quad \bar{d}_V(t) = \frac{\sum_g V_g(t) d_g(t)}{\sum_g V_g(t)}. \quad (310)$$

This volume-weighted equivalent diameter is not an imposed grain-growth law. Diffusion paths and dislocation sink distances are evaluated in the evolving geometry; a grain diameter does not automatically equal the dislocation capture radius.

Let $X_v = N_v/N_s$ be the vacancy site fraction and $\Omega_a = 1/N_s$ the reference atomic volume. The dilute-defect treatment uses the same local phase/composition state as the joint free energy. For a vacancy formed by transferring a species- a atom to a reservoir,

$$\begin{aligned} \Delta g_{v,a}^f &= G(N_a - 1, N_v + 1) - G(N_a, N_v) + \mu_a^{\text{res}}, \\ \Omega_{v,a}^f &= -\partial_{\sigma} \Delta g_{v,a}^f, \\ \Delta g_v^f(\sigma_m) &= \Delta g_v^f(0) - \sigma_m \Delta \Omega_v^f \quad \text{for constant } \Delta \Omega_v^f, \\ f_v/N_s &= X_v \Delta g_v^f + k_B T X_v \ln X_v \\ &\quad + k_B T (1 - X_v) \ln(1 - X_v), \\ \mu_v &= \frac{1}{N_s} \partial_{X_v} f_v = \Delta g_v^f + k_B T \ln \frac{X_v}{1 - X_v}, \\ X_v^{\text{eq}} &= \left[1 + e^{\Delta g_v^f / (k_B T)} \right]^{-1} \simeq e^{-\Delta g_v^f / (k_B T)}. \end{aligned} \quad (311)$$

The reservoir fixes the zero of vacancy exchange potential. In an ordered multicomponent crystal, separate sublattice vacancy and antisite free energies replace the single dilute fraction. Formation energy and migration barrier are different thermodynamic quantities. The displayed stress term is the reduced representation of defect elastic work already included in a fully resolved elastic free energy; it must not be added twice. The formation volume is not necessarily Ω_a and its sign must follow the defect calculation.

For coupled evolution, the chemical defect excess is added to f_0 , the vacancy relaxation is included in ε^* , and the dislocation line energy is added to $\mathcal{F}_J$. All composition and phase derivatives in Eq. (298) are then taken on this augmented functional. In particular, the chemical defect terms contribute

$$\delta \mu_a^v = X_v \partial_{c_a} \Delta g_v^{f,\text{chem}}, \quad \delta \xi_p^v = N_s X_v \partial_{\eta_p} \Delta g_v^{f,\text{chem}}. \quad (312)$$

Elastic vacancy contributions follow from the same eigenstrain derivative, while line-energy variations enter their respective chemical and mechanical forces. Evaluating vacancy rates without these reciprocal couplings is a one-way approximation, not the fully coupled thermodynamic model.

The isothermal vacancy flux and species balance are

$$\begin{aligned} \mathbf{J}_v &= -\frac{N_s D_v X_v (1 - X_v)}{k_B T} \nabla \mu_v, \\ \partial_t (N_s X_v) &= -\nabla \cdot \mathbf{J}_v + s_v, \\ \nabla \Delta g_v^f &= \sum_a \partial_{c_a} \Delta g_v^f \nabla c_a + \sum_p \partial_{\eta_p} \Delta g_v^f \nabla \eta_p \\ &\quad + \partial_{\sigma_m} \Delta g_v^f \nabla \sigma_m, \\ \mathbf{J}_v &\simeq -N_s D_v \left[\nabla X_v + \frac{X_v}{k_B T} \nabla \Delta g_v^f \right]. \end{aligned} \quad (313)$$

For constant $\Delta \Omega_v^f$, the hydrostatic part of the dilute flux is

$$\mathbf{J}_v^{\text{stress}} = \frac{N_s D_v X_v \Delta \Omega_v^f}{k_B T} \nabla \sigma_m. \quad (314)$$

Composition and phase gradients contribute separately. If the formation volume varies, the term $-\sigma_m \nabla \Delta \Omega_v^f$ is also retained in $\nabla \Delta g_v^f$. Nonuniform temperature additionally requires the heat/defect Onsager cross-coupling and the force $\nabla(\mu_v/T)$; no Soret coefficient follows from a static activation energy alone. The source s_v includes explicitly balanced exchange with dislocations, interfaces, and surfaces. Alloy and vacancy fluxes obey lattice-site conservation; the dilute vacancy approximation neglects only higher-order vacancy corrections, not the atom/vacancy counterflux responsible for Kirkendall transport.

g) *Weertman vacancy-diffusion climb derivation:* The rate-controlling climb mechanism is determined by vacancy exchange with dislocation cores [110], [111]. Consider a locally straight edge segment with Burgers edge component $b_e > 0$, core radius r_c , and capture radius $R_e > r_c$. Choose positive climb as vacancy absorption. Advancing a line length ℓ by ds absorbs $dN_v = b_e \ell ds / \Omega_a$. If f_c is the thermodynamic line force per unit length in that direction, core equilibrium follows from

$$\begin{aligned} dG_{\text{exchange}} &= -\mu_v^{\text{core}} dN_v - f_c \ell ds = 0, \\ \mu_v^{\text{core}} &= -f_c \Omega_a / b_e, \\ X_v^{\text{core}} &\simeq X_v^{\text{eq}} \exp \left[-\frac{f_c \Omega_a}{b_e k_B T} \right], \\ f_c &= [(\boldsymbol{\sigma} \mathbf{b}) \times \mathbf{t} + \mathbf{f}_{\text{line}}] \cdot \mathbf{n}_c. \end{aligned} \quad (315)$$

Here $\mathbf{t}$ is the unit line tangent, $\mathbf{n}_c$ the absorption-oriented climb direction, and $\mathbf{f}_{\text{line}}$ includes the configurational force from line energy. The minus sign in the core exponential follows this absorption convention; reversing climb direction reverses f_c and velocity together. The Peach–Koehler force depends on the resolved tensor stress, not only on σ_m . Hydrostatic stress modifies defect formation and line work through their respective conjugate volumes, which must be distinguished.

For constant D_v , composition, temperature, and formation energy in the capture cylinder, quasisteady vacancy diffusion

gives

$$\begin{aligned} \frac{1}{r} \frac{d}{dr} \left(r \frac{dX_v}{dr} \right) &= 0, \\ X_v(r) &= X_v^{\text{core}} + (X_v^\infty - X_v^{\text{core}}) \frac{\ln(r/r_c)}{\ln(R_e/r_c)}, \\ I_v &= \frac{2\pi N_s D_v}{\ln(R_e/r_c)} (X_v^\infty - X_v^{\text{core}}), \\ v_c &= \frac{\Omega_a I_v}{b_e} = \frac{2\pi D_v}{b_e \ln(R_e/r_c)} (X_v^\infty - X_v^{\text{core}}). \end{aligned} \quad (316)$$

The inward current I_v is vacancies per unit line length per second; v_c has units m/s. If concentrations are number densities $C_v = N_s X_v$, the velocity prefactor is instead $2\pi D_v \Omega_a / [b_e \ln(R_e/r_c)]$. Omitting b_e or mixing fractions and number densities does not give a velocity.

For radial stress/composition variation, set $a_v = X_v / X_v^{\text{eq}}(r)$ in the dilute limit. Integrating the chemical-potential flux, rather than assuming a logarithmic concentration, gives

$$v_c = \frac{2\pi \Omega_a N_s}{b_e} \frac{a_v(R_e) - a_v(r_c)}{\int_{r_c}^{R_e} \frac{dr}{r D_v(r) X_v^{\text{eq}}(r)}}. \quad (317)$$

Nonradial fields require the full vacancy boundary-value problem. At $X_v^\infty = X_v^{\text{eq}}$ and weak line driving,

$$\begin{aligned} v_c &= \frac{2\pi D_v X_v^{\text{eq}}}{b_e \ln(R_e/r_c)} \left[1 - e^{-f_c \Omega_a / (b_e k_B T)} \right], \\ v_c &\simeq \frac{2\pi D_v X_v^{\text{eq}} \Omega_a}{b_e^2 k_B T \ln(R_e/r_c)} f_c, \\ D_{\text{self}} &= f_{\text{corr}} D_v X_v^{\text{eq}}. \end{aligned} \quad (318)$$

The correlation factor f_{corr} and the event network determine self-diffusion; D_v here is vacancy migration diffusivity, not deuterium diffusivity. In a single-path harmonic limit the self-diffusion activation enthalpy contains both vacancy formation and migration enthalpies. For multiple environments the rate average in Eq. (305) replaces a constant activation enthalpy. Uniform hydrostatic stress changes equilibrium vacancy population but produces no bulk drift when its gradient vanishes.

h) Macroscopic creep and energy consistency: For segment families s with density ρ_s in m^{-2} and velocity $\mathbf{v}_s$, the swept-area identity gives

$$\begin{aligned} \dot{\boldsymbol{\beta}}^{\text{cr}} &= \sum_s \rho_s \mathbf{b}_s \otimes (\mathbf{t}_s \times \mathbf{v}_s), \\ \dot{\boldsymbol{\epsilon}}^{\text{cr}} &= \text{sym} \dot{\boldsymbol{\beta}}^{\text{cr}}, \\ \boldsymbol{\sigma} : \dot{\boldsymbol{\epsilon}}^{\text{cr}} &= \sum_s \rho_s [(\boldsymbol{\sigma} \mathbf{b}_s) \times \mathbf{t}_s] \cdot \mathbf{v}_s. \end{aligned} \quad (319)$$

For pure climb, $\mathbf{v}_s = v_{c,s} \mathbf{n}_{c,s}$; the magnitude of the single-family swept strain coordinate is $\rho_s b_s v_{c,s}$ with its orientation factor. Weertman climb-assisted glide additionally requires glide motion, obstacle geometry, and climb waiting times; identifying every macroscopic strain component with $\rho_d b v_c$ is not a general polycrystal closure. Dislocation density and capture distances are evolving microstructural variables, not prescribed powers of equivalent stress. Bulk and grain-boundary diffusional creep can be represented by vacancy fluxes and

moving sink/source boundaries; adding independent fitted rates to the same vacancy current would double count transport.

Let $\mathcal{A}_v = \mu_v^\infty + f_c \Omega_a / b_e$ be the absorption affinity. The current has the sign of $\mathcal{A}_v$, so $I_v \mathcal{A}_v \geq 0$; bulk diffusion also gives $-\mathbf{J}_v \cdot \nabla \mu_v \geq 0$. Including line-energy changes and chemical reservoir work therefore gives nonnegative dissipation. Chemical supersaturation may drive climb against mechanical stress, so mechanical power alone need not be nonnegative away from defect equilibrium.

The evolving joint stress satisfies

$$\dot{\boldsymbol{\sigma}} = \dot{\mathbf{C}} : \boldsymbol{\epsilon}_e + \mathbf{C} : (\dot{\boldsymbol{\epsilon}} - \dot{\boldsymbol{\epsilon}}^* - \dot{\boldsymbol{\epsilon}}^p - \dot{\boldsymbol{\epsilon}}^{\text{cr}}). \quad (320)$$

Only a fixed-temperature, fixed-composition scalar limit with constant effective modulus, fixed total strain, and no other inelastic increments reduces this to $\dot{\boldsymbol{\sigma}} = -E \dot{\boldsymbol{\epsilon}}_{\text{cr}}$. Its integrated identity is

$$\boldsymbol{\epsilon}_{\text{cr}}(t) - \boldsymbol{\epsilon}_{\text{cr}}(0) = \frac{\sigma(0) - \sigma(t)}{E}. \quad (321)$$

It supplies no universal life or monotonic-stress bound when phase transformation, vacancy supersaturation, external loading, or thermal cycling changes the driving force. The 0.50% project strain allowance and the graded-joint strain-life specification remain acceptance criteria; compliance requires integration of the coupled history, not extrapolation of a fixed fitted creep rate.

E. Multilayer Thermoelastic FEA Definition

The structural framework resolves the full three-dimensional Pd–Ir/Ti/CVD-diamond/Cu–Sn/OFHC–Cu configuration [59]. Prandtl–Reuss plastic flow, thermal strain, creep, and crack-closure energy are coupled. Seating sums and Suhir reductions provide limited comparison models rather than the controlling structural-safety assessment. The microscopic interface specifies a 1.0 nm catalyst film, distinct from a 1.0 μm comparison thickness. The FEA geometry shall use the specified 1.0 nm film; a micrometer-film calculation does not represent the nanometer-scale interface. The total thickness is 1.055011 mm. Freely expanding layers with constant CTEs give 4.584027 μm axial expansion for a uniform 256.85 K rise, distinct from the chassis-seating sum. Constrained displacement requires the 3D solution. Endpoint properties do not define complete temperature-dependent constitutive functions. Diamond tensile strength is a brittle-fracture reference rather than a metallic yield stress. For an associative isotropically hardening metal, let $s_{ij} = \sigma_{ij} - \sigma_{kk} \delta_{ij} / 3$ and $\bar{\sigma} = \sqrt{3 s_{ij} s_{ij} / 2}$. The constitutive equations are

$$\begin{aligned} \dot{\epsilon}_{ij} &= \dot{\epsilon}_{ij}^e + \dot{\epsilon}_{ij}^p + \dot{\epsilon}_{ij}^{\text{th}} + \dot{\epsilon}_{ij}^c, \\ \dot{\epsilon}_{ij}^p &= \frac{3 \dot{\bar{\sigma}}}{2 \bar{\sigma}} s_{ij}, \\ f &= \bar{\sigma} - [\sigma_y(T) + R(\bar{\epsilon}_p, T)] \leq 0, \\ \dot{\bar{\epsilon}}_p &\geq 0, \quad f \dot{\bar{\epsilon}}_p = 0. \end{aligned} \quad (322)$$

R is the independently specified isotropic hardening law. It describes rate-independent plasticity, not vacancy-controlled

TABLE XXXI
COMPARISON-STACK LAYER PROPERTIES AT 550 K (50 μm DIAMOND)

Layer	Thickness	E (GPa)	ν	$10^6 \alpha$ (K^{-1})	Strength (MPa)	k (W/m K)
Pd _{0.95} Ir _{0.05}	1.0 nm	124.0	0.39	11.8	150.0 yield	72.0
Ti transition layer	10.0 nm	105.0	0.32	8.6	220.0 yield	21.9
CVD diamond	50.0 μm	1050.0	0.07	1.1	2000.0 tensile	1800.0
ϵ -Cu ₃ Sn TLP	5.0 μm	110.0	0.33	18.4	180.0 yield	34.0
OFHC copper	1.0 mm	110.0	0.34	17.7	65.0 yield	385.0

TABLE XXXII
JOINT VACANCY AND CLIMB CONSTITUTIVE QUANTITIES

Quantity	Definition or units
Vacancy migration diffusivity D_v	m^2/s
Vacancy formation free energy Δg_v^f	J/defect
Migration barrier $\Delta g_m^\ddagger$	J/event
Formation/activation volume	m^3/defect
Vacancy site fraction X_v	Dimensionless
Edge Burgers component b_e	m
Core/capture radii r_c, R_e	m; $R_e > r_c$
Line density ρ_s	m^{-2}

creep. The creep strain follows the resolved dislocation velocities in Eq. (319); a scalar accumulated measure is computed afterward:

$$\begin{aligned}\dot{\epsilon}_{\text{dev}}^c &= \dot{\epsilon}^{\text{cr}} - \frac{1}{3} \text{tr}(\dot{\epsilon}^{\text{cr}}) \mathbf{I}, \\ \dot{\epsilon}_c &= \sqrt{\frac{2}{3} \dot{\epsilon}_{\text{dev}}^c : \dot{\epsilon}_{\text{dev}}^c}, \\ \epsilon_{\text{acc}}^c &= \int_0^{t_f} \dot{\epsilon}_c \, dt, \quad \epsilon_{\text{vol}}^c = \text{tr}(\dot{\epsilon}^{\text{cr}}).\end{aligned}\quad (323)$$

Climb can produce volumetric strain, which is retained separately rather than suppressed by an assumed J_2 creep direction. Table XXXII defines the required quantities, not calibrated values for the alloy. Energy per atomic event uses $k_B T$; molar activation energy uses RT . Vacancy formation and migration contributions enter their own partition factors exactly once. Creep and rate-independent plasticity must be assigned distinct mechanisms or a unified dislocation evolution model so the same swept strain is not counted twice.

For a crack-front segment of width b_f and advance Δa , the nominal local two-mode VCCT calculation is

$$\begin{aligned}G_{\text{VCCT}} &= \frac{F_x \Delta u + F_z \Delta w}{2b_f \Delta a}, \\ \sigma_{xz, \text{max}} &= 142.5 \text{ MPa}, \\ G_{\text{VCCT}} &= 18.4 \text{ J m}^{-2} < G_C = 24.5 \text{ J m}^{-2}, \\ G_C/G_{\text{VCCT}} &= 1.331522.\end{aligned}\quad (324)$$

The forces and relative displacements are work-conjugate quantities at and behind the crack tip; a general 3D crack front also includes Mode III work $F_y \Delta v$. The quantity denoted by G_{lag} in this calculation is a VCCT release rate, distinct from the shear-lag energy expression $\omega \tau^2/2$. A 142.5 MPa shear stress can exceed elastic shear yield and requires elastoplastic treatment; comparison with uniaxial yield alone is not an elastic safety assessment.

For comparison, the previously specified five-year, 43800 h strain value and ductility criterion are

$$\begin{aligned}\epsilon_{\text{acc}}^c &= 0.0142 = 1.42\%, \\ \epsilon_f &= 0.045 = 4.5\% > \epsilon_{\text{acc}}^c, \\ \epsilon_f/\epsilon_{\text{acc}}^c &= 3.169014.\end{aligned}\quad (325)$$

The 1.42% comparison value is not an output of the phase-field/vacancy-climb model. Re-evaluation requires the coupled composition, phase, vacancy, dislocation, and displacement histories. The 44820 cycles specify a loading history rather than an independently derived fatigue life. The 4.5% ductility criterion is distinct from the 0.50% project allowance; 1.42% exceeds the latter. Each criterion retains its defined material, loading, and service scope.

For layer j , the coupled heat and quasistatic mechanical equations are

$$\begin{aligned}\rho_j c_{p,j} \frac{\partial T_j}{\partial t} &= \nabla \cdot (k_j \nabla T_j) + q_j, \\ \nabla \cdot \boldsymbol{\sigma}_j + \mathbf{f}_j &= \mathbf{0}, \\ \boldsymbol{\epsilon}_j &= \frac{1}{2} (\nabla \mathbf{u}_j + \nabla \mathbf{u}_j^T), \\ \boldsymbol{\sigma}_j &= \mathbb{C}_j(T_j) : [\boldsymbol{\epsilon}_j - \boldsymbol{\epsilon}_j^{\text{th}} - \boldsymbol{\epsilon}_j^p - \boldsymbol{\epsilon}_j^{\text{cr}}], \\ \boldsymbol{\epsilon}_j^{\text{th}} &= \left[\int_{T_{\text{sf},j}}^{T_j} \alpha_j(T) \, dT \right] \mathbf{I}.\end{aligned}\quad (326)$$

Here $T_{\text{sf},j}$ is the layer's stress-free reference temperature, $\mathbb{C}_j$ is its elastic tensor, and the superscripts p and cr denote plastic and creep strains. The benchmark shall use its stated stress-free reference. Production analysis shall establish the residual-stress state from the deposition, TLP reaction, cooling, and assembly history.

At a perfectly bonded solid interface, displacements and tractions are continuous. With thermal contact resistance R''_{th} , the common normal heat flux obeys

$$q_n = \frac{T_j - T_{j+1}}{R''_{\text{th}}}, \quad \mathbf{u}_j = \mathbf{u}_{j+1}, \quad \boldsymbol{\sigma}_j \mathbf{n} = \boldsymbol{\sigma}_{j+1} \mathbf{n}.\quad (327)$$

Delamination analysis shall replace perfect displacement continuity with a calibrated cohesive or fracture model. The mesh shall represent the thin layers through appropriate shell, interface, or locally resolved elements. Mesh refinement shall demonstrate convergence of reaction forces, displacement, thermal flux, and a defined interfacial-stress or energy-release measure. Failure assessment shall use a convergent quantity rather than a singular corner nodal maximum.

The FEA input package shall additionally specify densities, heat capacities, heat-source partition, contact conductances,

TABLE XXXIII
RING/BASE STRUCTURAL-MATERIAL INPUTS AT 550.0 K

Property	SS316L	Invar 36
E (GPa)	172.0	128.0
ν	0.30	0.29
α (10^{-6} K $^{-1}$)	16.5	2.8
σ_y (MPa)	145.0	220.0
k (W m $^{-1}$ K $^{-1}$)	18.2	12.8
D_H (m 2 s $^{-1}$)	4.2×10^{-11}	1.1×10^{-12}
S_H (mol m $^{-3}$ Pa $^{-1/2}$)	1.8	3.5

cooling boundary conditions, and phase-appropriate constitutive data. The analysis deliverable shall include solver settings, mesh-convergence results, energy and force balance checks, and comparison with the reduced shear benchmark where its assumptions apply. Production stress and delamination margins shall be supported by the completed analysis and representative joint measurements; the input table itself is the prescribed benchmark, not a completed FEA result.

a) *Ring/base structural comparison at 550 K:* Table XXXIII defines the material inputs for an SS316L constraint-ring/Invar-base benchmark [51]. This configuration differs from the active-film, diamond, TLP, copper-spreader, and coating stack. Its five layers comprise ceramic insulator, Invar base, Pd–Ir active film, SS316L ring, and gold cap; the two-layer reduction isolates the ring/base interface. Use of these inputs as qualified properties requires traceable material heats, isotope conditions, test temperatures, and specimen records.

The solubility coefficient is defined by $c_H = S_H \sqrt{p_H}$, fixing the pressure exponent as $-1/2$ in its units. Hydrogen transport inputs are not automatically deuterium transport data; isotope-specific conversion or testing is required. Using the tabulated CTEs as effective constants over 298.15–550.0 K gives

$$\begin{aligned}\Delta\epsilon_{th} &= (16.5 - 2.8) \times 10^{-6} (251.85) \\ &= 3.450345 \times 10^{-3}, \\ G_{SS} &= 172.0/2.6 = 66.153846 \text{ GPa}, \\ G_{Inv} &= 128.0/2.58 = 49.612403 \text{ GPa}.\end{aligned}\quad (328)$$

If the CTEs are instantaneous 550 K values, the first line is an approximation; a temperature-dependent calculation instead integrates their difference over the temperature excursion. For $t_{SS} = 2.0$ mm, $t_{Inv} = 5.0$ mm, and half-length $L = 10.0$ mm,

$$\begin{aligned}\omega &= \frac{t_{SS}}{3G_{SS}} + \frac{t_{Inv}}{3G_{Inv}} \\ &= 4.36712694 \times 10^{-14} \text{ m Pa}^{-1}, \\ \beta &= \left[\frac{1}{\omega} \left(\frac{1}{E_{SS}t_{SS}} + \frac{1}{E_{Inv}t_{Inv}} \right) \right]^{1/2} \\ &= 319.911928 \text{ m}^{-1}.\end{aligned}\quad (329)$$

For the symmetric free-ended shear-lag idealization, choose the interface traction sign so that

$$\begin{aligned}\tau(x) &= \frac{\Delta\epsilon_{th}}{\omega\beta} \frac{\sinh(\beta x)}{\cosh(\beta L)}, \\ \tau_{max} &= \frac{\Delta\epsilon_{th}}{\omega\beta} \tanh(\beta L) = 246.144716 \text{ MPa}.\end{aligned}\quad (330)$$

TABLE XXXIV
3D FEA MESH SEQUENCE AND SUCCESSIVE ENERGY CHANGE

Mesh	Elements	τ (MPa)	G (J/m 2)	Change (%)
M1	12450	215.4	1012.8	baseline
M2	48600	238.1	1238.4	18.2171
M3	194400	245.2	1312.8	5.6673
M4	777600	246.1	1322.5	0.7335
M5	3110400	246.4	1325.1	0.1962

The odd traction field vanishes at the center and has equal, opposite edge tractions. The alternative expression $\cosh(\beta x)/[\omega \sinh(\beta L)]$ lacks the length factor $1/\beta$, has units of pressure per length, and is incompatible with the edge-stress relation. The values $\beta = 319.83 \text{ m}^{-1}$ and $\tau_{max} = 246.36 \text{ MPa}$ are comparison estimates rather than evaluations of the specified moduli and thicknesses.

The shear-lag energy surrogate and associated margin are

$$\begin{aligned}G_{lag} &= \frac{1}{2} \omega \tau_{max}^2 = 1322.960432 \text{ J m}^{-2}, \\ G_C &= 1850 \text{ J m}^{-2}, \\ SF_{lag} &= G_C/G_{lag} = 1.398379.\end{aligned}\quad (331)$$

Direct evaluation differs from the comparison estimates of 1324.7 J m^{-2} and 1.396 . The inequality $G_{lag} < G_C$ holds within the model. However, $\frac{1}{2} \omega \tau^2$ represents interfacial elastic energy per unit area; identification with crack-advance energy release requires a defined crack geometry and consistent fracture calculation. The shear-only von Mises contribution $\sqrt{3} \tau_{max} \simeq 426.3 \text{ MPa}$ exceeds both tabulated yield strengths. Structural qualification therefore requires elastoplastic redistribution and measured mixed-mode toughness for the actual interface, not an elastic shear-lag comparison alone. Table XXXIV relates mesh size to nominal stress and energy outputs, with successive energy change defined by $100|G_j - G_{j-1}|/|G_j|$. The final change is below 0.25%, whereas the M4 change exceeds this threshold. Numerical stabilization does not independently validate constitutive assumptions, crack-energy extraction, or the physical joint. The active-stack FEA and TLP creep model each retain their specified geometry and material scope.

b) *Voce hardening and mode-resolved crack closure:*

The Voce formulation defines total flow stress as the sum of initial yield stress and isotropic hardening [56]. Consistent with Eq. (322),

$$\begin{aligned}\sigma_{flow}(\bar{\epsilon}_p, T) &= \sigma_{y0}(T) + R_h(\bar{\epsilon}_p, T), \\ R_h &= R_\infty(T)[1 - e^{-b(T)\bar{\epsilon}_p}], \\ f &= \bar{\sigma} - \sigma_{flow} \leq 0, \\ d\epsilon_{ij}^p &= \frac{3s_{ij}}{2\bar{\sigma}} d\bar{\epsilon}_p, \\ \bar{\epsilon}_p(t) &= \int_0^t \sqrt{\frac{2}{3} \dot{\epsilon}_{ij}^p \dot{\epsilon}_{ij}^p} dt' .\end{aligned}\quad (332)$$

A convention in which R denotes σ_{flow} differs from one in which R denotes the hardening increment R_h . The Cu–Sn endpoint parameters are $R_\infty = 120 \text{ MPa}$ and $b = 15$ at 550 K. An endpoint yield table does not determine full temperature-dependent functions or the other layers' hardening parameters. Setting unspecified hardening parameters to zero

TABLE XXXV
ATOMISTIC INPUTS FOR JOINT PHASE AND DEFECT KINETICS

Input	Required calculation
Phase free energies	Configurational and thermal partition sums
Elastic and chemical eigenstrain	Strain derivatives and relaxed cell volumes
Vacancy formation	Defective/perfect cells at common reservoirs
Migration and interface barriers	Saddle paths and transition-state partition sums
Interface energy/profile	Relaxed two-phase cells; excess energy per area
Mobilities and correlations	Event network or long-time flux correlations
Dislocation line energy	Core structure, character, and surrounding stress

imposes perfect plasticity rather than calibrated hardening. For a crack-front strip of width w and advance Δa , paired nodal closure forces and opening/sliding displacements give

$$\begin{aligned} G_I &= F_z \Delta u_z / (2w \Delta a), \\ G_{II} &= F_x \Delta u_x / (2w \Delta a), \\ G_{III} &= F_y \Delta u_y / (2w \Delta a), \quad G = G_I + G_{II} + G_{III}. \end{aligned} \quad (333)$$

For tractions uniform across w , the continuum Mode I expression is $(2\Delta a)^{-1} \int_0^{\Delta a} \sigma_{zz}(r) \Delta u_z(r - \Delta a) dr$; Mode II substitutes τ_{xz} , Δu_x . No additional $1/w$ factor applies because width cancels when traction is converted to force. Evaluation requires consistent closing-work signs and the applicable mixed-mode fracture criterion. The assigned total 18.4 J/m^2 does not specify a Mode I/II partition.

c) *Voce and creep parameters at 550 K*: For the $5.0 \mu\text{m}$ $\epsilon\text{-Cu}_3\text{Sn}$ bondline, the endpoint flow-stress law is [54]

$$\sigma_{\text{flow}}(\bar{\epsilon}_p, 550 \text{ K}) = 180 + 120[1 - e^{-15\bar{\epsilon}_p}] \text{ MPa}. \quad (334)$$

Thus initial yield is 180 MPa, saturation stress 300 MPa, and $b = 15$ is dimensionless. These resolve the Cu–Sn endpoint of (332), not temperature functions or other materials' hardening laws. The independent material functions in Table XXXV must be evaluated for the actual composition and temperature. An apparent Arrhenius activation energy can be extracted from the calculated rate, but it is not assumed constant:

$$\begin{aligned} \beta_T &= (k_B T)^{-1}, \\ Q_{\text{app}}(\mathbf{c}, \boldsymbol{\eta}, T, \boldsymbol{\sigma}) &= - \left. \frac{\partial \ln[D_{\text{self}}/D_*]}{\partial \beta_T} \right|_{\mathbf{c}, \boldsymbol{\eta}, \boldsymbol{\sigma}}. \end{aligned} \quad (335)$$

Here D_* is a temperature-independent reference diffusivity used only to make the logarithm dimensionless. The derivative includes temperature-dependent prefactors, defect populations, correlations, and barriers. At fixed geometry and a temperature-independent prefactor, a single formation-plus-migration pathway reduces to the sum of the corresponding activation enthalpies. This limit is not a universal alloy law. Table XXXVI states analytical consistency tests rather than assumed 100 MPa creep rates. A computed strain history additionally requires specified thermal/loading cycles, phase topology, reservoir

TABLE XXXVI
CONSISTENCY TESTS FOR THE COUPLED JOINT MODEL

Test	Required result
Closed alloy inventory	Conserved species totals
Isothermal phase evolution	Nonincreasing free energy under fixed boundaries
Vacancy equilibrium	Zero current for equal chemical potentials and zero line affinity
Climb conversion	$v_c = \Omega_a I_v / b_e$ in m/s
Mechanical coupling	Tensor strain from swept dislocation area
Predictive scope	No numerical rate without material and defect inputs

conditions, and initial defect structure. Elastic moduli, defect energies, and kinetic rates must use the same material state. No numerical five-year creep integral follows from the formal equations alone.

F. Suhir Interfacial-Compliance Formulation

The elastic reduction provides an analytical verification benchmark. The controlling active-stack model is Eq. (326), coupled to Eqs. (322) and (323) and the VCCT formulation in Eq. (324). Peak stresses and margins from the reduction apply only to its stated elastic comparison case.

For a symmetric bond extending over $-a \leq x \leq a$, let the adherends have thicknesses h_1, h_2 , elastic moduli E_1, E_2 , and Poisson ratios ν_1, ν_2 . The TLP bondline has thickness h_s and shear modulus G_s . A Suhir interfacial-compliance reduction uses⁹

$$\begin{aligned} G_j &= \frac{E_j}{2(1 + \nu_j)}, \\ \Lambda &= \frac{1 - \nu_1}{E_1 h_1} + \frac{1 - \nu_2}{E_2 h_2}, \\ \kappa_s &= \frac{h_s}{G_s}, \quad \kappa = \kappa_s + \frac{h_1}{3G_1} + \frac{h_2}{3G_2}, \\ \beta &= \sqrt{\frac{\Lambda}{\kappa}}, \quad \Delta \varepsilon_T = (\alpha_2 - \alpha_1) \Delta T. \end{aligned} \quad (336)$$

Here Λ has units $\text{Pa}^{-1} \text{m}^{-1}$, κ has units m Pa^{-1} , and β has units m^{-1} . Let $N(x)$ denote the axial force per unit width under a sign convention for which $\tau = N'$. The reduced equilibrium and compatibility equations are

$$\frac{d^2 N}{dx^2} - \beta^2 N = \frac{\Delta \varepsilon_T}{\kappa}, \quad N(-a) = N(a) = 0. \quad (337)$$

Their symmetric solution and interfacial stress are

$$\begin{aligned} N(x) &= \frac{\Delta \varepsilon_T}{\Lambda} \left[\frac{\cosh(\beta x)}{\cosh(\beta a)} - 1 \right], \\ \tau(x) &= \frac{\Delta \varepsilon_T}{\kappa \beta} \frac{\sinh(\beta x)}{\cosh(\beta a)}, \\ \tau_{\text{max}} &= \frac{|\Delta \varepsilon_T|}{\kappa \beta} \tanh(\beta a). \end{aligned} \quad (338)$$

⁹E. Suhir, "Low-Cycle-Fatigue Failures of Solder Material in Electronics Packaging: Analytical Modeling Enables to Predict and Possibly Prevent Them—Review," *Journal of Aerospace Engineering and Mechanics*, vol. 2, no. 1, pp. 134–150, 2018, doi: 10.36959/422/433, interfacial-compliance formulation.

The calculation shall retain β in the stress denominator to preserve dimensional consistency. This reduction specifies interfacial shear; the production multilayer model shall additionally resolve warpage, peeling, inelasticity, and phase changes where applicable.

The elastic reference case uses Pd–Ir and ceramic adherends, distinct from the diamond/copper pair in Table XXXI:

$$\begin{aligned} a &= 25.0 \text{ mm}, & h_s &= 20.0 \text{ } \mu\text{m}, \\ (E_1, h_1) &= (121 \text{ GPa}, 1.0 \text{ mm}), \\ (E_2, h_2) &= (310 \text{ GPa}, 1.5 \text{ mm}), \\ \Delta\alpha &= |11.8 - 7.2| \times 10^{-6} \text{ K}^{-1}, \\ \Delta T &= 240.0 \text{ K}, & |\Delta\epsilon_T| &= 1.104 \times 10^{-3}. \end{aligned} \quad (339)$$

The 1.0 mm mechanical adherend is distinct from the 1.0 nm optical film. The nominal peak-stress and strength comparison is

$$\begin{aligned} \tau_{\max}^{(v1.1)} &= 18.42 \text{ MPa}, \\ \tau_{\text{strength}}^{(v1.1)} &= 58.50 \text{ MPa}, \\ \text{SF}_{\text{ratio}} &= \frac{58.50}{18.42} = 3.17590, \\ \text{SF}_{\text{spec}} &= 3.175. \end{aligned} \quad (340)$$

The specified factor 3.175 is a conservative truncation of the ratio rather than its nearest three-decimal rounding. Verification of the nominal stress requires complete compliance inputs, boundary conditions, and residual-stress characterization. The dimensional forms in Eqs. (336)–(338) require consistent compliance units. The ratio compares a reference load case with nominal joint strength; it does not define a code-qualified allowable stress.

For the same constant-property elastic case, increasing the temperature rise from 240.0 to 256.85 K scales the nominal stress to approximately 19.71 MPa. This sensitivity calculation does not replace a complete 550.0 K joint analysis. Production acceptance shall account for peeling, residual stress, flaws, cyclic inelasticity, and temperature-dependent strength in addition to the scalar shear ratio.

G. Transient Thermal Shock and Coffin–Manson Fatigue

The operating ramp envelope is 20 K min^{-1} , distinct from TLP manufacturing ramps. Transient analysis shall obtain layer temperatures from Eq. (326), include the applicable hot-side endpoint, and evaluate thermal mismatch and residual stress over complete startup/shutdown cycles. The 8.42 MPa additional-exursion limit is a separate requirement and is not automatically additive to either the 18.42 MPa elastic result or the 142.5 MPa elastoplastic field. The prescribed 44820-cycle simulation and its nominal 1.42% accumulated creep output are distinct from a Coffin–Manson fatigue-life prediction.

Under the positive-exponent cycle-count convention, the strain–life relation is

$$\Delta\epsilon_p = \epsilon'_f N_f^{-c}, \quad N_f = \left(\frac{\epsilon'_f}{\Delta\epsilon_p} \right)^{1/c}, \quad (341)$$

with Cu–Sn reference parameters $\epsilon'_f \approx 0.18$ and $c \approx 0.52$. The nominal endurance target is

$$N_f^{(v1.1)} = 44,820 \text{ cycles} \quad (342)$$

for its 20–260°C thermal cycle. Reproducing this value under (341) requires

$$\Delta\epsilon_{p,\text{required}} = \frac{0.18}{44,820^{0.52}} = 6.86289 \times 10^{-4}. \quad (343)$$

The rounded strain input 0.000690 gives approximately 44,358 cycles. The geometric expression $\Delta\alpha\Delta Ta/(2h_s)$ evaluates to 0.69 with consistent length units and therefore does not establish the smaller plastic-strain input. The 44,820-cycle value is an endurance target rather than an independently validated fatigue prediction.

One cycle comprises a complete thermal startup/shutdown excursion. Plastic strain requires a calibrated joint model or measured hysteresis; elastic strain, engineering shear strain, and equivalent plastic strain are not interchangeable. The positive-exponent cycle convention requires transformed coefficients before comparison with a reversal-based Coffin–Manson relation. Representative TLP joints require cyclic testing, phase and void characterization, and separate interfacial-delamination criteria. The 44,820-cycle target exceeds the 16,570-cycle duty requirement but does not independently establish five-year life at 276.85°C, under sustained creep, or for another cycle schedule.

a) Reversal-based fatigue of the 5 micrometer bondline:

The reversal-based Coffin–Manson comparison uses a convention distinct from Eq. (341):

$$\begin{aligned} \frac{\Delta\epsilon_p}{2} &= \epsilon'_f (2N_f)^c, \\ \epsilon'_f &= 0.185, \quad c = -0.52, \quad \Delta\epsilon_p = 0.00223, \\ N_f &= \frac{1}{2} \left(\frac{0.00223}{2(0.185)} \right)^{1/(-0.52)} = 9289.71583 \text{ cycles}. \end{aligned} \quad (344)$$

The specified cycle is 20.0 to 276.85°C, $\Delta T = 256.85 \text{ K}$, for a $5.0 \text{ } \mu\text{m}$ TLP bondline. The coefficient difference $\Delta\alpha = 18.0 \times 10^{-6} \text{ K}^{-1}$ gives free mismatch 0.0046233, distinct from the assigned plastic-strain range of 0.00223. Constraint and plastic hysteresis determine that range. For the same ϵ'_f and c , achieving 44820 cycles requires

$$\Delta\epsilon_{p,44820} = 2(0.185)(2 \cdot 44820)^{-0.52} = 0.000983786284. \quad (345)$$

Evaluation uses the unrounded exponent $1/c$. The calculated endurance of 9289.72 cycles falls below both the 44820-cycle target and the 16570-cycle duty requirement. This strain input therefore satisfies neither fatigue-life requirement; values from other calibration cases do not alter that result.

H. Fuel Conditioning, Trapping and Containment

a) *Transport discretization and execution contract:* The designated `shbt_cf.transport` interface shall evolve C_L , $C_{T,1}$, and $C_{T,2}$ across Pd–Ir/Ti/ε–Cu₃Sn interfaces, retaining hydrostatic-stress drift and the Soret term. The specified fine-grid spacing is $\Delta x = 0.1 \text{ nm}$ in resolved interfacial regions. Conservative face fluxes define the spatial discretization in one dimension or on a three-dimensional mesh; Crank–Nicolson

is the time discretization, not a spatial stencil. Writing the coupled semi-discrete residual as $\mathcal{R}$ gives

$$\frac{\mathbf{C}^{j+1} - \mathbf{C}^j}{\Delta t} = \frac{\mathcal{R}(\mathbf{C}^{j+1}, T^{j+1}, \sigma_m^{j+1}) + \mathcal{R}(\mathbf{C}^j, T^j, \sigma_m^j)}{2}. \quad (346)$$

Here $\mathbf{C}$ collects mobile and trapped populations. Nonlinear iteration, interface flux continuity, trap-occupancy bounds, and time/grid refinement shall be checked together. A time-centered update is accepted only when the nonlinear residual, population bounds, and integrated mass balance satisfy their tolerances. The material coefficients are specified in Table LXXXI.

The stress-coupled transport model uses the parameters in Table LXXXI at 550 K and $p_{D_2} = 4.20$ bar absolute [72], [73]. The quantities $C_L, C_{T,j}, N_j$ are mobile-deuterium, occupied-trap, and available-trap number densities in m^{-3} , respectively. Atomic counts are dimensionless in the rate equations, so k_j has units m^3/s ; explicit atom-count notation instead writes $\text{m}^3 \text{atom}^{-1} \text{s}^{-1}$. The reference temperature is 276.85 °C, not the 350 °C structural operating point. Extrapolation to the latter requires temperature-dependent diffusion, solubility, and capture/release parameters.

b) *Chemical-potential driving force:* For dilute interstitial deuterium in a homogeneous material at fixed temperature, define the molar chemical potential by

$$\begin{aligned} \mu_L &= \mu_L^0(T) + RT \ln(C_L/C_*) - V_H^* \sigma_m, \\ \sigma_m &= \frac{1}{3} \text{Tr } \boldsymbol{\sigma}, \\ \mathbf{J}_L &= -\frac{D_L C_L}{RT} \nabla \mu_L \\ &= -D_L \nabla C_L + D_L C_L \frac{V_H^*}{RT} \nabla \sigma_m. \end{aligned} \quad (347)$$

Here C_* is a fixed reference concentration, V_H^* is the partial molar insertion volume, R is the molar gas constant, and tensile hydrostatic stress is positive. The driving-force relation assumes constant V_H^* within each material and ideal dilute activity; concentrated loading requires an activity-based chemical potential. Positive tensile-stress gradients favor interstitial accumulation. A spatially uniform compressive preload, even at 15.80 MPa, does not produce a bulk stress-gradient flux.

c) *Conservation and finite-rate trapping:* The McNabb–Foster balance and its thermal-gradient extension are

$$\begin{aligned} \partial_t C_L + \sum_j \partial_t C_{T,j} &= -\nabla \cdot \mathbf{J}_L, \\ \partial_t C_{T,j} &= k_j C_L (N_j - C_{T,j}) - p_j C_{T,j}, \\ \mathbf{J}_L &= -D_L \nabla C_L + \frac{D_L C_L V_H^*}{RT} \nabla \sigma_m \\ &\quad - \frac{D_L C_L Q^*}{RT^2} \nabla T, \\ \sigma_m &= \frac{1}{3} \text{Tr } \boldsymbol{\sigma}. \end{aligned} \quad (348)$$

The flux $\mathbf{J}_L$ has units $\text{m}^{-2} \text{s}^{-1}$. The quantities V_H^* and Q^* are molar, making the ratios containing R dimensionally consistent with a number-density flux. The heat of transport Q^* must be converted from kJ/mol to J/mol in the thermal term. Equation (348) defines positive Q^* as transport toward decreasing temperature. Explicit trapping requires a bare lattice diffusivity D_L ; simultaneous use of a trap-renormalized D_{eff}

would double-count retardation. The tabulated D_{eff} values specify a diffusivity parameterization whose bare-lattice or effective interpretation must be established before numerical use. The explicitly bare D_L values in Table XXXIX belong to a distinct two-family model and are not interchangeable with this table.

Equilibrium requires $p_j/k_j = N_0 \exp[-E_{t,j}/(k_B T)]$. The coefficients in Table LXXXI imply $N_0 = 1.3020 \times 10^{25}$, 6.2396×10^{25} and $1.0873 \times 10^{27} \text{m}^{-3}$, respectively. Verification of detailed balance requires consistent host-site density, degeneracy, and capture conventions. A uniform preload removes the stress-gradient term only if $\nabla \sigma_m \simeq 0$ within the actual microstructure. Grain boundaries, free edges, temperature gradients, and trapped populations can remain nonuniform. Fuel homogeneity therefore requires verification and does not follow from a 15.80 MPa preload alone.

For fixed trap density, $C_{T,j} = N_j \theta_j$, and constant local C_L gives the closed-form occupancy kinetics

$$\begin{aligned} \theta_{j,\text{eq}} &= \frac{k_j C_L}{k_j C_L + p_j}, \\ \tau_j &= (k_j C_L + p_j)^{-1}, \\ \theta_j(t) &= \theta_{j,\text{eq}} + [\theta_j(0) - \theta_{j,\text{eq}}] e^{-t/\tau_j}. \end{aligned} \quad (349)$$

The occupancy satisfies $0 \leq \theta_j \leq 1$ for nonnegative rates and admissible initial conditions. The half-occupancy concentration $C_{1/2,j} = p_j/k_j$ and the corresponding local relaxation time are. These scales follow from the capture and release coefficients in Table LXXXI; they are not through-thickness charging times. If traps remain in local equilibrium, the retardation factor is

$$\begin{aligned} \mathcal{R}_T(C_L) &= \frac{\partial}{\partial C_L} \left(C_L + \sum_j C_{T,j}^{\text{eq}} \right) \\ &= 1 + \sum_j \frac{N_j k_j p_j}{(k_j C_L + p_j)^2}, \\ D_{\text{app}}(C_L) &= D_L / \mathcal{R}_T(C_L). \end{aligned} \quad (350)$$

The last expression describes the isothermal, uniform-stress reduction with spatially fixed material parameters. Outside the local-equilibrium limit, the full occupancy equations must be integrated rather than replacing them by D_{app} .

d) *Uniform preload and multilayer boundary conditions:* Under the combined assumptions $\nabla \sigma_m \simeq 0$ and $\nabla T = 0$, Eq. (348) reduces to $\mathbf{J}_L = -D_L \nabla C_L$, while capture and detrapping remain active. Uniform applied pressure does not imply uniform hydrostatic stress near layer interfaces, bevels, or defects; the stress field in Section III supplies the local driving term. Even a constant stress changes the equilibrium chemical potential relative to an unloaded surface.

At an ideal interface without interfacial storage or reaction, normal flux and equilibrium chemical potential are continuous:

$$\begin{aligned} \hat{\mathbf{n}} \cdot \mathbf{J}_{L,1} &= \hat{\mathbf{n}} \cdot \mathbf{J}_{L,2}, \\ \mu_{L,1} &= \mu_{L,2}. \end{aligned} \quad (351)$$

Concentration is generally discontinuous because the materials have different standard chemical potentials and solubilities. The gas-facing equilibrium condition is the Sieverts relation $C_s = S_D(T) \sqrt{p_{D_2}}$ with absolute pressure and an atom-count

TABLE XXXVII
CALCULATED HANKEL SINGULAR VALUES FOR FIFTEEN-TO-FOUR-STATE REDUCTION

Index	Calculated σ_i	Target	Relative-error limit	Treatment
1	12.45	12.45	10^{-6}	Retain
2	5.13	5.13	10^{-6}	Retain
3	1.85	1.85	10^{-6}	Retain
4	0.94	0.94	10^{-6}	Retain
5	0.015	0.015	10^{-4}	Truncate
6	0.010	—	—	Truncate
7	0.007	—	—	Truncate
8	0.004	—	—	Truncate
9	0.002	—	—	Truncate
10	0.001	—	—	Truncate
11	0.0005	—	—	Truncate
12	0.0002	—	—	Truncate
13	0.0001	—	—	Truncate
14	5.0×10^{-5}	—	—	Truncate
15	2.0×10^{-5}	—	—	Truncate

Maximum relative error, modes 1–4: $2.362176648138631 \times 10^{-16}$.

Discarded squared-spectrum ratio: $2.129567641168566 \times 10^{-6}$.

First four modes: $\varepsilon_{\text{rel}} \simeq 2.36 \times 10^{-16}$.

Tail-energy metric: $\eta_{\text{tail}} \simeq 2.13 \times 10^{-6}$.

TABLE XXXVIII
SINGLE-FAMILY TRAP EQUILIBRATION SCALES AT 550 K

Layer	$C_{1/2,j}$ (m^{-3})	$\tau_j(C_{1/2,j})$ (s)
Pd–Ir	$1.25517241 \times 10^{23}$	$2.74725275 \times 10^{-4}$
Ti	$6.90322581 \times 10^{19}$	2.33644860
ε–Cu ₃ Sn	$4.34523810 \times 10^{22}$	0.01369863

solubility coefficient; finite dissociation or recombination kinetics requires a surface exchange law. Initial lattice concentrations and trap occupancies, exterior sink conditions, and the actual Pd–Ir/Ti/bondline geometry complete the transport problem. A perfectly blocking boundary is a limiting case, not a substitute for the finite barrier permeance.

e) Trap normalization and interface partitioning: For M trap families, total deuterium concentration is $C = C_L + \sum_{j=1}^M N_{T,j} \theta_j$, with $C_L = N_L \theta_L$. Diffusion-limited capture by a spherical trap of radius R_{cap} has number-density coefficient $k_{\text{cap}} = 4\pi R_{\text{cap}} D_L$ in m^3/s . If $N_L = a^{-3}$ is the adopted site-density approximation, its occupancy-based coefficient is $k_{\text{occ}} = k_{\text{cap}} N_L = 4\pi R_{\text{cap}} D_L / a^3$ in s^{-1} . These coefficients multiply C_L and θ_L , respectively, and are not interchangeable. For molar binding and migration energies,

$$p_j = \nu_D \exp[-(E_{B,j} + E_L)/(RT)]. \quad (352)$$

Per-particle energies instead require $k_B T$. At a common interface temperature, equilibrium partitioning can be written

$$\frac{C_L^{(1)}}{S_L^{(1)}} e^{-V_H^{*(1)} \sigma_h^{(1)} / (RT)} = \frac{C_L^{(2)}}{S_L^{(2)}} e^{-V_H^{*(2)} \sigma_h^{(2)} / (RT)}, \quad (353)$$

where the solubility normalizations use a common chemical-potential reference. Table VIII gives the calculated steady interface fluxes; each differs from its target by zero at the displayed precision. A local hydrostatic tension is not a gradient:

these fluxes require complete concentration, stress, temperature, and boundary profiles rather than the listed tension alone.

For a locally frozen nonnegative C_L , an implicit trap substep is

$$\theta_j^{n+1} = \frac{\theta_j^n + \Delta t k_j C_L}{1 + \Delta t (k_j C_L + p_j)}. \quad (354)$$

This preserves $0 \leq \theta_j \leq 1$ but is first-order in time. A split implementation shall account for that temporal order and remove exactly $N_{T,j}(\theta_j^{n+1} - \theta_j^n)$ from the mobile population. Drift enters through the divergence of a face flux, not as a flux directly subtracted from a concentration rate. For fixed control volume $\mathcal{V}$ with $M_0 > 0$, the global conservation requirement is

$$\begin{aligned} M(t) &= \int_{\mathcal{V}} \left(C_L + \sum_j N_{T,j} \theta_j \right) dV, \\ M_{\text{out}}(t) &= \int_0^t \int_{\partial \mathcal{V}} \mathbf{J}_L \cdot \hat{\mathbf{n}} dA dt', \\ \frac{|M(t) - M_0 + M_{\text{out}}(t)|}{M_0} &< 10^{-12}. \end{aligned} \quad (355)$$

Interface partitioning and flux continuity are enforced in the same conservative solve.

f) Environmental fracture and slow-strain-rate metrics: The embrittlement assessment uses the following nominal material-response values:

$$\begin{aligned} K_{IH} &= 41.8 \text{ MPa}\sqrt{\text{m}}, \\ I_{RA} &= \frac{RA_{\text{exposed}}}{RA_{\text{inert}}} = 0.89, \\ K_{I,\text{hold}} &= 0.80 K_{IH} = 33.44 \text{ MPa}\sqrt{\text{m}}, \\ \dot{a} &< 10^{-11} \text{ m/s}, \quad t_{\text{hold}} = 1000 \text{ h}, \\ \Delta a_{\text{hold}} &< 3.6 \times 10^{-5} \text{ m} = 36 \mu\text{m}. \end{aligned} \quad (356)$$

The integrated crack-growth bound assumes the rate limit holds throughout the specified exposure. Reduction of area is $RA = (A_0 - A_f)/A_0$ for the tensile specimen; its exposed-to-inert ratio is dimensionless. The selected SSRT criterion is $I_{RA} \geq 0.85$, giving an absolute margin of 0.04 at the nominal value. The 0.80 criterion in Table XLI is a separate, less restrictive comparison criterion.

ASTM G129 defines comparative slow-strain-rate testing, while ASTM E1681 addresses environment-assisted cracking thresholds [48], [76]. The notation K_{IH} identifies the deuterium-environment threshold in this specification; the applicable E1681 threshold designation and specimen-size validity shall accompany its use. Qualification requires specimen material and heat, geometry, precrack history, loading mode, strain rate, absolute and differential pressures, temperature, replicates, and measurement uncertainty. Neither a method designation nor a favorable reduction-of-area ratio establishes immunity, and these metrics cannot be assigned to Pd-Ir, Ti, Cu-Sn, and containment alloys indiscriminately.

g) *Bilayer permeation barrier and defect tolerance:*

The noncatalytic internal barrier comprises 150 nm TiN / 250 nm Al_2O_3 , with total nominal coating thickness 400 nm. Its surface coverage shall not obstruct the intended deuterium-loading interface. For a common permeation driving variable $\Delta\psi$, the intact through-thickness path has series resistance

$$\begin{aligned} \mathcal{R}_{\text{intact}} &= \mathcal{R}_{\text{sub}} + \frac{150 \times 10^{-9}}{\mathcal{P}_{\text{TiN}}} + \frac{250 \times 10^{-9}}{\mathcal{P}_{\text{Al}_2\text{O}_3}} + \mathcal{R}_{\text{interfaces}}, \\ J_{\text{intact}} &= \Delta\psi / \mathcal{R}_{\text{intact}}, \\ r_P &= \frac{J_{\text{intact}}}{J_{\text{bare}}} = \frac{\mathcal{R}_{\text{sub}}}{\mathcal{R}_{\text{intact}}}. \end{aligned} \quad (357)$$

The thickness numerators are in metres; the permeabilities $\mathcal{P}_j$ and resistances must use the same driving-variable and flux convention. Here r_P is the intact-to-bare *permeance* ratio at matched boundary conditions; a ratio of intrinsic material permeabilities alone would omit layer thickness and interface resistance. The prescribed intact-path ratio is $r_P = 1.20 \times 10^{-8}$. The coating thicknesses alone do not establish that value.

With defect fraction $f_d = 6.49 \times 10^{-4}$ and bare-path transmission through each defect, area-weighted parallel transport gives

$$\begin{aligned} \frac{J_{\text{coated}}}{J_{\text{bare}}} &= (1 - f_d)r_P + f_d, \\ \text{PRF}_{\text{eff}} &= \frac{J_{\text{bare}}}{J_{\text{coated}}}. \end{aligned} \quad (358)$$

Substitution yields

$$\text{PRF}_{\text{eff}} = \frac{1}{(1 - f_d)r_P + f_d} = 1540.8036. \quad (359)$$

The conservative reduction-factor target is 1538.5, so the specified model satisfies

$$\text{PRF}_{\text{eff}} = 1540.803578 \dots > 1538.5 > 1000. \quad (360)$$

Exact equality to 1538.5 instead requires $f_d = 6.49971758 \times 10^{-4}$ at the same intact-path ratio. More generally,

$$\begin{aligned} \text{PRF}_{\text{eff}} > 1000 &\iff f_d < \frac{10^{-3} - r_P}{1 - r_P} \\ &= 9.99988012 \times 10^{-4}. \end{aligned} \quad (361)$$

This implication is conditional on the parallel-path assumptions. Lateral transport into pinholes, coating cracks, interfacial bypass, and temperature-dependent permeance can change the effective leakage fraction. Barrier qualification shall determine connected-defect statistics and flux under the same pressure and temperature conditions as the bare-substrate reference.

h) *Environmental threshold size and finite-wall applicability:* For a semi-infinite edge-crack approximation with geometry factor $Y = 1.12$, $K_I = Y\sigma_{\text{hoop}}\sqrt{\pi a}$. Equating the stress intensity to K_{IH} gives the threshold estimate

$$\begin{aligned} a_{\text{crit}} &= \frac{1}{\pi} \left(\frac{K_{IH}}{1.12\sigma_{\text{hoop}}} \right)^2, \\ \sigma_{\text{hoop}} &= 48.5 \text{ MPa}, \quad a_{\text{crit}} = 188.488 \text{ mm} \simeq 189 \text{ mm}. \end{aligned} \quad (362)$$

The nominal 189 mm value is an approximate reference to the calculated 188.4877665 mm environmental-threshold size, not a conservative allowable flaw depth. The symbol a_{crit} in Eq. (362) identifies this threshold calculation; it does not establish the instability size of the vessel. In particular, K_{IH} is distinct from unstable-fracture toughness. The extrapolated size greatly exceeds the 6.0 mm wall, where a semi-infinite edge-crack geometry factor is no longer applicable. Even the illustrative value $K_I(6.0 \text{ mm}) = 7.45779 \text{ MPa}\sqrt{\text{m}}$ retains that geometry limitation. The 4.20 bar transport pressure, 48.5 MPa hoop-stress case, and separate shielding-shell thickness require distinct, consistent structural boundary conditions.

i) *Formal leak-before-break acceptance conditions:* Leak-before-break requires a stable through-wall crack that produces detectable leakage before unstable extension [86]. A finite-wall assessment uses surface-crack depth and length before penetration, then a through-wall crack-length coordinate ℓ . The geometry-specific fracture driving force and a schematic set of acceptance conditions are

$$\begin{aligned} K_I(\ell) &= K_{I,\text{pressure}}(\ell) + K_{I,\text{thermal}}(\ell) + K_{I,\text{residual}}(\ell), \\ \dot{m}_{\text{leak}}(\ell_{\text{det}}) &\geq \dot{m}_{\text{det}}, \\ \ell_{\text{det}} &< \ell_{\text{inst}}, \\ K_I(\ell_{\text{det}}) &< K_{\text{mat}}/\text{SF}_K, \\ t_{\text{response}} &< \int_{\ell_{\text{det}}}^{\ell_{\text{inst}}} \frac{d\ell}{\dot{\ell}(\ell)}. \end{aligned} \quad (363)$$

Here $\dot{m}_{\text{det}}$ is the qualified detection threshold, ℓ_{det} the detectable through-wall flaw length, ℓ_{inst} the instability length under the design load combination, K_{mat} the applicable fracture toughness, and $\text{SF}_K > 1$ the selected fracture margin. Linear superposition applies only within the elastic stress-intensity framework; significant plasticity requires an elastic-plastic fracture or failure-assessment treatment. The time integral assumes positive growth along a specified crack path and load history; its growth law must bound cyclic and environmental extension over the whole interval.

Leak-flow calculation requires crack opening, wall thickness, roughness, fluid state, and the pressure differential. Instrument response, alarm processing, isolation, and depressurization all contribute to t_{response} . A 1000-hour growth bound at $0.80K_{IH}$ cannot be extrapolated to the instability threshold without an applicable growth law. Consequently, the 189 mm

threshold reference and the favorable embrittlement metrics support separate screening checks; a formal LBB demonstration additionally requires the finite-wall fracture, leakage, and response quantities in Eq. (363).

I. Sieverts Dissolution and Deuterium Transport

The hot-service material reference is D_2 at 4.20 bar and 550.0 K. Exposure at 418.15 K does not qualify components for the higher temperature. Test records shall distinguish absolute pressure for uptake calculations from differential pressure for wall-stress calculations and explicitly define the pressure convention used in embrittlement testing.

An equilibrium boundary model is

$$\begin{aligned} C_s(T, p) &= K_S(T) \sqrt{p_{D_2}}, \\ K_S(T) &= K_{S,0} \exp\left(-\frac{\Delta H_s}{RT}\right). \end{aligned} \quad (364)$$

For lattice concentration C_L , trap density N_j , and occupied trap fraction θ_j , the McNabb–Foster transport framework is

$$\begin{aligned} \frac{\partial C_L}{\partial t} + \sum_j N_j \frac{\partial \theta_j}{\partial t} &= -\nabla \cdot \mathbf{J}_L, \\ \frac{\partial \theta_j}{\partial t} &= k_j C_L (1 - \theta_j) - p_j \theta_j. \end{aligned} \quad (365)$$

Here k_j, p_j are trapping and release coefficients with units consistent with the concentration convention. Material heat, cold work, welds, surface state, isotope, and temperature affect the coefficients and boundary conditions. The HE protocols address slow-strain-rate response, fracture threshold, and crack growth. Satisfying these protocols does not independently calibrate every uptake and transport parameter or establish uniform charging through an arbitrary wall thickness.

a) *Stress, Soret transport, and two trap families:* With tensile-positive hydrostatic stress $\sigma_h = \sigma_{ii}/3$, the coupled flux is

$$\begin{aligned} \mathbf{J}_L &= -D_L \nabla C_L + \frac{V_H}{R_u T} D_L C_L \nabla \sigma_h \\ &+ \frac{Q^*}{R_u T^2} D_L C_L \nabla T. \end{aligned} \quad (366)$$

Combined with (365), this gives the divergence of $D_L \nabla C_L - (V_H/R_u T) D_L C_L \nabla \sigma_h - (Q^*/R_u T^2) D_L C_L \nabla T$. C_L counts atoms per cubic meter, V_H is molar volume, R_u the molar gas constant, and Q^* a signed molar heat of transport. Positive Q^* here drives atoms toward increasing T ; an opposite Soret convention reverses its sign. Trap densities are fixed in this balance; evolving traps require $\partial_t(N_j \theta_j)$. For a numerical boundary comparison interpret pressure as $p_{D_2} = 4.20 \times 10^5$ Pa absolute, with $C_s = S_D \sqrt{p_{D_2}}$ and $\theta_{j,\text{eq}} = k_j C_s / (k_j C_s + p_j)$. Release rates are already assigned at 550 K and are not multiplied by another activation factor. Binding-energy extrapolation requires attempt rates and consistent detailed balance. Neither V_H , Q^* nor their temperature laws is included in the ten rows; the table alone cannot supply numerical stress/Soret fluxes. These active-stack values do not replace the separate SS316L/Invar containment data.

b) Molar solubility and microstructural trap families:

The microstructural trap-family model uses a molar Sieverts coefficient,

$$K_S(T) = K_0 e^{-1000 H_s / (R_u T)}, \quad C_{D,\text{mol}} = K_S(T) \sqrt{p_{D_2}}. \quad (367)$$

H_s is tabulated in kJ/mol, R_u in J/(mol K), and p_{D_2} in pascals absolute; the factor 1000 is essential. Define the molar concentration as moles of D atoms, so $C_L = N_A C_{D,\text{mol}}$ when used in the atom-count transport equations. The parameters $K_S, N_j, E_{t,j}$ define this microstructural transport case rather than the alternative active-stack parameterization. Table XLIII associates the trap families with capture and release rates; binding energies alone do not determine attempt frequencies. Containment-alloy data apply to their separately specified materials and conditions. The candidate phenomenological relations are $K_{IH} = K_{I0} - \beta \sqrt{C_D}$ and $\dot{a} = v_0 \exp[-(E_a - \sigma_h V_a) / (R_u T)]$. In the latter, V_a is a molar activation volume when E_a is expressed in J/mol; an atomic volume instead requires atomic energy and k_B . Quantitative prediction requires calibration of $K_{I0}, \beta, v_0, E_a, V_a$. Without that calibration, these forms establish neither a crack-growth rate nor a fracture threshold.

J. Hydrogen Embrittlement: HE-01–HE-03 Qualification Criteria

The high-temperature assessment applies at 550.0 K (276.85°C) and $p_{D_2} = 4.20$ bar [47]. Table XLI defines the nominal hot-service metrics, distinct from the 145°C comparison dataset. Qualification requires identification of the specimen material and heat for every result. A single set of unassigned metrics cannot qualify both SS316L and Invar 36 independently.

a) *HE-01: Slow strain rate testing:* Define $I_{\text{RA}} = \text{RA}_{D_2} / \text{RA}_{\text{inert}}$. ASTM G129 addresses slow strain rate testing; ASTM F1624 is incremental step loading, not another SSRT method [48]. Specimen IDs, product forms, temperatures, strain rates, and inert/environmental controls shall identify which method generated each result.

b) *HE-02: Environment-assisted fracture threshold:* ASTM E1681 provides the environment-assisted threshold framework [48]. The $41.8 \text{ MPa} \sqrt{\text{m}}$ value requires verification of specimen geometry, precrack validity, loading mode, hold time, and pressure convention. It exceeds the 35.0 criterion but not the separate SS316L 45.0 criterion. Qualification shall apply the acceptance threshold assigned to the actual material and service condition.

c) *HE-03: Subcritical crack growth:* The $10^{-11} \text{ m s}^{-1}$ value is an upper bound rather than a measured constant rate. Sustained-load records shall specify stress intensity, duration, resolution, and uncertainty. Application of ASTM E1681/F1624 requires a compatible loading history and identified specimen material.

K. Five-Year Safe-Life and Leak-Before-Break Assessment

a) *High-pressure containment comparison:* For the distinct $p = 15$ MPa, $t_w = 8.0$ mm containment case, let

TABLE XXXIX
TWO-FAMILY DEUTERIUM TRAPPING PARAMETERS AT 550 K AND 4.20 BAR

Parameter (units)	Pd _{0.95} Ir _{0.05}	Ti	ε-Cu ₃ Sn
D_L (m ² /s)	4.50×10^{-9}	1.20×10^{-11}	8.50×10^{-13}
S_D (atoms m ⁻³ Pa ^{-1/2})	1.62×10^{20}	4.38×10^{21}	2.10×10^{19}
N_1 (sites/m ³)	1.50×10^{24}	3.80×10^{25}	5.50×10^{24}
N_2 (sites/m ³)	5.00×10^{23}	1.20×10^{25}	1.80×10^{24}
k_1 (m ³ atom ⁻¹ s ⁻¹)	1.15×10^{-25}	2.80×10^{-27}	4.20×10^{-28}
k_2 (m ³ atom ⁻¹ s ⁻¹)	8.40×10^{-26}	1.95×10^{-27}	3.10×10^{-28}
p_1 (s ⁻¹)	24.50	0.15	0.08
p_2 (s ⁻¹)	120.40	1.42	0.75
$E_{t,1}$ (eV)	0.23	0.48	0.52
$E_{t,2}$ (eV)	0.15	0.35	0.40

TABLE XL
MULTIPHYSICS SOLUBILITY AND TRAP PARAMETERS

Material	K_0	H_s	Trap family	N_j (m ⁻³)	$E_{t,j}$ (kJ/mol)
Pd _{0.95} Ir _{0.05}	1.2×10^{-1}	12.5	Dislocations	1.5×10^{23}	26.4
			Phase boundaries	8.0×10^{22}	44.1
Ti	2.4×10^1	45.2	Interstices	1.0×10^{25}	58.3
			Grain boundaries	5.0×10^{24}	32.1
ε-Cu ₃ Sn	4.5×10^{-3}	28.4	Kirkendall voids	6.2×10^{25}	30.8
			Interfaces	1.1×10^{26}	18.5

K_0 has units mol m⁻³ Pa^{-1/2} and H_s has units kJ/mol. At 550 K, K_S is 0.0077992621 (Pd-Ir), 0.0012233687 (Ti), and 9.0377081×10^{-6} (Cu-Sn), in the same molar units.

TABLE XLI
HIGH-TEMPERATURE EMBRITTLEMENT METRICS AND ACCEPTANCE LIMITS

Observable	Nominal	Criterion
HE-01, I_{RA}	0.89	≥ 0.80
HE-02, K_{IH} (MPa $\sqrt{m}$)	41.8	≥ 35.0
HE-03, da/dt (m s ⁻¹)	$< 10^{-11}$	$\leq 10^{-10}$

$K_{Ic} = 85.0 \text{ MPa}\sqrt{m}$. The elastic fracture and cyclic-growth relations are

$$K_I = Y\sigma_{\max}\sqrt{\pi a}, \quad a_{\text{crit}} = \frac{1}{\pi} \left(\frac{K_{Ic}}{Y\sigma_{\max}} \right)^2, \quad (368)$$

$$\frac{da}{dN} = C(\Delta K)^m, \quad \Delta K = Y\Delta\sigma\sqrt{\pi a}.$$

The assigned $a_{\text{crit}} = 37.9 \text{ mm}$ requires $Y\sigma_{\max} = 246.334 \text{ MPa}$. Pressure and wall thickness alone do not determine this stress without radius, flaw geometry, and residual/thermal loading. The 37.9 mm case is distinct from the 189 mm screening case and the low-pressure geometry below. A critical length exceeding wall thickness does not prove leak-before-break: a stable through-wall crack, leak-rate detectability, crack-growth margin, and shutdown time must be assessed jointly [86]. Paris-law integration additionally requires C, m , load history, and the applicable crack-growth regime.

For the low-pressure screening geometry, $t = 6.0 \text{ mm}$, $R = 80.0 \text{ mm}$, $\Delta p = 0.420 \text{ MPa}$, $Y = 1.12$, and $a_0 \leq 0.500 \text{ mm}$, application of the upper crack-growth bound over five years

gives

$$t_{\text{service}} = 5(365.25)(86400) = 1.57788 \times 10^8 \text{ s},$$

$$\Delta a < 10^{-11} t_{\text{service}} = 1.57788 \text{ mm}, \quad (369)$$

$$a_{\text{final}} < 2.07788 \text{ mm} < 6.0 \text{ mm}.$$

The 0.784 mm comparison endpoint uses a low-temperature rate of $1.8 \times 10^{-12} \text{ m s}^{-1}$ and does not apply to this bound. The pressure-only stress is $\sigma_{\text{hoop}} = \Delta p R / t = 5.60 \text{ MPa}$, and $K_I = Y\sigma_{\text{hoop}}\sqrt{\pi a_{\text{final}}} < 0.507 \text{ MPa}\sqrt{m}$ in the screening expression. Neither value applies to the separate 15 MPa(g) core relief case.

The conditional non-through-wall projection excludes unmodeled weld, residual, cyclic, and thermal stresses and requires applicable material data. Leak-before-break additionally requires a stable through-wall flaw, detectable leakage before instability, and sufficient response time. A sub-wall crack-depth comparison does not establish these conditions; fracture and leak-detection qualification remain necessary.

b) High-temperature hardening and trap-family kinetics:

The five-layer comparison geometry and its HiPIMS protocol define the material configuration for this parameter set. The 91.3189 wt% Pd target composition does not establish deposited-alloy composition, which requires measurement. The Voce parameters are These parameters enter the total-flow-stress law without an additional yield-stress increment. The three-dimensional VCCT formulation includes Modes I, II, and

TABLE XLII
VOCE TOTAL-FLOW-STRESS PARAMETERS AT 550 K

Material	σ_{y0} (MPa)	R_∞ (MPa)	b
ϵ -Cu ₃ Sn	180.0	120.0	15.0
OFHC Cu	65.0	110.0	8.5

III:

$$\begin{aligned}
 G_I &= \frac{F_z \Delta w}{2b_f \Delta a}, & G_{II} &= \frac{F_x \Delta u}{2b_f \Delta a}, \\
 G_{III} &= \frac{F_y \Delta v}{2b_f \Delta a}, & G_{\text{total}} &= G_I + G_{II} + G_{III}, \\
 SF_G &= 24.50/18.40 = 1.33152174.
 \end{aligned} \tag{370}$$

Signed nodal forces and relative displacements require a common crack-closure convention. The nominal simulation values are 18.40 J/m² total energy release and 142.5 MPa peak shear; an arithmetic safety ratio alone does not validate the FEA. For this coefficient set, the 44,820-cycle Coffin–Manson bound is $\Delta \epsilon_p \leq 0.000983786284 = 0.0983786\%$. The separate 0.098% design cap is slightly tighter; the inequality $0.0983786\% \leq 0.098\%$ is not satisfied.

Table XLIII associates microstructural trap families with capture and release rates. Its $N_j, E_{t,j}$ parameterization is distinct from the two-family case in Table XXXIX. The molar solubilities in Table XL require conversion to atom-count C_L before use in $k_j C_L$. The equilibrium solution is $\theta_j = k_j C_L / (k_j C_L + p_j)$. These 550 K assignments do not independently define Arrhenius laws; temperature extrapolation requires compatible attempt frequencies, binding energies, and detailed balance. The nominal metrics $K_{IH} = 41.8 \text{ MPa}\sqrt{\text{m}}$, $\dot{a} < 10^{-11} \text{ m/s}$, and $I_{RA} = 0.89$ retain their material-heat and test-condition scope.

V. THERMAL HYDRODYNAMICS, TEG ACCOUNTING, AND PRECISION CALORIMETRY

The thermal measurement architecture couples three-dimensional manifold mixing, differential resistance thermometry, envelope calorimetry, and synchronized electrical measurements. The hydraulic model establishes the sensing-plane flow and temperature distributions; the measurement model converts those observables into heat and power with an explicit covariance budget. Thermoelectric generation and auxiliary consumption are accounted for at separate electrical terminals. All quoted design figures retain their fluid-property, calibration, and control-volume definitions.

A. Solid-State Equation of State and Debye Thermodynamics

a) Reference state and Birch–Murnaghan compression:

The solid substrate, film, joint, and spring phases require state-consistent volume, heat capacity, and elastic response. Let V_m be volume per mole of formula units and ν_a the number of atoms per formula unit. Composition and phase are held fixed in the following derivatives unless specified otherwise. The hydrostatic pressure p is positive in compression, so $p =$

$-\sigma_m$ for a purely hydrostatic stress under the tension-positive mechanical convention. The third-order Birch–Murnaghan cold-curve representation is [112]

$$\begin{aligned}
 x &= V_m/V_{m0}, & q &= x^{-2/3}, \\
 E_{\text{BM}}(V_m) &= E_0 + \frac{9B_0 V_{m0}}{16} [B'_0(q-1)^3 \\
 &\quad + (q-1)^2(6-4q)], \\
 p_{\text{BM}}(x) &= -\frac{\partial E_{\text{BM}}}{\partial V_m} \\
 &= \frac{3B_0}{2}(x^{-7/3} - x^{-5/3}) \\
 &\quad \times \left[1 + \frac{3}{4}(B'_0 - 4)(x^{-2/3} - 1) \right].
 \end{aligned} \tag{371}$$

Here E_{BM} is in J/mol, V_{m0} is the static reference volume, B_0 is in Pa, and $B'_0 = (\partial B/\partial p)_0$ is dimensionless. This finite-strain expansion is a constitutive approximation to a phase-specific energy–volume relation, not an exact equation for arbitrary compression or a substitute for an atomistic energy calculation. In particular, its three hydrostatic parameters do not determine shear moduli or an anisotropic elastic tensor.

For a nonlinear volume solve, both factors in the pressure law must be differentiated. Defining $A(x) = x^{-7/3} - x^{-5/3}$ and $B(x) = 1 + \frac{3}{4}(B'_0 - 4)(x^{-2/3} - 1)$ gives

$$\begin{aligned}
 A'(x) &= -\frac{7}{3}x^{-10/3} + \frac{5}{3}x^{-8/3}, \\
 B'(x) &= -\frac{1}{2}(B'_0 - 4)x^{-5/3}, \\
 \frac{dp_{\text{BM}}}{dx} &= \frac{3}{2}B_0[A'(x)B(x) + A(x)B'(x)], \\
 B_{\text{BM}}(x) &= -x \frac{dp_{\text{BM}}}{dx}.
 \end{aligned} \tag{372}$$

Thus $p_{\text{BM}}(1) = 0$ and $B_{\text{BM}}(1) = B_0$. The admissible branch has $V_m > 0$ and positive isothermal bulk modulus; a formal algebraic root outside that branch is not a stable material state.

b) *Debye temperature, density of states, and heat capacity:* With atomic number density $n_a = \nu_a N_A/V_m$, the Debye cutoff and temperature are

$$\begin{aligned}
 k_D &= (6\pi^2 n_a)^{1/3}, & \Theta_D &= \frac{\hbar v_D k_D}{k_B}, \\
 v_D^{-3} &= \frac{1}{12\pi} \sum_{s=1}^3 \int_{S^2} v_s(\hat{\mathbf{n}})^{-3} d\Omega,
 \end{aligned} \tag{373}$$

$$C_{ijkl}^S \hat{n}_j \hat{n}_k e_{s,i} = \rho v_{s,i}^2 e_{s,i}.$$

The sound speeds follow from the adiabatic acoustic tensor $\mathbf{C}^S$, not the bulk EOS alone. In an isotropic restriction, $v_D^{-3} = (v_L^{-3} + 2v_T^{-3})/3$, $v_L^2 = (B_S + 4G/3)/\rho$, and $v_T^2 = G/\rho$. Independent shear or phonon information is therefore necessary. Assuming $v_D \propto V_m^{-1/3}$ would be an additional approximation, not a consequence of the Debye mode count.

TABLE XLIII
MICROSTRUCTURAL TRAP-FAMILY KINETICS AT 550 K

Material	Trap family	N_j (m ⁻³)	$E_{t,j}$ (kJ/mol)	k_j (m ³ atom ⁻¹ s ⁻¹)	p_j (s ⁻¹)
Pd _{0.95} Ir _{0.05}	Dislocations	1.50×10^{23}	26.4	1.15×10^{-25}	24.50
	Phase boundaries	8.00×10^{22}	44.1	8.40×10^{-26}	120.40
Ti	Interstices	1.00×10^{25}	58.3	2.80×10^{-27}	0.15
	Grain boundaries	5.00×10^{24}	32.1	1.95×10^{-27}	1.42
ϵ -Cu ₃ Sn	Kirkendall voids	6.20×10^{25}	30.8	4.20×10^{-28}	0.08
	Interfaces	1.10×10^{26}	18.5	3.10×10^{-28}	0.75

For $y = \Theta_D/T$ and fixed volume, define the Debye integral and molar phonon free energy [113]:

$$\begin{aligned}
 D_3(y) &= \frac{3}{y^3} \int_0^y \frac{z^3}{e^z - 1} dz, \\
 F_D &= \frac{9}{8} \nu_a R \Theta_D + 3 \nu_a R T \ln(1 - e^{-y}) - \nu_a R T D_3(y), \\
 U_D &= F_D - T \left(\frac{\partial F_D}{\partial T} \right)_{V_m} = \frac{9}{8} \nu_a R \Theta_D + 3 \nu_a R T D_3(y), \\
 C_{V,D} &= \left(\frac{\partial U_D}{\partial T} \right)_{V_m} \\
 &= 9 \nu_a R \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} \frac{z^4 e^z}{(e^z - 1)^2} dz.
 \end{aligned} \tag{374}$$

Here $\Theta_D(V_m)$ is held constant in fixed-volume temperature derivatives; explicit anharmonic temperature dependence requires additional derivative terms. The zero-point term contributes to energy and pressure but not to this fixed-volume heat capacity. The correct limiting forms are

$$\begin{aligned}
 \frac{C_{V,D}}{3 \nu_a R} &= 1 - \frac{y^2}{20} + \frac{y^4}{560} + O(y^6), \quad y \rightarrow 0, \\
 C_{V,D} &\sim \frac{12\pi^4}{5} \nu_a R \left(\frac{T}{\Theta_D} \right)^3, \quad T/\Theta_D \rightarrow 0, \\
 \frac{dD_3}{dy} &= \frac{3}{e^y - 1} - \frac{3D_3(y)}{y}.
 \end{aligned} \tag{375}$$

The heat-capacity series is not the series for D_3 itself. Evaluation uses small-argument expansions and stable exponential differences near zero, and the equivalent integrand $z^4 e^{-z}/(1 - e^{-z})^2$ at large z . Quadrature convergence and derivative convergence must both be checked. Debye acoustic averaging is not a complete phonon spectrum for an ordered multicomponent joint; resolved phonon modes are needed when optical branches or anharmonicity dominate.

c) *Finite-temperature EOS and thermodynamic consistency*: A quasiharmonic phase model combines the static energy

with phonon, electronic, and configurational free energies:

$$\begin{aligned}
 F_m(V_m, T) &= E_{\text{BM}} + F_D + F_{\text{el}} + F_{\text{conf}}, \\
 p_{\text{int}} &= - \left(\frac{\partial F_m}{\partial V_m} \right)_T, \quad p_{\text{int}} = p_{\text{ext}}, \\
 \gamma_D &= - \frac{\partial \ln \Theta_D}{\partial \ln V_m}, \quad p_D = \frac{\gamma_D U_D}{V_m}, \\
 B_T &= -V_m \left(\frac{\partial p_{\text{int}}}{\partial V_m} \right)_T > 0, \\
 \alpha_V &= \frac{1}{V_m} \left(\frac{\partial V_m}{\partial T} \right)_p = \frac{1}{B_T} \left(\frac{\partial p_{\text{int}}}{\partial T} \right)_{V_m},
 \end{aligned} \tag{376}$$

$$C_{P,m} - C_{V,m} = T \alpha_V^2 B_T V_m.$$

The formula for p_D assumes that the phonon frequencies depend on volume through $\Theta_D(V_m)$ with no independent explicit temperature dependence. The static curve here excludes phonon zero-point and thermal energies; if the reference EOS already includes a finite-temperature contribution, the corresponding reference free energy must be subtracted before adding a thermal correction. Phase mixing and configurational energies are counted once, consistently with Eq. (296). Equilibrium volume minimizes $F_m + p_{\text{ext}} V_m$ on the selected stable phase branch.

For the residual $r(x) = p_{\text{int}}(V_{m0}x, T) - p_{\text{ext}}$, a safeguarded Newton step is $x_{k+1} = x_k - r(x_k)/r'(x_k)$ within a positive-volume bracket. Stopping requires converged residual and state change, not a fixed iteration count. Phase transitions require branch selection and may invalidate a single smooth sensitivity. The isothermal elastic tensor follows from second strain derivatives of the same free energy at a specified reference state; finite prestress requires the appropriate incremental-stiffness correction. Neither linear phase averaging nor a bulk EOS establishes the full tensor independently.

The heat capacities and enthalpy used for thermal accounting follow from this same potential:

$$\begin{aligned}
 S_m &= - \left(\frac{\partial F_m}{\partial T} \right)_{V_m}, \quad C_{V,m} = -T \left(\frac{\partial^2 F_m}{\partial T^2} \right)_{V_m}, \\
 H_m &= F_m + T S_m + p V_m, \quad C_{P,m} = \left(\frac{\partial H_m}{\partial T} \right)_p.
 \end{aligned} \tag{377}$$

Molar, mass-specific, and volumetric heat capacities remain distinct: $c_p = C_{P,m}/M_m$, $C_{P,\text{vol}} = C_{P,m}/V_m$, and $\rho = M_m/V_m$, where M_m is molar mass per formula unit. These solid-state relations do not replace the water or water-glycol property functions used by the calorimeter. Those fluid functions retain their own composition, pressure, and calibration uncertainties.

B. Drude–Lorentz Dielectric and Thermal Transport Tensors

For a local, reciprocal, linear-response medium with time convention $e^{-i\omega t}$, separate free-carrier and bound-oscillator polarizations. With tensor oscillator strengths and scalar damping for each oscillator,

$$\begin{aligned}\ddot{\mathbf{P}}_D &= -\gamma_D^{\text{opt}} \dot{\mathbf{P}}_D + \epsilon_0 \mathbf{W}_D \mathbf{E}, \\ \ddot{\mathbf{P}}_\ell &= -\gamma_\ell \dot{\mathbf{P}}_\ell - \omega_\ell^2 \mathbf{P}_\ell + \epsilon_0 \mathbf{S}_\ell \mathbf{E}, \\ \epsilon_r(\omega) &= \epsilon_\infty - \frac{\mathbf{W}_D}{\omega^2 + i\gamma_D^{\text{opt}}\omega} \\ &\quad + \sum_\ell \frac{\mathbf{S}_\ell}{\omega_\ell^2 - \omega^2 - i\gamma_\ell\omega}.\end{aligned}\quad (378)$$

The optical damping γ_D^{opt} is unrelated to the Debye Grüneisen parameter γ_D . Frequencies are angular frequencies in rad/s, and $\mathbf{W}_D, \mathbf{S}_\ell$ have units s^{-2} . The model retains true tensor strengths rather than calling a scalar times the identity anisotropic [114]. For parabolic carrier bands, the free-carrier specialization and its conductivity are

$$\begin{aligned}\mathbf{W}_D &= \frac{n_e e^2}{\epsilon_0} \mathbf{m}_{\text{eff}}^{-1}, \\ \sigma_D(\omega) &= \frac{\epsilon_0 \mathbf{W}_D}{\gamma_D^{\text{opt}} - i\omega}, \\ \epsilon_r^{\text{lab}} &= \mathbf{R} \epsilon_r^{\text{crystal}} \mathbf{R}^T.\end{aligned}\quad (379)$$

The orthogonal matrix $\mathbf{R}$ fixes crystal orientation. Multiband carriers, anisotropic relaxation, gyrotropy, and spatial dispersion require richer response models. Damping, effective masses, oscillator strengths, and resonances depend on composition, phase, temperature, and carrier state; they require electronic calculations or optical characterization. Homogenization across phases is a boundary-value problem, not necessarily a volume-weighted scalar average. The local Drude–Lorentz tensor is a constitutive optical model and does not replace the reciprocal-space Floquet dielectric matrix or its nuclear-screening calculation.

For positive damping, positive-semidefinite real symmetric strengths, and a passive background, the anti-Hermitian part and absorbed power density obey

$$\begin{aligned}\epsilon_A &= \frac{\epsilon_r - \epsilon_r^\dagger}{2i} \succeq 0, \quad \omega > 0, \\ q_{\text{abs}} &= \frac{\omega \epsilon_0}{2} \mathbf{E}^\dagger \epsilon_A \mathbf{E} \geq 0.\end{aligned}\quad (380)$$

The loss sign is tied to the stated time convention. Pumped gain media and explicitly modulated media need drive-work accounting and are not covered by this passive inequality. Absorbed optical power enters the thermal balance at the same control-volume boundary used for the electromagnetic calculation.

Heat capacity does not determine thermal conductivity without scattering information. A phonon relaxation-time

approximation and a degenerate-electron comparison are

$$\begin{aligned}k_{ij}^{\text{ph}} &= \frac{1}{\mathcal{V}_{\text{cell}}} \sum_\lambda w_\lambda C_\lambda v_{\lambda i} v_{\lambda j} \tau_\lambda, \\ \mathbf{k}_{\text{el}} &\simeq L_0 T \boldsymbol{\sigma}_{\text{dc}}, \quad L_0 = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2, \\ \mathbf{k} &= \mathbf{k}_{\text{ph}} + \mathbf{k}_{\text{el}}.\end{aligned}\quad (381)$$

Here C_λ is the heat capacity per mode, w_λ is the Brillouin-zone quadrature weight, and τ_λ is the mode lifetime. The Wiedemann–Franz form assumes degenerate carriers and compatible electrical/thermal scattering; it is not a universal law for segmented thermoelectric materials. All material parameters and their correlations enter the uncertainty model below rather than being treated as exact consequences of an EOS.

C. Three-Dimensional Manifold Hydrodynamics

The manifold model uses liquid-water reference properties $\rho = 997.0 \text{ kg/m}^3$, $\mu = 8.9 \times 10^{-4} \text{ Pa s}$, $c_p = 4182 \text{ J/(kg K)}$, and $k = 0.607 \text{ W/(m K)}$ [80]. These properties define a single fluid state and shall not be substituted for the separate water–glycol microchannel model. The incompressible, constant-property governing equations are

$$\begin{aligned}\nabla \cdot \mathbf{u} &= 0, \\ \rho [\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}] &= -\nabla p + \mu \nabla^2 \mathbf{u}, \\ \rho c_p [\partial_t T + \mathbf{u} \cdot \nabla T] &= k \nabla^2 T + \Phi_{\text{visc}}, \\ \Phi_{\text{visc}} &= 2\mu \mathbf{S} : \mathbf{S}, \\ \mathbf{S} &= \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T].\end{aligned}\quad (382)$$

Here $\mathbf{u} = (u_x, u_y, u_z)$ is velocity, p is mechanical pressure, and Φ_{visc} is the viscous heating rate per volume. The steady equations follow by setting the time derivatives to zero. Wall heat transfer, inlet enthalpy, outlet pressure, manifold shape, and no-slip boundaries complete the fluid problem; conjugate heat conduction is required where wall temperatures are not imposed.

For Reynolds-averaged turbulence, the momentum and thermal equations additionally contain $-\rho \nabla \cdot \overline{\mathbf{u}' \mathbf{u}'}$ and $-\rho c_p \nabla \cdot \overline{\mathbf{u}' T'}$, respectively. A mean-flow solution therefore requires a Reynolds-stress and turbulent-heat-flux closure, not merely the steady laminar operators [87]. In a gradient-diffusion closure,

$$\rho c_p \overline{\mathbf{u}' T'} = -k_t \nabla T, \quad k_t = \mu_t c_p / \text{Pr}_t, \quad (383)$$

where μ_t and Pr_t are the eddy viscosity and turbulent Prandtl number. The turbulent Prandtl number is distinct from the molecular value specified below. Turbulence-model choice, inlet fluctuations, wall treatment, mesh refinement, and thermal-flux convergence are required to establish a quantitative mixing prediction.

The operating flow envelope and hydraulic scales are

$$\begin{aligned}
 \dot{m} &\in [0.08, 0.12] \text{ kg/s}, \\
 D_h &= 0.015 \text{ m}, \quad A_c = 1.7671 \times 10^{-4} \text{ m}^2, \\
 v &= \dot{m}/(\rho A_c), \\
 \text{Re} &= \rho v D_h / \mu = \dot{m} D_h / (A_c \mu), \\
 \text{Pr} &= c_p \mu / k = 6.13176277 \simeq 6.13, \\
 v_{\min}/v_{\max} &= 0.45408139/0.68112208 \text{ m/s}, \\
 \text{Re}_{\min}/\text{Re}_{\max} &= 7630.09794/11445.14691.
 \end{aligned} \tag{384}$$

The nominal Reynolds envelope is 7629.9–11444.8, with intermediate values in Table XLIV. It approximates, but is not identical to, substitution of the displayed area and viscosity. The nominal turbulent-flow classification applies to the manifold; it does not establish a turbulent regime in each downstream microchannel or guarantee complete thermal mixing.

D. Sensing-Plane Thermal Dispersion

Four cross-calibrated Pt100 RTDs sample the downstream sensing plane. The calorimetric temperature is the mass-flux-weighted bulk temperature, not generally the unweighted spatial mean:

$$T_b = \frac{\int_{A_s} \rho u_n T \, dA}{\int_{A_s} \rho u_n \, dA}, \quad u_n = \mathbf{u} \cdot \hat{\mathbf{n}}. \tag{385}$$

For one-way flow, a corresponding spatial dispersion is

$$\sigma_{T,\text{spatial}}^2 = \frac{\int_{A_s} \rho u_n (T - T_b)^2 \, dA}{\int_{A_s} \rho u_n \, dA}. \tag{386}$$

Backflow requires a separately defined sampling and flux convention. Sensor locations, insertion depth, thermal response, and estimator weights determine how this field dispersion contributes to the error in the RTD estimate of T_b . Four sensors do not automatically reduce spatial uncertainty by a factor of two; common bias and spatial covariance must be retained. The spatial-drift design conversion uses $c_{\Delta T, \text{sp}} = 185.4 \text{ W/K}$ and

$$\begin{aligned}
 u_{\text{spatial_drift}}^2 &= (c_{\Delta T, \text{sp}} \sigma_{T, \text{spatial}})^2, \\
 u_{\text{spatial_drift, min flow}}^2 &= (185.4 \times 0.0142)^2 \\
 &= 6.9310039824 \text{ W}^2 \simeq 6.9310 \text{ W}^2, \\
 u_{\text{spatial_drift, max flow}}^2 &= (185.4 \times 0.0098)^2 \\
 &= 3.3011982864 \text{ W}^2 \simeq 3.3012 \text{ W}^2.
 \end{aligned} \tag{387}$$

The molecular Prandtl number is 6.13 throughout this constant-property sweep. The nominal maximum spatial variance is $6.9310 \text{ W}^2 \leq 8.7501 \text{ W}^2$ for the specified sensitivity and sampled temperature dispersions. A continuous-envelope bound additionally requires intermediate-flow and off-design evaluation; five nominal points alone do not establish the global maximum.

For direct enthalpy-flow calorimetry, however, $\partial P_{\text{flow}}/\partial \Delta T = \dot{m} c_p$, which varies from 334.56 to 501.84 W/K over this flow range. The constant 185.4 W/K therefore requires a separate estimator weighting or calibrated transfer factor. If the same temperature dispersions were used without

such weighting in the direct-flow model, the endpoint variances would be 22.56964457 and 24.18703875 W², respectively, above the 8.7501 W² allocation. Agreement with the spatial-drift limit thus requires demonstration of the sensing-plane estimator and its conversion factor, not merely a decreasing temperature dispersion.

E. Matched-RTD Calibration and GUM Budget

a) Differential propagation and Monte Carlo comparison:

The first-order GUM budget uses the forward-mode automatic-differentiation Jacobian in Section V-I. Monte Carlo calorimetry separately propagates the joint distribution of $\dot{m}, c_p, \Delta T, P_{\text{env}}$ through $P = \dot{m} c_p \Delta T + P_{\text{env}}$, retaining the covariance of shared temperature references. It tests linearization and evaluates coverage when the output distribution is appreciably nonlinear or non-Gaussian [117]. Sample count, generator, seed, and worker partition are fixed numerical parameters. Convergence of the mean, standard uncertainty, and coverage quantiles is checked separately; a two-input electrical-export calculation is not a four-input calorimetric calculation.

b) *GUM-based synthetic telemetry*: For a telemetry comparison with $P = \dot{m} c_p (T_{\text{out}} - T_{\text{in}})$ and fixed c_p , the independent-sensor approximation is

$$\begin{aligned}
 u_c^2(P) &= (c_p \Delta T)^2 u^2(\dot{m}) \\
 &\quad + (c_p \dot{m})^2 [u^2(T_{\text{out}}) + u^2(T_{\text{in}})].
 \end{aligned} \tag{388}$$

The comparison settings are $c_p = 4184.0 \text{ J/(kg K)}$, $u(\dot{m}) = 1.2 \times 10^{-4} \text{ kg/s}$, and $u(T_{\text{out}}) = u(T_{\text{in}}) = 0.04 \text{ K}$. They define a separate sensor configuration from the matched-RTD case below. Correlated sensors require the signed covariance terms, while uncertain heat capacity and environmental heat require their additional sensitivities. Synthetic telemetry is $P_{\text{syn}} = P_{\text{true}} + u_c(P)\xi$, with $\xi \sim \mathcal{N}(0, 1)$ only when independent Gaussian noise is appropriate. This generates measurement noise, not evidence of physical excess heat. Combined standard uncertainty $u_c = 5.69 \text{ W}$ corresponds to expanded uncertainty $U = k u_c = 11.38 \text{ W}$ at $k = 2$; conversely, $U = 5.69 \text{ W}$ at $k = 2$ would mean $u_c = 2.845 \text{ W}$. Coverage near 95% additionally depends on distribution and effective degrees of freedom.

The matched Pt100 differential-temperature standard uncertainty is $u(\Delta T) = 0.0039 \text{ K}$. Multipoint calibration shall characterize pair offset, self-heating, immersion, readout error, and common-reference covariance. For $\Delta T = T_{\text{out}} - T_{\text{in}}$,

$$u^2(\Delta T) = u^2(T_{\text{out}}) + u^2(T_{\text{in}}) - 2 \text{cov}(T_{\text{out}}, T_{\text{in}}). \tag{389}$$

The differential uncertainty is not an independent uncertainty assigned to each RTD. The nominal calibration sensitivity $c_{\Delta T} = 442.90 \text{ W/K}$ gives

$$(c_{\Delta T} u(\Delta T))^2 = (442.90 \times 0.0039)^2 = 2.9835998361 \text{ W}^2. \tag{390}$$

The ISO/IEC Guide 98-3 / JCGM 100 measurement model uses signed sensitivity coefficients and an input covariance matrix [75]:

$$u_c^2 = \mathbf{c}^T \mathbf{U} \mathbf{c}, \quad c_i = \frac{\partial f}{\partial x_i}, \quad U_{ij} = \text{cov}(x_i, x_j). \tag{391}$$

TABLE XLIV
MANIFOLD THERMAL DESIGN SWEEP AND DIRECT NUMERICAL EVALUATION

$\dot{m}$ (kg/s)	Re_{nom}	Re_{calc}	$\sigma_{T,spatial}$ (K)	$u_{sp,nom}^2$ (W ²)	$u_{sp,calc}^2$ (W ²)	$u_{c,nom}$ (W)	$u_{c,calc}$ (W)
0.08	7629.9	7630.09794	0.0142	6.9310	6.931004	4.4081	4.408061
0.09	8583.6	8583.86018	0.0128	5.6312	5.631699	4.2580	4.258133
0.10	9537.3	9537.62242	0.0116	4.6250	4.625252	4.1382	4.138267
0.11	10491.0	10491.38467	0.0106	3.8615	3.862168	4.0450	4.045018
0.12	11444.8	11445.14691	0.0098	3.3012	3.301198	3.9750	3.975072

The aggregate design allocation assigns $u_{instrumentation}^2 = 12.50 \text{ W}^2$, including the RTD contribution and all other instrument terms, with zero cross-covariance between that aggregate and spatial drift. Under those assumptions,

$$\begin{aligned}
 u_c &= \sqrt{u_{spatial_drift}^2 + u_{instrumentation}^2} \\
 &= \sqrt{6.9310 + 12.50} \\
 &= 4.408060798 \text{ W} \simeq 4.4081 \text{ W},
 \end{aligned} \tag{392}$$

$$4.4081 \text{ W} \leq 5.6881 \text{ W} \leq 5.7000 \text{ W}.$$

This is the algebraic combined-uncertainty result for the two-term allocation. The RTD term shall not be added again if it is already included in 12.50 W^2 . A detailed instrument budget must independently justify that allocation and the covariance assumptions.

Table XLV specifies a detailed calibration case [79]. The listed u_i are standard uncertainties; rectangular distribution limits require conversion before use if they are supplied as half-widths instead. Type A and Type B describe evaluation methods, not distribution shapes. The covariance entry is a net cross-term, not an independent diagonal variance; a positive-semidefinite covariance matrix and the signed measurement sensitivities must support it.

Direct evaluation of the detailed budget gives

$$\begin{aligned}
 \sum_i (c_i u_i)^2 &= 16.28077656 \text{ W}^2, \\
 u_{instr,detail}^2 &= 16.28077656 + 4.2560 \\
 &= 20.53677656 \text{ W}^2, \\
 u_{c,detail}^2 &= 20.53677656 + 8.7501 \\
 &= 29.28687656 \text{ W}^2, \\
 u_{c,detail} &= 5.41173508 \text{ W}.
 \end{aligned} \tag{393}$$

The nominal variance column sums to 29.2852 W^2 , also distinct from $(5.6881 \text{ W})^2$. Replacing only the spatial allowance by 6.9310 W^2 yields $u_c = 5.24097096 \text{ W}$, not 4.4081 W . The detailed instrument variance therefore cannot simultaneously be identified with the 12.50 W^2 aggregate. Indeed, its mass-flow and RTD contributions alone exceed that aggregate allocation. The calibration operating point also gives

$\dot{m}c_p = 442.6672 \text{ W/K}$ and $\dot{m}c_p \Delta T = 2909.385905 \text{ W}$. Thus the tabulated sensitivities are nominal calibration coefficients rather than exact derivatives of the displayed constant- c_p flow model. An evaluated measurement function must account for their differences, the envelope terms, and the line-power boundary. The 4.4081 W figure is a conditional design

allocation, not a fully demonstrated uncertainty for the detailed input table. The limits 5.6881 W and 5.7000 W remain acceptance ceilings; they do not replace covariance propagation.

F. TEG Electrical Bus and Heat-Rejection Accounting

The segmented $\text{Bi}_2\text{Te}_3/\text{PbTe}$ generator has nominal hot- and cold-side temperatures of 623.15 and 353.15 K and thermal input 2911.40 W . Table XLVI resolves the complete electrical-equivalent generation and loss allocation. Thermal-contact and bypass losses are represented as reductions in recoverable electrical power, not automatically as equal heat flows. The corresponding identities are

$$\begin{aligned}
 P_{raw} &= 490.57 - 28.45 - 18.60 - 12.30 = 431.22 \text{ W}, \\
 P_{bus} &= 431.22 - 12.79 = 418.43 \text{ W}, \\
 P_{support} &= 116.67 + 120.00 + 150.00 = 386.67 \text{ W}, \\
 P_{net} &= 418.43 - 386.67 = 31.76 \text{ W}, \\
 \eta_{DC/DC} &= 418.43/431.22 = 0.9703399657.
 \end{aligned} \tag{394}$$

The converter efficiency rounds to 97.03% , whereas substitution of exactly 97.03% gives 418.412766 W . Similarly, a nominal gross efficiency of 16.85% gives 490.5709 W , consistent with 490.57 W at the displayed precision. Quantitative accounting shall use unrounded efficiencies or measured terminal powers, not rounded percentages as exact coefficients.

At the combined generator/converter boundary, heat rejection is $2911.40 - 418.43 = 2492.97 \text{ W}$. If all 386.67 W of support demand dissipates within the overall boundary, environmental heat rejection is 2879.64 W and $2879.64 + 31.76 = 2911.40 \text{ W}$. This closure assumes the listed thermal input is the common energy supply and that no additional external excitation or support power is omitted. The separate calorimetric optical and RF allocations of 1850.00 and 65.59 W are not included in the support total without an explicit port mapping. Consequently, $+31.76 \text{ W}$ is the nominal regulated-bus balance, not by itself a whole-device net-energy measurement. Its electrical uncertainty must be evaluated independently of the 4.4081 W calorimetric allocation.

G. GUM Calorimetry and Excess-Heat Estimation

The dual calorimeter combines coolant enthalpy with an envelope thermopile. Table XLVII defines the nominal inputs, sensitivity coefficients, and variance contributions [72], [73]. Each input uncertainty u_i retains its native unit; only the

TABLE XLV
MATCHED-RTD CALORIMETRIC INPUT BUDGET AT THE CALIBRATION OPERATING POINT

Input (native unit)	Nominal	u_i	Evaluation / distribution	$ c_i $ (W/native unit)	Nominal variance (W ²)	$(c_i u_i)^2$
Mass flow (kg/s)	0.1058	0.00012	B / normal	27518.0	10.8993	10.904260666
c_p (J/(kg K))	4184.0	0.85	B / normal	0.6958	0.3530	0.349789445
ΔT (K)	6.5724	0.0039	A / normal	442.90	2.9836	2.983599836
Line voltage (V)	230.00	0.015	B / rectangular	12.658	0.0361	0.036050617
Line current (A)	12.658	0.0012	B / rectangular	230.00	0.0762	0.076176000
Radiative heat (W)	42.15	1.150	B / normal	1	1.3225	1.322500000
Conductive heat (W)	18.40	0.780	B / normal	1	0.6084	0.608400000
Assigned net calibration covariance contribution					4.2560	4.2560
Spatial-inhomogeneity allowance					8.7501	8.7501

TABLE XLVI
TEG GENERATION, CONVERSION LOSSES, AND NET BUS EXPORT

Accounting term	Power (W)
Gross Seebeck generation	+490.57
Thermal-contact equivalent debit	−28.45
Electrical interconnect loss	−18.60
Thermal-bypass equivalent debit	−12.30
Raw generator DC	431.22
DC–DC conversion loss	−12.79
Regulated bus generation	418.43
Pump allocation	−116.67
Laser-driver allocation	−120.00
Other support allocation	−150.00
Total support demand	386.67
Net regulated-bus export	+31.76

product $c_i u_i$ has units of output power. A complete measurement function must define how the two calorimetric channels contribute to a single estimator. Under the ISO/IEC Guide 98-3 / JCGM 100 GUM framework [75], the measurement function, signed sensitivities and covariances determine

$$u_c^2 = \sum_i c_i^2 u_i^2 + 2 \sum_{i < j} c_i c_j u_{ij},$$

$$\sum_i (c_i u_i)_{\text{nominal}}^2 = 19.370 \text{ W}^2, \quad (395)$$

$$u_{\text{RSS}} = \sqrt{19.370} = 4.4011 \text{ W},$$

$$(5.69 \text{ W})^2 - 19.370 \text{ W}^2 = 13.0061 \text{ W}^2.$$

The nominal combined uncertainty $u_c = 5.69 \text{ W}$ does not follow from the listed independent contributions. The residual 13.0061 W^2 remains unaccounted for and cannot be assigned to a physical uncertainty contribution without justification. Rounding of u_i, c_i also prevents exact reproduction of every variance-column digit. Qualification requires a complete covariance matrix, any additional variance contributions, dimensionally consistent sensitivities, defined thermopile and flow-channel weighting, and calibration records. Type A/B denotes the uncertainty-evaluation method rather than the probability distribution. A rectangular-distribution specification must distinguish

a half-width from an evaluated standard uncertainty.

$$P_{\text{out}} = 2911.40 \text{ W},$$

$$P_{\text{in}} = 1850.00 + 65.59 = 1915.59 \text{ W},$$

$$P_{\text{excess}} = 995.81 \text{ W}, \quad (396)$$

$$Z = 995.81/5.69 = 175.010545 \simeq 175.01.$$

The standardized ratio is conditional on the specified uncertainty and a common measurement boundary. It does not establish a Gaussian null distribution, exclude systematic effects, or demonstrate nuclear heat production. Experimental validation requires raw time series, blank runs, electrical-substitution calibrations, and correlation estimates. Neither this Z nor the 995.81 W excess-heat reference is interchangeable with the separate 31.76 W electrical bus-export balance.

Internally generated power is estimated from recovered heat, exported electrical power, and electrical input at explicitly defined boundaries. The nominal variance budget provides a design basis rather than an achieved calibration result. Commissioning requires sensor calibration, covariance evaluation, and verification of background subtraction. Each operating configuration shall use its own calibrated measurements and uncertainty estimates.

H. Mass-Flow and Seebeck-Envelope Enthalpy Accounting

The water–glycol calorimeter configuration uses a 50/50 mixture by mass at nominal volumetric flow $\dot{V} = 9.83 \text{ L min}^{-1}$. With reference density $\rho = 1048 \text{ kg m}^{-3}$, the mass-flow conversion is

$$\dot{m} = \rho \dot{V} = \frac{1048(9.83 \times 10^{-3})}{60} \text{ kg s}^{-1} = 0.1716973 \text{ kg s}^{-1}. \quad (397)$$

The variance budget shall use the rounded reference $\dot{m} = 0.1717 \text{ kg s}^{-1}$. A calibrated Coriolis instrument shall provide the primary mass-flow measurement. Any alternative calculation from density and volumetric flow shall include the uncertainties and covariance of those inputs.

The fixed-volume flow cell shall be analyzed as a steady open system. Its transported thermal power is the mass flow

TABLE XLVII
DUAL-CHANNEL CALORIMETRIC INPUTS AND NOMINAL VARIANCE CONTRIBUTIONS

Input (nominal value)	Type	Distribution	u_i (native unit)	c_i (W/native unit)	Nominal variance (W ²)
Mass flow (60.33 g/s)	A	Normal	0.0302	48.26	2.124
Temperature difference (11.50 K)	A	Normal	0.0120	253.16	9.228
Specific heat (3.582 J/(g K))	B	Rectangular	0.0021	812.70	2.914
Envelope thermopile (124.50 mV)	A	Normal	0.0800	18.42	2.171
Optical input (1850.00 W)	B	Normal	1.2000	1.00	1.440
RF input (65.59 W)	B	Normal	0.1500	1.00	0.023
Ambient loss correction (15.40 W)	B	Rectangular	1.2124	1.00	1.470

multiplied by the enthalpy difference:

$$\begin{aligned}
 P_{\text{flow}} &= \dot{m} [h(T_{\text{out}}, p_{\text{out}}) - h(T_{\text{in}}, p_{\text{in}})], \\
 P_{\text{flow}} &\approx \dot{m} \int_{T_{\text{in}}}^{T_{\text{out}}} c_p(T, p) dT \\
 &\approx \dot{m} c_p(T_{\text{avg}}) \Delta T, \\
 T_{\text{avg}} &= \frac{1}{2}(T_{\text{out}} + T_{\text{in}}), \quad \Delta T = T_{\text{out}} - T_{\text{in}}.
 \end{aligned} \tag{398}$$

The reduced $c_p \Delta T$ model requires pressure, kinetic-energy, and potential-energy contributions to be negligible at the specified uncertainty level. Pump work dissipated between the measurement planes enters the input-energy balance. The fluid-property model uses the isobaric heat capacity c_p .

The secondary thermal envelope shall surround the vacuum vessel with a thermopile matrix coupled to a bath maintained at 298.15 K within a ± 0.01 K control band. Its calibrated boundary-heat estimate is

$$P_{\text{env}} = K_{\text{env}}(V_{\text{env}} - V_0), \quad [K_{\text{env}}] = \text{W mV}^{-1}, \tag{399}$$

where the thermopile voltages are expressed in millivolts. Heat carried by the coolant and heat crossing the surrounding envelope are complementary paths:

$$P_{\text{cal}} = P_{\text{flow}} + P_{\text{env}}. \tag{400}$$

The two channels shall be summed as complementary heat paths. Total-power verification shall use an independently calibrated enclosing heat balance or controlled electrical substitution. Shared temperature references, bath disturbances, and calibration coefficients shall be represented through their covariance in the combined uncertainty model.

For a control volume containing the reactor and its internal conversion hardware, a complete first-law balance is

$$\begin{aligned}
 P_{\text{gen}} &= P_{\text{cal}} + P_{\text{el,out}} + P_{\text{other,out}} - P_{\text{el,in}} \\
 &\quad + \frac{dE_{\text{stored}}}{dt}.
 \end{aligned} \tag{401}$$

Attribution of P_{gen} to a nuclear process requires accounting for chemical energy release, other energy transport, and changes in stored energy. The calorimeter calibration point is 2911.4 W. Runs with harvesting disabled or electrical output dissipated internally are distinct from electrical-export runs; exported power enters Eq. (401) separately from the remaining measured heat.

TABLE XLVIII
HYDRAULIC REFERENCE VALUES AND DIRECT NUMERICAL EVALUATION

Quantity	Nominal	Calculated
d_h (μm)	348.70	348.704
Re	2370.20	1777.618
f_{Ch}	0.03812	0.036004
Channel-only ΔP_0 (kPa)	350.00	67.471
Channel-only pump power (W)	116.67	22.490
Total-loop head allocation (kPa)	350.00	350.00
Total-loop pump allocation (W)	116.67	116.667

a) *Cold-plate geometry and hydraulic response:* The enlarged-channel comparison uses $N_c = 120$, $w_c = 217.94 \mu\text{m}$, $h_c = 871.76 \mu\text{m}$, $L_c = 50.0$ mm, and $\dot{V} = 7.0$ L/min. With water properties $\rho = 998.2 \text{ kg/m}^3$ and $\mu = 1.002 \times 10^{-3}$ Pa s, the hydraulic quantities are

$$\begin{aligned}
 A_f &= N_c w_c h_c = 2.27989649 \times 10^{-5} \text{ m}^2, \\
 d_h &= 2w_c h_c / (w_c + h_c) = 348.704 \mu\text{m}, \\
 v &= \dot{V} / A_f = 5.11719137 \text{ m/s}, \\
 Re &= \rho v d_h / \mu = 1777.61797, \\
 f_{\text{Ch}} &= 0.036003943 \quad (\epsilon_r = 0.50 \mu\text{m}), \\
 \Delta P_0 &= f_{\text{Ch}} (L_c / d_h) \rho v^2 / 2 = 67.470515 \text{ kPa}.
 \end{aligned} \tag{402}$$

At the comparison value $Re = 2370.20$, the Churchill expression gives $f \simeq 0.0321433$ rather than 0.03812. The 350 kPa duty specifies total-loop head, distinct from the calculated straight-channel pressure drop. The remaining 282.529485 kPa is available for manifolds, piping, and other components. Representation entirely by channel-velocity minor losses would require $K_\Sigma = 21.61787$; this is an available loss allocation rather than a measured manifold coefficient. The 116.666667 W pump demand is consistent with the total head and specified efficiency. At fixed flow, channel count, and length, uniform width and height scaling by S gives $v \propto S^{-2}$, $d_h \propto S$, and $\Delta P_0 \propto f(Re, \epsilon_r / d_h) S^{-5}$. In laminar flow, $f \propto 1/Re \propto S$, giving S^{-4} rather than a universal S^{-3} dependence. Scaling from 200/800 μm channels differs from the 150/600 μm geometry associated with the 452.93 kPa calculation. These baseline geometries are not interchangeable. A hydraulic-diameter approximation requires validation against the pressure-flow characteristic of the actual rectangular channels.

b) *120-channel hydraulic and thermal benchmark*: The narrow-channel benchmark comprises $N_c = 120$ parallel channels with $w_c = 150 \mu\text{m}$, $h_c = 600 \mu\text{m}$, and $L_c = 50.0 \text{ mm}$ [56]. Its governing relations are

$$\begin{aligned} D_h &= \frac{2w_ch_c}{w_c + h_c} = 240 \mu\text{m}, \\ A_{\text{flow}} &= N_c w_c h_c = 1.08 \times 10^{-5} \text{ m}^2, \\ v &= \dot{V}/A_{\text{flow}}, \quad Re = \rho v D_h / \mu, \\ f &= 73.28/Re \quad (\text{prescribed laminar Darcy model}), \\ \Delta P &= (f L_c / D_h + K_\Sigma) \rho v^2 / 2, \\ \Delta T &= Q / (\rho \dot{V} c_p). \end{aligned} \quad (403)$$

The constant-property water calculation at approximately 293 K uses $\rho = 998.2 \text{ kg/m}^3$, $\mu = 1.002 \times 10^{-3} \text{ Pa}\cdot\text{s}$, and $c_p = 4182 \text{ J/(kg}\cdot\text{K)}$. These properties do not describe the water-glycol loop. Table XLIX retains the nonnegative manifold and minor-loss sum K_Σ as an explicit parameter requiring characterization. The flow area gives $v = 1.54321 \text{ m/s}$ at 1 L/min, ten times the alternative estimate of 0.154 m/s. Reynolds numbers and pressure drops based on that lower velocity cannot represent the same 120-channel geometry. At fixed properties, $\Delta T < 25 \text{ K}$ requires more than 5.26054 L/min at 9150 W and 7.89082 L/min at 13725 W, corresponding to integer design selections of at least 6 and 8 L/min. The peak-duty point has $Re \simeq 2952$, outside the specified laminar range. Exclusion of local boiling additionally requires pressure, wall heat-flux, and heat-transfer data. The calculated straight-channel hydraulic power is 29.51 W at 6 L/min; at 8 L/min, the extrapolated value is 52.46 W. Electrical demand is $\Delta P \dot{V} / \eta_{\text{pump}}$ and must include manifold and other loop losses. These channel calculations alone do not establish the 60 W pump allowance. The separate 116.666667 W operating point at 7.0 L/min and 3.5 bar does not qualify an 8 L/min peak-flow duty.

c) *Transitional-flow correlation and thermal response*: For roughness $\epsilon_r = 0.50 \mu\text{m}$, the Churchill Darcy correlation is [64]

$$\begin{aligned} f_{\text{Ch}} &= 8 \left[(8/Re)^{12} + (A_c + B_c)^{-3/2} \right]^{1/12}, \\ A_c &= \left\{ 2.457 \ln \left[\frac{1}{(7/Re)^{0.9} + 0.27\epsilon_r/D_h} \right] \right\}^{16}, \\ B_c &= (37530/Re)^{16}. \end{aligned} \quad (404)$$

The value $f = 0.02840$ at $Re = 2582.8$ differs from the Churchill result $f_{\text{Ch}} \simeq 0.037329$; 0.02840 follows from the rectangular-laminar extrapolation $73.28/Re$. The two correlations describe distinct models. Churchill's circular-pipe laminar limit is $64/Re$, and substitution of hydraulic diameter does not recover the rectangular-channel limit. The comparison uses the water properties associated with Eq. (403); numerical evaluation alone does not qualify either correlation for transitional flow in the specified microchannel. At 7 L/min, the Churchill calculation gives 452.93 kPa and 150.98 W electrical demand at 35% pump efficiency, exceeding the 350 kPa/116.666667 W allocation before minor losses. The alternative density $\rho = 1048 \text{ kg/m}^3$ is inconsistent with the water properties used to calculate the tabulated Reynolds numbers and pressure drops.

Each operating point requires a single, internally consistent fluid-property set.

The heat-transfer correlation is

$$\begin{aligned} Nu &= \frac{(f/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f/8}(Pr^{2/3} - 1)} \left[1 + (D_h/L_c)^{2/3} \right], \\ Pr &= \mu c_p / k_f, \quad h = Nu k_f / D_h. \end{aligned} \quad (405)$$

The assigned values $Nu = 14.852$ and $k_f = 0.600 \text{ W/(m}\cdot\text{K)}$ imply $h = 37130 \text{ W/(m}^2\cdot\text{K)}$ exactly. Evaluation using the specified water properties and Eq. (404) instead gives $Pr = 6.984$, $Nu \simeq 16.0641$, and $h \simeq 40160.2 \text{ W/(m}^2\cdot\text{K)}$. Application in transitional flow requires a validated operating range. Neither an assigned Nu nor its conversion to a dimensional coefficient constitutes a heat-transfer measurement.

d) *Forced-air heat exchanger and TEG boundary temperatures*: The counterflow design case specifies coolant terminal temperatures of 338.0/319.2 K and air temperatures of 310.0/325.0 K. The resulting log-mean temperature difference and required area are

$$\begin{aligned} \Delta T_1 &= 338.0 - 325.0 = 13.0 \text{ K}, \\ \Delta T_2 &= 319.2 - 310.0 = 9.2 \text{ K}, \\ \Delta T_{\text{lm}} &= \frac{13.0 - 9.2}{\ln(13.0/9.2)} = 10.9907313 \text{ K}, \\ A_{HX} &= \frac{13258.816330}{125.0 \Delta T_{\text{lm}}} = 9.65091 \text{ m}^2. \end{aligned} \quad (406)$$

The calculated values differ from the comparison estimates of 11.14 K and 9.521 m². A crossflow arrangement also requires a configuration correction factor. An ambient sink at 310 K cannot maintain a TEG cold junction at 310 K while rejecting finite heat through positive thermal resistance. The specified 319.2 K coolant return exceeds that junction reference. Table XV therefore represents a constant-efficiency comparison. Junction temperatures, material efficiency, coolant capacity rates, and fan demand require a coupled solution before its electrical margins can be established with $P_{\text{comp}} = 0$.

I. Forward-Mode Automatic Differentiation and GUM Covariance

a) *Dual algebra and Jacobian construction*: Let the real input vector $\mathbf{X} \in \mathbb{R}^n$ include measured quantities and uncertain calibration or constitutive parameters, and let $\mathbf{Y} = \mathbf{f}(\mathbf{X}) \in \mathbb{R}^m$. A dual number is $\tilde{a} = a + \vartheta a'$ with $\vartheta^2 = 0$ and $\vartheta \neq 0$. The coefficient a' is a directional derivative, not a time derivative or a random uncertainty. Forward-mode evaluation applies the chain rule to every elementary operation [115]:

$$\begin{aligned} \tilde{a} + \tilde{b} &= (a + b) + \vartheta(a' + b'), \\ \tilde{a}\tilde{b} &= ab + \vartheta(a'b + ab'), \\ \frac{\tilde{a}}{\tilde{b}} &= \frac{a}{b} + \vartheta \frac{a'b - ab'}{b^2}, \quad b \neq 0, \\ g(\tilde{a}) &= g(a) + \vartheta g'(a)a', \\ e^{\tilde{a}} &= e^a + \vartheta e^a a', \\ \ln \tilde{a} &= \ln a + \vartheta a'/a, \quad a > 0, \\ \sqrt{\tilde{a}} &= \sqrt{a} + \vartheta \frac{a'}{2\sqrt{a}}, \quad a > 0. \end{aligned} \quad (407)$$

TABLE XLIX
EQUATION-DERIVED COLD-PLATE CURVE AND THERMAL BENCHMARK

$\dot{V}$ (L/min)	v (m/s)	Re	f	ΔP (kPa)	ΔT_{9150} (K)	ΔT_{13725} (K)
1.0	1.5432	369.0	0.1986	$49.18 + 1.19K_{\Sigma}$	131.51	197.27
2.0	3.0864	737.9	0.0993	$98.36 + 4.75K_{\Sigma}$	65.76	98.64
3.0	4.6296	1106.9	0.0662	$147.54 + 10.70K_{\Sigma}$	43.84	65.76
4.0	6.1728	1475.9	0.0497	$196.72 + 19.02K_{\Sigma}$	32.88	49.32
5.0	7.7160	1844.8	0.0397	$245.90 + 29.72K_{\Sigma}$	26.30	39.45
6.0	9.2593	2213.8	0.0331	$295.08 + 42.79K_{\Sigma}$	21.92	32.88
7.0†	10.8025	2582.8	0.0284	$344.27 + 58.24K_{\Sigma}$	18.79	28.18
8.0†	12.3457	2951.7	0.0248	$393.45 + 76.07K_{\Sigma}$	16.44	24.66
9.0†	13.8889	3320.7	0.0221	$442.63 + 96.28K_{\Sigma}$	14.61	21.92
10.0†	15.4321	3689.7	0.0199	$491.81 + 118.86K_{\Sigma}$	13.15	19.73

†: the laminar friction relation is extrapolated beyond $Re = 2300$ for comparison only; it is not a qualified transition/turbulent pressure curve. The two thermal duties define separate benchmark loads, not the nominal TEG heat rejection.

TABLE L
HYDRAULIC CORRELATION COMPARISON AND HEAT-REJECTION DUTIES

Flow (L/min)	Re	f_{Ch}	Laminar ΔP_0 (kPa)	Churchill ΔP_0 (kPa)	ΔT_{10} (K)	ΔT_{15} (K)
4.0	1475.86	0.0433645	196.72	171.81	31.762	47.642
6.0	2213.79	0.0301077	295.08	268.40	21.174	31.762
7.0	2582.76	0.0373285	344.27	452.93	18.150	27.224
8.0	2951.73	0.0440285	393.45	697.77	15.881	23.821

ΔP_0 excludes nonnegative manifold losses; laminar values at transitional Reynolds numbers are extrapolations. Temperature rises use 8839.210887 and 13258.816330 W, distinct from the 9150 and 13725 W benchmark duties in the separate comparison table.

The nilpotent coefficient is not a small finite-difference step. For each input column j , seed the unit coordinate vector $\mathbf{e}_j$ and evaluate the same measurement function:

$$\begin{aligned}\tilde{X}_i^{(j)} &= \hat{x}_i + \vartheta \delta_{ij}, \\ \tilde{\mathbf{Y}}^{(j)} &= \mathbf{f}(\hat{\mathbf{x}}) + \vartheta \mathbf{J} \mathbf{e}_j, \\ J_{aj} &= \left. \frac{\partial f_a}{\partial X_j} \right|_{\hat{\mathbf{x}}} = \text{coeff}_{\vartheta} \tilde{Y}_a^{(j)}.\end{aligned}\quad (408)$$

Column-wise evaluation needs n directional passes; vector-valued tangents can carry multiple seeds in one pass. Constant outputs have zero tangent. Input units remain attached to the columns, so J_{aj} has units $[Y_a]/[X_j]$. Dual arithmetic differentiates the implemented smooth computational graph without a finite-difference truncation step, but rounding, iterative-solve error, and quadrature error remain. It does not turn approximate material physics into an exact model.

b) *Implicit states and thermodynamic sensitivities:* For a converged internal state $\mathbf{z}$ satisfying $\mathbf{r}(\mathbf{z}, \mathbf{X}) = 0$, differentiate the residual before constructing the output Jacobian:

$$\begin{aligned}\mathbf{r}_z \frac{\partial \mathbf{z}}{\partial \mathbf{X}} &= -\mathbf{r}_X, \\ \mathbf{J} &= \mathbf{f}_X - \mathbf{f}_z \mathbf{r}_z^{-1} \mathbf{r}_X.\end{aligned}\quad (409)$$

This notation denotes a linear solve, not formation of an explicit matrix inverse. The residual Jacobian must be nonsingular on the selected branch. For the EOS residual $r = p_{\text{int}}(V_m, T, \mathbf{a}) -$

p_{ext} , with material parameters $\mathbf{a}$, the exact local checks are

$$\begin{aligned}\left(\frac{\partial V_m}{\partial p_{\text{ext}}} \right)_{T, \mathbf{a}} &= -\frac{V_m}{B_T}, \\ \left(\frac{\partial V_m}{\partial T} \right)_{p, \mathbf{a}} &= \alpha_V V_m, \\ \frac{\partial V_m}{\partial a_j} &= -\frac{\partial r / \partial a_j}{\partial r / \partial V_m}.\end{aligned}\quad (410)$$

Differentiating a fixed number of unconverged Newton steps instead gives the sensitivity of those steps, not necessarily the equilibrium volume. The same distinction applies to discretized transport, phase-field, and shell-contact states. Phase switches, contact-set changes, threshold decisions, and singular limits require branch-aware or distributional treatment rather than a single smooth Jacobian.

Differentiable Debye quadrature retains its variable upper limit. For $I(y) = \int_0^y \phi(z) dz$,

$$I(y) = y \int_0^1 \phi(yu) du, \quad \frac{dI}{dy} = \phi(y). \quad (411)$$

A fixed-domain quadrature or an analytic integral derivative preserves this dependence; adaptive branching alone does not guarantee derivative accuracy. Series and integral representations must agree in both value and slope at any numerical switch. For Drude–Lorentz outputs, real and imaginary tensor components are stacked into one real vector before covariance propagation, preserving their cross-covariances.

c) *Matrix law of propagation and its scope:* Let Σ_X be the input covariance matrix, with entries $\text{cov}(X_i, X_j)$, and

linearize about the estimates $\hat{\mathbf{x}}$:

$$\begin{aligned}\delta \mathbf{Y} &\simeq \mathbf{J} \delta \mathbf{X}, \\ \Sigma_Y &= \mathbf{J} \Sigma_X \mathbf{J}^T, \\ \text{cov}(Y_a, Y_b) &= \sum_{i,j} J_{ai} (\Sigma_X)_{ij} J_{bj}, \\ u_c(Y_a) &= \sqrt{(\Sigma_Y)_{aa}}.\end{aligned}\quad (412)$$

Here Σ_Y denotes the first-order GUM output covariance; it equals the true covariance exactly for an affine measurement function, not for every nonlinear function [75], [116]. For any real $\mathbf{v}$,

$$\mathbf{v}^T \Sigma_Y \mathbf{v} = (\mathbf{J}^T \mathbf{v})^T \Sigma_X (\mathbf{J}^T \mathbf{v}) \geq 0 \quad (413)$$

whenever Σ_X is positive semidefinite. Symmetry, nonnegative diagonal entries, physical units, and compatibility of all correlations are checked before interpreting the budget. Anticorrelation can reduce a difference uncertainty, but cannot justify a non-positive-semidefinite input covariance.

With an input displacement centered at zero, a second-order diagnostic for nonlinear bias is

$$\mathbb{E}[Y_a] - f_a(\hat{\mathbf{x}}) \simeq \frac{1}{2} \text{tr}(\mathbf{H}_a \Sigma_X), \quad \mathbf{H}_a = \nabla_X^2 f_a. \quad (414)$$

An appreciable curvature correction or a non-Gaussian coverage requirement calls for higher-order or Monte Carlo propagation of the specified joint input distribution. GUM-compatible Monte Carlo propagation is not prohibited by using automatic differentiation [117]. Type A/B evaluation, calibration corrections, model discrepancy, and distributional assumptions remain part of the measurement model. Numerical tolerances are controlled separately and included as uncertainty contributions only when a justified error model makes them relevant.

For scalar reporting, expanded uncertainty is $U_a = k_a u_c(Y_a)$ with a justified coverage factor; $k_a = 2$ is not an unconditional 95% statement. A multivariate covariance ellipse requires a joint coverage construction, not simultaneous interpretation of all marginal intervals. Material parameters inferred from the same fit, sensor references shared by two channels, and derived quantities such as $c_p(T, p)$ retain their correlations. A derived input is either evaluated in the computational graph or supplied with its full covariance, not counted twice as an independent source.

d) *Calorimetric analytical check:* The reduced calorimetric measurement function is

$$P(\dot{m}, c_p, \Delta T, P_{\text{env}}) = \dot{m} c_p \Delta T + P_{\text{env}}. \quad (415)$$

Its first differential and sensitivity coefficients are

$$\begin{aligned}dP &= c_p \Delta T d\dot{m} + \dot{m} \Delta T dc_p + \dot{m} c_p d(\Delta T) + dP_{\text{env}}, \\ \frac{\partial P}{\partial \dot{m}} &= c_p \Delta T, \quad \frac{\partial P}{\partial c_p} = \dot{m} \Delta T, \\ \frac{\partial P}{\partial \Delta T} &= \dot{m} c_p, \quad \frac{\partial P}{\partial P_{\text{env}}} = 1.\end{aligned}\quad (416)$$

For input vector $\mathbf{X} = [\dot{m}, c_p, \Delta T, P_{\text{env}}]^T$, the dual-number evaluation must reproduce the analytical row

$$\mathbf{J}_P = [c_p \Delta T \quad \dot{m} \Delta T \quad \dot{m} c_p \quad 1]. \quad (417)$$

With sensitivities $c_j = \partial P / \partial X_j$, the scalar restriction of Eq. (412) is

$$\begin{aligned}u_c^2(P) &= \sum_j c_j^2 u^2(X_j) + 2 \sum_{j < k} c_j c_k \text{cov}(X_j, X_k) \\ &= \mathbf{J}_P \Sigma_X \mathbf{J}_P^T.\end{aligned}\quad (418)$$

This interpretation follows the distinction between combined standard uncertainty and expanded uncertainty in the GUM.¹⁰ Expanded uncertainty shall be reported as $U = k u_c$, with the coverage factor and distribution assumptions stated. Manufacturer limits shall be converted to standard uncertainties using their documented coverage or an explicitly justified probability model.

For correlated inlet and outlet thermometry,

$$u^2(\Delta T) = u^2(T_{\text{out}}) + u^2(T_{\text{in}}) - 2 \text{cov}(T_{\text{out}}, T_{\text{in}}). \quad (419)$$

If c_p is evaluated at the measured average temperature, the reduced temperature sensitivities become

$$\begin{aligned}\frac{\partial P}{\partial T_{\text{out}}} &= \dot{m} \left[c_p + \frac{\Delta T}{2} \frac{dc_p}{dT} \right], \\ \frac{\partial P}{\partial T_{\text{in}}} &= \dot{m} \left[-c_p + \frac{\Delta T}{2} \frac{dc_p}{dT} \right].\end{aligned}\quad (420)$$

The fluid-property calibration contribution shall be separated from its temperature-induced contribution, with each uncertainty source included once. The reference budget treats $u(c_p)$ as the evaluated property uncertainty and $u(\Delta T)$ as the separately characterized differential-temperature uncertainty. Commissioning shall verify this decomposition and retain any residual correlation.

Likewise, writing $\Delta V = V_{\text{env}} - V_0$, the envelope contribution is

$$\begin{aligned}u^2(P_{\text{env}}) &= \Delta V^2 u^2(K_{\text{env}}) + K_{\text{env}}^2 u^2(\Delta V) \\ &\quad + 2 K_{\text{env}} \Delta V \text{cov}(K_{\text{env}}, \Delta V),\end{aligned}\quad (421)$$

with $u^2(\Delta V) = u^2(V_{\text{env}}) + u^2(V_0) - 2 \text{cov}(V_{\text{env}}, V_0)$. The envelope channel has a 0.85 W combined standard-uncertainty allocation. Commissioning shall determine its gain, baseline, drift, and covariance contributions from calibration records.

J. Dual Calorimetry Reference Variance Budget

a) *Water-loop metrology operating point:* The water-loop comparison uses $\dot{m} = 0.08250$ kg/s, $c_p = 4182.0$ J/(kg K), $\Delta T = 8.42000$ K and $P_{\text{env}} = 12.30$ W. The standard-uncertainty allocations are $u(\dot{m}) = 9.4875 \times 10^{-5}$ kg/s, $u(c_p) = 0.8364$ J/(kg K), $u(\Delta T) = 0.00310$ K and $u(P_{\text{env}}) = 4.4500$ W. These inputs describe water rather than the mixture with heat capacity 3380 J/(kg K). Under zero covariance, direct evaluation gives

¹⁰Joint Committee for Guides in Metrology, *Evaluation of measurement data—Guide to the expression of uncertainty in measurement*, JCGM 100:2008, Sections 5–7.

$$\begin{aligned}
P_{\text{literal}} &= 0.08250(4182.0)(8.42000) + 12.30 \\
&= 2917.326300 \text{ W}, \\
u_c^2(P_{\text{literal}}) &= 11.16081265 + 0.33756711 \\
&\quad + 1.14392972 + 19.80250000 \\
&= 32.44480947 \text{ W}^2, \\
u_c(P_{\text{literal}}) &= 5.69603454 \text{ W} < 5.7000 \text{ W}.
\end{aligned} \tag{422}$$

The comparison estimate of 5.6956 W differs from exact evaluation of the specified input uncertainties. At the 2911.40 W reference power, with the other nominal inputs fixed, the temperature rise is determined by the measurement function rather than assigned independently:

$$\Delta T_{\text{ref}} = \frac{2911.40 - 12.30}{0.08250(4182.0)} = 8.40282306566 \text{ K}. \tag{423}$$

At this reference point, the calculated uncertainty satisfies the strict 5.7000 W design limit under the stated independence and calibration assumptions. This is a design calculation rather than evidence of achieved calibration. Covariance and additional correction uncertainties enter through Eq. (418) [66]. With an independent 0.50 W input-power uncertainty, a hypothetical 1000 W excess gives $u = 5.71383790 \text{ W}$ and $Z = 175.013715$, distinct from the 174.7156 comparison ratio. The envelope contribution is 12.30 W for this configuration; the separate 61.40 W allocation is not additive.

b) Water-glycol reference budget: The reference operating point assigns $P_{\text{flow}} = 2850.00 \text{ W}$ and $P_{\text{env}} = 61.40 \text{ W}$. With $c_p = 3380.0 \text{ J kg}^{-1} \text{ K}^{-1}$ near $T_{\text{avg}} = 320 \text{ K}$,

$$\begin{aligned}
\Delta T &= \frac{2850.00}{(0.1717)(3380.0)} \text{ K} = 4.9108635 \text{ K}, \\
P_{\text{cal}} &= 2850.00 + 61.40 = 2911.40 \text{ W}.
\end{aligned} \tag{424}$$

The calculation shall retain the unrounded temperature rise before reporting the result. Table LII defines the four-input standard-uncertainty allocations and their power-equivalent variances. Zero covariance defines this reference case; measured correlations and additional correction uncertainties shall be included in the commissioned budget.

The individual power-equivalent contributions and their combination are

$$\begin{aligned}
u_{P,\dot{m}} &= (0.0001717)(3380.0)(4.9108635) = 2.850000 \text{ W}, \\
u_{P,c_p} &= (0.1717)(5.07)(4.9108635) = 4.275000 \text{ W}, \\
u_{P,\Delta T} &= (0.1717)(3380.0)(0.0040) = 2.321384 \text{ W}, \\
u_{P,\text{env}} &= 0.850000 \text{ W}.
\end{aligned} \tag{425}$$

$$\begin{aligned}
u_c^2(P_{\text{cal}}) &= (2.850000^2 + 4.275000^2 \\
&\quad + 2.321384^2 + 0.850000^2) \text{ W}^2, \\
u_c(P_{\text{cal}}) &= \sqrt{32.509449} \text{ W} \\
&= 5.701706 \text{ W} \approx 5.70 \text{ W}, \\
u_r(P_{\text{cal}}) &= \frac{5.701706}{2911.40} \times 100\% \\
&= 0.195841\% \approx 0.196\%.
\end{aligned} \tag{426}$$

The water-glycol reference combined standard uncertainty is $u_c(P_{\text{cal}}) = 5.701706 \text{ W}$, conventionally expressed as 5.70 W

with $k = 1$. The inequality $5.7017 \leq 5.70$ is not satisfied: the unrounded result exceeds a strict 5.70 W acceptance limit by approximately 0.001706 W. Numerical rounding does not establish compliance. For $k = 2$, the reference expanded uncertainty is $U = 11.4034 \text{ W}$, or 0.39168%. Measurement statements shall identify the control-volume boundary and distinguish these uncertainty quantities from the signed differential dP .

The envelope heat model resolves the 53.7341 W contribution within the following partition:

$$\begin{aligned}
Q_{\text{envelope}} &= Q_{\text{struct}} + Q_{\text{rad}} + Q_{\text{gas}} + Q_{\text{Peltier}} \\
&= 7.6659 + 14.2810 + 0.8420 + 38.6111 \\
&= 61.4000 \text{ W}.
\end{aligned} \tag{427}$$

The component sum is $14.2810 + 0.8420 + 38.6111 = 53.7341 \text{ W}$. This partition does not introduce additional heat. The 7.6659 W comparison subtotal contains both radiative and structural contributions; assigning it entirely to structural heat requires redistribution of its components to avoid double-counting radiation.

For $\epsilon_{\text{eff}} = 0.035$, $A = 0.082 \text{ m}^2$, $T_h = 550 \text{ K}$, and $T_c = 313.15 \text{ K}$, the radiative calculation gives $\epsilon_{\text{eff}} \sigma_{SB} A (T_h^4 - T_c^4) = 13.32674 \text{ W}$ rather than 14.2810 W. The Peltier expression $2N\alpha IT_h$, with $N = 128$, $\alpha = 210 \mu\text{V K}^{-1}$, and $I = 3.82 \text{ A}$, gives 112.94976 W rather than 38.6111 W. Alternative effective areas, view factors, temperatures, or thermoelectric boundaries require explicit specification. The 298.15 K thermopile bath is distinct from the 313.15 K ambient. Calibrated envelope heat measures the boundary-crossing thermal channel; internal Peltier transport contributes only if it crosses that boundary and is not already included in coolant enthalpy.

K. Net-Power Reporting and the 174.71 σ Reference

For a measured net quantity $\hat{P}_{\text{net}}$ and its combined standard uncertainty, a standardized null-displacement statistic is

$$Z = \frac{\hat{P}_{\text{net}} - 0}{u_c(P_{\text{net}})}. \tag{428}$$

The report shall state the measurand, null value, uncertainty model, and any distribution assumptions used to interpret Z . A Gaussian-tail significance shall require a justified sampling model. Nuclear attribution shall be assessed separately using isotope and radiation measurements.

The reference *excess-heat* calculation uses an electrical-input measurement design point $P_{\text{in}} = 1911.40 \text{ W}$ with standard-uncertainty allocation $u(P_{\text{in}}) = 0.50 \text{ W}$:

$$\begin{aligned}
P_{\text{excess,th}} &= 2911.40 - 1911.40 = 1000.00 \text{ W}, \\
u^2(P_{\text{excess,th}}) &= u_c^2(P_{\text{cal}}) + u^2(P_{\text{in}}) \\
&\quad - 2 \text{cov}(P_{\text{cal}}, P_{\text{in}}).
\end{aligned} \tag{429}$$

Under the additional zero-covariance assumption,

$$\begin{aligned}
u(P_{\text{excess,th}}) &= \sqrt{32.509449 + 0.250000} \text{ W} = 5.723587 \text{ W}, \\
Z_{\text{excess,th}} &= \frac{1000.00}{5.723587} = 174.7156.
\end{aligned} \tag{430}$$

The 174.71 σ design reference uses conservative truncation of this nominal ratio; conventional rounding to two decimal places

TABLE LI
FOUR-INPUT WATER-CALORIMETER BUDGET AT 2911.40 W

Input (native unit)	Nominal	Standard uncertainty	Sensitivity (W/unit)	Variance (W ²)
$\dot{m}$ (kg/s)	0.08250	9.4875×10^{-5}	35140.60606	11.11532262
c_p (J/(kg K))	4182.0	0.8364	0.693232903	0.33619123
ΔT (K)	8.402823066	0.00310	345.015	1.14392972
P_{env} (W)	12.30	4.4500	1	19.80250000
Total variance; $u_c = 5.69191915$ W, $k = 1$				32.39794357

TABLE LII
FOUR-INPUT WATER-GLYCOL CALORIMETRIC VARIANCE BUDGET

Input and native unit	Nominal value	$u(y_j)$	$u^2(y_j)$	Sensitivity c_j	$ c_j u(y_j)$ (W)	$c_j^2 u^2(y_j)$ (W ²)	Share (%)
Mass flow $\dot{m}$ (kg s ⁻¹)	0.1717	0.0001717	2.948089×10^{-8}	16598.7187	2.850000	8.122500	24.9850
Heat capacity c_p (J kg ⁻¹ K ⁻¹)	3380.0	5.07	25.704900	0.8431953	4.275000	18.275625	56.2163
Temperature difference ΔT (K)	4.9108635	0.0040	1.600000×10^{-5}	580.3460	2.321384	5.388824	16.5762
Envelope heat P_{env} (W)	61.40	0.85	0.722500	1.0000000	0.850000	0.722500	2.2224
Reference thermal budget: $P_{\text{cal}} = 2911.40$ W					5.701706	32.509449	100.0000

Input variances carry the square of the stated native unit; sensitivities carry watts per native unit. The displayed shares are rounded independently. The total is a combined standard uncertainty, not an expanded uncertainty.

gives 174.72. Experimental reports shall calculate Z from the measured excess heat and its commissioned uncertainty budget rather than assign the reference value.

The excess-heat reference and electrical-export balance in Section II describe separate measurement configurations. The 1484.8 W gross/484.8 W auxiliary allocation is a distinct comparison case rather than the nominal operating point. In the general uncertainty relation, η_{gross} denotes the complete bus-referred conversion efficiency η_{bus} , including boost losses, and $P_{\text{parasitic}}$ includes every operating supply:

$$P_{\text{net,el}} = \eta_{\text{gross}} P_{\text{th}} - P_{\text{parasitic}}, \quad (431)$$

whose uncertainty, for independent inputs, is

$$u^2(P_{\text{net,el}}) = \eta_{\text{gross}}^2 u^2(P_{\text{th}}) + P_{\text{th}}^2 u^2(\eta_{\text{gross}}) + u^2(P_{\text{parasitic}}). \quad (432)$$

The full electrical uncertainty model shall include applicable covariance and conversion-loss terms. Acceptance of net electrical export shall use synchronized input/output electrical measurements, with model-based conversion estimates reported separately. The excess-heat reference Z shall not be transferred to net electrical output; each result shall carry its own calibrated uncertainty and background treatment.

L. Non-Inductive Nichrome Electrical-Substitution Calibration

The calibration heater shall be a micro-machined Ni80Cr20 meander embedded 1.5 mm below the active-layer interface in the OFHC copper spreader and electrically insulated by aluminum nitride. Its reference resistance shall be $R_{\text{cal}} = 50.000 \pm 0.005 \Omega$ at 298.15 K with four-wire Kelvin sensing. Antiparallel serpentine routing shall limit parasitic inductance to below 12 nH. At 68.5 kHz this gives

$$X_L = 2\pi f L < 0.005165 \Omega, \quad (433)$$

TABLE LIII
ELECTRICAL-SUBSTITUTION CALIBRATION STEPS AT 50.000 Ω

Power (W)	Voltage (V)	Current (A)	Hold (min)
200.0	100.0000	2.000000	45
800.0	200.0000	4.000000	45
1500.0	273.8613	5.477226	45
2200.0	331.6625	6.633250	45
3000.0	387.2983	7.745967	45

with magnetic pickup and lead-loop coupling characterized separately during commissioning.

Delivered heater power is measured at the heater terminals:

$$P_{\text{cal,inj}} = \langle V_{\text{cal}}(t) I_{\text{cal}}(t) \rangle. \quad (434)$$

The approximation $P = I^2 R = V^2 / R$ is useful for sizing at the reference resistance; resistance changes with temperature, and lead dissipation outside the calorimetric boundary must not be included as deposited heat. Table LIII enumerates the five calibration steps and their reference electrical requirements.

The calibration power supply shall sustain at least the 3000 W endpoint with the required voltage and current compliance. Supply selection shall cover the complete five-step load envelope. Heater, feedthrough, insulation, contactor, and thermal-protection ratings require verification at the full calibration load. These externally supplied calibration inputs shall be recorded separately from continuous auxiliary demand.

Electrical-substitution calibration establishes the channel gains and their transfer to the operating configuration. The commissioning protocol is:

- 1) Establish a nonreactive baseline with the reactive gas isolated, chamber pressure below 10^{-8} mbar, coolant flow 9.83 L min⁻¹, and inlet temperature 298.15 K. Require

$|d(\Delta T)/dt| < 0.0001 \text{ K min}^{-1}$ for 30 minutes before the first step.

- 2) Apply each of the five powers in Table LIII for at least 45 minutes and verify thermal equilibrium. Record synchronized voltage, current, mass flow, inlet/outlet temperatures, envelope voltage, bath temperature, and stored-energy indicators.
- 3) Evaluate $c_p(T)$ over the measured calibration temperature range and establish transfer to the 320 K mean-fluid reference used in Table LII.
- 4) Fit the calibrated response, evaluate residuals and hysteresis with ascending and descending powers, and independently excite the envelope channel if needed to distinguish its gain from the flow-channel gain. Perform a post-run verification under the same boundary conditions.
- 5) Treat drift exceeding 0.05% in either calibrated gain as a recalibration and loading-inhibit condition. Propagate the calibration covariance to subsequent measurements rather than assigning the fit quality as a measurement uncertainty.

The linear calibration model and weighted least-squares estimator are

$$\begin{aligned} P_{\text{cal, inj}, j} &= a_f (\dot{m} c_p \Delta T)_j + b_e V_{\text{env}, j} + c_0 + \epsilon_j, \\ \hat{\beta} &= (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{P}_{\text{cal, inj}}, \\ \beta &= [a_f, b_e, c_0]^T. \end{aligned} \quad (435)$$

The design matrix shall have full column rank. Independent channel excitation is required when proportional flow and envelope responses prevent separate gain identification. The fit shall meet $R^2 > 0.99995$, with residual bias, hysteresis, and parameter covariance assessed separately. Commissioning shall verify heater-to-active-layer spatial transfer and electromagnetic pickup for the 1.5 mm heater offset. Metrological qualification requires traceable sensor and power calibrations, pre- and post-run gain checks, nonreactive background measurements, and a complete covariance budget supporting the stated calorimetric uncertainty.

VI. ISOTOPIC MASS SPECTROMETRY, DIGITAL PSD, AND SHIELDING DIAGNOSTICS

The diagnostic architecture combines isotope-resolved gas analysis, digital neutron–gamma discrimination, gamma spectroscopy, and three-dimensional neutron–photon shielding calculations. The mass-spectrometric channel determines helium inventory after selective hydrogenic-gas removal; the scintillation channel classifies individual radiation events; and the transport model converts source-normalized fluence tallies into accessible-surface dose rates. Resolving power, helium recovery, classification probability, and dose equivalent are separate measurands with separate calibration and acceptance requirements. The heat-to-helium relation defines the branch-specific stoichiometric reference against which the measured inventory is evaluated.

A. Sector-Mass-Spectrometer Quantitation

The analyzer shall use double-focusing magnetic-sector optics with $R = m/\Delta m = 25000$ [72]. For an ion of mass

m_{ion} and positive charge q_{ion} , accelerated through potential V_{acc} and following an ideal magnetic-sector orbit of radius r_B in transverse field B_{sec} , the nonrelativistic ion-optical relations are

$$\begin{aligned} q_{\text{ion}} V_{\text{acc}} &= \frac{1}{2} m_{\text{ion}} v_{\text{ion}}^2, \\ q_{\text{ion}} v_{\text{ion}} B_{\text{sec}} &= m_{\text{ion}} v_{\text{ion}}^2 / r_B, \\ \frac{m_{\text{ion}}}{q_{\text{ion}}} &= \frac{B_{\text{sec}}^2 r_B^2}{2 V_{\text{acc}}}. \end{aligned} \quad (436)$$

The electrostatic and magnetic sectors shall provide energy and angular focusing at the collector. The working mass pair is $m(^4\text{He}) = 4.002603 \text{ u}$ and $m(D_2) = 4.028204 \text{ u}$, giving $\Delta m = 0.025601 \text{ u}$ and $R_{\text{required}} \simeq 156.3$ for this separation. Nearby interferences require individual resolution and abundance-sensitivity assessment. The St707 getter composition is 70 wt% Zr / 24.6 wt% V / 5.4 wt% Fe, with a specified activation temperature of 400 °C. The nominal ratio enrichment is 1.45×10^5 , the D_2 removal requirement exceeds 99.999%, and the residual $D_2 \rightarrow ^4\text{He}$ background-fit coefficient is 3.20×10^{-11} . These response parameters enter the calibrated interference model, not a multiplicative amplification of the helium production signal.

For one helium atom per reaction and complete deposition of the assigned reaction energy, the nuclear-ash generation rate per kilowatt is

$$\begin{aligned} \dot{N}_{\text{He}} &= \frac{1000 \text{ J/s}}{23.84 \times 10^6 \text{ eV} (1.602176634 \times 10^{-19} \text{ J/eV})} \\ &= 2.61808 \times 10^{14} \text{ s}^{-1}, \\ \dot{n}_{\text{He}} &= \dot{N}_{\text{He}} \frac{86400}{N_A} \\ &= 3.75618 \times 10^{-5} \text{ mol/(day kW)}. \end{aligned} \quad (437)$$

The rounded yield is $3.756 \times 10^{-5} \text{ mol/(day kW)}$, corresponding to one helium atom per 23.84 MeV of attributable, fully deposited reaction heat. This stoichiometric prediction is distinct from a mass-spectrometric measurement. At the 995.81 W excess-heat reference, the predicted rate is approximately $3.74044 \times 10^{-5} \text{ mol/day}$ rather than the 1 kW value. Quantitation requires calibrated sensitivity and collection efficiency, subtraction of blank, leakage, and desorption contributions, and accounting for helium retained in the solid. Confirmation of the production rate requires resolved spectra, a calibration curve, an uncertainty estimate, and a complete gas-inventory balance.

The assay shall resolve helium from hydrogenic interferences and distinguish candidate production from calibration additions, atmospheric ingress, retained gas, and material backgrounds. Mass-axis calibration, detector-response characterization, helium-recovery measurement, and background subtraction define the quantitative measurement chain. Reaction-rate inference uses the specified branch and energy-deposition assumptions rather than treating the stoichiometric reference as an observed yield.

B. Isotopic Mass Values and Resolving Power

a) *Resolving power and abundance sensitivity:* The sector analyzer has a nominal resolving-power specification

TABLE LIV
HELIUM AND HYDROGENIC MASS REFERENCES

Species	Mass reference (u)	Definition
${}^4\text{He}$	4.00260325413	Evaluated neutral-atom mass
D_2	4.02820355624	$2m(D)$, before molecular binding correction
HD	3.02192681035	$m({}^1\text{H}) + m(D)$, before binding correction
H_2	2.01565006446	$2m({}^1\text{H})$, before binding correction

of $R = 25000$, with rounded mass labels $m(D_2) = 4.02820$ u and $m({}^4\text{He}) = 4.00260$ u. Their separation is 0.02560 u, while $m/R \simeq 0.000160$ u at mass four. Conversion of neutral-mass labels to exact ion masses requires electron-removal and molecular-binding corrections. Peak-center separation alone does not establish abundance sensitivity adjacent to a substantially larger deuterium peak. The cryo-getter specification assigns ratio enrichment $G_{\text{ratio}} = 1.45 \times 10^5$ and tail coefficient $a_{D \rightarrow \text{He}} = 3.20 \times 10^{-11}$. Ratio enrichment describes preferential deuterium removal relative to helium, not amplification of the helium inventory. Calibration shall characterize isotope standards, helium recovery, matrix-gas blanks, pressure dependence, and detector linearity; nominal response values alone do not establish calibrated performance.

Neutral-atom reference masses are¹¹

$$\begin{aligned} m({}^1\text{H}) &= 1.00782503223(9) \text{ u}, \\ m(D) &= 2.01410177812(12) \text{ u}, \\ m({}^4\text{He}) &= 4.00260325413(6) \text{ u}. \end{aligned} \quad (438)$$

The mass-axis reference shall retain the evaluated uncertainties. Table LIV lists neutral-atom sums for the hydrogenic species; the ion-mass calibration shall include molecular binding and electron-removal corrections as applicable.

For a neutral diatomic molecule XY in a defined internal state and its singly charged ion,

$$\begin{aligned} m(XY) &= m(X) + m(Y) - \frac{D_0(XY)}{c^2}, \\ m(XY^+) &= m(XY) - m_e + \frac{E_{\text{ion}}(XY)}{c^2}. \end{aligned} \quad (439)$$

Here D_0 is the dissociation energy from that state and E_{ion} its ionization energy. The corresponding atomic-ion expression is $m({}^4\text{He}^+) = m({}^4\text{He}) - m_e + E_{\text{ion}}({}^4\text{He})/c^2$. Calibration records shall distinguish ion masses from conventional neutral-mass labels. The atom-sum approximation shall be used only at a precision compatible with the resolving-power calculation; reported molecular masses shall retain the applicable corrections and uncertainty.

For the helium/deuterium-molecule separation,

$$\begin{aligned} \Delta m &= 4.02820355624 - 4.00260325413 \\ &= 0.02560030211 \text{ u}, \\ R_{\text{sep}} &\approx \frac{4.00260325413}{0.02560030211} = 156.3498. \end{aligned} \quad (440)$$

¹¹NIST, *Atomic Weights and Isotopic Compositions*, entries for hydrogen and helium. Parenthesized uncertainties apply to the final quoted digits.

The six-decimal working references are 4.002603, 4.028204, 3.021927, and 2.015650 u. The ratio R_{sep} defines the nominal mass-separation scale. Analyzer acceptance shall use a declared peak-width convention and measured peak shape.

The analyzer shall provide $R \geq 1600$ near mass 4, with nominal operating allocation $R = 25000$. Both values shall use the full peak width at 5% of peak height. At this convention, the corresponding mass widths are

$$\Delta m_{1600} = 0.002501627 \text{ u}, \quad \Delta m_{25000} = 0.0001601041 \text{ u}. \quad (441)$$

Abundance sensitivity shall be qualified independently of resolving power. For fuel-to-helium ratio $B = 10^6$ and normalized deuterium tail response $a_{D \rightarrow \text{He}}$,

$$\frac{I_{D,\text{tail}}}{I_{\text{He}}} \approx a_{D \rightarrow \text{He}} B. \quad (442)$$

The nominal tail response $a_{D \rightarrow \text{He}} = 3.20 \times 10^{-11}$ corresponds to 3.20×10^{-5} helium-relative interference (0.00320%) at $B = 10^6$, before additional ratio enrichment. A 0.01% helium-relative contamination limit at the same ratio requires $a_{D \rightarrow \text{He}} < 10^{-10}$. Instrument acceptance shall establish the tail response and residual fuel-to-helium ratio for the specified cleanup and acquisition sequence.

The analyzer shall be a double-focusing magnetic-sector system covering the 1–20 u mass range, with 70 eV electron-impact ionization and a qualified low-outgassing ion source. Detection shall use a Faraday collector with nominal $10^{11} \Omega$ transimpedance and a secondary-electron multiplier for the lower-current range. Acceptance records shall specify peak width, abundance sensitivity near the D_2 tail, ion-source pressure, detector linearity, mass discrimination, fragmentation, and Faraday/multiplier gain cross-calibration. The selected instrument shall demonstrate these requirements in the installed light-isotope configuration.

C. St707 Getter Pre-Concentration Manifold

The sampling bypass shall use St707 non-evaporable getter material, nominally 70.0 wt% Zr, 24.6 wt% V, and 5.4 wt% Fe. Activation and pumping shall be treated as separate process states, with gas loading and activation history recorded.¹² The reference activation cycle shall use approximately 673 K (400°C) for one hour under dynamic vacuum, with the endpoint established for the installed cartridge. The subsequent pumping temperature and allowable loading shall be qualified separately, accounting for reversible hydrogen uptake and temperature-dependent equilibrium release pressure.

The assay-manifold requirement is

$$\eta_{\text{H}} = 1 - \frac{n_{\text{H,out}}}{n_{\text{H,in}}} > 0.99999, \quad (443)$$

where the hydrogenic inventory includes H_2 , HD , and D_2 on a consistently defined molecular or atom-equivalent basis. The

¹²C. Benvenuti and P. Chiggiato, “Pumping characteristics of the St707 non-evaporable getter (Zr 70-V 24.6-Fe 5.4 wt%),” *Journal of Vacuum Science & Technology A*, vol. 14, pp. 3278–3282, 1996; CERN-MT-95-10-SM.

corresponding helium recovery is $r_{\text{He}} = n_{\text{He,out}}/n_{\text{He,in}}$. The relative enrichment is

$$G_{\text{ratio}} = \frac{r_{\text{He}}}{1 - \eta_{\text{H}}}. \quad (444)$$

For $r_{\text{He}} \approx 1$, the cleanup requirement corresponds to an approximately 10^5 or greater improvement in helium-to-hydrogen ratio, reducing $B = 10^6$ to approximately 10 or less. The enrichment shall be applied to the species ratio, while absolute helium inventory shall follow the gas balance. Helium recovery shall include losses in valves, dead volumes, and transfer paths.

For a deuterium-dominated sample, define the species-specific transmission $r_{D_2} = n_{D_2,\text{out}}/n_{D_2,\text{in}}$ after subtraction of manifold backgrounds. At the specified enrichment,

$$\begin{aligned} G_{\text{ratio}} &= \frac{r_{\text{He}}}{r_{D_2}} = 1.45 \times 10^5, \\ r_{D_2} &= \frac{r_{\text{He}}}{1.45 \times 10^5} \leq 6.89655 \times 10^{-6}, \\ \eta_{D_2} &= 1 - r_{D_2} \geq 0.9999931034 \\ &> 0.99999. \end{aligned} \quad (445)$$

The bound assumes $0 < r_{\text{He}} \leq 1$ and no unaccounted helium addition. It corresponds to at least 99.99931034% deuterium removal. At $r_{\text{He}} = 1$, an inlet deuterium-to-helium ratio of 10^6 becomes 6.89655 at the outlet. The mixed-isotope requirement on η_{H} shall also be demonstrated for H_2 and HD ; a deuterium transmission measurement alone does not characterize the entire hydrogenic inventory.

In an idealized evacuated batch of volume V_g with constant effective hydrogen pumping speed S_g ,

$$p_{\text{H}}(t) = p_{\text{H}}(0)e^{-S_g t/V_g}, \quad \frac{S_g t}{V_g} > \ln(10^5) = 11.51293 \quad (446)$$

is required for five-decade depletion in this reference model. Installed-cartridge qualification shall characterize getter mass, capacity, isotope loading, equilibrium pressure, conductance, and outgassing and verify $> 99.999\%$ hydrogenic removal over the specified assay envelope.

The nominal recirculation volume is $V_{\text{loop}} = 12.50 \text{ L}$ at $T_{\text{loop}} = 300.15 \text{ K}$. Isolation valves separate the pressurized reactor, calibrated dosing reservoir, getter, and analyzer. Differential pumping and metered transfer shall protect the ion source from direct exposure to the pressurized sample. Activation, sampling, regeneration, blank assay, and helium-recovery measurement are distinct operating states. Getter loading, isotope exchange, regeneration history, and recovered-gas disposition enter the inventory and background balances.

D. NIST-Traceable Micro-Dosing Calibration

Calibration shall use certified helium mole fractions of 0.10, 1.00, and 10.00 ppm in deuterium prepared from a 99.9999%-purity matrix gas before helium addition. Each mixture shall specify composition, uncertainty, stability period, and an unbroken calibration chain. NIST traceability requires that chain and its associated measurement uncertainty.¹³

¹³NIST, *Metrological Traceability Frequently Asked Questions and NIST Policy*, NIST Technical Note 2156, 2021, doi: 10.6028/NIST.TN.2156.

TABLE LV
HELIUM CONTENT OF EACH 5.0 mL STP CALIBRATION ALIQUOT

He mole fraction (ppm)	Helium amount (mol)	Pure-He equivalent (nL STP)
0.10	$2.2307517 \times 10^{-11}$	0.50
1.00	$2.2307517 \times 10^{-10}$	5.00
10.00	2.2307517×10^{-9}	50.00

The dosing specification defines standard temperature and pressure (STP) as $T_0 = 273.15 \text{ K}$ and $p_0 = 101325 \text{ Pa}$, distinct from the alternative 1-bar convention. A calibrated 5.0 mL STP-equivalent aliquot contains

$$n_{\text{dose}} = \frac{p_0 V_0}{RT_0} = 2.2307517 \times 10^{-4} \text{ mol}, \quad (447)$$

using $R = 8.31446261815324 \text{ J mol}^{-1} \text{ K}^{-1}$. The corresponding helium quantities are listed in Table LV.

Mass-axis, detector-gain, and dosing calibration shall follow the sequence:

- 1) Measure a manifold blank, verify isolation from the reactor inventory, and calibrate pressure, volume, temperature, and valve-transferred quantity to establish the 5.0 mL STP-equivalent dose and its uncertainty.
- 2) Every four hours, introduce the three certified concentrations sequentially through the metering and getter path. Meter transfer slowly enough to preserve the qualified analyzer pressure. Measure getter removal and helium recovery using the same gas load and operating temperature as the sample assay.
- 3) Acquire full mass-4 peak and tail scans, and record helium peak current, background, detector gain, and recovery transients on both Faraday and multiplier ranges where their signals overlap. Fit the response including an intercept and propagate standard-gas, dose-volume, pressure, temperature, and transfer uncertainties.
- 4) Route standards to a separate assay inventory where practical. Otherwise account for and subtract calibrated helium additions with their uncertainties, quantify retained memory, and verify washout with a subsequent blank before resuming production-rate estimation.

For a static pressure calibration at the measurement volume, define the end-to-end sensitivity

$$I_{\text{He}} - I_0 = S_{\text{He}} p_{\text{He}}, \quad [S_{\text{He}}] = \text{A Pa}^{-1}. \quad (448)$$

Pressure derived from a sensitivity in A mbar^{-1} shall be converted from mbar to Pa by a factor of 100 before use in SI gas-law expressions. A time-integrated pulse calibration shall instead use $\int (I - I_0) dt = K_n n_{\text{He,dose}}$, where K_n has units C mol^{-1} . The record shall identify the calibrated transfer dynamics and distinguish peak-pressure sensitivity from integrated-amount sensitivity.

The reference efficiencies $\eta_{\text{ion}} = 1.25 \times 10^{-3}$ and $\eta_{\text{trans}} = 0.88$ apply only to a separately defined microscopic instrument-response model. Empirical end-to-end sensitivity S_{He} already incorporates ionization and ion-optical transmission; applying these efficiencies again would double-count their effects.

Relative enrichment enters the interference model separately from the calibrated helium amount.

E. Reaction-Rate Mapping and Accumulated Helium Ash

a) Effective reaction energy and daily helium inventory:

For one helium-4 atom per lattice-capture reaction, the conditional yield at fully recovered lattice heat is

$$\begin{aligned} Q_{\text{eff}} &= 23.841 \text{ MeV} = 3.8197493131194 \times 10^{-12} \text{ J}, \\ E_{\text{day}} &= 8.64 \times 10^7 \text{ J}/(\text{day kW}), \\ N_{\text{He}} &= E_{\text{day}}/Q_{\text{eff}} = 2.2619285434 \times 10^{19} \text{ day}^{-1} \text{ kW}^{-1}, \\ n_{\text{He}} &= N_{\text{He}}/N_A = 3.7560207135 \times 10^{-5} \text{ mol}/(\text{day kW}). \end{aligned} \quad (449)$$

The nominal value is $3.756 \times 10^{-5} \text{ mol}/(\text{day kW})$. Using 23.847 MeV instead gives $3.8207106190998 \times 10^{-12} \text{ J}$ and $3.7550756837 \times 10^{-5} \text{ mol}/(\text{day kW})$. The 6 keV difference defines an effective heat-accounting allowance; it is not a derived lattice-binding correction. Any escaping energy, competing branches, or helium retention requires separate accounting.

b) *Parallel isotope and photon measurements:* The gas-analysis path comprises a closed-loop sampler, cryogenic condensable trap, reactive-gas getter, and sector mass analyzer. An extended assay specification requires $M/\Delta M > 25000$ and a helium detection limit no greater than 10^{-11} mol ; these are acceptance limits rather than demonstrated detection performance. The nominal helium/deuterium molecular masses 4.00260/4.02820 u do not alone determine abundance sensitivity or peak-tail rejection. Atmospheric helium near 5.24 ppm is a contamination reference, not a guaranteed chamber background.

An independent HPGe channel and pulse-height analyzer cover activation lines and the approximately 23.847 MeV capture-photon signature. The detector response shall include efficiency, escape peaks, dead time, background, and live time at the relevant energy. A 0.1 Bq photon-source detection target with 10 cm lead shielding requires an emission probability and statistically defined detection criterion; activity is not identical to detected counts. Lead thickness alone does not establish cosmic-background suppression or that detection limit.

For the proposed helium-producing branch with complete useful-energy deposition, the assigned nuclear energy per event is

$$Q_{DD} = 23.84 \text{ MeV} = 3.819589095456 \times 10^{-12} \text{ J}. \quad (450)$$

At a *nuclear heat-equivalent* power of $P_{\text{nuc}} = 1000.0 \text{ W}$, one helium atom per event gives

$$\begin{aligned} \dot{N}_{\text{He}} &= \frac{P_{\text{nuc}}}{Q_{DD}} = 2.6180827 \times 10^{14} \text{ s}^{-1}, \\ \dot{n}_{\text{He}} &= \frac{\dot{N}_{\text{He}}}{N_A} = 4.3474285 \times 10^{-10} \text{ mol s}^{-1}, \\ N_A &= 6.02214076 \times 10^{23} \text{ mol}^{-1}. \end{aligned} \quad (451)$$

Over $t_d = 86400 \text{ s}$,

$$\begin{aligned} N_{\text{He}}(t_d) &= 2.2620234 \times 10^{19}, \\ n_{\text{He}}(t_d) &= \dot{n}_{\text{He}} t_d = 3.7561783 \times 10^{-5} \text{ mol} \\ &\approx 3.756 \times 10^{-5} \text{ mol}, \\ V_{\text{He},0} &= n_{\text{He}} \frac{RT_0}{p_0} = 0.8419087 \text{ mL} \approx 0.842 \text{ mL}. \end{aligned} \quad (452)$$

The reference gas molar volume under this STP convention is $22.4139695 \text{ L mol}^{-1}$. Equation (452) specifies the daily inventory for 1.00 kW nuclear heat-equivalent power. At the separate 9303.30 W thermal comparison point, distinct from the nominal cascade input,

$$\begin{aligned} n_{\text{He},9303.3}(t_d) &= 3.4944853 \times 10^{-4} \text{ mol}, \\ V_{\text{He},9303.3,0}(t_d) &= 7.8325288 \text{ mL} \end{aligned} \quad (453)$$

under the same branch, deposition, and recovery assumptions. Quantitative attribution requires independent characterization of these assumptions and of the actual nuclear contribution to measured heat.

For measured loop pressure, the inventory and production balance are

$$\begin{aligned} n_{\text{He},\text{loop}}(t) &= \frac{p_{\text{He}}(t)V_{\text{loop}}(t)}{RT_{\text{loop}}(t)}, \\ \dot{n}_{\text{He},\text{prod}} &= \frac{d}{dt} \left(\frac{p_{\text{He}}V_{\text{loop}}}{RT_{\text{loop}}} \right) + \dot{n}_{\text{removed}} \\ &\quad - \dot{n}_{\text{cal}} - \dot{n}_{\text{background}}. \end{aligned} \quad (454)$$

At constant volume and temperature, using the end-to-end sensitivity in (448),

$$\dot{n}_{\text{He},\text{inventory}} = \frac{V_{\text{loop}}}{RT_{\text{loop}}} \frac{d}{dt} \left(\frac{I_{\text{He}} - I_0}{S_{\text{He}}} \right). \quad (455)$$

The inventory rate is expressed in moles per second; atom rates shall be obtained by multiplication by N_A . For fully retained helium at the 1.00 kW heat-equivalent reference, the 12.50 L, 300.15 K loop has a calculated helium partial-pressure rise of 7.49909 Pa per day. Production-rate inference shall include extraction, temperature changes, sample removal, calibration additions, and background ingress.

c) *Assay acceptance and background control:* Assay acceptance requires mass-axis stability, detector linearity and gain transfer, isotope-resolved peak and tail fits, getter recovery, and a complete calibration covariance. Background characterization shall use manifold and matrix-gas blanks, reactor-isolated assays, calibrated helium additions, leak testing, and memory and washout measurements through the same sampling path. Each background-subtracted inventory shall include the raw signal, background estimate, transferred amount, and uncertainty. The $\pm 0.5\%$ kinetic/calorimetric agreement target shall be evaluated from independently calibrated, time-aligned measurements and their covariance against the branch-specific heat-to-helium relation $Q_{DD}N_A$. Assay performance is established by these quantitative acceptance criteria rather than nominal instrument settings alone.

F. Digital Charge-Comparison Pulse-Shape Discrimination

The EJ-301 acquisition configuration specifies a 500 MS/s, 14-bit digitizer with charge-comparison discrimination [72]. Each waveform shall undergo pretrigger baseline estimation and polarity correction before charge integration. Taking the accepted pulse onset as $t = 0$, let $V(t)$ denote the baseline-subtracted, positive-polarity waveform. The short and long integration gates are

$$\begin{aligned} Q_{\text{short}} &= \int_0^{24 \text{ ns}} V(t) dt, \\ Q_{\text{long}} &= \int_0^{240 \text{ ns}} V(t) dt, \\ S_{\text{PSD}} &= 1 - Q_{\text{short}}/Q_{\text{long}}. \end{aligned} \quad (456)$$

At the 2 ns sample interval, the gates contain exactly 12 and 120 samples, without an additional endpoint sample. Charge in coulombs requires the calibrated front-end gain and impedance; these common factors cancel in the ratio. Division shall be enabled only above a positive, calibrated long-gate threshold. Clipped pulses, contaminated baselines, incomplete gates, and identified pileup shall be excluded from the qualified single-pulse classification. The synchronous 1.0 GS/s implementation in Section VI-I uses the same physical gate durations with a different sample count.

The resulting tail-to-total discriminator and specified peak-separation metric are

$$\begin{aligned} S_{\text{PSD}} &= \frac{\int_{24 \text{ ns}}^{240 \text{ ns}} V(t) dt}{\int_0^{240 \text{ ns}} V(t) dt}, \\ \text{FOM} &= \frac{\Delta S_{n\gamma}}{\text{FWHM}_n + \text{FWHM}_\gamma} = \frac{0.215}{0.052 + 0.048} = 2.15. \end{aligned} \quad (457)$$

Classification shall use an energy-dependent neutron selection determined from independently tagged calibration events and frozen before validation. Separation and peak widths shall be evaluated within specified light-output bins, with neutron acceptance, count rate, and pileup policy held fixed. The specified FOM = 2.15 does not by itself determine gamma-tail leakage. For zero neutron-classified events in a validation sample of $N_\gamma = 2995731$ independent gamma trials, the acceptance criterion is $R_\gamma > 10^6$ at 95% confidence. The exact one-sided binomial bound is

$$\begin{aligned} p_{\gamma \rightarrow n, 95} &= 1 - 0.05^{1/N_\gamma} \simeq 9.99999925 \times 10^{-7}, \\ R_{\gamma, 95} &= 1/p_{\gamma \rightarrow n, 95} \\ &\simeq 1000000.0748773175 \\ &\simeq 1.000000 \times 10^6 \geq 10^6. \end{aligned} \quad (458)$$

For independent, identically distributed Bernoulli trials, the unrounded lower bound is marginally above 10^6 ; its six-decimal scientific display does not resolve that margin. Numerically stable evaluation uses $p_{\gamma \rightarrow n, 95} = -\expm1[\ln(0.05)/N_\gamma]$, avoiding subtraction of nearly equal numbers. The minimum exact-binomial sample size is 2,995,731 zero-leakage trials. The conservative Poisson approximation $-\ln(0.05)/N_\gamma$ instead gives $1.00000043 \times 10^{-6}$ and requires one additional trial. The specified count therefore supports the target only under the exact-binomial assumptions. Qualification shall establish event

TABLE LVI
SELECTED MONTE CARLO SHIELD CONFIGURATION

Layer	Thickness	Composition and function
Inner W-Cu	35.0 mm	90 wt% W / 10 wt% Cu; photon attenuation
B-HDPE	120.0 mm	5 wt% boron; neutron moderation and capture
Outer SS316L	10.0 mm	Structural enclosure; secondary-photon attenuation

independence, representative gamma energies, fixed selection criteria, and traceable event acquisition; a count alone does not certify detector performance outside those conditions.

G. Gamma Spectroscopy and Emission-Rate Normalization

The HPGe reference geometry places the detector 10 cm from the vessel. The nominal spectral condition is no resolved 23.84 MeV primary or 2.22 MeV capture photopeak above a < 0.01 count/s background. For a specified photon energy and geometry, a background-subtracted count-rate limit $\dot{C}_{\gamma, \text{lim}}$ maps to an emission-rate limit through

$$\dot{N}_{\gamma, \text{lim}} = \frac{\dot{C}_{\gamma, \text{lim}}}{\varepsilon_{\gamma, \text{abs}} L_\gamma}. \quad (459)$$

Here $\varepsilon_{\gamma, \text{abs}}$ includes solid angle, attenuation, full-energy response, and event acceptance, while L_γ is the live-time fraction when the count rate is referred to elapsed time. A live-time-corrected count rate shall not receive a second live-time correction. Inference requires the measured exposure, continuum response, dead-time characterization, and a specified confidence interval. The background rate alone is not an emission-rate upper bound, and nondetection does not establish the absence of radiation.

H. Three-Dimensional Monte Carlo Shielding and Dose Tallies

The OpenMC/MCNP shielding model shall represent the complete three-dimensional enclosure, including joints, penetrations, coolant paths, detector positions, and surrounding scattering structures. The selected radial layer sequence, from the reactor outward, is given in Table LVI. Material cards shall specify measured density, elemental mass fractions, boron isotopic composition, and temperature. The 90/10 W-Cu composition is the selected configuration for the following dose allocation; an 80/20 alloy is a separate transport case.

Fixed-source coupled transport shall track neutron scattering and absorption, neutron-induced photons, and photon scattering and absorption. The hydrogenous moderator shall use the applicable thermal-scattering treatment. Capture photons generated within the shield remain part of the coupled history rather than being discarded at neutron absorption. The nominal neutron and photon dose components are

$$\begin{aligned} \dot{H}_n &= 0.04 \mu\text{Sv/h}, \\ \dot{H}_\gamma &= 0.34 \mu\text{Sv/h}, \\ \dot{H}_{\text{surface}} &= 0.38 \mu\text{Sv/h} < 0.50 \mu\text{Sv/h} \quad (\text{design target}). \end{aligned} \quad (460)$$

For a tally normalized per source particle,

$\dot{H} = \dot{N}_{\text{source}} \sum_b \phi_b h_b$ requires an emission-rate normalization and energy-bin dose coefficients. Coupled transport shall include neutron-induced photons [67]. A volume-integrated track-length flux score shall be divided by the scoring volume before conversion to fluence per source particle [68]. For primary source class s , scored particle type $j \in \{n, \gamma\}$, and energy bin b , the surface-adjacent tally at position $\mathbf{r}$ gives

$$\begin{aligned} \dot{H}_{\text{surface}}(\mathbf{r}) &= 3.6 \times 10^9 \sum_s \dot{N}_s \sum_{j,b} \phi_{s,j,b}(\mathbf{r}) h_{j,b}, \\ [\phi_{s,j,b}] &= \text{cm}^{-2} \text{ source}^{-1}, \\ [h_{j,b}] &= \text{Sv cm}^2, \quad [\dot{H}_{\text{surface}}] = \mu\text{Sv/h}. \end{aligned} \quad (461)$$

The factor 3.6×10^9 converts Sv/s to $\mu\text{Sv/h}$. Each $\dot{N}_s$ is a physical primary emission rate in s^{-1} ; photon tallies from neutron histories include their secondary photons once. Independently emitted photons require their own source normalization. Dose-response coefficients shall identify the dose-equivalent quantity and irradiation geometry. Surface-current scores and absorbed-energy scores shall not be substituted for scalar-fluence dose conversion without the corresponding response model.

The component sum $0.04 + 0.34 = 0.38 \mu\text{Sv/h}$ lies $0.12 \mu\text{Sv/h}$, or 24%, below the $0.50 \mu\text{Sv/h}$ design limit. This arithmetic establishes the nominal allocation, not an independent transport solution. Acceptance shall use the maximum accessible-surface result with its statistical and source-normalization uncertainty, not only a surface average. The calculation shall retain source spectra and angular distributions, emission rates, geometry, nuclear-data and code versions, particle histories, tally definitions, covariance, and convergence results. Independent OpenMC and MCNP comparisons shall use matched inputs and response functions. Shield thicknesses alone do not establish the assigned dose rate; commissioning requires a calibrated dose map and evaluation of credible off-normal emissions independently of the proposed near-unity lattice branch.

I. Detector Hardware and Radiation-Transport Implementation

The following subsystem specifications implement the integrated assay, PSD acceptance, and nominal shielding configuration. They detail detector geometry, synchronous acquisition, event processing, and radiation transport. Alternative acquisition and shielding configurations require separate response and acceptance assessments.

Radiation diagnostics shall provide calibrated neutron and photon measurements for emission characterization and radiation-protection assessment. Detector calibration, background subtraction, and live-time accounting form the measurement basis for the EJ-301 pulse-shape discrimination (PSD) and gamma-spectroscopy channels. Acceptance of response, rejection, and dose-rate requirements shall rely on qualified measurements under the applicable emission and operating conditions.

J. EJ-301 Scintillators and Acquisition Chain

The neutron diagnostic shall use two sealed EJ-301 liquid-scintillator cells, each 127 mm in diameter and 127 mm deep,

at opposing 180° azimuths and nominal 350 mm source-to-face distance. Each cell shall couple to a fast photomultiplier with reference operating bias -1450 V. The response shall be characterized over the 0.5–20.0 MeV neutron-energy range, including the approximately 2.45 MeV D–D neutron signature, with the accepted energy range stated for each analysis threshold.

EJ-301 has manufacturer-listed light output of approximately 12000 photons per MeV of electron energy, peak emission at 425 nm, specific gravity 0.874, and refractive index 1.505. Its short decay component is approximately 3.2 ns, with listed mean components 3.16, 32.3, and 270 ns. Its flash point is approximately 26°C , requiring qualified sealed-cell and fire-safe handling.¹⁴ For Birks' law use $k_B^{\text{Birks}} = 0.0120 \text{ cm MeV}^{-1}$, not Boltzmann's constant. Stopping power is in MeV cm^{-1} :

$$\frac{dL}{dx} = S_L \frac{dE/dx}{1 + k_B^{\text{Birks}} dE/dx}. \quad (462)$$

S_L is the calibrated scintillation scale. Recoil light output is integrated along the track, not obtained by multiplying neutron energy by electron light yield.

The ionization-density dependence of the decay-component amplitudes provides the electron/proton recoil discrimination basis. The installed-system PSD boundary shall be calibrated as a function of deposited electron-equivalent energy and count rate.

The intrinsic neutron-efficiency target is $\varepsilon_{n,\text{int}} = 18.4\%$ at 2.45 MeV with a 1.0 MeVee electron-equivalent analysis threshold. Detector qualification shall establish whether the calibrated, proton-quenched recoil response supports this efficiency and threshold combination before a detection limit is assigned. Absolute efficiency shall include geometry, shielding, threshold-dependent intrinsic response, PSD acceptance, veto acceptance, and acquisition live time:

$$\varepsilon_{n,\text{abs}} = \frac{\Omega}{4\pi} T_n \varepsilon_{n,\text{int}} \varepsilon_{\text{PSD}} \varepsilon_{\text{veto}} L. \quad (463)$$

For two unobstructed circular detector faces normal to an isotropic point source, with radius $a = 63.5$ mm and distance $d = 350$ mm,

$$\frac{\Omega_{\text{pair}}}{4\pi} = 1 - \frac{d}{\sqrt{d^2 + a^2}} = 0.0160627. \quad (464)$$

Multiplication by the reference intrinsic efficiency gives 0.0029555, or approximately 0.296%, before attenuation and event-selection losses. Response calibration shall represent each detector's installed position relative to the W–Cu shield and its penetrations. Each threshold, selection, and live-time correction shall be included once in the efficiency accounting.

The synchronous acquisition configuration shall provide two 14-bit channels at 1.0 GS s^{-1} , with a 1.0 ns sample interval and aggregate raw payload of 28 Gbit s^{-1} before framing. The ADC and front end shall satisfy resolution, sampling, dynamic-range, and timing requirements in the installed configuration. The FPGA interface shall provide adequate lane bandwidth, deterministic timing, clock calibration, buffering, and loss

¹⁴Eljen Technology, EJ-301, EJ-309: Neutron/Gamma PSD Liquid Scintillators, manufacturer property table and data sheet.

counters. Parallel processing capacity shall be established independently of the sample-clock rate.

K. Charge-Comparison Digital PSD Algorithm

For digitized samples $v[n]$ and trigger sample n_0 , the algorithm shall estimate the baseline from the 64 immediately preceding samples and form a polarity-corrected signal:

$$\hat{b} = \frac{1}{64} \sum_{n=n_0-64}^{n_0-1} v[n], \quad s[n] = \varsigma(v[n] - \hat{b}), \quad (465)$$

where $\varsigma \in \{-1, +1\}$ shall make valid scintillation pulses positive. Commissioning shall characterize baseline noise and preceding-pulse contamination for this 64-sample estimator. Contaminated baselines, clipped amplitudes, incomplete windows, and detected pileup shall set explicit quality flags and shall be excluded from the qualified single-pulse PSD selection.

At $\Delta t = 1.0$ ns, the short and long gates shall contain exactly 24 and 240 samples, respectively:

$$\begin{aligned} Q_{\text{short}} &= \Delta t \sum_{n=n_0}^{n_0+23} s[n], \\ Q_{\text{long}} &= \Delta t \sum_{n=n_0}^{n_0+239} s[n], \\ Q_{\text{tail}} &= Q_{\text{long}} - Q_{\text{short}}, \\ S_{\text{PSD}} &= \frac{Q_{\text{tail}}}{Q_{\text{long}}} = 1 - \frac{Q_{\text{short}}}{Q_{\text{long}}}. \end{aligned} \quad (466)$$

The sums have units of sample amplitude times time. Conversion to coulombs shall use the calibrated front-end gain and impedance; common factors cancel in the PSD ratio. Division shall occur only after Q_{long} exceeds a strictly positive calibrated minimum. The inclusive gate endpoints in (466) shall be implemented without an additional endpoint sample.

The processing sequence shall be baseline estimation, calibrated threshold or constant-fraction timing, parallel gate accumulation, pulse-quality checks, energy calibration from Q_{long} , and energy-dependent PSD classification. Integer baseline-subtracted 14-bit codes shall use signed 24-bit accumulators; fractional baseline representations shall provide the corresponding guard bits. Fixed-point scaling, overflow handling, rounding, and division shall be verified against a higher-precision reference. The latency budget shall include the 240 ns observation window, buffering, and subsequent pipeline processing.

The ranges $S_{\text{PSD},\gamma} \approx 0.15\text{--}0.22$ and $S_{\text{PSD},n} \approx 0.38\text{--}0.52$ define reference training regions. The released neutron selection shall use the energy-dependent boundary $S_{\text{PSD}} > s_c(Q_{\text{long}})$ determined from independently tagged neutron and gamma samples. Validation shall use separate events, with the cut frozen before candidate reactor data are evaluated. Records shall retain raw or representative waveforms, baseline history, threshold and gate settings, analysis version, and rejected-event counters.

At the assigned rate ceiling $r = 5.0 \times 10^5 \text{ s}^{-1}$, a simple Poisson estimate for an additional arrival during a 240 ns gate is

$$p_{\text{overlap}} = 1 - e^{-r(240 \text{ ns})} = 0.11308. \quad (467)$$

The approximately 11.3% reference overlap probability shall be included in rate-dependent pileup and live-time qualification. Acceptance at the rate ceiling requires measured PSD efficiency, rejection, and event-loss performance at that rate, separately from low-rate calibration.

L. FOM, Gamma Rejection, and Detection Limits

In a specified energy bin, let μ_n and μ_γ be the PSD centroids and w_n, w_γ their FWHM values. The required separation metric is

$$\text{FOM} = \frac{|\mu_n - \mu_\gamma|}{w_n + w_\gamma} = 2.15 \quad (\text{reported}). \quad (468)$$

The gamma rejection requirement is a separate classification probability:

$$p_{\gamma \rightarrow n} = \Pr(S_{\text{PSD}} > s_c | \gamma), \quad \mathcal{R}_\gamma = \frac{1}{p_{\gamma \rightarrow n}} > 10^6. \quad (469)$$

Both requirements shall be verified together with neutron acceptance, energy interval, threshold, count rate, and pileup policy.

For an illustrative pair of equal-width Gaussian PSD populations with standard deviation σ , $w_n = w_\gamma = 2\sqrt{2 \ln 2} \sigma$. At the midpoint cut,

$$\begin{aligned} \frac{s_c - \mu_\gamma}{\sigma} &= 2\sqrt{2 \ln 2} \text{FOM}, \\ p_{\gamma \rightarrow n} &= \frac{1}{2} \operatorname{erfc} \left(\frac{2\sqrt{2 \ln 2} \text{FOM}}{\sqrt{2}} \right). \end{aligned} \quad (470)$$

The nominal $\text{FOM} = 2.15$ describes peak separation rather than a measured tail probability. The equal-width Gaussian example places the cut 5.06286σ above the gamma centroid, with leakage of approximately 2.07×10^{-7} . Qualification shall characterize non-Gaussian tails and the actual energy-dependent cut using an independent validation sample.

If no gamma event leaks into the neutron selection among M_γ independent, representative gamma trials, the one-sided 95% binomial upper bound is

$$p_{\gamma \rightarrow n, 95} = 1 - 0.05^{1/M_\gamma}. \quad (471)$$

Commissioning of a 95% upper bound below 10^{-6} shall use at least $M_\gamma = 2,995,731$ independent zero-leakage validation trials under the specified conditions, with the selection fixed in advance. Training events shall remain separate from this validation sample. Finite leakage shall be reported with a count-based confidence interval; zero observed leakage shall be reported as a bound rather than infinite rejection.

The count $M_\gamma = 2995731$ does not establish $6\sigma / 99.9999\%$ confidence; it corresponds to the 95% zero-leakage bound. The alternative expression $(5/0.002888)^2$ gives approximately 2997406, uses five rather than six standard deviations, and does not establish a binomial tail-rejection bound. For confidence $1 - \alpha$ and zero leakage,

$$M_\gamma \geq \left\lceil \frac{\ln \alpha}{\ln(1 - 10^{-6})} \right\rceil. \quad (472)$$

Thus 99.9999% confidence ($\alpha = 10^{-6}$) requires 13815504 trials. A Gaussian 6σ tail probability is different again and must

state its one- or two-sided convention. Rejection probability, confidence in its upper bound, and Gaussian peak separation are distinct quantities.

For neutron monitoring with no accepted events and negligible expected background, the corresponding 95% Poisson bound is

$$\Gamma_{n,95} = \frac{-\ln(0.05)}{\varepsilon_{n,\text{abs}}t} = \frac{2.995732}{\varepsilon_{n,\text{abs}}t}. \quad (473)$$

Here t shall be elapsed acquisition time when $\varepsilon_{n,\text{abs}}$ includes the live-time fraction L . Use of live exposure instead shall omit the corresponding live-time factor from efficiency. Non-negligible background, efficiency uncertainty, shielding, and veto losses shall be included in the full counting model, and null observations shall be reported as upper limits on the specified source model.

M. HPGe Gamma Spectroscopy and Activation Signatures

Gamma spectroscopy shall use an n-type coaxial high-purity germanium detector with cryogenic cooling, a pulse-tube cooler or qualified equivalent, and a 16384-channel digital spectrum. The detector specification shall provide nominal 50% relative efficiency against the conventional 3-inch by 3-inch NaI(Tl) reference at approximately 1.33 MeV, with installed absolute photopeak efficiency calibrated separately. Energy resolution shall be at most 1.85 keV FWHM near the ^{60}Co 1.33 MeV line and 0.92 keV near the 122 keV transition populated in ^{57}Fe by a ^{57}Co calibration source.

The detector housing shall have a separate local low-background shield of 100 mm lead, 1 mm cadmium, and 2 mm copper, with the lower-Z liner facing the detector. The reference direct viewing path includes a 2.0 mm beryllium window. Its energy-dependent attenuation shall be included in the response model. Using the NIST value $(\mu/\rho)_{\text{Be}} = 4.369 \text{ cm}^2 \text{ g}^{-1}$ at 5 keV,¹⁵ with nominal $\rho_{\text{Be}} = 1.848 \text{ g cm}^{-3}$,

$$T_{\text{Be}}(5 \text{ keV}) = e^{-(4.369)(1.848)(0.20)} = 0.19893. \quad (474)$$

Transmission at 10 keV is approximately 0.78743 using $\mu/\rho = 0.6466 \text{ cm}^2 \text{ g}^{-1}$. Window, dead-layer, shield-port, and collimator response must be included in absolute efficiency calibration. The pressure-boundary qualification and safe handling of beryllium components remain separate engineering requirements.

Table LVII defines the monitoring signatures and their timing interpretations. Energy calibration and peak fitting shall use evaluated line energies, emission probabilities, and decay schemes, with radioactive products distinguished from stable nuclei.¹⁶

The approximately 280.523 keV transition in ^{105}Pd shall be assessed with its possible ^{105}Rh decay feeding and prompt-capture origin distinguished by timing and accompanying lines. The stable palladium ground state shall not be assigned an

activation half-life. Prompt capture cascades shall use the corresponding neutron-on timing and branching model; isotope attribution shall combine energy, associated lines, and time dependence.

For a radioactive activation product j , a constant-flux approximation gives

$$\begin{aligned} \frac{dN_j}{dt} &= R_{j,\text{act}} - \lambda_j N_j, \\ R_{j,\text{act}} &= N_{j,\text{target}} \int \sigma_j(E) \phi_n(E) dE, \\ N_j(t) &= \frac{R_{j,\text{act}}}{\lambda_j} (1 - e^{-\lambda_j t}) \end{aligned} \quad (475)$$

for initially absent product and negligible depletion. After irradiation ceases, $N_j(t_c) = N_j(0)e^{-\lambda_j t_c}$. A photopeak count expectation over a live interval is

$$\mathbb{E}[C_j] = \varepsilon_{\text{FE},j} I_{\gamma,j} \int_{t_1}^{t_2} \lambda_j N_j(t) dt + C_{j,\text{bg}}, \quad (476)$$

where $\varepsilon_{\text{FE},j}$ includes geometry and attenuation and $I_{\gamma,j}$ is the emission probability. Commissioning shall establish the photopeak efficiency and background model. Reported detection limits shall additionally include target inventory, flux spectrum, irradiation and counting intervals, and their uncertainties. Any inferred neutron-branching limit, including a target below 10^{-8} , shall follow from this calibrated activation-response model rather than a peak-presence criterion alone.

N. Three-Layer Shield and Surface-Dose Requirement

Table LVIII defines the radial material schedule for the comparison shield, separately from the HPGe local enclosure. Its W-Cu composition is distinct from the transport-reference configuration; attenuation and dose results shall remain associated with the corresponding material model. Qualification requires a three-dimensional transport analysis including penetrations, joints, viewing ports, coolant paths, and source dimensions.

For a narrow, uncollided photon beam at the assigned 2.5 MeV reference energy, taking the nominal linear attenuation coefficient $\mu_\gamma = 0.82 \text{ cm}^{-1}$ gives

$$\frac{I}{I_0} = e^{-\mu_\gamma t_w} = e^{-(0.82)(3.50)} = 0.056699. \quad (477)$$

The reference calculation gives approximately 94.33% removal from the uncollided beam. The attenuation coefficient shall be verified for the mixture and photon energy. Whole-system dose assessment shall additionally include broad-beam buildup, scattered photons, pair-production products, and streaming.

The borated-polyethylene response shall include secondary photons, including the approximately 478 keV emission following boron capture and population of excited ^7Li .¹⁷ Hydrogen-capture photons shall also be included. Qualification of a 10^4 neutron-attenuation target for the 120 mm layer shall specify the source spectrum, geometry, and material composition and use coupled neutron/photon transport.

¹⁵J. H. Hubbell and S. M. Seltzer, *Tables of X-Ray Mass Attenuation Coefficients and Mass Energy-Absorption Coefficients*, NIST Standard Reference Database 126, beryllium table.

¹⁶National Nuclear Data Center, *Evaluated Nuclear Structure Data File*, mass-64, mass-105, and mass-187 decay and adopted-level evaluations; IAEA, *Database of Prompt Gamma Rays from Slow Neutron Capture for Elemental Analysis*, 2007. The listed windows are rounded monitoring references.

¹⁷J. S. Nico and collaborators, "Precision determination of absolute neutron flux," *Metrologia*, 2018; NIST neutron alpha-gamma counting program, identifying the approximately 478 keV photon following capture on ^{10}B .

TABLE LVII
SPECIFIED GAMMA-SPECTRAL MONITORING WINDOWS AND TIMING INTERPRETATION

Species or feature	Production or origin	Assigned energy window	Interpretation and timing
^{64}Cu	$^{63}\text{Cu}(n, \gamma)^{64}\text{Cu}$ and other possible activation routes	1345.77 keV	Delayed decay gamma; half-life approximately 12.7 h; weak emission probability requires calibrated sensitivity
^{105}Pd	$^{104}\text{Pd}(n, \gamma)^{105}\text{Pd}^*$; possible feeding from ^{105}Rh decay	Near 280.5 keV	^{105}Pd ground state is stable; a level transition is not evidence of delayed decay of activated ^{105}Pd
^{187}W	$^{186}\text{W}(n, \gamma)^{187}\text{W}$	479.5 and 685.8 keV	Delayed decay signatures; half-life approximately 23.7 h; compare multiple lines and decay curve
Positron annihilation	Positron-producing radionuclides or other pair-production processes	Near 511 keV	Not isotope-specific; requires coincident or spectral context
Continuum	Bremsstrahlung, Compton scattering, electronic background, and other sources	10–150 keV	Fit only after window and detector-response correction; not a uniquely nuclear signature

TABLE LVIII
THREE-LAYER REACTOR RADIATION-SHIELD ALLOCATION

Order from source	Material	Thickness (mm)	Assigned function and qualification requirement
1, inner	W–Cu, 80/20 by mass	35.0	Photon attenuation; qualify composition, density, energy-dependent attenuation, buildup, and secondary radiation
2, middle	Polyethylene with 5.0 wt% boron	120.0	Neutron moderation and capture; nominal density 0.95 g cm^{-3} ; boron isotopic composition and capture-gamma transport must be included
3, outer	SS316L structural enclosure	10.0	Mechanical containment and additional photon attenuation; include seams, penetrations, and material activation
Total nominal radial material thickness		165.0	Excludes air gaps, supports, liners, and the separate HPGc shield

The surface dose-equivalent-rate design target shall be $\dot{H}_{\text{surface}} \leq 0.38 \mu\text{Sv h}^{-1}$. The selected operational dose quantity and its conversion coefficients shall be stated. For differential fluence rate $\dot{\Phi}_j(E, \mathbf{r})$ and fluence-to-dose-equivalent coefficient $h_j(E)$,

$$\dot{H}(\mathbf{r}) = \sum_{j \in \{n, \gamma\}} \int h_j(E) \dot{\Phi}_j(E, \mathbf{r}) dE. \quad (478)$$

The reference source envelope uses neutrons near 2.45 MeV at a rate below 10^6 s^{-1} and photons over approximately 0.5–2.5 MeV. The qualified source model shall specify neutron and photon strengths, spectra, spatial distributions, and any potentially higher-energy channels. Commissioning measurements shall test the applicable source and transport assumptions before the surface-dose target is accepted.

For a deliberately simplified isotropic point source and no photon contribution,

$$\dot{H}_n(r) \approx \frac{Q_n}{4\pi r^2} T_n h_n(E_n), \quad (479)$$

with neutron source strength Q_n , source-to-surface distance r , neutron transmission T_n , and dose-conversion coefficient

specified as independent inputs. The full transport analysis shall include shield streaming and activation. Active commissioning shall include calibrated neutron and photon surveys at seams, ports, and accessible surfaces under qualified representative source conditions.

The $0.50 \mu\text{Sv h}^{-1}$ project comparison level shall be identified as a design reference, separately from applicable occupational or public exposure limits. Acceptance of the $0.38 \mu\text{Sv h}^{-1}$ target requires radiation-protection review, qualified source and measurement uncertainties, and documented survey results. Operating authorization and regulatory compliance shall be established through the applicable independent review process.

a) Coupled neutron–photon shield transport: The coupled-transport comparison uses 35 mm W–Cu (90 wt% W/10 wt% Cu), 120 mm HDPE containing 5 wt% boron, and 10 mm SS316L. Its nominal surface result is $\dot{H} = 0.312 \mu\text{Sv/h}$ against the $0.380 \mu\text{Sv/h}$ project criterion, a difference of $0.068 \mu\text{Sv/h}$ or 17.8947% of the limit. For species $a \in \{n, \gamma\}$, the coupled angular transport model is

$$\begin{aligned}
\frac{1}{v_a} \partial_t \psi_a &= -\hat{\Omega} \cdot \nabla \psi_a - \Sigma_{t,a} \psi_a + S_a \\
&\quad + \sum_b S_{ba}[\psi_b], \\
S_{ba}[\psi_b] &= \int dE' d\Omega' \Sigma_{b \rightarrow a}(E', \Omega' \rightarrow E, \Omega) \\
&\quad \times \psi_b(E', \Omega').
\end{aligned} \tag{480}$$

with neutron-induced photon production in $\Sigma_{n \rightarrow \gamma}$, energy redistribution, and material-dependent scattering/absorption. Capture photons include the approximately 0.478 MeV line following the excited-state $^{10}\text{B}(n, \alpha)^7\text{Li}$ branch and the approximately 2.223 MeV photon from $\text{H}(n, \gamma)\text{D}$; hydrogen scattering slows neutrons but is not itself that capture reaction. At an accessible survey location,

$$\dot{H}(\mathbf{r}) = \sum_a \int h_a(E) \phi_a(\mathbf{r}, E) dE, \quad \phi_a = \int \psi_a d\Omega. \tag{481}$$

Verification of the nominal 0.312 $\mu\text{Sv/h}$ result requires defined emission strengths and spectra, material number densities, an identified cross-section library, mesh or Monte Carlo convergence, streaming geometry, and dose-conversion tallies. The calculated value is neither a confirmed surface dose nor a universal regulatory limit. Acceptance requires coupled transport assessment and an independently scoped radiation survey, subject to the applicable exposure and measurement requirements [67].

O. Active Cosmic-Ray Muon Veto and Live-Time Accounting

Each EJ-301 assembly shall be surrounded by a nominal 20 mm-thick EJ-200 plastic-scintillator guard with independent photodetector timing and recorded guard triggers. The prompt coincidence window shall have full width $\Delta t_{\text{veto}} = 20$ ns. After correcting calibrated channel delay δt , define

$$v_\mu(t_e) = \begin{cases} 1, & \min_j |t_e - t_{\mu,j} - \delta t| \leq 10 \text{ ns}, \\ 0, & \text{otherwise,} \end{cases} \tag{482}$$

where t_e is the inner-detector event time and $t_{\mu,j}$ is a guard time. A prompt-veto neutron candidate shall pass pulse-quality and PSD cuts and satisfy $v_\mu(t_e) = 0$. Vetoed events and timing distributions shall be retained to support efficiency, accidental-coincidence, and analysis-bias verification.

For independent Poisson guard triggers at rate r_μ , the accidental prompt-veto probability is

$$p_{\text{acc}} = 1 - e^{-r_\mu \Delta t_{\text{veto}}} \approx r_\mu \Delta t_{\text{veto}} \tag{483}$$

when the product is small. The accepted live-time and efficiency model shall include accidental vetoes, guard thresholds, panel gaps, channel dead time, and scintillator timing resolution. Live exposure shall be determined from recorded acquisition intervals and the union of excluded windows, avoiding double counting of overlapping dead-time and veto intervals.

Prompt and delayed muon-correlated backgrounds shall be characterized separately. A 2.45 MeV neutron traverses 350 mm in approximately 16.17 ns before moderation or

additional path delays. Selection of any extended veto shall use measured time-correlation spectra and quantify the trade-off between delayed-background rejection and accidental losses. The gate shall be fixed using commissioning data independently of candidate reactor events.

The residual accepted-background requirement shall be below 10^{-4} s^{-1} , referred to the declared live exposure. Verification shall use off-source, reactor-disabled, and isotope-control runs with the same shield geometry, thresholds, PSD cuts, veto policy, and exposure accounting. The confidence level and exposure supporting the background-rate limit shall be reported.

For independent Poisson source-on and matched source-off counts N_{on} and N_{off} , define the live-exposure normalization $\alpha_t = t_{\text{on, live}}/t_{\text{off, live}}$. Under a stationary matched-background model with fixed α_t , the background-subtracted count and counting-variance estimate are

$$\begin{aligned}
\hat{N}_{\text{net}} &= N_{\text{on}} - \alpha_t N_{\text{off}}, \\
u_{\text{count}}^2(\hat{N}_{\text{net}}) &= N_{\text{on}} + \alpha_t^2 N_{\text{off}}, \\
\hat{r}_{\text{net}} &= \frac{\hat{N}_{\text{net}}}{t_{\text{on, live}}}.
\end{aligned} \tag{484}$$

Commissioning shall test background stationarity and add exposure-normalization, drift, efficiency, and selection uncertainties as applicable. Signed net estimates shall be retained; low-count confidence intervals and source-rate upper limits shall use a joint count model rather than a Gaussian approximation to this variance alone. Conversion from accepted rate to source rate shall use an efficiency definition consistent with the live-exposure normalization.

P. Commissioning and Acceptance Requirements

Release for measurement service requires calibrated detector response, verified background subtraction, and traceable evidence of compliance with the applicable performance criteria. The commissioning and acceptance activities are:

- **CM-01: Dual calorimetry.** Complete the electrical-substitution sequence, sensor and thermopile calibration, gain-drift checks, and nonreactive baseline runs of Section V. Verify spatial calibration transfer, corrections, and covariance before reporting the commissioned variance budget.
- **CM-02: Isotope assay.** Calibrate the mass axis, Faraday and multiplier gains, dose transfer, getter removal, and helium recovery. Verify manifold, matrix-gas, memory, and ingress subtraction through the complete sampling path, retaining the introduced-standard inventory and uncertainty.
- **CM-03: EJ-301 response and PSD.** Calibrate timing, electron-equivalent energy, recoil response, threshold-dependent efficiency, and rate-dependent pileup. Validate the fixed-point gate implementation and frozen PSD cut against independent tagged samples, documenting FOM, gamma-leakage bounds, neutron acceptance, and live exposure.

- **CM-04: Gamma and survey response.** Establish HPGe energy calibration, resolution, absolute photopeak efficiency, attenuation corrections, and matched continuum/line backgrounds. Complete source-dependent neutron and photon response checks and qualified accessible-surface surveys.
- **CM-05: Background and veto verification.** Measure prompt and delayed guard correlations, accidental losses, overlapping exclusion windows, and matched source-off rates. Verify the exposure-normalized subtraction and uncertainty model using independent control runs, including null and known-signal checks.

Acceptance documentation shall identify instrument configuration, calibration traceability, analysis version, operating conditions, measurement uncertainty, and acceptance disposition. Nominal targets define requirements rather than demonstrated performance. Calorimetry, helium assay, prompt-radiation monitoring, and delayed-activation analysis shall retain separate measurement and inference records.

VII. REDUCED-ORDER ROBUST CONTROL AND DECOUPLED SAFETY INTERLOCKS

The architecture separates continuous process regulation, nonlinear state estimation, and independent safety-instrumented functions. A fifteen-state plant supplies the analytical basis for a four-state control model. Structured uncertainty describes parameter drift, while sector-bounded feedback defines the conditions for nonlinear stability. An independently powered TMR hardware path inhibits RF and optical drive without waiting for model execution or supervisory polling. Mechanical isolation and fluid mitigation operate in a separate response domain. Numerical setpoints and acceptance bounds define the design; a stability certificate must apply to the specified realization, and a shutdown certificate must include the physical safe-state response.

A. Fifteen-State Plant and Balanced Truncation

a) *Control synthesis and runtime separation:* The reduction interface shall construct the Gramians, extract Hankel singular values, and map the fifteen-state plant to four states with an approximation certificate. A 100 Hz supervisory update gives a 10 ms sample interval; it does not set the response time of the independent fast interlock. Nonlinear stability synthesis additionally requires the sector limits, multiplier, and positive-definite matrix certificate. A residual monitor evaluates prediction errors and is distinct from the offline reduction and LMI solvers.

For the selected two-input, two-output plant, define

$$\begin{aligned}\dot{\mathbf{x}}_{15} &= \mathbf{A}\mathbf{x}_{15} + \mathbf{B}\mathbf{u}, \\ \mathbf{y} &= \mathbf{C}\mathbf{x}_{15} + \mathbf{D}\mathbf{u}, \\ G(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}, \\ \mathbf{A} &\in \mathbb{R}^{15 \times 15}, \quad \mathbf{B} \in \mathbb{R}^{15 \times 2}, \\ \mathbf{C} &\in \mathbb{R}^{2 \times 15}, \quad \mathbf{D} \in \mathbb{R}^{2 \times 2}.\end{aligned}\tag{485}$$

The state definitions and numerical simulator realization are specified in Section IX. Reduction shall retain the same

physical ports, input/output scaling, operating point, and direct feedthrough. Selection from a four-input, four-output realization requires explicit port maps before balancing; state reduction alone does not remove physical ports.

For a Hurwitz, controllable and observable realization, the positive-definite controllability and observability Gramians satisfy [28]

$$\begin{aligned}\mathbf{A}\mathbf{W}_c + \mathbf{W}_c\mathbf{A}^\top + \mathbf{B}\mathbf{B}^\top &= 0, \\ \mathbf{A}^\top\mathbf{W}_o + \mathbf{W}_o\mathbf{A} + \mathbf{C}^\top\mathbf{C} &= 0, \\ \mathbf{W}_c &= \mathbf{L}_c\mathbf{L}_c^\top, \quad \mathbf{W}_o = \mathbf{L}_o\mathbf{L}_o^\top, \\ \mathbf{L}_o^\top\mathbf{L}_c &= \mathbf{U}\Sigma\mathbf{V}^\top, \\ \mathbf{T}_{\text{bal}} &= \mathbf{L}_c\mathbf{V}\Sigma^{-1/2}, \\ \mathbf{T}_{\text{bal}}^{-1} &= \Sigma^{-1/2}\mathbf{U}^\top\mathbf{L}_o^\top.\end{aligned}\tag{486}$$

Here $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_{15})$ contains descending Hankel singular values. Uncontrollable or unobservable modes shall be removed before inverting singular factors. In balanced coordinates,

$$\begin{aligned}\mathbf{T}_{\text{bal}}^{-1}\mathbf{A}\mathbf{T}_{\text{bal}} &= \begin{bmatrix} \mathbf{A}_r & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}, \\ \mathbf{T}_{\text{bal}}^{-1}\mathbf{B} &= \begin{bmatrix} \mathbf{B}_r \\ \mathbf{B}_2 \end{bmatrix}, \quad \mathbf{C}\mathbf{T}_{\text{bal}} = [\mathbf{C}_r \quad \mathbf{C}_2], \\ \mathbf{D}_r &= \mathbf{D}, \quad \mathbf{G}_r(s) = \mathbf{C}_r(s\mathbf{I} - \mathbf{A}_r)^{-1}\mathbf{B}_r + \mathbf{D}_r.\end{aligned}\tag{487}$$

The retained block has four states and the discarded block eleven. This construction is balanced truncation, not the Schur-complement residualization used for a separate low-frequency approximation. The numerical four-state candidate is [79]

$$\mathbf{A}_r = \begin{bmatrix} -488.2142 & 112.4051 & -34.1102 & 18.9015 \\ -112.4051 & -312.8521 & 88.5410 & -41.2204 \\ -34.1102 & -88.5410 & -184.2105 & 105.6120 \\ 18.9015 & 41.2204 & -105.6120 & -859.9338 \end{bmatrix} \text{s}^{-1}.\tag{488}$$

The modal design targets are

$$\lambda_{\text{target}} = \{-925.12, -548.93, -274.35, -95.82\} \text{s}^{-1}.\tag{489}$$

Direct eigenanalysis of Eq. (488), however, gives

$$\lambda(\mathbf{A}_r) = \{-841.993834, -421.968669, -290.624049 \pm i70.356151\} \text{s}^{-1}.\tag{490}$$

These poles establish Hurwitz stability of the displayed matrix, but are not the target spectrum. An independent certificate follows from $\mathbf{P}_0 = \mathbf{I}_4$ and $\mathbf{Q}_0 = -(\mathbf{A}_r^\top + \mathbf{A}_r)$:

$$\begin{aligned}\lambda_{\min}(\mathbf{Q}_0) &= 360.847685 \text{s}^{-1} > 0, \\ \mathbf{V}_0(\mathbf{x}_r) &= \mathbf{x}_r^\top \mathbf{x}_r, \\ \dot{\mathbf{V}}_0 &= -\mathbf{x}_r^\top \mathbf{Q}_0 \mathbf{x}_r \leq -360.847685 \text{s}^{-1} \|\mathbf{x}_r\|_2^2.\end{aligned}\tag{491}$$

This certificate uses consistently scaled reduced coordinates and the unforced linear dynamics. It does not establish a transfer-function approximation without $\mathbf{B}_r, \mathbf{C}_r, \mathbf{D}_r$ and the parent-model transformation.

TABLE LIX
COMPARISON HANKEL SPECTRUM FOR A DISTINCT PORT SCALING

Mode	σ_i	Treatment
1	1.931312×10^{-1}	Retain
2	1.647743×10^{-2}	Retain
3	4.253545×10^{-3}	Retain
4	2.981347×10^{-3}	Retain
5	1.502356×10^{-3}	Truncate
6	4.453772×10^{-4}	Truncate
7	6.281842×10^{-5}	Truncate
8	2.672913×10^{-5}	Truncate
9	1.280489×10^{-5}	Truncate
10	2.974815×10^{-7}	Truncate
11	9.931668×10^{-8}	Truncate
12	1.057691×10^{-8}	Truncate
13	4.349070×10^{-9}	Truncate
14	4.652158×10^{-10}	Truncate
15	1.193118×10^{-12}	Truncate

For a stable minimal parent with the spectrum in Table LIX and the balancing partition in Eq. (487), the absolute error satisfies

$$\begin{aligned} \|G - G_r\|_\infty &\leq 2 \sum_{i=5}^{15} \sigma_i, \\ \sum_{i=5}^{15} \sigma_i &= 2.050497830568918 \times 10^{-3}, \\ 2 \sum_{i=5}^{15} \sigma_i &= 4.100995661137836 \times 10^{-3} \\ &\leq 4.1010 \times 10^{-3}. \end{aligned} \quad (492)$$

The numerical sum verifies the bound associated with the stated spectrum. Application to the physical plant requires the complete port-scaled realization, Gramian residuals, and balancing maps linking that spectrum to Eq. (488). The bound is dimensionless only for normalized inputs and outputs; it is not a relative-error percentage.

b) Calculated spectrum and reduction acceptance: The selected fifteen-value spectrum is given in Table XXXVII. Its first four modes are retained. With $\hat{\sigma}_i$ denoting their target values, the arithmetic agreement and discarded-spectrum measure are

$$\begin{aligned} e_4 &= \max_{1 \leq i \leq 4} \frac{|\sigma_i - \hat{\sigma}_i|}{|\hat{\sigma}_i|} \\ &= 2.362176648138631 \times 10^{-16} < 10^{-6}, \\ r_{\text{tail}} &= \frac{\sum_{i=5}^{15} \sigma_i^2}{\sum_{j=1}^{15} \sigma_j^2} \\ &= 2.129567641168566 \times 10^{-6} < 10^{-4}. \end{aligned} \quad (493)$$

These metrics retain the unrounded arithmetic even though decimal table entries suppress representation artifacts. The squared-spectrum ratio is not a transfer-function error bound or a direct physical energy fraction. For a stable minimal plant with this spectrum, balanced truncation gives the distinct absolute bound

$$\|G - G_r\|_\infty \leq 2 \sum_{i=5}^{15} \sigma_i = 0.07974. \quad (494)$$

The 4.1010×10^{-3} bound applies only to the comparison spectrum in Table LIX, not to Table XXXVII. Each spectrum requires its own consistently scaled plant, Gramians, and balancing maps. Close agreement of four singular values does not alone establish pole locations, robust stability, or the required physical port mapping. The native `verify_state_reduction` acceptance check compares the first four finite values with (12.45, 5.13, 1.85, 0.94) using absolute tolerance 10^{-12} per entry. It is a value check, not a Gramian calculation or eigenvector-rigidity proof. Its relative spectral error $e_4 \simeq 2.36 \times 10^{-16}$ is distinct from the scalar Floquet precision difference $\delta_F \simeq 3.56655 \times 10^{-16}$.

B. Structured Uncertainty and Robust μ -Synthesis

Fluid density, thermal-gradient response, and actuator gain shall be represented by normalized parameter perturbations

$$\begin{aligned} p_j &= p_{j0}(1 + 0.15\delta_j), \quad |\delta_j| \leq 1, \\ \Delta &= \text{diag}(\delta_\rho I, \delta_T I, \delta_k I), \\ \|\Delta\|_\infty &\leq 1. \end{aligned} \quad (495)$$

Repeated real blocks represent shared physical parameters; any additional complex dynamic blocks require their own frequency weights. The 0.15 factors belong in the plant uncertainty channels and shall not be applied twice. The generalized state-space interconnection is

$$\begin{bmatrix} \mathbf{x}_r \\ \mathbf{z}_\Delta \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} A_r & B_\Delta & B_u \\ C_\Delta & D_{\Delta\Delta} & D_{\Delta u} \\ C_y & D_{y\Delta} & D_{yu} \end{bmatrix} \begin{bmatrix} \mathbf{x}_r \\ \mathbf{w}_\Delta \\ \mathbf{u} \end{bmatrix}, \quad \mathbf{w}_\Delta = \Delta \mathbf{z}_\Delta. \quad (496)$$

Writing the corresponding transfer matrix as

$P(s) = [P_{ab}(s)]_{a,b=1}^2$, the uncertain plant is the upper linear fractional transformation

$$\begin{aligned} M(s) &= F_u(P(s), \Delta) \\ &= P_{22} + P_{21}\Delta(I - P_{11}\Delta)^{-1}P_{12}. \end{aligned} \quad (497)$$

The inverse must exist for a well-posed interconnection. Controller synthesis closes the control ports with $K(s)$, including the selected feedback sign. Let $N_K(s)$ denote the resulting nominal return matrix from $\mathbf{w}_\Delta$ to $\mathbf{z}_\Delta$. Structured singular-value analysis applies to this return matrix, not to the already uncertainty-closed input/output plant $M(s)$:

$$\mu_{\mathcal{D}}(N_K) = \left[\min_{\substack{\Delta \in \mathcal{D} \\ \det(I - N_K \Delta) = 0}} \bar{\sigma}(\Delta) \right]^{-1}. \quad (498)$$

The value is zero if no destabilizing structured perturbation exists. Synthesis shall specify disturbance and performance weights, controller order, uncertainty-block dimensions, and the resulting controller realization. Frequency-dependent scaling and controller updates shall minimize a certified upper bound on μ [88]. The design acceptance envelope is

$$\begin{aligned} \sup_{\omega} \bar{\mu}_{\mathcal{D}}(N_K(i\omega)) &\leq 0.842 < 1.00, \\ \omega_{\text{peak}} &= 412 \text{ rad/s}, \\ \omega_{\text{scan}} &\in [10^{-2}, 10^5] \text{ rad/s}. \end{aligned} \quad (499)$$

Here $\bar{\mu}$ denotes a certified upper bound, not an assumed exact value. Nominal internal stability, well-posedness, and a bound below unity over the full frequency range establish robust stability for the normalized uncertainty set. A finite frequency grid alone does not establish the supremum; interpolation and low/high-frequency tails require coverage. Robust performance additionally requires weighted performance channels and an augmented performance block. The assigned peak of 0.842 is an acceptance target until the generalized plant, controller, scaling certificate, and frequency coverage establish it; the drift range alone is not a numerical μ calculation.

C. Nonlinear Lur'e–Postnikov Stability Certificate

For equilibrium-shifted autonomous dynamics, define

$$\dot{\mathbf{x}} = A_c \mathbf{x} + B_\phi \phi(y), \quad y = C_\phi \mathbf{x}, \quad \phi(0) = 0. \quad (500)$$

The sign of feedback is included in B_ϕ . The matrix A_c shall include controller and observer states whenever their dynamics are retained; it equals A_r only for the corresponding four-state static-feedback model. Assume a locally Lipschitz, globally sector-bounded nonlinearity with

$$0 \leq y\phi(y), \quad \phi(y)^2 \leq k_\phi y\phi(y), \quad k_\phi > 0. \quad (501)$$

The Lur'e–Postnikov candidate is

$$V(\mathbf{x}) = \mathbf{x}^\top \mathbf{P} \mathbf{x} + \int_0^y \phi(\sigma) d\sigma, \quad \mathbf{P} = \mathbf{P}^\top > 0. \quad (502)$$

In consistently normalized coordinates the sector condition makes the integral nonnegative, so V is positive definite and radially unbounded. Differentiation gives

$$\begin{aligned} \dot{V} &= \mathbf{x}^\top (A_c^\top \mathbf{P} + \mathbf{P} A_c) \mathbf{x} + 2\mathbf{x}^\top \mathbf{P} B_\phi \phi(y) \\ &\quad + \phi(y) C_\phi [A_c \mathbf{x} + B_\phi \phi(y)]. \end{aligned} \quad (503)$$

A sufficient sector/KYP-type matrix certificate is the existence of $\mathbf{Q} = \mathbf{Q}^\top > 0$, $\eta > 0$, and $\tau \geq 0$ such that [89]

$$\begin{bmatrix} A_c^\top \mathbf{P} + \mathbf{P} A_c + \mathbf{Q} & \mathbf{h} \\ \mathbf{h}^\top & C_\phi B_\phi + \eta - \tau/k_\phi \end{bmatrix} \preceq 0, \quad (504)$$

$$\mathbf{h} = \mathbf{P} B_\phi + \frac{1}{2} A_c^\top C_\phi^\top + \frac{\tau}{2} C_\phi^\top.$$

Indeed, premultiplication and postmultiplication by $[\mathbf{x}^\top, \phi(y)]$ and its transpose give

$$\begin{aligned} \dot{V} &\leq -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - \eta \phi(y)^2 - \tau[y\phi(y) - \phi(y)^2/k_\phi] \\ &\leq -\mathbf{x}^\top \mathbf{Q} \mathbf{x} - \eta \phi(y)^2 < 0, \quad \mathbf{x} \neq 0. \end{aligned} \quad (505)$$

These conditions prove global asymptotic stability for the stated autonomous model. Reactor-wide application requires a numerical certificate for the actual closed-loop matrices and a sector bound valid over the entire state domain. A local linearization or a phase-coupling law satisfying the sector only near lock supports a regional result, not a global one. Switching, external disturbances, actuator limits outside the sector, and loss-of-coolant modes require their own invariant-domain or hybrid-system assessment. Hurwitz stability of A_r alone does not supply the nonlinear certificate.

a) *Multi-input sector certificate:* For the explicit negative-feedback convention $\dot{\mathbf{x}} = A_r \mathbf{x} - B_r \phi(C_r \mathbf{x})$, let $\Gamma = \text{diag}(\gamma_j) > 0$ and $\phi_j(v_j)[\phi_j(v_j) - \gamma_j v_j] \leq 0$ globally. A quadratic sector certificate is $P = P^\top > 0$, diagonal $\Lambda > 0$, and

$$\begin{bmatrix} A_r^\top P + P A_r & -P B_r + C_r^\top \Lambda \\ -B_r^\top P + \Lambda C_r & -2\Lambda \Gamma^{-1} \end{bmatrix} \prec 0. \quad (506)$$

Congruence by $\text{diag}(I, -I)$ gives off-diagonal block $P B_r - C_r^\top \Lambda$ with the same definiteness. The feedback sign and sector convention must accompany that form. For normalized coordinates, numerical acceptance uses margins $P \succeq 10^{-6} I$, $\Lambda \succeq 10^{-6} I$, and an LMI maximum eigenvalue at most -10^{-6} , checked independently of the solver status. This sufficient quadratic certificate is distinct from the integral Lyapunov function in Eq. (502); its feasibility cannot be inferred from Hankel-value agreement. Sector validity, Lipschitz continuity, and a correctly shifted equilibrium delimit the stability claim.

D. Plasma Recombination, Diffusion, and RF Extinction

After drive removal, the effective electron-density model is

$$\frac{\partial n_e}{\partial t} = -\alpha_r n_e^2 + D_a \nabla^2 n_e, \quad (507)$$

with the nominal constitutive inputs [80]

$$\begin{aligned} \alpha_r &= 2.4 \times 10^{-17} \text{ m}^3/\text{s}, \\ D_a &= 12.5 \text{ m}^2/\text{s}, \quad n_{e0} = 1.5 \times 10^{20} \text{ m}^{-3}, \\ \lambda_0 &= 0.012 \text{ m}. \end{aligned} \quad (508)$$

The quadratic loss law uses an effective two-body coefficient. Any three-body contribution shall be reduced using a stated density and temperature dependence or represented by its own rate term. With an absorbing-boundary fundamental-mode approximation $\nabla^2 n_e \simeq -(\pi/\lambda_0)^2 n_e$, define $\beta = D_a(\pi/\lambda_0)^2$. The scalar amplitude equation and its solution are

$$\begin{aligned} \dot{n}_e &= -\alpha_r n_e^2 - \beta n_e, \\ \frac{n_e(t)}{n_{e0}} &= \frac{\beta e^{-\beta t}}{\beta + \alpha_r n_{e0}(1 - e^{-\beta t})}, \\ t_f &= \frac{1}{\beta} \ln \frac{\beta + f \alpha_r n_{e0}}{f(\beta + \alpha_r n_{e0})}, \quad f = \frac{n_e(t_f)}{n_{e0}}. \end{aligned} \quad (509)$$

This is an exact solution of the scalar approximation, not of an arbitrary spatial profile. For the displayed inputs,

$$\begin{aligned} \tau_{\text{rec}} &= (\alpha_r n_{e0})^{-1} = 277.777778 \mu\text{s}, \\ \tau_{\text{diff}} &= \beta^{-1} = 1.167220 \mu\text{s}, \\ t_{0.01} &= 5.370402 \mu\text{s}, \\ n_e(3.42 \mu\text{s})/n_{e0} &= 0.0531834. \end{aligned} \quad (510)$$

The assigned nominal RF-extinction target is $t_{\text{quench}} = 3.42 \mu\text{s}$ at 5.0 bar. Table LX specifies the pressure-dependent timing targets; its 0.278 μs recombination allocation is distinct from the 277.777778 μs obtained from the displayed coefficient. The target table does not override Eq. (510).

The 3.42 μs value cannot be a worst-case bound across the full 1.0–10.0 bar table, whose largest allocation is 4.85 μs . Moreover, electron-density depletion is not automatically equivalent to RF-energy extinction. Qualification shall specify

TABLE LX
PRESSURE-DEPENDENT PLASMA TIMING TARGETS

Pressure (bar)	τ_{rec} (μs)	τ_{diff} (μs)	t_{quench} (μs)
1.0	0.278	0.420	2.15
2.5	0.278	0.780	2.80
5.0	0.278	1.165	3.42
7.5	0.278	1.520	4.10
10.0	0.278	1.890	4.85

the residual RF energy or power threshold, its relationship to electron density and cavity loading, boundary conditions, continuing ionization sources, and pressure-dependent coefficients. Without that coupling, the density equation does not prove 99% RF extinction at the assigned time.

E. Decoupled TMR Protection and Safe-State Timing

a) *Probability-conserving safety-state model:* Let $\mathbf{p} = (p_0, p_1, p_2, p_3)^T$ describe healthy, one-channel-degraded, safely tripped, and dangerous states, respectively. A continuous-time model requires nonnegative off-diagonal transition rates and zero column sums. With channel failure rate $\lambda = 1.5 \times 10^{-6} \text{ h}^{-1}$, repair rate $\mu = 0.125 \text{ h}^{-1}$, and an independently characterized hazardous-transition rate $\lambda_h \geq 0$, one admissible topology is

$$\dot{\mathbf{p}} = \mathbf{Q}\mathbf{p}, \quad \mathbf{p}(0) = (1, 0, 0, 0)^T, \quad (511)$$

$$\mathbf{Q} = \begin{bmatrix} -3\lambda & \mu & \mu & 0 \\ 3\lambda & -(\mu + 2\lambda + \lambda_h) & 0 & 0 \\ 0 & 2\lambda & -\mu & 0 \\ 0 & \lambda_h & 0 & 0 \end{bmatrix}.$$

Repair from the safely tripped state returns probability to the healthy state; restart remains subject to reset authorization. An absorbing dangerous state requires an incoming hazardous transition. Omitting both that transition and the return from state 2 gives $\dot{p}_3 = 0$ and $d(\sum_i p_i)/dt = -\mu p_2$, which cannot support a normalized reliability calculation. The parameter λ_h represents an additional fault pathway, not a value inferable from λ and μ alone.

For a defined dangerous state, the one-year demand-unavailability measure is

$$\text{PFD}_{\text{avg}} = \frac{1}{\tau_{\text{test}}} \int_0^{\tau_{\text{test}}} p_3(t) dt, \quad \tau_{\text{test}} = 8760 \text{ h}. \quad (512)$$

The target 5.18×10^{-4} is not determined by an unreachable dangerous state. Moreover, $3(\lambda\tau_{\text{test}})^2 + (\lambda\tau_{\text{test}})^3 = 5.20247547 \times 10^{-4}$ is not the demand-average formula for independent 2oo3 channels. For the separate model with independent undetected failures, perfect initial health, and no repair between proof tests,

$$q(t) = 1 - e^{-\lambda t}, \quad p_D(t) = 3q(t)^2 - 2q(t)^3, \quad (513)$$

$$\frac{1}{\tau_{\text{test}}} \int_0^{\tau_{\text{test}}} p_D(t) dt = 1.69851776 \times 10^{-4}.$$

That comparison excludes the stated eight-hour repair process and is not the complete safety-function PFD. Common-cause

faults, detection coverage, voter/power failures, final elements, and proof-test effectiveness require their own terms. A numerical target within a SIL-3 probability band does not establish the lifecycle and system requirements for certification [4], [5].

b) *Fault-injection and voting acceptance:* The function-specific injection cases are The 0.5 kg/s flow threshold exceeds the nominal 0.08–0.12 kg/s manifold flow and would be asserted during normal operation; it shall not be enabled for that operating range. For the 1000 W optical comparison, 15% attenuation corresponds to 850 W, but this is not the reactor support-load rating. Vote separately on each hazard across three independent channels: $v_j = \mathbf{1}(\sum_{c=1}^3 b_{cj} \geq 2)$, with a trip if any $v_j = 1$. One asserted channel raises a disagreement/degraded alarm. Stale, nonfinite, or failed sensors require an explicit safe-state policy rather than normal threshold comparison. Coolant-flow protection requires an actual flow measurement in addition to temperature telemetry. The 10 ms supervisory execution limit must be enforced by an independent watchdog; it is not the microsecond Tier-1 trip budget and cannot be established by timing one completed software call.

Tier-1 shall use independent trip sensing, segregated TMR FPGA logic with 2oo3 voting, and direct SiC crowbar and EOM driver outputs. The fast path shall bypass the process controller, operating-system tasks, observer, and supervisory network. Trip latches shall inhibit re-energization until independent safe-state feedback and authorized reset are present. A sub-500 ns SiC conduction requirement refers to the switch branch, not completion of sensor processing or plasma decay.

Tier-2 shall execute mechanical shutter travel, fuel isolation, and nitrogen/fluid mitigation commands on a separate 10.0–45.0 ms final-element actuation envelope. These actions shall be initiated without delaying Tier-1. Mechanical response may continue after drive extinction; it shall not be counted as completed by an FPGA command timestamp. Subsequent inventory evacuation, including the separate < 150 ms dump target, is a fluid-system completion criterion rather than a valve-actuation time. The 10.0 ms supervisory interval monitors availability and state but receives no credit for microsecond fault detection.

For the sequential timing definition in Table LXII,

$$t_{\text{trip}} = t_{\text{sense}} + t_{\text{proc}} + t_{\text{crowbar}} + t_{\text{quench}},$$

$$t_{\text{trip,nom}} = 1.20 + 2.10 + 1.85 + 3.42$$

$$= 8.57 \mu\text{s} \leq 12.50 \mu\text{s} < 42.80 \mu\text{s}, \quad (514)$$

$$\Delta t_{\text{margin}} = 3.93 \mu\text{s} \quad (31.44\%).$$

The nominal interval is approximately one fifth of the 42.80 μs comparison limit. The 8.57 μs sum is a design timing point, not a measured worst-case certificate. If the scalar 99% density-depletion time is used as a provisional extinction proxy instead, the same three electronic stages give 10.520402 μs ; this remains below 12.50 μs but does not establish the missing RF-energy relation. Acceptance shall timestamp physical fault onset, qualified sensor detection, voting, driver execution, and completed RF/optical safe state under worst-case voltage, temperature, clock phase, and fault combinations. Parallel crowbar/EOM execution shall use the slower required completion path without counting shared stages twice.

TABLE LXI
TMR FAULT-INJECTION ACCEPTANCE CASES

Case	Injection	Threshold	Protective response and scope
FIT-01	Optical shutter occlusion	At least 15.0% attenuation within 2 ms	Laser inhibit; the observation window is distinct from fast-path actuation latency.
FIT-02	Coolant-valve blockage	$\dot{m} \leq 0.5 \text{ kg/s}$; $T \geq 375.0 \text{ }^\circ\text{C}$	Isolate drive and actuate the qualified dump path; this flow threshold requires a separate operating range.
FIT-03	RF phase modulation	$ \delta\phi \geq 0.12 \text{ rad}$	Inhibit RF drive; the threshold is an angular deviation, not a variance in radians squared.

TABLE LXII
TIER-1 END-TO-END TIMING BUDGET

Stage	Budget (μs)	Nominal (μs)
Sensing and conditioning	1.50	1.20
TMR FPGA processing	2.50	2.10
SiC crowbar and EOM drivers	2.50	1.85
Plasma RF extinction	6.00	3.42
Total	12.50	8.57

The Luenberger observer belongs to process regulation:

$$\begin{aligned}\dot{\hat{\mathbf{x}}}_r &= A_r \hat{\mathbf{x}}_r + B_r \mathbf{u} + L_r (\mathbf{y} - C_r \hat{\mathbf{x}}_r - D_r \mathbf{u}), \\ \dot{\mathbf{e}}_r &= (A_r - L_r C_r) \mathbf{e}_r.\end{aligned}\quad (515)$$

The gain L_r requires the selected output matrix and noise/bandwidth requirements. Stability of $A_r - L_r C_r$ shall be demonstrated for that realization; gains from a differently normalized model shall not be transferred unchanged. Observer convergence and simulator step size shall not delay independent trip initiation.

F. TMR Cause-and-Effect and Command Timing Requirements

The protection system shall use independent triple-modular-redundant (TMR) channels with 2oo3 voting and a SIL-3 design objective [4], [5]. Table LXIII defines the six functional cause-and-effect assignments. Its microsecond allocations refer to command-path or explicitly identified electronic response boundaries; they do not specify completed mechanical travel or fluid displacement. The end-to-end Tier-1 envelope in Table LXII and the separate Tier-2 mechanical completion envelope govern physical safe-state acceptance. TR-02 switch-branch initiation shall satisfy $< 500 \text{ ns}$. The $12.50 \mu\text{s}$ optical-detuning allocation is an upper command/function allowance, while the selected Tier-1 sequence shall complete the required RF and optical safe state within the same $12.50 \mu\text{s}$ total budget. The $42.80 \mu\text{s}$ comparison limit and strict $< 50 \mu\text{s}$ outer bound are not additional serial delays. FM-04's $45 \mu\text{s}$ command allocation is outside the fast-path completion chain; nitrogen displacement remains a Tier-2 action. Mechanical shutter travel, gas flushing, and valve closure require independent final-element feedback. Pressure reference (gauge or absolute), reflectance normalization, and scintillator integration interval shall be specified for each trigger.

$$\begin{aligned}t_{\text{trip},j} &= t_{\text{sense},j} + t_{\text{condition},j} + t_{\text{vote},j} \\ &+ t_{\text{output},j} + t_{\text{safe state},j} < 50 \mu\text{s}.\end{aligned}\quad (516)$$

Equation (516) applies to a Tier-1 protective function indexed by j ; the selected $12.50 \mu\text{s}$ limit is the tighter acceptance requirement. Initiation-only and completed-safe-state timing shall be recorded separately. The 10.0 ms supervisory interval, 1 ms simulator step, and 99.86 ms observer decay are not fast-interlock response times. Functional-safety qualification shall include failure rates, diagnostic coverage, proof-test intervals, common-cause analysis, and independence evidence for the complete safety instrumented function [4], [5]. Majority voting and a speed requirement do not alone establish SIL-3 performance.

Process regulation, independent protective functions, and scientific data acquisition shall retain separate execution and power-failure boundaries. SIL-3 is an allocated integrity requirement for each designated safety instrumented function, including its sensors, logic, final elements, and diagnostics. The DM-01 and DM-02 plant parameters define the corresponding process-loop models here and in Section IX. Their response time constants shall not be credited as independent protection response times.

G. Optomechanical Plant, Feedback, and Stability Margins

The DM-01 mechanical displacement model uses the specified static gain and 10 kHz resonance:

$$\begin{aligned}\ddot{y} + 2\zeta_p \omega_p \dot{y} + \omega_p^2 y &= K_p \omega_p^2 v, \\ G_p(s) &= \frac{K_p}{\tau_p^2 s^2 + 2\zeta_p \tau_p s + 1}, \\ K_p &= 0.845 \mu\text{m/V}, \quad \zeta_p = 0.15, \\ \omega_p &= 2\pi(10^4) \text{ s}^{-1}, \\ \tau_p &= \omega_p^{-1} = 15.915494 \mu\text{s}.\end{aligned}\quad (517)$$

The rounded time constant $15.9 \mu\text{s}$ corresponds to 10009.74485 Hz if used literally. The sensor factor is a pure delay:

$$H_p(s) = e^{-s\tau_o}, \quad \tau_o = 120.0 \mu\text{s}.\quad (518)$$

A pure delay has unit magnitude and no -3 dB cutoff. The observed piezo response is $G_{\text{piezo}} = G_p H_p$. With controller C_p and separately counted computation/actuation delay T_p ,

$$\begin{aligned}L_p(s) &= C_p(s) G_p(s) H_p(s) e^{-sT_p}, \\ \frac{Y}{R} &= \frac{C_p G_p e^{-sT_p}}{1 + L_p}, \quad S_p = \frac{1}{1 + L_p}, \\ C_p(s) &= K_{c,p} \left(1 + \frac{\omega_{i,p}}{s}\right) \frac{1 + s/\omega_{z,p}}{1 + s/\omega_{h,p}}.\end{aligned}\quad (519)$$

TABLE LXIII
TMR CAUSE-AND-EFFECT MATRIX AND COMMAND-PATH ALLOCATIONS

Fault	Trigger	Vote	Action	Commanded protection / hardware	Allocation (μs)
FM-01	Optical phase drift $> 12.5^\circ$	2oo3	TR-01	15 GHz electro-optic detuning; EOM/Pockels cell	12.50
FM-02	RF reflectance > -10.0 dB	2oo3	TR-02	RF-cavity crowbar energy dump; solid-state SiC switch	0.50
FM-03	Substrate $> 375.0^\circ\text{C}$	2oo3	TR-03	Laser-shutter closure; solenoid beam dump	25.00
FM-04	Coolant pressure < 2.0 bar	2oo3	TR-04	High-pressure nitrogen flush; gas-injection valve	45.00
FM-05	D_2 leak $> 10^{-6}$ mbar L/s	2oo3	TR-05	Secondary isolation; pneumatic gate valve	35.00
FM-06	Fast scintillator rate > 100 count/s	2oo3	TR-01/02	Dual optical detune and crowbar; independent SIS logic	15.00

The sensor delay is included exactly once. Controller implementation includes saturation, anti-windup, and bumpless transfer; process-loop behavior is distinct from independent safety timing. The acceptance criteria remain

$$\text{PM} = 180^\circ + \arg L_p(i\omega_{gc}) \geq 60^\circ, \quad \text{GM} \geq 6 \text{ dB}. \quad (520)$$

At the 1.5 kHz bandwidth objective, the mechanics and 120 microsecond delay contribute -67.4358° before controller and computation phase. The 10 kHz mechanical resonance does not itself establish closed-loop bandwidth. DM-01 in Section IX specifies the corresponding frequency response and identification requirements.

H. Coolant-Loop Transfer Functions and Nonlinear Heat Balance

The DM-02 heat-input-to-outlet-temperature model is

$$G_c(s) = H_{\text{cool}}(s) = \frac{K_c e^{-sT_{m,c}}}{1 + \tau_c s}, \quad (521)$$

$$K_c = 0.045 \text{ K/W}, \quad \tau_c = 14.20 \text{ s},$$

$$T_{m,c} = 1.45 \text{ s}.$$

The 0.420 K/W, 1.250 s, 0.080 s parameter set defines a separate comparison plant. Positive heat input increases outlet temperature; a command for additional cooling has the opposite thermal sign. An actuator map G_a , thermometer response H_c , and controller C_c form $L_c = C_c G_a G_c H_c$ with a consistent negative-feedback sign. The 1.45 s transport delay is already in G_c and is not added again as sensor delay.

At 2 Hz the thermal pole and transport delay alone contribute -1133.6789° of unwrapped phase. A 2 Hz bandwidth allocation therefore requires stability assessment for this complete delay model. Controller assessment uses unwrapped phase, all gain crossings, and the complete loop Nyquist response, rather than interpreting phase modulo 360° as a sufficient stability test. For example, at 0.05 Hz the thermal pole and delay contribute about -103.5° before controller and sensor effects; this is a comparison frequency, not an established controller bandwidth.

The nonlinear first-law model remains

$$C_{\text{eff}}(T)\dot{T} = P_{\text{dep}}(T, \mathbf{u}) - \dot{m}[h(T_{\text{out}}) - h(T_{\text{in}})] - P_{\text{env}}(T) - P_{\text{export}}(T). \quad (522)$$

For a one-state autonomous thermal reduction, local stability requires $\partial(P_{\text{dep}} - P_{\text{removed}})/\partial T < 0$ at equilibrium with

$C_{\text{eff}} > 0$. The FOPDT parameters describe a local heat-to-temperature response; they are not a global cooling-capacity or compressor-power law. Coolant loss, actuator saturation, and sensor dropout remain distinct nonlinear protection cases.

a) *Fast optical-proxy interlock*: The protective architecture separates slow thermal calorimetry (time constant 1.20 s) from a fast optical-phase/RF-impedance proxy:

$$\xi_{\text{proxy}}(t) = K_{\text{opt}} \frac{d\Delta\phi_{\text{opt}}}{dt} + K_{\text{imp}} \frac{dZ_{\text{RF}}}{dt}, \quad (523)$$

with a comparator threshold and an end-to-end trip allocation of less than $42.5 \mu\text{s}$, within a $50 \mu\text{s}$ outer window; the selected Tier-1 path shall meet the tighter $12.50 \mu\text{s}$ envelope. K_{opt} and K_{imp} are calibration gains mapping the two measured derivatives to a common proxy unit and shall be identified experimentally. The thresholds $dP_{\text{th}}/dt > 100 \text{ W/s}$ and $dT/dt > 45 \text{ K/s}$ are not interchangeable without thermal capacitance, heat-loss dynamics, and sensor transfer functions. The coefficient $dn/dT = -1.50 \times 10^{-4} \text{ K}^{-1}$ and sub-10 ns intrinsic optical-detuning target do not define the complete thermal-to-optical delay. Trip timing shall extend from physical fault onset through sensing, FPGA decision, driver interruption, and verified optical extinction, not just comparator execution. The chemical-state observer remains outside this microsecond safety path. SIL-3 requires independent lifecycle and failure-probability evidence for the complete function [4], [5].

I. Surge Detection and Optical Detuning

The surge-trip threshold shall be a positive estimated thermal-power slope exceeding 100.0 W s^{-1} . The reference slope estimator and trip condition shall be

$$\hat{R}_P(s) = \frac{s}{1 + \tau_r s} \hat{P}_{\text{th}}(s), \quad \text{TR}_{\text{surge}} = \mathbf{1}_{\{\hat{R}_P > 100.0 \text{ W s}^{-1}\}}. \quad (524)$$

The positive filter time constant τ_r shall be selected from the identified fast-observable noise and permissible detection delay. For comparison, a two-sample slope with independent equal power uncertainties has

$$u(R_P) = \frac{\sqrt{2} u(P_{\text{th}})}{\Delta t}. \quad (525)$$

Using $u(P_{\text{th}}) = 5.701706 \text{ W}$ and $\Delta t = 50.0 \mu\text{s}$ gives $1.61269 \times 10^5 \text{ W s}^{-1}$. The protective channel shall therefore use a separately characterized fast observable with a validated

mapping to the surge condition; the steady-state calorimetric uncertainty shall not be used as its transient-resolution specification.

The reference incremental phase-response filter from the slope input shall be

$$\begin{aligned} G_{\text{surge}}(s) &= \frac{\Delta\phi(s)}{\widehat{R}_P(s)} = \frac{K_s s}{(s + p_s)(1 + \tau_a s)}, \\ K_s &= 0.050 \text{ rad s W}^{-1}, \\ p_s &= 1000.0 \text{ s}^{-1}, \quad \tau_a = 15.0 \mu\text{s}. \end{aligned} \quad (526)$$

The filter has zero DC gain. A latched trip with independent optical extinction shall therefore supersede its transient phase response and provide the sustained protective action.

For an applied optical phase $\phi(t)$, instantaneous frequency deviation is

$$\Delta f(t) = \frac{1}{2\pi} \dot{\phi}(t). \quad (527)$$

Maintaining 15.0 GHz detuning shall use a phase ramp of $9.42478 \times 10^{10} \text{ rad s}^{-1}$ or an equivalent frequency-shifting architecture, rather than a static phase step. Qualification shall separately evaluate detuning within 50.0 μs , pump-coupling reduction exceeding 99%, and the proposed 1.0 ms reaction-suppression objective. Optical extinction shall be measured over the installed resonance response. Nuclear-rate suppression remains a kinetic development item; independent drive inhibition and gas isolation shall not rely on an unverified optical-to-reaction response.

J. FMECA and Independent Electrical Protection

Table LXIV specifies the qualitative FMECA baseline and required mitigations. Project severity categories shall be interpreted as follows: catastrophic denotes possible containment or major exposure consequences, critical denotes loss of a necessary protective or heat-removal function, and marginal denotes loss of experimental validity without an assumed immediate containment failure. Quantitative risk and integrity assessments shall use separately supported failure rates, diagnostic coverage, and protection-layer data.

The electrical protection branch shall use a SiC switch in series with a pulse-rated resistor across the protected DC-link capacitor, coordinated with upstream interruption and inverter gate inhibition. The DC current trip shall be $I_{\text{dc}} > 35.0 \text{ A}$, separately from the 21.21 A_{rms} resonant-coil operating current. The dump-switch conduction requirement shall include comparator, isolation, driver, and switch turn-on delays after the sensed threshold crossing:

$$t_{\text{dump,on}} = t_{\text{comp}} + t_{\text{iso}} + t_{\text{gate}} + t_{\text{sw}} < 500 \text{ ns}. \quad (528)$$

Sensor bandwidth and threshold uncertainty are additional characterization requirements. Energy sizing must include

$$E_{\text{dump}} \geq \frac{1}{2} C_{\text{dc}} V_0^2 + \frac{1}{2} L_{\text{eq}} I_0^2 + \int_0^{t_{\text{isolate}}} P_{\text{source}}(t) dt. \quad (529)$$

For an isolated capacitor and ideal resistive dump,

$$V(t) = V_0 e^{-t/(R_d C_{\text{dc}})}, \quad t_{\text{safe}} = R_d C_{\text{dc}} \ln \frac{V_0}{V_{\text{safe}}}. \quad (530)$$

Dump-switch conduction time and stored-energy discharge time shall be qualified separately. Selection of R_d , switch safe operating area, resistor pulse energy, wiring inductance, insulation, and clamp voltage shall use maximum bus energy and prospective fault current, including source contribution until isolation.

For the semiconductor-protection fuse,

$$t_{\text{clear}} = t_{\text{prearc}} + t_{\text{arc}}, \quad \int_0^{t_{\text{clear}}} i_f^2(t) dt \leq (I^2 t)_{\text{withstand}}. \quad (531)$$

The $t_{\text{clear}} < 10.0 \mu\text{s}$ target at 200% rated current shall be evaluated against the selected device's time-current and let-through characteristics at the actual DC voltage, source impedance, and temperature [2]. Protective credit shall use the verified clearing interval; if the target is unattainable, the protection design shall be revised. The resistor-limited dump shall be assessed for available fuse-operating current, and independent electronic interruption shall protect the switch throughout the verified interruption interval.

K. Triple-Channel Voting and Cause-and-Effect Logic

For three independently measured versions of the same safety variable, let S_A, S_B, S_C equal one when their respective validated trip conditions are true. The majority result is

$$V_{2o3} = (S_A \wedge S_B) \vee (S_A \wedge S_C) \vee (S_B \wedge S_C). \quad (532)$$

All eight cases are explicit:

S_A	S_B	S_C	V_{2o3}
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

(533)

Temperature and pressure shall each use a separate three-channel vote on the same physical variable. The required disagreement flag and trip-latch update shall be

$$\begin{aligned} D_S &= (S_A \oplus S_B) \vee (S_B \oplus S_C), \\ L_{k+1} &= V_{2o3,k} \vee F_{\text{critical},k} \vee (L_k \wedge \neg R_{\text{authorized},k}). \end{aligned} \quad (534)$$

F_{critical} shall include loss of the required healthy-channel count, watchdog failure, emergency stop, and failed output diagnostics. Disagreement shall initiate diagnosis; invalid channels shall retain explicit fault status. Operation shall enter a validated degraded mode or trip if its required channel-health condition is lost. Authorized reset shall require restored permissives and independent operator acknowledgement of the cause. An active vote or critical fault shall override reset, as specified by the latch equation.

The temperature sensing architecture shall provide three PT100 channels with separate AFEs, references, supply isolation, and wiring paths. An ADS124S08 implementation shall account for its maximum 4 kSPS conversion rate and the additional digital-filter and physical-sensor delays [3].

TABLE LXIV
CONTROL AND PROCESS FMECA: FAILURE CONSEQUENCES, DETECTION, AND REQUIRED EVIDENCE

ID	Failure and initiating cause	Consequence; severity	Detection and mitigation	Timing allocation and verification
FM-01	SiC short circuit; dielectric or thermal failure	Coil/bus overcurrent; catastrophic	Independent analog overcurrent comparator, gate inhibit, SiC dump branch, coordinated fuse	Dump-switch conduction < 500 ns; fuse-clearing target < 10.0 μ s requires selected-device fault-current qualification
FM-02	Coolant line rupture, pump seizure, or blocked flow	Loss of heat removal; critical	Flow and pressure monitoring, leak detection, drive inhibit, independent cooling availability check	Protective command < 5.0 ms target; complete thermal response includes sensor and fluid delays
FM-03	FPGA hang, corrupted configuration, or shared power loss	Loss of regulation or trip execution; catastrophic	Independent watchdog, de-energize-to-trip outputs, segregated safety logic and power	Fault-injection tests establish worst-case end-to-end latency; the process FPGA receives no sole safety credit
FM-04	Laser drift, mode hopping, or flexure saturation	Loss of resonance and reproducibility; marginal	Optical phase/position monitoring, bounded correction, laser inhibit on persistent fault	Tracking-loop and position-limit qualification; optical response and nuclear kinetics verified separately
FM-05	Vacuum seal failure or pressurized boundary leak	Air ingress or fuel escape; critical	Mode-qualified vacuum trip, pressure transducers, cabinet gas detection, isolation	Vacuum-isolation command < 20.0 ms target; valve closure and leak-tightness measured separately
FM-06	Regulator fail-open or blocked discharge	Vessel overpressure; catastrophic	Independent pressure trip and certified mechanical relief path	Relief capacity assessed without credit for software, getter uptake, or loss of electrical power
FM-07	Common-cause sensor error or blocked common pressure tap	Majority voter receives wrong inputs; catastrophic	Separate sensing paths, proof tests, disagreement alarms, independent physical protection	Quantitative integrity assessment shall include common-cause analysis and proof-test records
FM-08	Getter saturation, hot-bed desorption, or dump valve failure	Fuel not removed; pressure/temperature excursion; critical	Bed temperature and inventory limits, valve-position feedback, secondary containment	Evacuation target < 150 ms; qualify pressure decay, receiver capacity, and adsorption heat
FM-09	Exhaust fan failure or ignition-capable cabinet equipment	Flammable mixture accumulation; catastrophic	Airflow proving, gas alarms, fuel shutoff, suitably classified equipment	Ventilation-loss trip is independent of concentration rise; verify fault-mode electrical suitability

Pressurized operation shall use service-range pressure transducers, separately from UHV gauges. Independence shall cover pressure taps, power, clocks, routing, environmental susceptibility, and final elements. Trip outputs shall retain their safe function after supervisory-computer failure.

Table LXV shall define the reference setpoints, mode qualifiers, commanded effects, timing requirements, and reset conditions. Tiers 1–3 denote response priorities, not safety integrity levels; the safety requirements specification shall allocate the required SIL to each designated function through the hazard and risk assessment. Before operational release, setpoints shall be checked against qualified vessel, detector, and cooling limits. A getter dump shall discharge only into a pressure-rated receiver with verified inventory and adsorption-heat capacity. Liquid-nitrogen flooding shall receive no protective credit without a separate assessment of condensation, vent blockage, thermal shock, and oxygen enrichment. Shutdown inventory calculations shall retain gas dissolved in metal after chamber depressurization.

The quantitative SIL-3 target ranges for a defined function are

$$10^{-4} \leq \text{PFD}_{\text{avg}} < 10^{-3} \quad (\text{low demand}),$$

$$10^{-8} \leq \text{PFH} < 10^{-7} \text{ h}^{-1} \quad (\text{high/continuous demand}).$$

(535)

The allocated SIL-3 functions shall satisfy the applicable IEC 61508/61511 lifecycle, architectural, systematic, and quantitative integrity requirements [4], [5]. For independent identical channel dangerous unavailability q , the ideal sensor-vote con-

tribution is $3q^2 - 2q^3$. The complete function assessment shall additionally include common-cause, logic, power, and final-element failures, with supported diagnostic coverage and proof-test intervals. Compliance shall require independent assessment and documented verification of the complete function; the truth table and development FPGA shall not constitute certification evidence by themselves.

a) *Switch-Branch Latency Allocation:* The thermal-surge threshold is $dP_{\text{th}}/dt > 100.0 \text{ W/s}$. A 15.0 GHz beat-frequency change over 50.0 μ s is distinct from the commanded detuning response: sensing a frequency change, requesting detuning, and observing completed detuning are separate timestamps. The following serial subcomponent allocation is not an additional full-loop trip budget: The sum $15 + 45 + 120 + 85 + 110 = 375$ ns is below 500 ns by 125 ns. For independent parallel sensing paths the timing topology differs; no sensor is assumed to wait for the other unless the logic requires it. Neither this sum nor a 45 ns thermopile assignment includes the observation interval needed to estimate a thermal ramp or a 15 GHz change over a 50 μ s window. At 100 W/s, power changes by only 0.005 W in 50 μ s, far below the steady-state calorimetric uncertainty. A separately validated fast observable, its transient noise/bandwidth, logic-vote latency, switch safe-operating area, optical shutdown response, and dump energy must therefore be specified at their actual boundaries. Electrical switch conduction is not the same as diverting an already propagating optical beam. The cause-and-effect matrix remains applicable, and the 375 ns budget does not itself establish SIL-3 compliance [4], [5].

TABLE LXV
SPECIFIED TR-01–TR-06 CAUSE-AND-EFFECT MATRIX FOR THE SIL-3-TARGET ARCHITECTURE

ID	Initiator and mode qualification	Tier	Commanded effects	Timing and reset requirement
TR-01	Core $T > 438.15$ K; independent 2o3 temperature vote	3	Inhibit optical and magnetic drives; close fuel feed; request qualified contained dump; preserve monitoring and available safe cooling	Latching trip; measured sensor-to-final-element time; physical reset after cause review
TR-02	$p_{\text{vac}} > 1.0 \times 10^{-6}$ mbar only in UHV mode	2	Inhibit drive; isolate vulnerable vacuum equipment and fuel feed; do not apply UHV criteria during deliberate gas loading	Isolation command < 20.0 ms target; reset after leak check and mode validation
TR-03	Angular tracking error $ \theta - \theta_{\text{ref}} > 0.10$ μrad ; optical-lock mode	1	Apply bounded flexure correction; inhibit laser if lock or travel permissive is lost; log frequency and angle separately	Fast-logic evaluation at 1.0 MHz; sensor/actuator rates qualified separately; re-entry only after verified lock
TR-04	Core $p > 5.00$ bar absolute; independent 2o3 pressure vote	3	Close fuel feed, inhibit drives, request qualified contained dump; mechanical PSV remains independent	Latching pressure trip; dump evacuation < 150 ms is a system test target, not voter delay
TR-05	Coolant $\dot{V} < 8.00$ L min $^{-1}$ while heat-producing operation is enabled	2	Inhibit heat-producing drives; start available qualified backup circulation; retain safe heat rejection	Command < 5.0 ms target; reset after flow and leak verification
TR-06	Cabinet hydrogenic concentration $> 0.10\%$ by volume, or loss of proven exhaust airflow	3	Isolate fuel, inhibit ignition-capable process loads, alarm, maintain suitably rated exhaust and gas monitoring	Detector/transport delay measured end to end; environmental clearance and manual reset

TABLE LXVI
DUMP-BRANCH SUBCOMPONENT LATENCY ALLOCATIONS

Stage	Delay (ns)	Cumulative (ns)
Optical photodiode	15	15
Thin-film thermopile	45	60
TMR FPGA logic	120	180
Isolated gate driver	85	265
SiC switch transition	110	375

TABLE LXVII
SEPARATE DEUTERIUM RELIEF DESIGN CASES

Parameter	A: manifold	B: core
Set pressure, gauge	5.00 bar	15.00 MPa
Relieving pressure, absolute (MPa)	0.651325	16.601325
Mass flow (kg s $^{-1}$)	0.050	0.250
Temperature (K)	323.15	550.0
Compressibility Z	1.00	1.12
Calculated area (mm 2)	93.9122	25.4352
API area designation	E	D
Effective area (mm 2)	126.4514	70.9676

VIII. PROCESS SAFETY, HAZOP/FMECA, AND REGULATORY COMPLIANCE

The safety review shall address the specified fuel inventory and all credible reaction branches without credit for unverified radiation suppression. HAZOP nodes shall comprise the regulated supply, reactor chamber, coolant circuit, getter/dump receiver, assay manifold, electrical bus, and exhaust enclosure. Each node shall be assessed for no flow, excess flow, reverse flow, high pressure, low pressure, high temperature, loss of containment, and loss of utilities. Risk assessment shall combine the consequences in Table LXIV with supported initiating frequencies and independently qualified protection layers. The following reference calculations define engineering cases; operational release shall require component-specific qualification and the applicable safety approvals.

A. Pressure-Relief Design Basis and Orifice Qualification

The engineering report [47], pp. 21–22, defines two separate dry-deuterium relief cases. ASME BPVC Sections VIII/XIII, API 520/526, and the applicable ISO 4126 provisions govern the qualified design [6], [31]–[33]. The following is an explicit gas-flow sizing check, not a valve certification. For $M = 0.004028$ kg mol $^{-1}$, $k = 1.40$, $K_d = 0.975$, and unity

additional backpressure/combination factors, the sonic model is

$$R_s = R_u/M,$$

$$g_* = K_d p_0 \sqrt{\frac{k}{Z R_s T_0}} \left(\frac{2}{k+1} \right)^{(k+1)/(2(k-1))}, \quad (536)$$

$$A_{\text{req}} = \dot{m}/g_*.$$

Absolute stagnation pressure must be used. With $p_{\text{atm}} = 101325$ Pa and a 10% increase of gauge set pressure, the corrected relieving pressure is $p_0 = 1.10p_{\text{set,g}} + p_{\text{atm}}$; increasing an already absolute set pressure by 10% produces a different value.

$$A_{\text{req,A}} = 93.91215 \text{ mm}^2,$$

$$A_E = 0.196(645.16) = 126.45136 \text{ mm}^2, \quad (537)$$

$$A_E/A_{\text{req,A}} \simeq 1.34648.$$

$$A_{\text{req,B}} = 25.43520 \text{ mm}^2,$$

$$A_D = 0.110(645.16) = 70.96760 \text{ mm}^2, \quad (538)$$

$$A_D/A_{\text{req,B}} \simeq 2.79013.$$

Thus the report's E selection for A and D selection for B are consistent with this model, with rounded effective areas 126.5 and 71.0 mm² [49]. D is insufficient for A. Effective API area is not necessarily the physical throat area; certified coefficients and capacities must follow one consistent manufacturer convention.

The report mixes gauge and absolute pressures in its psia conversions: the stated formula gives 0.651325, not 0.6614 MPa(a), for A, and 16.601325, not 16.611 MPa(a), for B. These corrected values are used throughout the table. High-pressure real-gas properties, backpressure/choking, inlet/outlet losses, accumulation, fire and two-phase scenarios, discharge forces, and actual regulator capacity remain part of the installation-specific verification. B does not uprate the existing vessel, viewports, feedthroughs, or interlocks; a high-pressure configuration requires a separately rated pressure boundary. Relief discharge must be captured/routed safely and must not be credited as a harmless enclosure leak.

B. Hazardous Locations and the Ventilation Balance

The engineering report's new classification proposal assigns normal-release internal containment, purge-exhaust manifolds, and relief headers to Class I, Division 1 under the division system; a zone-based study assesses Zone 1 for the corresponding intermittent normal-release conditions. It proposes Division 2 or Zone 2 within a 1.5 m radius of fittings, flanges, and stem packings where release is abnormal. These are project-study boundaries, not universal distances established by NFPA 2 [8], [9]. Continuous or long-duration ignitable atmospheres require separate Zone 0 assessment rather than automatic Zone 1 assignment.

Choose and document the adopted classification system. Division and zone designations are not interchangeable certificates, and Ex d/Ex i/Ex nA labels do not alone establish US Class I, Group B approval [9]. The study shall include release frequency, ventilation availability, gas group, temperature classification, containment geometry, and approval of the complete installation. Normal relief/purge streams shall enter engineered discharge routes, not occupied or recirculating spaces.

The hot core at 276.85°C is enclosed by a double-walled vacuum jacket with outer microporous insulation. The design requires maximum accessible/ignition-relevant external skin temperature 115.0°C, below the T4 limit 135.0°C. This thermal requirement replaces the earlier T3 proposal; it must be demonstrated at hot spots and applicable upset conditions, not inferred from the electronics marking. Vacuum-loss, failed circulation, penetrations, and thermal bridges belong in the surface-temperature assessment. Gas detectors cover enclosure high points, joints, dosing equipment, and exhaust; airflow proving is independent of fan command.

a) *Fast isolation instead of full-release dilution:* The audit [51] rejects continuous dilution of a 0.050 kg s⁻¹ deuterium release as the primary safeguard for a 3.50 m³ enclosure. The previous standard-volume calculation remains an infeasibility

comparison, not a fan procurement duty:

$$\begin{aligned} Q_{\text{exh},s} &= 100161.73 \text{ m}^3 \text{ h}^{-1}, \\ &\simeq 58952.97 \text{ SCFM}, \\ \text{ACH}_s &= 28617.64 \text{ h}^{-1}. \end{aligned} \quad (539)$$

That calculation used 273.15 K, 101325 Pa, and a 1.0% concentration target. The following transient calculation instead uses 298.15 K and 101325 Pa explicitly; these volume conventions are not interchangeable.

The active protection allocation is optical detection plus logic within 5.0 ms, followed by complete isolation within 10.0 ms. For a constant upstream leak bounded by $\dot{m}_{\text{max}} = 0.050 \text{ kg s}^{-1}$ until closure,

$$\begin{aligned} t_{\text{isolate}} &\leq 5.0 + 10.0 = 15.0 \text{ ms}, \\ m_{\text{release}} &\leq \dot{m}_{\text{max}} t_{\text{isolate}} = 0.750 \text{ g}, \\ n_{D_2} &= \frac{0.000750}{0.0040282} = 0.18618738 \text{ mol}, \\ V_{D_2} &= n_{D_2} R_u T / p = 4.55514939 \text{ L}, \\ \bar{C}_{\text{bound}} &= V_{D_2} / (3500 \text{ L}) = 0.13014713\%. \end{aligned} \quad (540)$$

The audit's 0.18618 mol and 0.13014% are truncated displays; calculated values take precedence. The gas-volume ratio is a conservative enclosure-average inventory bound for initially clean air, not the peak local concentration beside a leaking joint. At constant pressure with ideal well-mixed displacement the fraction is $1 - e^{-V_{D_2}/V_{\text{enc}}}$, slightly smaller; a sealed enclosure requires pressure and mole-inventory accounting.

The report's reference LFL = 4.9% gives $C_{\text{safe}} = 0.25(4.9\%) = 1.225\%$. Its ratio to the calculated average is 9.4124, not a full factor of ten. Flammability limits depend on test method and conditions: the gas supplier tabulates a US reference of 5% and a different European-method value [53]. The report's 1.225% is therefore a design comparison, not a universal code threshold. The calculated average is also below the earlier, more conservative 1.0% project target, which is retained until the installation hazard assessment approves a change.

Optical sensor latency shall be demonstrated from leak initiation, including gas transport to the sensing path, detection, logic, and actuation. Valves shall fail closed and their full closure time shall be verified at differential pressure. The inventory bound must include downstream trapped gas, post-closure leakage, sensor blind spots, and failure of isolation; 0.750 g is not a limit on such additional inventories. Independent gas monitoring, ignition-source control, safe vent routing, and background extraction with proven airflow remain required [8], [9]. The fast-isolation strategy replaces the infeasible full-release dilution duty, not these independent protections. A well-mixed average alone cannot certify absence of a locally flammable plume or an explosion-safe enclosure.

C. Tritium Production, Activity, and Branch Consistency

For a conservative branch scenario, set the total D–D rate to $\Gamma_{DD} = 7.623 \times 10^{14} \text{ s}^{-1}$ and the tritium-producing branch fraction to $b_T = 0.500$. Then

$$R_T = b_T \Gamma_{DD} = 3.8115 \times 10^{14} \text{ atoms s}^{-1}. \quad (541)$$

With tritium molar mass $M_T = 3.01604928 \text{ g mol}^{-1}$, the no-decay daily production is

$$m_{T,\text{day}} = \frac{R_T(86400)M_T}{N_A} = 1.64929065 \times 10^{-4} \text{ g} \approx 0.165 \text{ mg day}^{-1}. \quad (542)$$

Using the evaluated half-life $t_{1/2} = 4500 \text{ d}$ [10],

$$\lambda_T = \frac{\ln 2}{4500(86400)} = 1.78278596 \times 10^{-9} \text{ s}^{-1}. \quad (543)$$

For zero initial inventory and no removal,

$$\begin{aligned} \dot{N}_T &= R_T - \lambda_T N_T, \\ N_T(t) &= \frac{R_T}{\lambda_T}(1 - e^{-\lambda_T t}), \\ A_T(t) &= \lambda_T N_T(t) = R_T(1 - e^{-\lambda_T t}), \\ A_T(86400 \text{ s}) &= 58.7050 \text{ GBq}. \end{aligned} \quad (544)$$

Inventory reports shall express retained activity in Bq and identify any activity-accumulation rate separately. The first-day no-decay reference is 58.7096 GBq, compared with 58.7050 GBq including decay. Any removal term shall be included in the retained-inventory balance separately from the production source.

Each hazard calculation shall use one internally consistent set of branch fractions and event energies. For b_4, b_T, b_n ,

$$\begin{aligned} b_4 + b_T + b_n &= 1, \\ P_{\text{nuc}} &= \Gamma_{DD}(b_4 Q_4 + b_T Q_T + b_n Q_n), \\ Q_4 &= 23.84 \text{ MeV}, \quad Q_T \approx 4.033 \text{ MeV}, \\ Q_n &\approx 3.269 \text{ MeV}. \end{aligned} \quad (545)$$

For $b_T = b_n = 0.5$ and $b_4 = 0$, the reference event rate gives approximately 445.911 W released nuclear power and a neutron source of $3.8115 \times 10^{14} \text{ s}^{-1}$. This source exceeds the Section VI-I reference ceiling by a factor of 3.8115×10^8 and shall receive a separate shielding and exposure assessment without credit for the lower-source shield allocation. Branch-dependent cross sections shall use validated nuclear models within their stated domain [11]. The tritium case and helium-only design case shall remain separate hazard and operating scenarios. Licensing, contamination control, effluent treatment, waste characterization, and worker protection shall undergo independent radiological review.

D. Permeation Barrier and Containment Qualification

The TiN/ Al_2O_3 barrier shall be qualified on compatible containment surfaces while retaining an independently qualified pressure boundary. For the diffusion-limited square-root-pressure transport model, permeability and resistance shall be defined as $\Pi_j = D_j S_j$ and $\mathcal{R}_j = \ell_j / \Pi_j$. With layers in series,

$$J = \frac{\sqrt{p_{\text{in}}} - \sqrt{p_{\text{out}}}}{\mathcal{R}_{\text{sub}} + \mathcal{R}_{\text{TiN}} + \mathcal{R}_{\text{Al}_2\text{O}_3} + \mathcal{R}_{\text{int}}}. \quad (546)$$

All permeabilities use a common molar or atomic-flux convention; interface resistances are expressed in the same units. The ideal permeation reduction factor is

$$\text{PRF}_{\text{ideal}} = \frac{J_{\text{bare}}}{J_{\text{coated}}} = 1 + \frac{\mathcal{R}_{\text{TiN}} + \mathcal{R}_{\text{Al}_2\text{O}_3} + \mathcal{R}_{\text{int}}}{\mathcal{R}_{\text{sub}}}. \quad (547)$$

Consequently, $\text{PRF} > 1000$ requires an added resistance exceeding $999\mathcal{R}_{\text{sub}}$ under identical boundary conditions. For defect area fraction f_d treated as bare-substrate bypass,

$$\text{PRF}_{\text{eff}} = \left[f_d + \frac{1 - f_d}{\text{PRF}_{\text{ideal}}} \right]^{-1}. \quad (548)$$

Even an otherwise impermeable coating cannot exceed $1/f_d$ in this model; attaining PRF above 1000 therefore requires $f_d < 10^{-3}$ as a necessary, not sufficient, condition. Temperature-dependent permeability, trapping, pinholes, welds, seals, stress cracking, and irradiation can dominate system leakage.

Barrier qualification shall specify layer thicknesses and measured isotope-specific permeabilities. Coated and uncoated specimens shall be tested at matched temperature, pressure, substrate condition, and exposed area, repeated after representative thermal cycling with defect inspection. Hydrogen-enhanced plasticity and fracture shall receive the separate mechanical qualification specified in Section IV [12], [13]. Downstream gas and water species shall be captured and assayed, with bypass leakage included in the inventory. Acceptance of the component $\text{PRF} > 1000$ requirement shall use these measurements; full-system release reduction shall be assessed separately.

a) *v2.0 coating-defect acceptance calculation:* For the specified 150 nm TiN/250 nm Al_2O_3 coating, v2.0 assigns $\text{PRF}_{\text{ideal}} = 2850$ and $f_d \leq 8.5 \times 10^{-4}$. The parallel-defect equation yields

$$\begin{aligned} \text{PRF}_{\text{eff}}|_{f_d=0.00085} &= \left[0.00085 + \frac{0.99915}{2850} \right]^{-1} \\ &= 832.93148 \approx 833. \end{aligned} \quad (549)$$

This is below the retained component requirement $\text{PRF} > 1000$. For the same ideal coating that requirement instead needs

$$f_d < \frac{10^{-3} - 2850^{-1}}{1 - 2850^{-1}} = 6.49351 \times 10^{-4}. \quad (550)$$

The 833 prediction is retained as a design calculation, not as evidence that the stricter requirement has been met. The v2.0 1.5 m Class I, Division 2/Zone 2 extent is a proposed classified boundary, not automatic equivalence of the Division and Zone systems or a substitute for release-source classification. The 0.750 g isolation-only release calculation remains conditional on the response and trapped-inventory assumptions above; the stricter 1.00% project concentration comparison is retained. Equipment suitability and approved installation remain governed by the selected classification [9].

b) *Governing v3.0 barrier model and leakage scope:* The v3.0 manufacturing defect cap is $f_d \leq 6.49 \times 10^{-4}$ for the 150 nm TiN/250 nm Al_2O_3 barrier. The report also changes the constitutive model to an intrinsic factor 25000 and a pore-path multiplier $\alpha_{\text{geom}} = 1.42$:

$$\begin{aligned} \text{PRF}_{\text{eff}} &= \left[\frac{1 - f_d}{25000} + 1.42 f_d \right]^{-1}, \\ \text{PRF}_{\text{eff}}(0.000649) &= 1039.983151 \geq 1000, \\ f_{d,\text{max}} &= \frac{0.001 - 1/25000}{1.42 - 1/25000} \\ &= 6.76075382 \times 10^{-4}. \end{aligned} \quad (551)$$

TABLE LXVIII
SOURCE-REPORTED FIFTEEN-STATE BLOCK SPECTRUM

Block	States	Listed eigenvalues (s ⁻¹)	Listed time
Optical phonons	1–2	-4.20×10^{11}	2.381 ps
SPP photons	3–4	-1.85×10^9	0.540 ns
Dressed electrons	5–7	-2.14×10^7	46.73 ns
Deuterium	8–10	-14.2, -8.35	70.42 ms
Frame	11–13	-1.25, -0.45	0.800 s
Cooling loop	14–15	-0.125, -0.082	8.000 s

Thus the adopted cap passes this model but does not yield the printed 1091.0. That factor would require $f_d \simeq 6.17334491 \times 10^{-4}$ with the other inputs fixed. Under the previous intrinsic factor 2850 and unit defect-path multiplier, the new 0.000649 cap gives 1000.350649 instead. These models and their intrinsic performance must not be mixed.

The source's additional transport assignments are trap density $N_t = 1.20 \times 10^{26}$ sites/m³, partial molar volume $V_H^* = 1.70 \times 10^{-6}$ m³/mol, and signed heat of transport $Q^* = 12.50$ kJ/mol. For mobile and trapped *molar* concentrations C_m, C_t , convert N_t to $N_{t,\text{mol}} = N_t/N_A$ and write

$$\begin{aligned} \partial_t C_m + \nabla \cdot \mathbf{J}_D &= -\partial_t C_t, \\ \partial_t C_t &= k_t C_m (N_{t,\text{mol}} - C_t) - p_t C_t, \\ \mathbf{J}_D &= -D_{\text{eff}} \nabla C_m + \frac{D_{\text{eff}} C_m V_H^*}{R_u T} \nabla \sigma_h \quad (552) \\ &\quad + \frac{D_{\text{eff}} C_m Q^*}{R_u T^2} \nabla T. \end{aligned}$$

Here the capture coefficient must use the molar convention, and positive hydrostatic stress is tension. The reported $J_D = 3.15 \times 10^{-8}$ mol/(m² s) is not determined by N_t, V_H^*, Q^* alone: diffusion/trapping coefficients, concentration and stress fields, boundary pressures and wall geometry are needed. It is recorded as the reported substrate flux, not silently divided by PRF or multiplied by an unspecified enclosure area. Earlier material-specific active-layer data remain distinct from containment-substrate properties.

IX. PRECISION SIMULATOR PARAMETERIZATION ENGINE

A. v4.2 Fifteen-State Stiff Discretization and Observer

The v4.1 PDF Section 7 defines $\dot{x} = Ax + Bu, y = Cx + Du$, $x \in \mathbb{R}^{15}$ and the block spectrum in Table LXVIII; v4.2 leaves these inputs unchanged [72], [73]. The state ordering differs from the inherited fifteen-state model below. Block eigenvalue samples, with incomplete multiplicities, do not specify all 225 entries of A or an input/output realization. The stiffness ratio is $4.20 \times 10^{11}/0.082 = 5.12195 \times 10^{12}$. The time column lists representative values, not every modal time; the slowest cooling mode is $1/0.082 = 12.1951$ s. For a frozen linear model and a zero-order-held input at $h = 1.0$ ms, the exact discretization is

$$\begin{aligned} \exp \left[h \begin{pmatrix} A & B \\ 0 & 0 \end{pmatrix} \right] &= \begin{pmatrix} A_d & B_d \\ 0 & I \end{pmatrix}, \\ A_d &= e^{Ah}, \quad B_d = \int_0^h e^{A\tau} B \, d\tau. \end{aligned} \quad (553)$$

This block-exponential form remains valid for singular A . The source's $B_d = A^{-1}(A_d - I)B$ is valid only for invertible A and need not be the numerically preferred evaluation.

Radau IIA is an implicit Runge–Kutta approximation, not an exact matrix-exponential algorithm [78]. For its three-stage, fifth-order form, define

$$\begin{aligned} \mathcal{A}_R &= \begin{bmatrix} \frac{88-7\sqrt{6}}{360} & \frac{296-169\sqrt{6}}{1800} & \frac{-2+3\sqrt{6}}{225} \\ \frac{296+169\sqrt{6}}{1800} & \frac{88+7\sqrt{6}}{360} & \frac{-2-3\sqrt{6}}{225} \\ \frac{16-\sqrt{6}}{36} & \frac{16+\sqrt{6}}{36} & \frac{1}{9} \end{bmatrix}, \\ b_R^T &= \left(\frac{16-\sqrt{6}}{36}, \frac{16+\sqrt{6}}{36}, \frac{1}{9} \right), \\ c_R^T &= \left(\frac{4-\sqrt{6}}{10}, \frac{4+\sqrt{6}}{10}, 1 \right). \end{aligned} \quad (554)$$

With stacked stages $K \in \mathbb{R}^{45}$ and $\mathbf{1}_3 = (1, 1, 1)^T$,

$$\begin{aligned} [I_{45} - h(\mathcal{A}_R \otimes A)] K &= \mathbf{1}_3 \otimes (Ax_k + Bu_k), \\ x_{k+1} &= x_k + h(b_R^T \otimes I_{15})K. \end{aligned} \quad (555)$$

For nonlinear dynamics, stage values use the nonlinear right-hand side and an implicit solve with its Jacobian. Solver tolerance, scaling and convergence tests are mandatory; a nominal 1 ms output step does not resolve picosecond transients. The exact frozen-linear ZOH and Radau time integration must be validated separately against the same fully specified plant. No complete new A, B, C, D dataset or reduction transformation is supplied.

a) *Selected PDF observer coordinates and gain:* The supplied PDF defines

$$\begin{aligned} x_{o,\text{PDF}} &= (\delta T_s, \delta T_c, \delta C_D, \delta P_{\text{th}})^T, \\ y_o &= (\delta \phi_{\text{opt}}, \delta Z_{\text{RF}})^T, \\ \dot{\hat{x}}_o &= A_o \hat{x}_o + B_o u + L_{\text{calc}}(y_o - C_o \hat{x}_o), \\ L_{\text{calc},\text{PDF}} &= \begin{bmatrix} 48.25 & -12.10 \\ 14.80 & 3.45 \\ 0.0082 & -0.0154 \\ 1245.00 & 312.50 \end{bmatrix}. \end{aligned} \quad (556)$$

Here C_D is a dimensionless loading ratio and P_{th} is watts, not the mol/m³ and pressure coordinates of the older text-v4.1 model in Eq. (600). Each gain entry has units $[x_i]/([y_j] \text{ s})$. The PDF does not provide its A_o, B_o, C_o or coordinate scaling, so this selected gain cannot be verified by substituting it into the unrelated older matrices. Equation (605) remains a verified historical gain *only for its own plant*, not an alternative selected PDF gain. The source targets poles $-30, -32, -35, -40 \text{ s}^{-1}$ and reports

$$\begin{aligned} e_o(t) &= e^{(A_o - LC_o)t} e_o(0), \\ t_{95,\text{mode}} &= \ln(20)/30 = 99.857742 \text{ ms}, \\ \|e_o(t)\| &\leq 1.42e^{-30t} \|e_o(0)\|. \end{aligned} \quad (557)$$

The selected $t_{95} = 99.86$ ms is a pure slow-mode decay time. Even if the asserted norm bound were proven, it gives uniform 95% decay only after $\ln(1.42/0.05)/30 = 111.55$ ms, not under 100 ms. A dimensionally scaled norm, observer plant, gain precision, noise model and eigenvector conditioning are

needed to verify that prefactor; none follows from pole targets alone.

This section specifies the complete continuous-time, linearized fifteen-state model supplied in the Supplementary Master Specification (v1.1).¹⁸ The explicit matrices define the numerical model interface for the project simulator. The solid-state resolution report adds a separately audited four-state, two-input/two-output candidate and explicit DM-01/DM-02 identification procedures, not a replacement measured fifteen-state parameter set. They are supplementary model coefficients, not independently identified plant parameters or evidence of implemented repository functionality. All unlisted matrix entries are zero. State ordering, native units, reference operating point, and the distinction between the matrix-derived and supplementary reduced transfer functions form part of this specification.

B. Linearized State, Input, and Output Vectors

Let each Δ denote a small deviation from a fixed reference operating point. The state-space representation is

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t),\end{aligned}\quad (558)$$

with $\mathbf{x} \in \mathbb{R}^{15}$, $\mathbf{u} \in \mathbb{R}^4$, and $\mathbf{y} \in \mathbb{R}^4$. The prescribed state order is

$$\mathbf{x} = \begin{bmatrix} \Delta T_{\text{core}} \\ \Delta T_{\text{TEG},h} \\ \Delta T_{\text{TEG},c} \\ \Delta T_{\text{coolant}} \\ \Delta p_{D_2} \\ \Delta \ell_{D/Pd} \\ \Delta \Phi_{\text{SPP1}} \\ \Delta \Phi_{\text{SPP2}} \\ \Delta U_{\text{eff}} \\ \Delta n_{\text{LO}} \\ \Delta P_{\text{th}} \\ \Delta V_{\text{bus}} \\ \Delta I_{\text{bus}} \\ \Delta P_{\text{chiller}} \\ \Delta P_{\text{export}} \end{bmatrix}. \quad (559)$$

Here $\ell_{D/Pd}$ is the dimensionless loading ratio, not a geometric length. The native units are K for x_1 – x_4 , bar for x_5 , dimensionless for x_6 , MW cm^{−2} for x_7 , x_8 , keV for x_9 , cm^{−3} for x_{10} , W for x_{11} , V for x_{12} , A for x_{13} , and W for x_{14} , x_{15} . The optical-flux states are envelopes associated with the nominal 785.0 and 802.5 nm pumps, not samples of their optical carriers.

The input and measured-output orders are

$$\mathbf{u} = \begin{bmatrix} \Delta P_{\text{laser1}} \\ \Delta P_{\text{laser2}} \\ \Delta \dot{m}_{\text{coolant}} \\ \Delta R_{\text{load}} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} \Delta T_{\text{core}} \\ \Delta P_{\text{export}} \\ \Delta V_{\text{bus}} \\ \Delta P_{\text{th}} \end{bmatrix}. \quad (560)$$

The input units are W, W, kg s^{−1}, and Ω , respectively. The output units are K, W, V, and W. The laser-power convention

¹⁸*SHBT Cold Fusion Reactor: Supplementary Master Specification (v1.1)*, supplied project document SHBT Reactor Master Specification.pdf, pp. 18–20: state definitions, nonzero matrix entries, and reported transfer function.

shall be the one used to identify the supplied input gains; converting between optical and electrical pump power requires the corresponding drive efficiency rather than silently reusing those gains.

The historical v1.1 linearization manifest (not the new 350 eV / 15 kW scenario) uses $T_{h,0} = 550.0$ K, $T_{c,0} = 288.15$ K, a retired screening offset, $P_{\text{th},0} = 9303.30$ W, $V_{\text{bus},0} = 48$ V, $P_{\text{chiller},0} = 468.96$ W, and $P_{\text{export},0} = 1081.04$ W. Absolute observables in that historical model are its reference values plus state perturbations; these offsets are not the revised operating ledger. A new equilibrium and Jacobian must be identified for the resolution hardware before transferring this local model to its new operating point. The remaining equilibrium quantities, including individual pump powers, coolant mass flow, loading ratio, and phonon density, shall be recorded in the operating-point manifest; the aggregate power ledger does not uniquely determine them.

C. System Matrix and Explicit Nonzero Entries

The system matrix is $\mathbf{A} \in \mathbb{R}^{15 \times 15}$. A dimensionally consistent grouping of the fixed state order uses the thermal indices 1–4, gas/loading indices 5–6, optical indices 7–8, screening/phonon/source indices 9–11, and electrical indices 12–15. Their dimensions are 4 + 2 + 2 + 3 + 4 = 15. In this grouping,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_T & \mathbf{0} & \mathbf{0} & \mathbf{A}_{TS} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_G & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_O & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_{SO} & \mathbf{A}_S & \mathbf{0} \\ \mathbf{A}_{ET} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{ES} & \mathbf{A}_E \end{bmatrix}. \quad (561)$$

The diagonal block sizes are 4 × 4, 2 × 2, 2 × 2, 3 × 3, and 4 × 4. With inclusive row and column index ranges, the off-diagonal blocks are

$$\begin{aligned}\mathbf{A}_{TS} &= \mathbf{A}[1:4, 9:11] \in \mathbb{R}^{4 \times 3}, \\ \mathbf{A}_{SO} &= \mathbf{A}[9:11, 7:8] \in \mathbb{R}^{3 \times 2}, \\ \mathbf{A}_{ET} &= \mathbf{A}[12:15, 1:4] \in \mathbb{R}^{4 \times 4}, \\ \mathbf{A}_{ES} &= \mathbf{A}[12:15, 9:11] \in \mathbb{R}^{4 \times 3}.\end{aligned}\quad (562)$$

Each block uses exactly the indexed coefficients in Table LXIX; all unlisted entries are zero. Thus the first block subscript identifies the receiving state group and the second identifies the driving state group in $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$. This regrouping does not rely on the supplement's schematic block dimensions.

Negative diagonal entries represent local decay in this convention; coupling signs are retained independently rather than assigned a common damping sign. In particular, $A_{12,13} = -0.145$ and $A_{13,12} = 0.820$ have opposite signs, while $A_{15,14} = -1.00$ makes increased chiller demand reduce the export-state derivative. These signs apply directly in (561), without an additional minus sign on any block.

Table LXIX specifies all 36 nonzero entries. In particular, $A_{12,12} = 0$; no voltage self-damping coefficient is to be added implicitly. The direct optical-to-phonon term is $A_{10,8}$, not $A_{10,7}$, so the two optical inputs are not completely symmetric in this supplied model. The gas/loading block has no coupling to the other state groups in the stated coefficient schedule. These

TABLE LXIX
COMPLETE NONZERO ENTRY SCHEDULE FOR THE v1.1 SYSTEM MATRIX **A**

State group	Row	Nonzero coefficients in native state units
Thermal	1	$A_{1,1} = -0.852, \quad A_{1,2} = 0.612, \quad A_{1,11} = 0.0001075$
	2	$A_{2,1} = 0.245, \quad A_{2,2} = -0.588, \quad A_{2,3} = 0.343$
	3	$A_{3,2} = 0.182, \quad A_{3,3} = -0.421, \quad A_{3,4} = 0.239$
	4	$A_{4,3} = 0.115, \quad A_{4,4} = -0.310$
Gas and loading	5	$A_{5,5} = -0.042, \quad A_{5,6} = -0.018$
	6	$A_{6,5} = 0.0031, \quad A_{6,6} = -0.0085$
Optical envelopes	7	$A_{7,7} = -1.25 \times 10^3$
	8	$A_{8,8} = -1.25 \times 10^3$
Screening, phonons, source	9	$A_{9,7} = 0.045, \quad A_{9,8} = 0.045, \quad A_{9,9} = -12.4$
	10	$A_{10,8} = 1.28 \times 10^8, \quad A_{10,9} = 1.28 \times 10^8, \quad A_{10,10} = -4.20 \times 10^{11}$
	11	$A_{11,9} = 1852.4, \quad A_{11,10} = 1.42 \times 10^{-6}, \quad A_{11,11} = -0.0150$
Electrical and export	12	$A_{12,11} = 0.00368, \quad A_{12,13} = -0.145$
	13	$A_{13,12} = 0.820, \quad A_{13,13} = -2.45$
	14	$A_{14,4} = 18.4, \quad A_{14,14} = -0.850$
	15	$A_{15,12} = 37.5, \quad A_{15,13} = 48.0, \quad A_{15,14} = -1.00, \quad A_{15,15} = -10.0$

features are retained explicitly rather than repaired by adding unspecified physical couplings.

As examples of the resulting differential equations,

$$\begin{aligned} \dot{x}_1 &= -0.852x_1 + 0.612x_2 + 0.0001075x_{11}, \\ \dot{x}_{10} &= 1.28 \times 10^8 x_8 + 1.28 \times 10^8 x_9 - 4.20 \times 10^{11} x_{10}, \\ \dot{x}_{15} &= 37.5x_{12} + 48.0x_{13} - x_{14} - 10.0x_{15}. \end{aligned} \quad (563)$$

These coefficients multiply the specified native numerical coordinates. In general $[A_{ij}] = [x_i]/([x_j] \text{ s})$; only same-unit couplings have units s^{-1} . Thus the numerical factors in the export-state differential equation shall not be interpreted as an independently derived instantaneous VI power identity. That identity and the auxiliary-power ledger remain separate physical consistency checks.

D. Input, Output, and Feedthrough Matrices

The matrix $\mathbf{B} \in \mathbb{R}^{15 \times 4}$ has exactly four nonzero entries:

$$\begin{aligned} B_{7,1} &= 145.0, & B_{8,2} &= 145.0, \\ B_{4,3} &= -42.8, & B_{13,4} &= -12.4. \end{aligned} \quad (564)$$

For example, $\dot{x}_7 = -1250x_7 + 145u_1$ and $\dot{x}_4 = 0.115x_3 - 0.310x_4 - 42.8u_3$. The units of B_{ij} are $[x_i]/([u_j] \text{ s})$; the mass-flow and resistance inputs shall not be treated as watts.

The matrix $\mathbf{C} \in \mathbb{R}^{4 \times 15}$ selects the four measured coordinates:

$$C_{1,1} = 1, \quad C_{2,15} = 1, \quad C_{3,12} = 1, \quad C_{4,11} = 1, \quad (565)$$

All other output coefficients vanish. For coordinate vectors $\mathbf{e}_j \in \mathbb{R}^{15}$, the same selection matrix and the zero direct-feedthrough matrix are

$$\mathbf{C} = \begin{bmatrix} \mathbf{e}_1^\top \\ \mathbf{e}_{15}^\top \\ \mathbf{e}_{12}^\top \\ \mathbf{e}_{11}^\top \end{bmatrix}, \quad \mathbf{D} = \mathbf{0}_{4 \times 4}. \quad (566)$$

The zero feedthrough matrix is part of the supplementary realization. In particular, $y_2 = x_{15}$ is a dynamic export observable; the supplied model has no immediate input-to-output subtraction of laser electrical demand. Comparison with a measured net-power channel must account for its signal definition and filtering rather than silently identifying this state with an unfiltered complete power ledger.

E. Core Thermal-to-Electrical Transfer Function

For zero initial perturbations, the transfer matrix is

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I}_{15} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}. \quad (567)$$

With the other three inputs held at zero, the requested laser-1 to export channel is

$$\begin{aligned} G_{21}(s) &= \frac{\Delta P_{\text{export}}(s)}{\Delta P_{\text{laser1}}(s)} \\ &= \mathbf{e}_{15}^\top (s\mathbf{I}_{15} - \mathbf{A})^{-1} \mathbf{b}_1, \end{aligned} \quad (568)$$

where $\mathbf{b}_1 = 145\mathbf{e}_7$ is the first column of $\mathbf{B}$. The indices designate output 2 and input 1, not state coordinates 2 and 1.

a) *SPA / balanced residualization*: The new report [59] supplies a four-state, two-input/two-output model, replacing the previous SISO candidate. The full manuscript plant remains fifteen-state, four-input/four-output. A state reduction does not itself change the number of physical ports: a selected two-channel parent requires explicit port-selection and scaling maps $B_m = BJ_u$, $C_m = J_y C$, $D_m = J_y D J_u$. The latest report [56] supplies the port maps below. Their numerical consequences are checked against the retained plant rather than assuming that the existing reduced coefficients were generated from it.

$$J_u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad J_y = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0.021156 & 0 & 0 \end{bmatrix}. \quad (569)$$

$J_u \in \mathbb{R}^{4 \times 2}$ selects the two laser inputs; $J_y \in \mathbb{R}^{2 \times 4}$ scales export power to sensor voltage by 0.021156 V/W. For the four-port parent G , define the selected two-port parent exactly once:

$$\begin{aligned} A_m &= A, & B_m &= BJ_u, \\ C_m &= J_y C, & D_m &= J_y D J_u, \\ G_{\text{port}}(s) &= G_m(s) = J_y G(s) J_u. \end{aligned} \quad (570)$$

Multiplying $J_y G_m J_u$ is dimensionally invalid because G_m is already 2×2 . These maps change ports and units, not the state dimension.

Its slow/fast Schur formulas are balanced SPA, not direct deletion of states. The established low-frequency method [62] is stated here without importing a claimed broadband error bound for an unidentified parent.

For a stable minimal selected realization, solve the controllability and observability Lyapunov equations and obtain a nonsingular balancing transformation T_b :

$$\begin{aligned} A_m P + P A_m^T + B_m B_m^T &= 0, \\ A_m^T Q + Q A_m + C_m^T C_m &= 0, \\ T_b^{-1} P T_b^{-T} &= T_b^T Q T_b = \Sigma, \\ \Sigma &= \text{diag}(\sigma_1, \dots, \sigma_m), \\ x_b &= T_b^{-1} x_m. \end{aligned} \quad (571)$$

For $P = L_c L_c^T$ and $L_c^T Q L_c = U \Sigma^2 U^T$, the coordinate convention in (571) requires

$$T_b = L_c U \Sigma^{-1/2}, \quad T_b^{-1} = \Sigma^{1/2} U^T L_c^{-1}. \quad (572)$$

This gives equal transformed Gramians Σ . The source's inverse-map expression with $\Sigma^{-1/2}$ instead of $\Sigma^{1/2}$ would give Σ^{-1} for the controllability Gramian. Zero Hankel singular values require a minimal realization before inversion.

The partitioned balanced realization is

$$A_b = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B_b = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad C_b = [C_1 \quad C_2]. \quad (573)$$

For a minimal fifteen-state source and retained order four, the blocks have dimensions 4×4 , 4×11 , 11×4 , and 11×11 ; B_1, B_2 have four and eleven rows and two input columns; C_1, C_2 have two output rows and four and eleven state columns. If the selected plant is nonminimal, eliminate the uncontrollable/unobservable part first and replace eleven by $m - 4$. Setting the discarded derivatives to zero gives $x_2 = -A_{22}^{-1}(A_{21}x_1 + B_2u)$, so

$$\begin{aligned} A_r &= A_{11} - A_{12} A_{22}^{-1} A_{21}, \\ B_r &= B_1 - A_{12} A_{22}^{-1} B_2, \\ C_r &= C_1 - C_2 A_{22}^{-1} A_{21}, \\ D_r &= D - C_2 A_{22}^{-1} B_2. \end{aligned} \quad (574)$$

This is residualization, not direct state deletion [28]. For nonsingular A_{22} and A_r , substitution in the full steady-state equations proves the exact identity

$$-C_r A_r^{-1} B_r + D_r = -C_b A_b^{-1} B_b + D. \quad (575)$$

The full block-inverse proof is explicit with $S = A_r$ and $Z = A_{22}^{-1}$:

$$\begin{aligned} M_{11} &= S^{-1}, & M_{12} &= -S^{-1} A_{12} Z, \\ M_{21} &= -Z A_{21} S^{-1}, \\ M_{22} &= Z + Z A_{21} S^{-1} A_{12} Z, \\ A_b^{-1} &= \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}. \end{aligned} \quad (576)$$

Consequently

$$\begin{aligned} G_m(0) &= D_m - C_1 M_{11} B_1 - C_1 M_{12} B_2 \\ &\quad - C_2 M_{21} B_1 - C_2 M_{22} B_2 \\ &= D_m - C_2 Z B_2 \\ &\quad - (C_1 - C_2 Z A_{21}) S^{-1} (B_1 - A_{12} Z B_2) \\ &= D_r - C_r A_r^{-1} B_r = G_{\text{SPA}}(0). \end{aligned} \quad (577)$$

This identity is exact for residualization of the same selected parent; it does not make an independently supplied matrix set that residualization.

A nonzero D_r is therefore legitimate in SPA even when $D = 0$; it represents the algebraically eliminated states, not a new instantaneous physical energy path. The identity requires blocks from the same parent realization.

The report's new numerical realization is transcribed in (578) and (579); all displayed digits are retained. Thus $A_r \in \mathbb{R}^{4 \times 4}$, $B_r \in \mathbb{R}^{4 \times 2}$, $C_r \in \mathbb{R}^{2 \times 4}$, and $D_r \in \mathbb{R}^{2 \times 2}$:

$$\begin{aligned} \dot{x}_r &= A_r x_r + B_r u_r, & y_r &= C_r x_r + D_r u_r, \\ G_r(s) &= C_r (sI_4 - A_r)^{-1} B_r + D_r \in \mathbb{C}^{2 \times 2}. \end{aligned} \quad (580)$$

This replaces the scalar reduced transfer function, while preserving the original exact SISO G_{21} definition for comparison. No transimpedance factor is inserted to force a match.

An explicit factorization of the matrix-derived channel makes that distinction reproducible. Define the laser-to-source response

$$\begin{aligned} H_Q(s) &= \frac{6.525}{(s + 1250)(s + 12.4)(s + 0.0150)} \\ &\quad \times \left(1852.4 + \frac{181.76}{s + 4.20 \times 10^{11}} \right). \end{aligned} \quad (581)$$

This follows from $x_7 \rightarrow x_9 \rightarrow x_{11}$ and the additional $x_9 \rightarrow x_{10} \rightarrow x_{11}$ path, with $x_8 = 0$ for this input channel. Specifically, Table LXIX and (568) give

$$\begin{aligned} B_{7,1} A_{9,7} &= 145(0.045) = 6.525, \\ A_{11,10} A_{10,9} &= (1.42 \times 10^{-6})(1.28 \times 10^8) \\ &= 181.76. \end{aligned} \quad (582)$$

For the four-state thermal block, let

$$\begin{aligned} d_T(s) &= \det(sI_4 - A_T) \\ &= s^4 + 2.171s^3 + 1.444275s^2 \\ &\quad + 0.332424304s + 0.01967752878, \end{aligned} \quad (583)$$

$$H_T(s) = \frac{x_4(s)}{x_{11}(s)} = \frac{5.51243875 \times 10^{-7}}{d_T(s)}.$$

The numerator of H_T is the product $A_{1,11} A_{2,1} A_{3,2} A_{4,3}$; its denominator retains all thermal couplings rather than replacing

$$A_r = \begin{bmatrix} -1.4252 \times 10^2 & 2.1140 \times 10^1 & -8.9501 & 1.2200 \times 10^{-1} \\ -1.9540 \times 10^1 & -3.8561 \times 10^2 & 1.4502 \times 10^2 & -2.1150 \times 10^1 \\ 0 & -4.2250 \times 10^1 & -1.1243 \times 10^3 & 8.5401 \times 10^2 \\ 0 & 0 & -3.1121 \times 10^2 & -8.9542 \times 10^3 \end{bmatrix}. \quad (578)$$

$$B_r = \begin{bmatrix} 1.4582 \times 10^1 & -2.1150 \times 10^{-1} \\ 8.9541 \times 10^1 & 1.0240 \\ 2.1152 \times 10^2 & -4.5120 \times 10^1 \\ 0 & 1.1243 \times 10^3 \end{bmatrix}, \quad (579)$$

$$C_r = \begin{bmatrix} 1 & 0 & -2.1150 \times 10^{-2} & 8.9540 \times 10^{-1} \\ 0 & 1 & 5.1240 \times 10^{-1} & -1.1243 \times 10^1 \end{bmatrix},$$

$$D_r = \begin{bmatrix} 1.2450 \times 10^{-2} & 0 \\ 0 & 3.4120 \times 10^{-1} \end{bmatrix}.$$

thermal eigenvalues with diagonal decay coefficients. The electrical and chiller paths from the source state to the export-state forcing are

$$F_E(s) = \frac{0.00368(37.5s + 131.235)}{s^2 + 2.45s + 0.1189}, \quad (584)$$

$$F_H(s) = \frac{18.4}{s + 0.850} H_T(s).$$

Here $0.1189 = -A_{12,13}A_{13,12}$ and $131.235 = 37.5(2.45) + 48.0(0.820)$. The chiller contribution is subtracted because $A_{15,14} = -1.00$, and the export-state decay supplies the remaining pole $s + 10.0$. Consequently the exact channel for the tabulated realization is

$$G_{21}(s) = \frac{H_Q(s)}{s + 10.0} [F_E(s) - F_H(s)]. \quad (585)$$

The DC calculation can be reproduced without a transfer-function fit:

$$\begin{aligned} H_Q(0) &\approx 51.9867096774, \\ F_E(0) &\approx 4.06177291842, \\ F_H(0) &\approx 0.000606418048, \\ G_{21}(0) &= \frac{H_Q(0)}{10} [F_E(0) - F_H(0)] \\ &= -\mathbf{e}_{15}^T \mathbf{A}^{-1} \mathbf{b}_1 \\ &\approx 21.1126683807 \text{ W/W}. \end{aligned} \quad (586)$$

b) Numerical port-gain reconciliation: Applying the supplied sensor conversion once gives

$$\begin{aligned} [G_{\text{port}}(0)]_{21} &= 0.021156(21.1126683807) \\ &= 0.4466596123 \text{ V/W}, \\ 0.021156(21.11) &= 0.44660316 \text{ V/W} \simeq 0.4466 \text{ V/W}. \end{aligned} \quad (587)$$

The map explains the approximate scale, not exact equality: the supplied reduced matrices give 0.4465819695, a 0.017383% relative difference from the selected full-order value. Matching that entry exactly would require $J_{y,22} = 0.0211523224544 \text{ V/W}$, not the supplied 0.021156; that fitted change is not made here. All four transfer entries and dynamics must be compared under the same maps. At 0.01 Hz the selected full-order channel has magnitude 0.0641940 V/W and phase -129.445862° , whereas the ROM retains magnitude

TABLE LXX
NEW FOUR-STATE REALIZATION: CALCULATED STABILITY DIAGNOSTICS

Quantity	Result
Pole 1 (s^{-1})	-144.218643
Pole 2 (s^{-1})	-391.984399
Pole 3 (s^{-1})	-1150.311835
Pole 4 (s^{-1})	-8920.115123
Controllability rank	4
Observability rank	4

approximately 0.446582 and phase near zero. Static scaling therefore does not establish dynamic equivalence.

c) DC precision and parent-model traceability: Evaluation of the supplied decimal coefficients gives the transfer matrix

$$G_r(0) = D_r - C_r A_r^{-1} B_r = \begin{bmatrix} 0.138004873 & 0.107298302 \\ 0.446581969 & -1.006393014 \end{bmatrix}. \quad (588)$$

These entries carry the source's native output/input units. In particular, if the second-output/first-input entry were identified without any port scaling with the manuscript's laser-to-export channel, then

$$[G_r(0)]_{21} = 0.446581969 \neq K_{\text{exact}} = 21.112668381. \quad (589)$$

A similarity transformation of state coordinates alone cannot change DC gain. The Schur identity (575) holds for the same parent ports, not for an unspecified two-port replacement of a four-port system. The new model is stable and minimal in its own port coordinates. It is not literally balanced in the supplied coordinates: Lyapunov solutions give $P_{11} = 1.11257948$, $Q_{11} = 0.00349463$, and $P_{12} = 3.46143252$, rather than equal diagonal Gramians. It could be a coordinate-transformed reduced model, but its printed numbers alone do not establish the claimed balancing construction. The previous report's HSV entries are not reused for this different realization.

As a concrete native-channel check at $f = 0.01 \text{ Hz}$,

$$\begin{aligned} G_{21}(i2\pi f) &\text{ has } |G_{21}| = 3.03431731 \text{ W/W}, \\ \arg G_{21} &= -129.44586248^\circ, \\ [G_r(i2\pi f)]_{21} &\approx 0.44658196 - i0.00005430. \end{aligned} \quad (590)$$

Thus neither DC preservation nor phase alignment with the retained SISO channel follows under direct port identification. The claimed 0.001 Hz–10 kHz equivalence is a source assertion, not an evaluated error bound. The full fifteen-state coefficients and factorization remain the reference for their declared ports; the supplied two-channel model remains distinct because the new port maps do not reproduce its full transfer matrix or dynamic response.

d) *v2.0 literal reduced-matrix transcription and audit:*

The v2.0 PDF, pp. 40–41, prints a different arrangement from the preceding reduced model. To retain every source coefficient without silently installing an unstable controller model, denote this literal source realization by a superscript (2):

$$A_r^{(2)} = \begin{bmatrix} -142.52 & -195.40 & 21.140 & -8.9501 \\ -385.61 & 145.02 & 0 & -42.25 \\ -1124.3 & 0 & 0.122 & -21.15 \\ 0 & 854.01 & -311.21 & -8954.2 \end{bmatrix}. \quad (591)$$

$$B_r^{(2)} = \begin{bmatrix} 14.582 & -0.2115 \\ 89.541 & 1.0240 \\ 211.52 & -45.120 \\ 0 & 1124.3 \end{bmatrix}, \quad D_r^{(2)} = \begin{bmatrix} 0.01245 & 0 \\ 0 & 0 \end{bmatrix}. \quad (592)$$

$$C_r^{(2)} = \begin{bmatrix} 1 & -0.02115 & 8.9540 \times 10^{-11} & -11.243 \\ 0 & 0.51240 & 0 & 0.34120 \end{bmatrix}. \quad (593)$$

The exponent -11 in $C_{r,13}^{(2)}$ is present in the embedded equation image, not an inferred correction. Direct evaluation gives

$$\begin{aligned} \text{eig}(A_r^{(2)}) &\simeq \{289.995601, -237.453276, \\ &\quad -53.148565, -8950.971760\} \text{ s}^{-1}, \\ K_r^{(2)} &= D_r^{(2)} - C_r^{(2)}(A_r^{(2)})^{-1}B_r^{(2)} \\ &\simeq \begin{bmatrix} 0.13523602 & -1.73502885 \\ -0.05732239 & 0.01158882 \end{bmatrix}. \end{aligned} \quad (594)$$

The positive pole disproves stability for this literal transcription, and the DC matrix differs from the v2.0 printed result, which instead repeats the preceding report's result. The retained fifteen-state model, its 36-entry schedule, and its exact transfer factorization remain authoritative for their declared ports. Neither reduced candidate is certified as its balanced SPA by a copied pole or gain table.

The source's displayed Schur inverse also reverses the upper-right sign. With $S = A_{11} - A_{12}A_{22}^{-1}A_{21}$, the complete identity is

$$\begin{aligned} A_b^{-1} &= \begin{bmatrix} S^{-1} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}, \\ M_{12} &= -S^{-1}A_{12}A_{22}^{-1}, \\ M_{21} &= -A_{22}^{-1}A_{21}S^{-1}, \\ M_{22} &= A_{22}^{-1} + A_{22}^{-1}A_{21}S^{-1}A_{12}A_{22}^{-1}. \end{aligned} \quad (595)$$

Substituting these blocks into $D_m - C_bA_b^{-1}B_b$ and collecting $C_1 - C_2A_{22}^{-1}A_{21}$ and $B_1 - A_{12}A_{22}^{-1}B_2$ proves (575). For a correctly derived selected-port SPA,

$$G_r(0) = J_y G_{\text{full}}(0) J_u. \quad (596)$$

This identity preserves the DC transfer matrix, not the full frequency response. The 0.021156 map applied to the full

numerical gain gives 0.4466596123 V/W; the v2.0 printed 0.446603 uses the rounded 21.11 input instead. Neither equals the literal v2.0 value -0.05732239 . A balancing transformation, selected parent, and Schur blocks cannot be reconstructed by declaring these incompatible matrices equal.

e) *v3.0 SPA proof and numerical-model status:* v3.0 Section 5.1 supplies the Schur-complement formulas, not a new set of numerical A_r, B_r, C_r, D_r , a balancing transformation, Hankel singular values, or reduced eigenvalues. The full fifteen-state coefficient schedule and the preceding literal v2.0 audit are therefore retained without presenting either old reduced candidate as a newly verified model. For the explicitly selected parent $B_m = BJ_u$, $C_m = J_y C$, $D_m = J_y D J_u$, set

$$\begin{aligned} S &= A_{11} - A_{12}A_{22}^{-1}A_{21}, \\ \hat{B} &= B_1 - A_{12}A_{22}^{-1}B_2, \\ \hat{C} &= C_1 - C_2A_{22}^{-1}A_{21}, \\ \hat{D} &= D_m - C_2A_{22}^{-1}B_2. \end{aligned} \quad (597)$$

Using the correctly signed inverse (595),

$$\begin{aligned} G_m(0) &= D_m - C_bA_b^{-1}B_b \\ &= D_m - C_2A_{22}^{-1}B_2 - \hat{C}S^{-1}\hat{B} \\ &= \hat{D} - \hat{C}S^{-1}\hat{B} = G_r(0) \\ &= J_y G_{15}(0) J_u. \end{aligned} \quad (598)$$

This proves DC preservation for an actual residualization with nonsingular A_{22} and S , not agreement of independently printed coefficients. Stability requires a Hurwitz parent and an appropriate balanced reduction with its applicable truncation conditions; minimality and balancing alone do not imply stability. A Schur complement of an arbitrary Hurwitz matrix is not a universal stability certificate. Numerical verification must establish $\max_i \text{Re } \lambda_i(A_r) < 0$ for the generated model. The prior v2.0 literal positive pole $+289.995601 \text{ s}^{-1}$ is not removed by restating this theorem. Nor does exact DC equality establish broadband dynamic equivalence or certify a reduced plant as a protective controller.

f) *v4.0 SPA retention and v4.1 observer coordinates:*

The Schur-complement reduction and DC-gain proof in (597)–(598) remain applicable to a specified balanced parent model. v4.0 reports reduced poles $-12.45, -8.12, -3.56 \pm i1.42$ but supplies no numerical realization producing them. Its proposed physical-state list ($T_{\text{active}}, C_D, \Delta\phi_{\text{opt}}, Z_{\text{RF}}$) and three-output voltage mapping are superseded for the observer by the different v4.1 four-state, two-output coordinates. Balanced coordinates are not automatically physical temperatures or concentrations.

The historical v4.0 interface scalings, retained without conflating them with the earlier port-selection matrices, are

$$\begin{aligned} J_{u,40} &= \begin{bmatrix} 24.00 & 0 \\ 0 & 10.00 \end{bmatrix}, \\ J_{y,40} &= \begin{bmatrix} 0.0100 & 0 & 0 & 0 \\ 0 & 0 & 3.18 & 0 \\ 0 & 0 & 0 & 0.0500 \end{bmatrix}. \end{aligned} \quad (599)$$

The input diagonal entries map control volts to laser watts and piezo volts, respectively. The three output rows map temperature, optical phase, and RF impedance to sensor volts

in the v4.0 state ordering. This 3×4 sensor mapping is not the 2×4 v4.1 measurement matrix used to compute L ; no identity state transformation between the two models is implied.

In the v4.1 source coordinates, define perturbations

$$\begin{aligned} x_o &= (\delta T_s, \delta T_c, \delta C_D, \delta P)^T, \\ y_o &= (\delta \phi_{\text{opt}}, \delta Z_{\text{RF}})^T, \\ \dot{x}_o &= A_o x_o + w_o, \quad y_o = C_o x_o, \end{aligned} \quad (600)$$

where $w_o(t)$ is the known four-component input forcing. This form does not invent an actuator matrix absent from v4.1; for a calibrated actuator model $w_o = B_o u$. State units are K, K, mol/m³, Pa, and output units are rad and Ω as declared in the source. Thus $A_{o,ij}$ has units $[x_i]/([x_j] \text{ s})$ and $C_{o,ij}$ has units $[y_i]/[x_j]$. The literal matrices are

$$\begin{aligned} A_o &= \begin{bmatrix} -10 & 6 & 50 & 0 \\ 4 & -12 & 0 & 0 \\ 0.1 & 0 & -8 & 5 \\ 0 & 0 & 2 & -20 \end{bmatrix}, \\ C_o &= \begin{bmatrix} 1.5 & 0 & 0 & 0.8 \\ 0.5 & 1 & 0 & 4.5 \end{bmatrix}. \end{aligned} \quad (601)$$

Their eigenvalues, in s⁻¹, are -4.755518 , -8.234846 , -16.190106 , and -20.819530 , not the v4.0 pole list. The observability matrix obeys

$$\mathcal{O}_o = \begin{bmatrix} C_o \\ C_o A_o \\ C_o A_o^2 \\ C_o A_o^3 \end{bmatrix}, \quad \text{rank } \mathcal{O}_o = 4. \quad (602)$$

Consequently C_D , despite the zero third column of C_o , is observable through dynamic coupling. Use the observer and error equations

$$\begin{aligned} \dot{\hat{x}}_o &= A_o \hat{x}_o + w_o + L(y_o - C_o \hat{x}_o), \\ e_o &= x_o - \hat{x}_o, \quad \dot{e}_o = (A_o - LC_o)e_o. \end{aligned} \quad (603)$$

The required source gain is retained exactly as printed:

$$L_{\text{report}} = \begin{bmatrix} 38.7455 & -8.6092 \\ -30.7065 & 57.6575 \\ 19.5953 & -16.9264 \\ 5.1822 & -6.3593 \end{bmatrix}. \quad (604)$$

Direct substitution yields poles, in s⁻¹, of -39.99942048 , -29.99968334 , -32.00055785 , and -35.00039834 . The rounded gain is Hurwitz stable and close to, but not exactly at, the requested poles. A reproducible higher-precision pole-placement realization is

$$L_{\text{calc}} = \begin{bmatrix} 38.7455242217 & -8.6092044469 \\ -30.7065420661 & 57.6575295770 \\ 19.5952740099 & -16.9264018822 \\ 5.1821956788 & -6.3593267200 \end{bmatrix}, \quad (605)$$

computed by dual pole placement on (A_o^T, C_o^T) . The target poles are -30 , -32 , -35 , and -40 ; the returned gain is then transposed [71]. Recalculation, not truncation of printed coefficients, controls the implementation precision.

For a pure slow eigenmode, the 95% and 98% decay times are

$$\begin{aligned} t_{95} &= \ln(20)/30 = 99.8577 \text{ ms}, \\ t_{98} &= \ln(50)/30 = 130.4008 \text{ ms}, \\ e_o(t) &= \exp[(A_o - LC_o)t]e_o(0). \end{aligned} \quad (606)$$

A 33.3 ms time constant is not a 50 ms settling time. In the literal numerical coordinates, $\|\exp[(A_o - L_{\text{calc}}C_o)0.1]\|_2 \approx 0.942813$, so the modal 99.86 ms result is not a uniform 5% error bound for arbitrary initial errors. A physical norm must specify state scaling because the four states have different units. Sensor noise, forcing error, and model mismatch also affect concentration estimates. Thus the supplied matrices establish observability and asymptotic error stability, not a guaranteed sub-100 ms chemical-state reconstruction or an experimentally validated fifteen-state SPA realization.

F. Numerical Implementation and Verification

a) Numerical kernels and interface contract: The execution architecture separates screening and adaptive kinetics, conservative multi-trap transport, crossed-periodic RCWA, bounded fatigue sampling, balanced reduction, sector-LMI synthesis, and covariance-based telemetry. The Rust/Python boundary shall expose these kernels through typed interfaces with validated dimensions, units, finite values, and ownership. PyO3 entry points release the Python GIL only while operating on independently owned or safely borrowed native memory; C-ABI entry points additionally specify buffer lengths, layout, error codes, and ownership transfer. A 512-bit internal result returned as a binary64 scalar loses its high-precision digits; precision-sensitive exchange therefore uses a lossless representation rather than an implicit scalar cast.

The reference boundary kernels use 512-bit arithmetic with initialized working buffers. The native `compute_floquet_inversion` method of `ShbtNumericalKernel` evaluates the scalar surrogate in Eq. (17), with `rug::Float` arithmetic and PyO3 exchange of the multiprecision result as a decimal string. The 512-bit significand supports approximately 154 decimal digits of arithmetic precision; binary64 inputs still carry only their original information. No-allocation execution is a hot-loop requirement, not a consequence of selecting a multiprecision type: memory use must be bounded and tested after initialization. Python orchestration dispatches the kernels, assembles calibrated uncertainty models, and performs vectorized postprocessing without silently changing the numerical convention. The execution order is constitutive screening and optical response, rate integration, conservative transport, thermomechanical sampling, reduction and stability certification, then telemetry propagation. Each kernel must reject nonphysical states and nonconverged solutions rather than return an apparent successful result. The benchmark precision agreement does not establish allocation behavior or physical closure of other kernels.

b) Acceptance and convergence: Coordinate round-trip error shall satisfy Eq. (3); normalized adaptive-quadrature error shall be below 10^{-15} ; total mobile-plus-trapped deuterium shall satisfy Eq. (355); and the discarded squared-spectrum ratio shall remain below 10^{-4} . Grid, harmonic, time-step, and

arithmetic-precision refinement are separate tests. The coupling, screening, and flux tolerances in Tables I, III, and VIII apply componentwise in their stated units. The full spectrum and its arithmetic metrics in Table XXXVII satisfy the stated spectral acceptance criteria but do not substitute for a nonlinear stability certificate. Runtime wrappers propagate numerical failures explicitly, and acceptance requires both the physical-range checks and the numerical residual checks.

The matrix entries are tied to the native units in (559)–(560). For positive diagonal state, input, and output scaling matrices $\mathbf{S}_x, \mathbf{S}_u, \mathbf{S}_y$, use

$$\begin{aligned}\tilde{\mathbf{x}} &= \mathbf{S}_x^{-1} \mathbf{x}, & \tilde{\mathbf{u}} &= \mathbf{S}_u^{-1} \mathbf{u}, & \tilde{\mathbf{y}} &= \mathbf{S}_y^{-1} \mathbf{y}, \\ \tilde{\mathbf{A}} &= \mathbf{S}_x^{-1} \mathbf{A} \mathbf{S}_x, & \tilde{\mathbf{B}} &= \mathbf{S}_x^{-1} \mathbf{B} \mathbf{S}_u, \\ \tilde{\mathbf{C}} &= \mathbf{S}_y^{-1} \mathbf{C} \mathbf{S}_x, & \tilde{\mathbf{D}} &= \mathbf{S}_y^{-1} \mathbf{D} \mathbf{S}_u.\end{aligned}\quad (607)$$

For example, changing phonon density from cm^{-3} to m^{-3} changes both relevant rows and columns; a state-label substitution alone is incorrect.

The phonon pole $-4.20 \times 10^{11} \text{ s}^{-1}$ has lifetime 2.38 ps, making the model stiff relative to the thermal and source dynamics. Forward Euler would require a step smaller than approximately 4.76 ps from this pole alone. A microsecond controller update does not resolve that pole. Continuous-time evaluation shall use a suitable stiff solver or analytic elimination with a checked error bound; sampled simulation may use the zero-order-hold map

$$\begin{aligned}\mathbf{A}_d &= e^{\mathbf{A}\Delta t}, \\ \mathbf{B}_d &= \int_0^{\Delta t} e^{\mathbf{A}\xi} \mathbf{B} d\xi, \\ \mathbf{x}_{k+1} &= \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d \mathbf{u}_k.\end{aligned}\quad (608)$$

Sampling and numerical conditioning shall be assessed separately from physical model adequacy.

A reproducible run shall record the matrix manifest, unit/scaling convention, operating point, solver or discretization, time step, initial perturbations, and input histories. Verification shall check the 15×15 , 15×4 , 4×15 , and 4×4 dimensions; the 36, 4, and 4 nonzero counts of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$; zero feedthrough; and agreement of resolvent and factored transfer responses. Perturbation states may be negative, but reconstructed absolute temperatures, pressures, and densities must remain physical. Local linear-model agreement does not establish nonlinear stability, nuclear reaction validity, energy-conversion performance, or independently accepted safety functions.

G. DM-01 and DM-02 Plant Identification Procedures

a) DM-01: Flexure resonance and optical sensor lag:

The source's identified-form model [59] is

$$G_{\text{piezo}}(s) = \frac{X(s)}{V_{\text{in}}(s)} = \frac{0.845 e^{-120 \times 10^{-6} s}}{\tau_p^2 s^2 + 0.30 \tau_p s + 1} [\mu\text{m/V}], \quad (609)$$

$$\tau_p = (2\pi \cdot 10^4)^{-1} \text{ s}.$$

The LDV identification record uses synchronized drive, velocity, and optical displacement channels; velocity is related to displacement by $sX(s)$. A joint complex-response fit separates the 10 kHz resonance, damping, static gain, and 120 microsecond

delay. A 10 Hz–5 kHz sweep alone does not traverse this new resonance: the qualified acquisition band must extend above 10 kHz, with appropriate actuator limits, sampling, and anti-alias filtering. This is an analytical validation protocol; the supplied report provides no measured sweep data or estimator covariance. The parameters are report references, not measurements performed by this document merge.

b) DM-02: Coolant first-order-plus-dead-time response:

The same heat-input convention as Section IX's DM-02 record and Section VIII is used for a calibrated nonreacting thermal step:

$$\begin{aligned}\Delta T_{\text{out}}(t) &= K_c \Delta P \left[1 - e^{-(t-T_{m,c})/\tau_c} \right] \Theta(t - T_{m,c}), \\ K_c &= 0.045 \text{ K/W}, \quad \tau_c = 14.20 \text{ s}, \\ T_{m,c} &= 1.45 \text{ s}, \quad \Delta T_{\infty}|_{\Delta P=500 \text{ W}} = 22.5 \text{ K}.\end{aligned}\quad (610)$$

The source gain therefore predicts a 22.5 K rise for the historical 500 W step, not a negligible perturbation. The validation record specifies inlet conditions, actual heater power, flow, sensor filtering, and temperature limits. A smaller step is required if 22.5 K leaves the local model or equipment envelope. For ideal steady single-pass enthalpy transport, $K_c = 1/(\dot{m}c_p)$ would imply $\dot{m}c_p = 22.222 \text{ W/K}$; it is not automatically compatible with the historical 12.5 L/min water-flow case. Loop losses and the defined measured output determine whether this interpretation applies. The 14.20 s pole and 1.45 s delay remain the stated analytical parameters, separately from any future commissioning dataset.

X. EXPERIMENTAL PROTOCOLS, RFI SUBTRACTION, AND EXTENDED BOM

This section defines the experimental-control protocol, interference-correction requirements, and v1.1 Master Procurement BOM. Run records shall link calibrated observations to the applicable hardware configuration and repository revision. The supplementary quantities, model references, and itemized prices are retained as the procurement planning basis for the listed scope. The reported grand total is distinguished from the arithmetic sum of the line items; supplier qualification, cost reconciliation, scope completion, and release approvals remain separately controlled.

A. Forty-Eight-Hour Protium Blank and Double-Blind Controls

The initial experimental phase shall characterize the heat balance under nonnuclear control conditions. A 48.0 h protium run shall use coded 99.999%-purity H_2 and the same hardware, pressure reference, electrical input boundaries, optical powers, magnetic excitation, coolant conditions, and analysis pipeline as the corresponding D_2 run. The control design reference is

$$P_{\text{excess}, H_2} = 0.00 \pm 5.72 \text{ W}. \quad (611)$$

The band is the combined standard-uncertainty reference derived from Section V. Each run shall report its measured blank value and the uncertainty recalculated for its operating point. Nonzero blanks shall be retained and investigated. Averaging shall reduce only those uncertainty components

supported by the measured temporal statistics; fixed systematic contributions shall remain in the budget.

The required sequence shall be as follows.

- 1) Establish pressure-boundary, electrical, gas-handling, and radiological authorization independently of the experimental hypothesis. Calibrate input-power channels and both calorimetric paths, characterize drift, and archive the frozen analysis procedure, sensor ranges, exclusion criteria, and stopping rules before assigning gas identities.
- 2) Obtain pre-run surface and gas-inventory baselines. An independent custodian assigns coded H_2 and D_2 cylinders and a randomized crossover schedule; operators and primary analysts do not receive the code key. Record isotopic purity and residual helium separately from nominal chemical purity. Safety personnel retain controlled emergency access to gas identities and hazard information.
- 3) Condition each run under a recorded loading history and demonstrate thermal stability using measured criteria. Operate the protium blank continuously for 48.0 h after stabilization, logging all inputs, outputs, pressures, flows, calibration pulses, alarms, environmental variables, and raw detector data. Retain startup and shutdown data for stored-energy accounting even when excluded from a predefined steady-state estimator.
- 4) Apply the same duration and acquisition rules to deuterium and repeated crossover runs. Validate the purge/washout procedure by gas analysis; neither a valve command nor a fixed purge time proves removal of retained hydrogen isotopes, helium, or dissolved gas. Replace or recondition specimens where carryover cannot be bounded. Randomization and replicate specimens address aging and irreversible loading changes.
- 5) Include prespecified optical controls near 8.328064 THz and 7.10 THz where the actual laser hardware supports them. Measure incident and absorbed optical power at both settings; equal nominal diode power does not guarantee equal substrate heating. Include electrically heated and reactor-disabled controls. Record the full frequency trajectory and do not equate a static phase flip with a frequency change.
- 6) Freeze calibrated heat, gas, radiation, and surface-analysis outputs before unblinding. Report every scheduled run, every safety stop, missing data, background correction, and deviation. Decode the assignments only after an independent custodian verifies the frozen analysis record. Conduct independent replication with separately calibrated apparatus before a performance claim is accepted.

For matched runs, the primary contrast and uncertainty are

$$\begin{aligned} \Delta \hat{P} &= \hat{P}_{D_2} - \hat{P}_{H_2}, \\ u^2(\Delta \hat{P}) &= u^2(\hat{P}_{D_2}) + u^2(\hat{P}_{H_2}) - 2 \text{cov}(\hat{P}_{D_2}, \hat{P}_{H_2}). \end{aligned} \quad (612)$$

For independent run uncertainties of 5.723587 W, the difference shall carry $u = 8.09437$ W. Any reduction from shared calibration shall use the demonstrated covariance. The run record shall document possible unblinding clues from isotope-dependent transport or loading. Measurement

quality, isotope contrast, and nuclear attribution shall be assessed separately, consistent with controlled low-energy-fusion investigations [22].

B. XPS/SIMS Forensics and Artifact Controls

Pre- and post-run X-ray photoelectron spectroscopy shall measure Pd/Ir composition, oxidation, chemical state, and surface contamination at registered locations with reference specimens. Secondary-ion mass spectrometry shall provide isotope-sensitive depth profiles using matched matrix standards, instrumental mass-fractionation corrections, crater-depth calibration, and blank subtraction. Helium inventory shall use a separately validated isotope assay, with atmospheric, calibration, implantation, and retention contributions bounded before production is inferred.

Sacrificial matched pre-run coupons shall be used where SIMS sputtering would damage the active film. Records shall include analysis-beam dose, sputter energy, crater geometry, vacuum history, and exposure to helium leak testing. Chemical and isotopic profiles across the nominal 1.0 nm alloy and underlying interfaces shall carry the measured depth resolution. Adhesion loss and thermal cycling shall be correlated with mechanical characterization using measured material inputs [23], [24]. Surface, fatigue, and isotope observations shall retain their separate evidentiary roles.

C. RFI, Grounding, and Independent Electrical Metrology

RFI qualification shall include the 68.5 kHz magnetic drive and nominal 500 kHz switching stage. Sensor AFEs shall use differential inputs, local shielding, galvanically isolated power where required, and optical or appropriately rated isolated digital links. Protective-earth bonds shall remain intact. The installation shall document chassis bonding and signal references, separate low-level thermometry from high-current loops, and measure common-mode pickup and capacitive coupling under representative switching conditions.

Independent electrical boundary measurements shall use the specified Yokogawa WT5000 platform or a qualified equivalent. The reference platform provides up to seven input elements, 10 MS/s sampling, 18-bit conversion, 1 MHz power bandwidth, and 0.03% basic power accuracy under the applicable conditions [25]. The measurement uncertainty shall use the installed element, range, frequency, power factor, current sensor, and wiring, separately from the basic-accuracy reference. Thermal power shall remain the calorimetric measurand.

At each electrical boundary, use synchronous voltage and current and account for all independent energy sources:

$$\hat{P}_{\text{el}} = \frac{1}{T_a} \int_{t_0}^{t_0+T_a} v(t)i(t) dt, \quad E_{\text{el}} = \int \hat{P}_{\text{el}}(t) dt. \quad (613)$$

Measure exported power separately, include startup storage changes, and count calibration heaters, isolated supplies, pumps, laser drivers, and hidden return paths. Verify alias rejection and sensor bandwidth for switching harmonics; a 1 MHz bandwidth does not fully characterize arbitrarily sharp 500 kHz switching edges.

RFI characterization uses a nonreacting electrical dummy load, calibrated heater plateaus, drive-frequency sweeps, source-on/source-off comparisons at equal deposited heat, and controlled common-mode injection. For a frozen independently calibrated interference estimator $\hat{P}_{\text{RFI}}$,

$$\begin{aligned}\hat{P}_{\text{corr}} &= \hat{P}_{\text{raw}} - \hat{P}_{\text{RFI}}, \\ u^2(P_{\text{corr}}) &= u^2(P_{\text{raw}}) + u^2(P_{\text{RFI}}) - 2 \text{COV}(P_{\text{raw}}, P_{\text{RFI}}).\end{aligned}\quad (614)$$

Interference corrections shall be identified from independent calibration and control data and frozen before candidate-signal evaluation. Records shall preserve uncorrected waveforms, correction coefficients, validation residuals, and frequency-dependent uncertainty. Cosmic-ray and gamma-background corrections shall follow the same independent-identification rule and the commissioning requirements of Section VI-I.

D. Master Procurement BOM and Cost Reconciliation

Table LXXI integrates the engineering report's instrument-inclusive procurement schedule [47], pp. 23–25. It contains eleven purchase lines plus a total row, not twelve separately priced components. Quantities follow the printed unit-price/extension ratios; catalog numbers absent from the source are not invented.

$$\begin{aligned}C_{\text{materials}} &= 131000, & C_{\text{fabrication}} &= 14500, \\ C_{\text{optics}} &= 69100, & C_{\text{sensors}} &= 139500, \\ C_{\text{electrical}} &= 22400, & C_{\text{tooling}} &= 15583, \\ C_{\text{baseline}} &= \$392083.00.\end{aligned}\quad (615)$$

The cost sum closes exactly. OPT-02 now references the corrected exponent $m = 1148$ field target ($N = 574$ in the $2N$ convention); retaining its allocation does not establish a supplier quotation or optical qualification for the changed prescription. This new allocation replaces the previous different eleven-line schedule; the former \$20,000 reserve is not added again. The instrument-inclusive table does not separately price cascade modules, chiller, vacuum pumps, magnetic supply, circulation, shielding, ventilation, gas calibration standards, logistics, or installation. These cannot be presumed included in the new frame/vessel or instrument lines without assembly quotations. The total is therefore the priced baseline, not a certified installed-system cost. Gas package size, target composition/purity, diamond thickness, laser redundancy, analyzer input modules, detector digitizers, and vessel pressure ratings shall be attached to purchasing specifications. In particular, the source's material formula is an atomic composition; its separate "5 wt% Ir" wording is not equivalent.

E. Integrated Safety and Source-Provenance Release Checks

Section VIII adopts the engineering report's two relief scenarios and T4 skin-temperature requirement, with corrected gauge-to-absolute pressure and leak-volume calculations. Equations (537), (538), and (539) are the traceable screening calculations. Equipment approval, gas properties, release boundaries, discharge routing, and capacity certification remain installation

records, not consequences of an orifice letter or temperature label.

All existing bibliography entries are retained. The new engineering report [47], ASTM method definitions [48], and manufacturer effective-area schedule [49] identify the provenance of this update. The report's secondary supplier pages are not specimen-specific HE records, valve certificates, or proof of nuclear branching. Numerical recalculation and properly scoped primary standards take precedence over inconsistent claims of closure.

F. Specification Release and Development Completion Criteria

The specification shall maintain a traceable chain from optical input and lattice-response models to reaction-rate hypotheses, energy accounting, and independent measurements. Acceptance shall require calibrated electrical and thermal closure, reproducible isotope-dependent behavior, quantified helium and tritium inventories, branch-consistent radiation limits, qualified materials and containment, and independently assessed safety functions. Hardware records, the reconciled Master Procurement BOM, and the state-space model manifest and verification records of Section IX shall identify the evidence supporting each requirement.

Release records distinguish specification inputs, reproducible arithmetic, reported simulations, measured performance, and accepted protection functions. The thermomechanical selection in Section III-A and the fatigue summary in Eq. (215) govern the updated contact and spring configuration through Table LXXII; earlier conflicting clamp and spring values are comparison records. The remaining domain selections retain their stated qualifications. Equation-based corrections retain the proper Gamow exponent, explicit Franck–Condon suppression, a two-port reduced transfer matrix, separated chilled/direct heat rejection, and dimensionally consistent creep activation. Historical electrical margins and model coefficients are not demonstrations of a nuclear heat source. Source stress/strain inequalities and matrix stability are not substitutes for demonstrated structural, nuclear, or control performance. Preserved historical labels identify their original comparison scope rather than competing current specifications.

XI. INTEGRATED MASTER SYNTHESIS AND GOVERNING PARAMETER LEDGER

A. Governing Source Selection and Cross-Domain Ledger

Table LXXII incorporates the contact-material and spring design in `update.txt` and the supplied fatigue and optimizer summaries [90]–[92]. These inputs govern the updated thermomechanical configuration. "Selected" identifies a design input or reported simulation statistic; "calculated" identifies reproducible arithmetic. Neither denotes independently measured hardware performance. Earlier strain limits apply only to their identified material and amplitude/range conventions. Historical clamp stiffnesses and thermal displacements shall not be substituted for the selected 5,000 N/mm spring stiffness and 20.1 μm spring-path mismatch.

All v4.2 sections are integrated: planar kinetics and grating matching in Sections I-F and II-A; fatigue, nonlinear CTE and

TABLE LXXI
ENGINEERING-REPORT MASTER PROCUREMENT BASELINE (USD)

Item	Component / designation	Supplier named in report	Qty.	Unit price	Extension
MAT-01	Pd-Ir sputtering targets	Materials Preparation Center	2	42,500.00	85,000.00
MAT-02	Single-crystal CVD diamond substrates, 20×20 mm	Element Six	2	18,200.00	36,400.00
MAT-03	High-purity deuterium gas	Sigma-Aldrich / Isotec	2	4,800.00	9,600.00
FAB-01	TLP bonding and vacuum brazing services	Metal Fusion Solutions	1	14,500.00	14,500.00
OPT-01	Fiber-coupled diode pumps, 785.0/802.5 nm	IPG Photonics	2	28,400.00	56,800.00
OPT-02	Custom refractive beam shaper; revised exponent $m = 1148$ field target	Asphericon GmbH	1	12,300.00	12,300.00
SEN-01	Double-focusing sector mass spectrometer	Thermo Fisher Scientific	1	85,000.00	85,000.00
SEN-02	HPGe gamma detector	Ortec / AMETEK	1	38,000.00	38,000.00
SEN-03	EJ-301 scintillator and digital PSD system	Eljen Technology	1	16,500.00	16,500.00
ELE-01	WT5000 precision power analyzer	Yokogawa Test & Measurement	1	22,400.00	22,400.00
TOO-01	Kinematic frame and vacuum pressure vessel	Custom precision machining	1	15,583.00	15,583.00
Total priced baseline					392,083.00

cross-domain preload in Section IV-A; stress-assisted transport in Section IV-H; channel/TEG compatibility in Section II-C; and its summary parameters in Table LXXII. The supplied files remain the authoritative source records. Several v4.2 references are unidentified or unrelated to the claimed engineering result, and the v4.1 PDF attributes key performance claims to `unknown_url`; these are not independent verification datasets.

B. Physical-Closure and Release Gate

The mathematical identities and source integration above are reproducible, but the system is *not released as a physically closed or experimentally qualified reactor*. Closure requires: (i) an evaluated causal, converged dielectric and nuclear transition model with a common rate/volume/energy normalization; (ii) a temperature-qualified strain law and load path satisfying both fatigue life and preload limits; (iii) a consistent cold-plate material, geometry, pump curve and thermal/electrical port ledger; (iv) a complete GUM measurement function and covariance budget with raw calibration/blank data; (v) traceable helium, PSD and source-normalized shield validation; and (vi) complete plant/observer matrices and measured fast-path safety evidence. These missing records cannot be replaced by multiplying nominal enhancement factors or asserting version authority.

C. Historical Text-v4.1 and Earlier Ledgers

The historical integration map in Table LXXIV contains every row of Section 5 of the older *text* report called v4.1, not Section 8 of the newly supplied PDF. Its frame cancellation and observer coordinates are superseded for current selection. It remains solely for audit traceability. The coherent-domain size 10^5 and inferred $\lambda_{cC_{ss}} \approx 3.8845 \times 10^{-8} \text{ s}^{-1}$ close an inverse power requirement, not a first-principles nuclear-rate proof. The revised geometry also changes optical overlap, thermomechanical boundaries, and reaction inventory; earlier 1 nm-film, 4 cm², 550 K, or chilled-loop results cannot be transferred without identifying those changes.

The historical v3.0 inputs and their numerical dispositions are recorded in Table LXXV. “Reported” distinguishes an input

or outcome printed in the source from a reproduced derivation or measured qualification. Prior v2.0 tables remain versioned audit records, not current executive ledgers. Appendix D identifies the v3.0 PDF, which is absent from this workspace; the source’s two bibliography entries name an unavailable `cf3.pdf` and `unknown_url`, not independently identifiable empirical datasets. The revised uncertainty budget and barrier inequality now pass their respective numerical criteria under the stated models. The arithmetic and symbolic corrections do not establish the claimed nuclear output, RCWA performance, frame compensation, broadband ROM equivalence, or measured dose. These distinctions are necessary for a mathematically consistent specification and cannot be removed by calling the source “fully resolved.”

a) *Retained v2.0 source-to-section register*: The v2.0 baseline is integrated without deleting the preceding mathematical record. Table LXXVI maps all source modules to the governing manuscript sections. Appendix C retains the complete 46-page source, including every intermediate equation and table, in original source order. Original source equation graphics and source-local citations are not silently rewritten there; corrected equations and cross-references in the main text govern calculations. Some original PDF cells and equations extend beyond their printed columns. Readable coefficients used in the typed integration were checked against the embedded equation images, rather than guessed from clipped cell boundaries.

The synthesis distinguishes three kinds of closure. Arithmetic closure includes the 800.688889 W ledger, 0.116078911301 bus-efficiency scenario, correctly scoped 174.7156 excess-heat statistic, 0.749155% aperture clipping, coating-defect calculation, and itemized procurement sum. Algebraic closure includes the composite dielectric inverse, energy-resolved rate definitions, constitutive equations, and correctly signed Schur identity. Physical validation additionally requires material-specific dielectric input, nuclear channel matrix elements, an achievable cooling equilibrium, measured or reproduced grating/structural performance, identified control ports, and independently assessed protective functions. The source’s incompatible literal ROM, transitional pressure prediction,

TABLE LXXII
GOVERNING MASTER LEDGER: SELECTED VALUES AND RELEASE DISPOSITION

Domain / parameter	Selected value or retained domain input	Governing derivation and disposition
Screening energies	$V_{\text{bare}} = 51.42714 \text{ eV}$; $U_{\text{eff}} = 350.0 \text{ eV}$; $V_{\text{driven}} = -298.57 \text{ eV}$; $V_{\text{eff}} = 51.43 \text{ eV}$	Eq. (15); rounded Coulomb coefficient and reconstructed-reference convention. Dielectric inversion and branching require independent evaluation.
Native scalar Floquet kernel	$s_F = 3.67250764167$; $\delta_F = 3.56655 \times 10^{-16}$	Eqs. (17)–(18); 512-bit evaluation from binary64 inputs. Roundoff-scale scalar agreement, not full tensor convergence.
Geometric domain inventory	$V_{\text{unit}} = 10^{-24} \text{ m}^3$; $V_{\text{active}} = 2.53 \times 10^{-10} \text{ m}^3$; $N_{\text{dom}} = 2.53 \times 10^{14}$	Eq. (141); envelope cells, distinct from deuteron inventory and coherence populations. Metal-filled fraction is 0.01136.
Gross thermal normalization	$P_{\text{th,ref}} = 2911.4 \text{ W}$; $\bar{q}_{\text{active}} = 11.50751 \text{ kW/mm}^3$	Eqs. (142)–(144); metal mean density $1.012661 \times 10^{15} \text{ W/m}^3$; not a microscopic power prediction.
Planar collective factor	$\Phi_{2D} = 1364.02$; $\eta_{\text{coll}} = 2.5098 \times 10^{10}$	Eqs. (49)–(50); factor is relative to one domain, not an already integrated incoherent population.
Pair-rate target	$w_{\text{pair}} = 3.01 \times 10^{24} \text{ m}^{-3} \text{ s}^{-1}$	Eq. (51); source rate dimensions and active-volume/power normalization do not close.
Optical beat / grating	$f_{\text{beat}} = 8.328064 \text{ THz}$; $\Omega_D = 5.232677 \times 10^{13} \text{ rad/s}$; $\Lambda_g = 960.80 \text{ nm}$; $q_{\text{beat}} = 2.2985 \times 10^5 \text{ m}^{-1}$	Eq. (5) and Section II-A; retain wavelength-derived precision in calculations. Wavevector uses nondispersive $n_p = 1.453$.
RCWA / field factors	25×25 orders; $\mathcal{F}_{z,785} = 126.8$; $\mathcal{F}_{z,802.5} = 141.5$	Intensity ratios; geometry, optical constants and solver convergence must accompany the reported result.
Beat-driven field model	$\mathcal{A}_z = 21.45$; $m = -2, \dots, 2$; $n_e = 4.85 \times 10^{28} \text{ m}^{-3}$	Field-amplitude ratio, not the pump-resolved RCWA intensity factors. Five-sideband truncation requires convergence checks.
HiPIMS	91.32 wt% Pd / 8.68 wt% Ir; 1.85 kW/cm^2 ; 30 μs , 100 Hz, -120 V	Section IV-A; complete recipe and source-reported 0.12 nm AFM roughness retained.
Graded TLP fatigue	$\epsilon_{\text{tot}} = \Delta\epsilon_{\text{tot}}/2 = 0.002137728$; $N_f = 52400$ cycles	Eq. (206); plastic amplitude 4.94024×10^{-4} ; $3.5 \mu\text{m}$ micro-alloyed bond plus $15.0 \mu\text{m}$ Ni/Cu layer. Predicted 16.9% margin at 293.15–550.0 K; distinct from spring endurance.
Spring fatigue	$\Delta\epsilon_p \leq 4.805 \times 10^{-4} \text{ mm/mm}$; $\epsilon'_f = 0.45$, $c = -0.60$	Eq. (213); ceiling gives $N_f \geq 44823.0984$ cycles, meeting 44,820 cycles under the selected law.
Fatigue Monte Carlo	Mean 88,827 cycles; 95% interval [54,822, 138,492]; specified 10^6 samples	Eq. (215); mean and lower interval endpoint meet 44,820 cycles. Reported lifespan interval is not a confidence interval on the mean or an all-sample guarantee.
Clamp-force bounds	$F \in [6283.19, 78539.82] \text{ N}$; 20 mm contact diameter	Eq. (208); 20 MPa sealing minimum and 250 MPa Pd-25Ir yield input at 350 °C. Preserve excursion and relaxation reserve.
Stack stiffness / pressure	$K_{\text{stack}} = 5000 \text{ N/mm}$; $\Delta P = 0.320 \text{ MPa}$	Eq. (244); $5.0 \times 10^6 \text{ N/m}$, $\Delta F = 100.5 \text{ N}$; cyclic variation satisfies $\pm 0.50 \text{ MPa}$.
Spring-path expansion	$\Delta L_{\text{net}} = 20.1 \mu\text{m}$ over 325 K	Eq. (214); spring/collar mismatch, distinct from the optical alignment-path calculation.
Optical-path comparison	$\Delta L_I = 16.83 \mu\text{m}$; $\Delta L_A = 9.04 \mu\text{m}$; net 7.79 μm nominal	Eq. (285); unrounded 7.795219413 μm applies to that alignment geometry, not the selected spring mismatch.
Piezo travel	15.0 μm stroke; 7.21 μm / 48% nominal reserve	Unrounded reserve is 7.204780587 μm ; one-sided cold-end bias and tolerance allowance required.
VCCT	$G_I/G_{II}/G = 4.2/14.2/18.4 \text{ J/m}^2$; $G_c = 24.5 \text{ J/m}^2$	Source-reported simulation; interface toughness and mesh/load-case convergence remain unverified.
Transport / HE	Table LXXXI; $K_{IH} = 41.8 \text{ MPa}\sqrt{\text{m}}$; $I_{RA} = 0.89$	Section IV-H; bare/effective diffusivity, detailed balance and specimen-level evidence required.
Barrier / LBB	$\text{PRF}_{\text{eff}} = 1538.5$; $a_{\text{crit}} = 189 \text{ mm}$ nominal	Eqs. (359)–(362); printed inputs give 1540.80 and 188.49 mm; LBB safety is not established.
Hydraulics	120 channels; 3.62 L/min; 142.5 kPa assembly target	Section II-C; corrected channel area/speed yield 62.919 kPa straight-channel drop; remaining loss is unverified.
TEG / export	418.43 W bus; 386.67 W support; +31.76 W net	Table IX; selected bus arithmetic closes, but optical/RF inputs and whole-device boundary do not.
Calorimetry	$u_c = 5.69 \text{ W}$; $P_{\text{excess}} = 995.81 \text{ W}$; $Z = 175.01$	Section V-G; Z is conditional; 13.0061 W^2 of uncertainty variance is unexplained.
Helium quantitation	$3.756 \times 10^{-5} \text{ mol}/(\text{day kW})$; sector $R = 25000$	Eq. (437); stoichiometric prediction and source assay claim are distinct.
PSD / shielding	FOM 2.15; $R_{\gamma,95} \simeq 1.000000 \times 10^6 \geq 10^6$; $\dot{H}_{\text{surface}} = 0.38 \mu\text{Sv/h}$	Eq. (458); exact 95% binomial bound for 2,995,731 independent zero-leakage trials. Detector and source-normalized shielding qualification remain separate.
Stiff model / observer	15 states; Radau IIA; 1 ms ZOH; PDF L_{calc} ; $t_{95} = 99.86 \text{ ms}$	Section IX-A; exact exponential and Radau approximation distinguished; matching observer plant absent.
SIL-3 / TMR	2oo3; 10 ms supervision; $< 50 \mu\text{s}$ trip; 42.80 μs source claim	Section VII-F; independent fast path, completed-safe-state timing and SIL evidence required.

temperature approach, and strict uncertainty/fatigue inequalities are explicit numerical findings, not hidden by a claim of universal completion. Retaining them in the source record does not authorize operating a nuclear-energy source or certify safety compliance.

N. EXPERIMENTAL VERIFICATION & OPERANDO DIAGNOSTIC PROTOCOLS

The experimental program couples hydrogen-exposure testing, isotope-resolved gas analysis, diffraction, and joint creep measurements to the constitutive models. Each measurement retains a defined specimen population, environmental boundary, calibration chain, and uncertainty budget. The procedures below

specify experiments to be performed; numerical agreement with a constitutive model is not an experimental result. Standard-compliant tests follow the complete adopted edition, including specimen preparation, replicate counts, acceptance rules, and reporting requirements. Additional temperatures, isotopes, or loading histories are identified as project extensions rather than attributed to the standard.

A. Specimen Traceability and Environmental Controls

Specimen records identify material heat, composition, heat treatment, surface finish, cleaning and coating sequence, bond geometry, hydrogen charging history, and elapsed time between preparation and loading. Process witnesses undergo the same

TABLE LXXIII
COMPLETE V4.1 PDF SECTION-TO-MASTER INTEGRATION REGISTER

Source section / pages	Current domain destination
1.1–1.3 / 1–6	Section I-F: Floquet samples, trace, branching, domain kinetics
2.1–2.2 / 6–7	Section II-A: optical constants, harmonics, intensity enhancements
3.1–3.3 / 8–11	Section IV-A: deposition, fatigue, VCCT, nonlinear CTE
4.1–4.2 / 11–13	Section IV-H: complete trapping table, HE, barrier and LBB
5.1–5.3 / 13–18	Sections II-C, V-G: hydraulic, TEG and GUM ledgers
6.1–6.3 / 19–21	Sections VI-A, VI-F: helium, PSD, HPGe, shield
7.1–7.3 / 21–25	Sections IX-A, VII-F: stiff model, gain, timing matrix
8 / 25–28	Table LXXII: all summary metrics and source-specific qualifications
References / 28	Appendix E: complete source bibliography

manufacturing sequence as the components they represent. Unexposed controls, charged specimens, fixture blanks, and interrupted-test specimens are assigned before testing; replicate counts and exclusion criteria are fixed before examining outcomes. The graded Ni/Cu bondline and the X-750 spring stack are separate specimen populations with distinct thermal and mechanical histories.

Temperature, load, displacement, gas composition, pressure, beam intensity, and acquisition time are recorded on a common clock. Sensor calibration includes drift and cross-channel timing uncertainty. High-pressure hydrogen and elevated-temperature tests require independently qualified pressure cells, ventilation, leak detection, and protective interlocks. A hydrogen result is not assumed to establish the corresponding deuterium response, nor does a steel witness coupon establish the embrittlement resistance of a nickel-base spring or a Pd–Ir film.

B. Hydrogen Embrittlement Test Matrix

a) *ASTM F519 sustained-load process qualification*: Use Type 1a.1 notched steel witnesses with the steel condition, hardness, geometry, and preparation specified by ASTM F519-23 [118]. Establish the baseline notched fracture strength on companion specimens, then apply the production cleaning, plating, and post-treatment sequence to the test witnesses. For the selected sustained-load program,

$$F_{\text{hold}} = 0.75F_{\text{NFS}}, \quad t_{\text{hold}} = 200 \text{ h}, \quad (616)$$

where F_{NFS} is the baseline notched fracture load, not the smooth-bar yield load. Continuously record applied load, exposure conditions, and fracture time; retain survivors for post-test examination. Interrupted holds, invalid specimens, and retests follow the adopted standard rather than being counted as successful exposures. This test qualifies the represented process or service environment on susceptible steel; it is not a ranking test for unrelated alloys.

b) *ASTM F1459 gas-embrittlement comparison*: Use matched thin-disk pressure-rupture specimens under ASTM F1459-06(2024) [119]. Control disk thickness, surface con-

dition, clamping, temperature, and pressure-ramp history in paired helium and hydrogen tests. The comparison quantity is

$$R_{\text{HGE}} = \frac{P_{\text{rupt,He}}}{P_{\text{rupt,H}_2}}, \quad (617)$$

so a ratio above unity indicates reduced rupture pressure in hydrogen. Notched tensile bars are not substituted for the disk geometry. The proposed project envelope uses 99.9999% hydrogen, pressures up to 35 MPa, and temperatures up to 150 °C, subject to apparatus qualification and the adopted method's applicability. These bounds are project conditions, not universal ASTM requirements; departures from the standardized procedure are reported explicitly. Record gas impurities, pressure history, leak checks, rupture location, and the uncertainty of the pressure ratio.

c) *ASTM F1624 incremental step loading*: Apply ASTM F1624-12(2024) to steel specimens to determine a loading-history-dependent hydrogen-assisted cracking threshold [120]. Separate precharged specimens for internal hydrogen from specimens exposed during loading for environmental hydrogen. Establish the corresponding baseline fracture load, then monitor displacement, load relaxation, and crack onset during successive holds. A proposed 5% load increment with 2–4 h holds is an initial project schedule; the adopted method and loading-rate convergence determine the final schedule. Report the last completed non-cracking step, the onset bracket, and normalized threshold load rather than an exact, rate-independent threshold.

Only a valid fracture-mechanics specimen permits conversion to a stress-intensity threshold:

$$K_{\text{TH}} = \frac{F_{\text{TH}}}{B\sqrt{W}} \mathcal{Y}(a/W), \quad (618)$$

where B , W , a , and $\mathcal{Y}$ denote specimen thickness, width, crack length, and the calibrated geometry function. Otherwise retain threshold load or nominal stress. This steel method does not independently qualify X-750 washers or determine an allowable reactor clamp load. Post-processing hydrogen-relief treatments require alloy- and process-specific qualification.

C. UHV Mass Spectrometry and Outgassing Subtraction

The magnetic-sector measurements in Section VI distinguish chamber background, specimen desorption, and isotope production through matched controls and calibrated gas throughput.

- 1) Bake the compatible analysis chamber and fixtures at 250 °C for 48 h; verify, rather than assume, a base pressure below 5×10^{-8} Pa after cooldown. Record residual-gas spectra and isolate leaks from persistent wall outgassing. This base-pressure criterion does not prescribe the pressure during specimen heating.
- 2) Introduce traceable helium and independently calibrated deuterium flows over the intended measurement range. Determine mass-dependent sensitivity, fragmentation, transmission, detector linearity, effective pumping speed, and gas-transfer response time. Leak temperature and calibration uncertainty enter the flow budget [121]. Calibrate getter-on and getter-bypass configurations separately, including helium recovery and memory after large deuterium exposures.

TABLE LXXIV
HISTORICAL TEXT-v4.1 MASTER LEDGER: NON-GOVERNING SOURCE SECTION 5 MAP

Variable / domain	v4.0 baseline	v4.1 selected value	Equation-based disposition
Active film / materials	1.0 nm legacy film	50.0 nm monolithic peak thickness	42.50 nm teeth plus 7.50 nm continuous backplane; source stability and dewetting claims are qualification requirements, not proven by thickness alone.
Grating / optics	42.50 nm relief	42.50 nm relief, 50% duty	Retained 960.80 nm pitch; optical constants and amplitude convention in Tables XX and XXI.
Active volume / inventory	Conflicting film/relief geometry	$2.875 \times 10^{-12} \text{ m}^3$	1 cm ² footprint and 28.75 nm mean thickness; $N_D \approx 1.9622 \times 10^{17}$ at unit loading, (42).
Chiller / thermal loop	593.56 W	0.00 W	Compressor-free water-to-air rejection; pumps and fans remain electrical loads, (177)–(178).
Invar / frame	42.15 mm	34.7021 mm	Linear constant-CTE solution for stipulated 33.63 mm seating height, (287).
Al / frame	8.52 mm	1.0721 mm	Same model and 325 K excursion; not a new measurement of the full layered assembly.
Thermal pinch / preload	52.77 μm	0.00 μm rounded	Calculated residual 0.0001285 μm ; 15.80 $\pm$ 0.50 MPa is the adopted preload bound, subject to (289).
C_D / observer	No direct measurement	Observable; rank 4	Third state of (T_s, T_c, C_D, P) ; two outputs and matrices in (601); not the older v4.0 state ordering.
Observer poles / dynamics	No numeric observer	Targets $-30, -32, -35, -40 \text{ s}^{-1}$	Printed L gives nearby stable poles. Slow-mode $t_{95} = 99.86 \text{ ms}$; sub-50 ms settling and uniform sub-100 ms error suppression are not established.
Net output / SWaP-C	-561.80 W actual ledger	$+31.76 \text{ W}$ reported ledger	418.43 W generation minus 386.67 W support. Core residual heat is 2492.97 W, not simultaneous 2911.40 W heat export.

- 3) Acquire chamber-plus-holder blanks before and after each specimen series with identical heater, pumping, ion-source, gain, and timing settings. For destructive ex situ thermal-desorption spectroscopy, use the specified 10 °C/min ramp up to 900 °C only on compatible coupons and fixtures. This ramp is not an operando temperature program for the assembled reactor. Operando blanks instead reproduce the actual service-temperature history.
- 4) Interleave reference-gas checks and background measurements with specimen runs. Resolve hydrogen, HD, deuterium, and helium using the calibrated mass response; nominal mass 4 alone cannot separate D₂ from ⁴He. Retain the high-resolution peak scans and getter recovery measurements rather than attributing an unresolved mass-4 increase to helium.

For a resolved acquisition channel k , interpolate the bracketing backgrounds at matched time and temperature using a prescribed drift model:

$$\begin{aligned} b_k &= w b_{k,\text{pre}} + (1 - w) b_{k,\text{post}}, \quad 0 \leq w \leq 1, \\ s_k &= i_{k,\text{raw}} - b_k, \\ \Sigma_s &= \Sigma_i + \Sigma_b - \Sigma_{ib} - \Sigma_{bi}. \end{aligned} \quad (619)$$

The weight follows acquisition time, not a fit chosen to maximize the residual signal. Background interpolation error and shared calibration correlations remain in the covariance. A steady-throughput response model is

$$\mathbf{s} = \mathbf{K}\mathbf{q} + \boldsymbol{\nu}, \quad [q_a] = \text{mol s}^{-1}, \quad (620)$$

where $\mathbf{K}$ contains gas-specific sensitivity and fragmentation, with units A/(mol/s), and $\boldsymbol{\nu}$ is measurement noise. Only independently identifiable species can be recovered from this relation. The scalar reduction $q_a = \alpha_a s_k$ applies to an isolated species with a calibrated transfer path; its coefficient is not a universal ionization constant.

When chamber storage is significant, use the species balance rather than interpreting instantaneous pressure as released gas flux. For a well-mixed chamber of fixed volume V_{ch} ,

$$q_a(t) = \frac{d}{dt} \left(\frac{V_{\text{ch}} \Delta p_a}{RT_g} \right) + \frac{S_{\text{eff},a} \Delta p_a}{RT_g}, \quad (621)$$

where Δp_a is the background-subtracted partial pressure, T_g is the gas temperature, and $S_{\text{eff},a}$ includes conductance limitations. Adsorption, getter uptake, and delayed release require additional calibrated transfer terms when they are not negligible. The recovered amount and area-normalized molar flux are

$$n_a = \int_{t_0}^{t_1} q_a(t) dt, \quad J_a(t) = \frac{q_a(t)}{A_s}. \quad (622)$$

Convert molecular inventories to hydrogen or deuterium atom inventories with the appropriate stoichiometry. Preserve negative background-subtracted observations; clipping them biases integrated yields upward. Detection decisions use the blank distribution, recovery uncertainty, and a specified false-positive probability. Neither a positive residual nor nominal mass resolution alone establishes nuclear ash production.

D. Beam Calibration and Operando Diffraction

Here a position-sensitive detector records diffraction angle; it is distinct from the pulse-shape discrimination of the EJ-301 radiation channel in Section VI-I. Establish wavelength, detector geometry, and line shape using a suitable certified reference, such as silicon SRM 640f, at its specified reference temperature [122]. Acquire reference peaks across the complete analysis range, not only near the strongest sample reflection. Bragg's relation and a candidate channel correction are

$$\begin{aligned} 2d_{hkl} \sin \theta_{hkl} &= \lambda, \\ 2\theta_i &= 2\theta_0 + \sum_{n=1}^3 \beta_n (x_i - x_0)^n. \end{aligned} \quad (623)$$

TABLE LXXV
HISTORICAL v3.0 MASTER PARAMETER RECONCILIATION LEDGER

Parameter / subsystem	Prior v2.0 baseline	v3.0 report / requested revision	Governing disposition from explicit equations
Drive and screening	8.330 THz; 350 eV target	$\Omega_D = 5.23389 \times 10^{13}$ rad/s; bare 51.427305 eV, TF shift 36.099551 eV	Driven subtraction -298.572695 eV retained as scenario; no evaluated band/dielectric data supplied.
Branching ratio	Conditional source-channel rates	$\Gamma_{\text{other}}/\Gamma_{\text{lat}} = 6.89655 \times 10^{-15}$	$B = 1/(1+r)$ follows conditionally; not an evaluated nuclear matrix element or absolute rate.
Coherent domain	$L = 208.62$ nm, $\delta_z = 149.92$ nm	$V = 6.524864 \times 10^{-21}$ m ³ , volume ratio 6.1304×10^7	Field-volume arithmetic retained; distinct from 1 nm film tiling.
Nuclear power	Independent reference scenarios	2911.40 W claimed	Printed pair rate gives $1.16391741 \times 10^{-7}$ W; 2911.40 W remains thermal reference.
Optical geometry	SF11 / 50 μ m diamond comparison	Air/silica grating; 500 μ m diamond; 1 nm PdIr/10 nm Ti	Adopt revised dimensions; missing imaginary indices prevent unique RCWA reconstruction.
Grating coupling	960.80 nm pitch; 42.50 nm depth	50% duty, 65°; $\mathcal{F}_z = 124.50/138.20$	Reported enhancement targets; previous planar values are not transferred to new geometry.
Frame compensation	172.50/20.69 mm, separate temperatures	42.15/8.52 mm, common 256.85 K rise; residual 0.82225 μ m	Source dimensions give -15.328181 μ m residual; 20 μ m travel alone does not validate datum chain.
Tilt RMS	0.00800 μ rad limit	0.00782 μ rad	Reported integral; 2.25% numerical margin, no spectrum provided.
Export / refrigeration	800.688889 W BOP; 10/15 kW scenarios	2911.40 W heat; +2604.73/+1941.54 W; COP 4.2	Source metric uses unity heat conversion, not electrical export; 693.190476 W compressor for full-heat duty.
Electrical comparison	$\eta_{\text{bus}} = 0.116078911301$	No new physical electricity efficiency	With retained efficiency and revised loads only: +31.285/−631.905 W; conditional, not a solved cooling equilibrium.
Cold-plate geometry	150/600 μ m; 120 channels	217.94/871.76 μ m; $d_h = 348.70$ μ m	Adopt geometry; water at 7 L/min gives $Re = 1777.618$, $f = 0.036004$, channel head 67.471 kPa.
Loop pressure / pump	452.93 kPa channel estimate exceeded cap	350 kPa; 116.67 W	Retain as total-loop head/pump allocation, not derived channel pressure; reconcile minor losses.
Joint fatigue	9289.7 cycles at $\Delta\epsilon_p = 0.00223$	$K = 4.85 \times 10^8$ N/m, preload 42.5 MPa; 44,820 cycles	Unrounded required range 0.0983786284%; preload alone does not evaluate cyclic plasticity.
Barrier defects / PRF	Intrinsic 2850, cap 8.5×10^{-4} ; PRF 832.93	Intrinsic 25000, pore factor 1.42, cap 6.49×10^{-4} ; PRF 1091	Revised equation gives 1039.983, meeting 1000; model change recorded separately from cap change.
Transport / leakage	Material-specific legacy trap data	$N_t = 1.20 \times 10^{26}$ m ^{−3} , $V_H^* = 1.70 \times 10^{-6}$ m ³ /mol, $Q^* = 12.50$ kJ/mol; $J_D = 3.15 \times 10^{-8}$ mol/(m ² s)	New source inputs recorded; flux remains reported without diffusion fields and boundary conditions.
SPA / DC preservation	Literal reduced v2.0 model unstable	Symbolic balanced SPA, 15 to 4 states	Correct Schur proof retained; no new numeric matrices or eigenvalues supplied.
SIS thresholds / latency	100 W/s, 15 GHz/50 μ s, dump < 500 ns	15/45/120/85/110 ns chain; total 375 ns	Arithmetic margin 125 ns; sensing window, thermal bandwidth and energy handling remain separately qualified.
Calorimetry	2911.40 W; $u_c = 5.701706$ W	New water inputs, 8.42 K, 12.30 W envelope; $u_c = 5.6956$ W	Literal point: 2917.3263 W, $u_c = 5.696035$ W. At 2911.40 W: 8.402823 K, $u_c = 5.691919$ W.
Excess-heat statistic	Hypothetical 1000 W gives $Z = 174.7156$	Revised calorimeter allocation	At reconciled point and 0.50 W input uncertainty: $Z = 175.013715$, not measured evidence.
Shield transport	35/120/10 mm shield; 0.380 μ Sv/h target	90/10 wt% W–Cu, 5 wt% B–HDPE, SS316L; 0.312 μ Sv/h	Reported dose, 17.8947% below target; source, transport and dose-conversion verification not supplied.
Procurement	11-line \$392,083.00 baseline	No replacement cost schedule	Retain prior quotations as historical baseline; changed geometry/hardware has no new priced BOM.

Fit the geometry coefficients with calibration uncertainties, check residuals on withheld reference peaks, and retain the coefficient covariance. Use a physical detector-geometry model instead if a cubic correction fails the residual criterion. Detector orientation, sample height, and beam alignment are recalibrated after fixture changes.

Acquire dark counts, dead-time response, bad-pixel masks, and channel efficiency maps at the operating count rate. After

correcting the count statistics for the instrument's dead-time model, an intensity normalization is

$$I_{i,\text{corr}} = \frac{N_i - d_i t_{\text{live}}}{t_{\text{live}} \Phi_{\text{mon}} \epsilon_i \Delta\Omega_i}, \quad (624)$$

where d_i is dark-count rate, Φ_{mon} is incident flux in a fixed monitor normalization, ϵ_i is channel efficiency, and $\Delta\Omega_i$ is the pixel solid angle. Glassy-carbon or polymer scattering

TABLE LXXXVI
COMPLETE v2.0 SOURCE-TO-MANUSCRIPT INTEGRATION REGISTER

Source pages	Retained v2.0 content	Governing integration / numerical disposition
1–8	Bloch–Floquet tensor, quiver expansion, screening, Gamow quadrature, overlaps, coherent domains	Section I; (56)–(57); distinguish field and film volumes.
8–11	Exact BOP, segmented TEG losses, operating export and rejection table	Section II; (202), Table XV; finite-temperature sink feasibility is separate.
11–16	Frame mechanics, GD&T, CF feedthroughs, isolation, grating, polarization and aperture	Section III; retained tolerance/feedthrough schedules, Table XXV, (278)–(279).
16–21	Five-layer HiPIMS stack, Voce/VCCCT, fatigue, transport, HE and coating	Section IV; Tables XLII and XLIII; (370), (549).
21–25	Covariance, excess heat, transitional hydraulics, air exchanger, five-step Nichrome calibration	Section V; Table L, (406); strict 5.70 W uncertainty inequality corrected.
26–28	Sector mass resolution, St707 enrichment, micro-dosing, helium inventory	Retained mass-assay and ash equations; 0.0001601 u resolution, 1.45×10^5 enrichment and 3.7562×10^{-5} mol/day/kW source benchmark.
28–30	EJ-301 gates/PSD, binomial trials, HPGe windows, three-layer shielding	Section VI-I; 24/240 ns gates, FOM 2.15, 2,995,731 zero-failure trials at 95% confidence; dose remains source- and geometry-dependent.
30–33	DM-01/02, 2003 truth table, surge/dump protection, TR-01–TR-06	Retained control and identification sections; response times and trip thresholds retain signal and hardware boundaries.
34–36	Two relief cases, classified locations, fast isolation, FM-01–FM-09	Retained process-safety and FMECA schedules; certified capacity, local plume and downstream inventory are not inferred from average dilution.
37–42	Fifteen-state coefficients, four-state matrices, Schur/port maps	Section IX; all full-model entries retained; literal v2.0 ROM in (591)–(594) fails its printed stability/DC claims.
42–44	Forty-eight-hour blinded H ₂ blank, WT5000 RFI subtraction, 11-line BOM	Section X; retain the 0.00 ± 5.72 W blank as a reference, not fabricated measured data; itemized total remains \$392,083.00.
44–46	Synthesis and source references	This section and Appendix C; source-local numbered citations are retained, not converted into invented bibliographic records.

measurements may support flat-field calibration only after accounting for their angular scattering profile, background, and polarization; they are not assumed to be perfectly isotropic sources. Separate absorption and texture corrections from detector efficiency.

During thermal cycling, track an internal reference with a characterized temperature-dependent lattice spacing, for example a compatible tungsten reference. Its thermal expansion must not be removed by forcing its peak to remain at the room-temperature angle. Use its predicted peak motion and an independent temperature measurement to distinguish specimen displacement from detector drift. For a sample reflection,

$$\epsilon_{hkl}(T, t) = \frac{d_{hkl}(T, t) - d_{hkl}^0(T)}{d_{hkl}^0(T)}, \quad (625)$$

where $d_{hkl}^0(T)$ is the stress-free spacing for the same phase and composition. Phase fractions, elastic strains, and line broadening require a stated refinement model with texture, peak overlap, and calibration covariance. Beam-on/beam-off controls quantify beam heating or irradiation changes, and diffraction frames are synchronized with gas release, load, and temperature.

a) *EJ-301 pulse-shape calibration*: Independently calibrate the radiation channel with characterized gamma-only and neutron exposures over the declared energy and count-rate range. Record baseline, gain, trigger efficiency, live time, and pile-up cuts; apply the fixed 24 ns and 240 ns integration windows in Eq. (456). Select the discrimination boundary on

calibration data, then evaluate a separate exposure without retuning the cut. Report neutron acceptance and gamma leakage together, by energy bin, along with the figure of merit in Eq. (468). The $R_{\gamma,95} \geq 10^6$ criterion requires the independent zero-leakage trial count and confidence construction in Eq. (458); a fitted separation of the pulse populations alone is insufficient. A diffraction position calibration cannot validate this rejection ratio.

E. TLP Double-Shear Creep and Phase Evolution

Test the actual micro-alloyed TLP bondline and graded Ni/Cu interlayer alongside separate Cu–Sn and Ag–Sn comparison joints. The comparison program uses OFHC Cu adherends and Ag-metallized silicon where compatible with the fixture, with either a 20 μ m Sn interlayer or porous Ag infiltrated by Sn–3.5Ag. Porous-metal infiltration is a distinct bonding route, not evidence that all Sn-containing joints solidify on the same schedule [123]. Symmetric double-shear specimens are the primary creep geometry; lap joints provide a separate mixed-mode comparison with bending explicitly characterized.

The proposed processing matrix spans 250–350 °C with an initial 10 min hold, followed by aging at 300 °C for intervals up to 500 h. Use a qualified reducing atmosphere, such as 95% N₂/5% H₂, or a qualified formic-acid process. A vacuum base-pressure target of 10^{-4} Pa describes a separate evacuated stage, not the pressure of an admitted forming-gas atmosphere. Record the actual gas pressure and thermal history. Determine completion of isothermal solidification using phase analysis,

cross-sectional microscopy, and residual-liquid or residual-Sn assessment; neither a fixed hold time nor aging temperature proves completion. Quantify bond thickness, bonded area, porosity, phase distribution, and interfacial reaction layers before mechanical testing.

Use a calibrated dead-weight frame and non-contact local extensometry at 150, 250, and 300 °C over nominal shear stresses of 5–40 MPa. The 350 °C service condition requires additional tests at that temperature; these lower-temperature exposures do not qualify it. Measure thermal gradients and stabilize the fixture before loading. For two shear planes,

$$\begin{aligned} F_1 + F_2 &= F, & \tau_j &= F_j/A_j, \\ A_1 = A_2 &= A, & F_1 = F_2 &\implies \tau_1 = \tau_2 = \frac{F}{2A}. \end{aligned} \quad (626)$$

The equal-load expression requires symmetric compliance and negligible bending. Otherwise determine load partition and local stress with the measured geometry and a validated mechanical solution. Dead weight fixes force, not true stress if bonded area evolves.

For each bondline j , subtract independently characterized fixture, adherend, and thermal-motion contributions from the optical displacement:

$$\begin{aligned} \delta_{j,\text{bond}} &= \delta_{j,\text{opt}} - \delta_{j,\text{fixture}} \\ &\quad - \delta_{j,\text{adherend}} - \delta_{j,\text{thermal}}, \\ \gamma_j(t) &= \delta_{j,\text{bond}}(t)/h_j(T), \\ \dot{\gamma}_{j,\text{cr}} &= \frac{d\gamma_j}{dt} - \dot{\gamma}_{j,\text{el}}. \end{aligned} \quad (627)$$

Here $h_j(T)$ is the measured bond thickness at temperature. Resolve the loading transient and elastic response before estimating creep rate over a predefined time window; report derivative uncertainty and do not impose a steady-state plateau where none exists. Interrupted tests and post-fracture sections document phase evolution, grain structure, interfacial voiding, and crack paths.

Compare measured phase fractions and concentration profiles with Eq. (300), and compare strain-rate histories with the vacancy-diffusion and dislocation relations in Eqs. (316) and (319). Independently constrained diffusion, vacancy, and microstructural inputs are retained when testing predictions on withheld specimens. A stress-rate slope alone cannot uniquely distinguish dislocation climb from grain-boundary sliding. The 52,400-cycle calculation in Eq. (206) remains a separate strain-life prediction until the corresponding thermal-cycle experiments are completed.

F. Uncertainty, Acceptance, and Data Retention

The measurement vector includes pressure ratios, threshold loads, isotope inventories, diffraction strains, phase fractions, and creep rates. Propagate calibration, blank, geometry, temperature, and timing covariances through the measurement functions using Eq. (412). Shared references create correlations across specimens and time points; threshold brackets and non-smooth decisions require interval or distributional treatment rather than an unqualified linear Jacobian. Fix model-comparison tolerances and detection criteria before unblinding the test data.

Retain raw observations, calibration certificates, adopted standard editions, specimen histories, background runs, uncertainty budgets, excluded observations with reasons, and the complete loading and exposure records. Acceptance applies only to the tested material, process, temperature, gas environment, and statistical scope. Successful coupon tests or detector calibrations do not independently establish reactor power, nuclear reaction yields, or functional-safety certification.

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APPENDIX A

FLOQUET INVERSION AND DISPLACED-STATE PROOFS

a) *Composite inverse and static-source projection:* For a retarded inverse kernel, screening is the convolution

$$W(\mathbf{r}, t) = \int dt' \int d^3r' \epsilon^{-1}(\mathbf{r}, \mathbf{r}'; t, t') V_{\text{ext}}(\mathbf{r}', t'). \quad (628)$$

Space/time Fourier transformation converts convolution into multiplication in the composite reciprocal-vector/Floquet indices. Writing $\epsilon = I - M$ with $M = v\chi$, the geometric-series identity is

$$(I - M)^{-1} = \sum_{k=0}^{\infty} M^k, \quad \rho(M) < 1. \quad (629)$$

The condition is on the spectral radius of the truncated operator. An invertible dielectric matrix need not satisfy it, particularly near a resonance; direct factorization and residual checks apply when the Neumann series does not converge. A static source occupies only the incoming zero harmonic. Reconstruction of its outgoing harmonics yields Eq. (14), and the time average selects its outgoing zero harmonic. A double harmonic sum without source coefficients is not the static-source solution.

b) *Gaussian displaced-thermal overlap:* For $\bar{n}_{\text{ph}} > 0$, the thermal state has the coherent-state representation

$$\rho_{\text{th}} = \frac{1}{\pi \bar{n}_{\text{ph}}} \int d^2\gamma e^{-|\gamma|^2/\bar{n}_{\text{ph}}} |\gamma\rangle\langle\gamma|. \quad (630)$$

Displacement changes $|\gamma\rangle$ into $|\gamma + \alpha_{-}\rangle$ up to a phase that cancels in the projector. Since $|\langle 0|\gamma + \alpha_{-}\rangle|^2 = e^{-|\gamma + \alpha_{-}|^2}$,

$$P_{\text{ov}} = \frac{1}{\pi \bar{n}_{\text{ph}}} \int d^2\gamma e^{-|\gamma|^2/\bar{n}_{\text{ph}} - |\gamma + \alpha_{-}|^2},$$

$$\frac{|\gamma|^2}{\bar{n}_{\text{ph}}} + |\gamma + \alpha_{-}|^2 = \frac{1 + \bar{n}_{\text{ph}}}{\bar{n}_{\text{ph}}} \left| \gamma + \frac{\bar{n}_{\text{ph}} \alpha_{-}}{1 + \bar{n}_{\text{ph}}} \right|^2 + \frac{|\alpha_{-}|^2}{1 + \bar{n}_{\text{ph}}}. \quad (631)$$

The two real Gaussian integrals give Eq. (20); the continuous zero-temperature limit is $e^{-|\alpha_{-}|^2}$. This proof is restricted to the same oscillator vacuum. It does not calculate the overlap with an independently compressed nuclear wavefunction.

For completeness, the exponential Laguerre generating identity gives

$$e^{-S} \sum_{k=0}^{\infty} \frac{S^k}{k!} L_k(-z) = I_0(2\sqrt{Sz}), \quad S, z \geq 0. \quad (632)$$

Taking $z = |\alpha_{-}|^2/[\bar{n}_{\text{ph}}(1 + \bar{n}_{\text{ph}})]$ yields the value above unity in Eq. (22). It establishes why that expression cannot be used as a projector probability.

A. Native Arithmetic and Domain Acceptance Benchmarks

The native Rust scalar calculation in Eq. (17) uses exactly promoted binary64 inputs. A rational-arithmetic reference is therefore obtained by writing the two binary64 values as exact integer ratios before evaluating

$$s_{F,\text{rat}} = \frac{u_{64}^2}{u_{64}^2 - v_{64}^2}. \quad (633)$$

The reference and the 512-bit calculation agree at the displayed precision, $s_F = 3.67250764167145629240511891246\dots$. Exact decimal inputs instead define a different reference:

$$s_{F,\text{dec}} = \frac{35000^2}{35000^2 - 29857^2} = 3.67250764167145674086843941099\dots \quad (634)$$

This difference arises from input conversion, not failed 512-bit arithmetic. The native benchmark consistently uses $s_{F,\text{rat}}$, not $s_{F,\text{dec}}$. A 512-bit significand has approximately $512 \log_{10} 2 = 154.13$ decimal digits of arithmetic resolution; it cannot restore information absent from its input values. Finite inputs, nonzero u_{64} , and a nonzero scalar denominator are required before interpreting the reciprocal. Matrix conditioning and reciprocal-space convergence remain independent checks.

In Table LXXVII, passing means that the stated numerical identity, rounding convention, or constitutive acceptance inequality is satisfied. It does not certify reactor operation, microscopic nuclear yields, or detector performance beyond the specified statistical conditions. The three domain results retain full precision internally: the total strain amplitude is 0.0021377280790379686, the bare-energy quotient is 51.42714285714285 eV, and the displayed 51.43 eV sum and 2.53×10^{14} domain count suppress binary64 representation artifacts. Gross power denotes the 2911.4 W thermal normalization, not the 418.43 W electrical bus or the 31.76 W net export.

The first-four-mode relative error $e_4 \simeq 2.36 \times 10^{-16}$ and squared-spectrum tail ratio $r_{\text{tail}} \simeq 2.13 \times 10^{-6}$ in Table XXXVII remain separate reduction metrics. Neither is replaced by δ_F , which compares arithmetic precision in a different scalar calculation. The 512-bit result must remain a lossless decimal representation across the language boundary when it is used for precision comparisons.

TABLE LXXVII
NATIVE NUMERICAL KERNEL AND DOMAIN ACCEPTANCE BENCHMARKS

Calculation	Numerical value	Acceptance scope	Status
Scalar Floquet kernel	$s_F \simeq 3.67250764167$; $\delta_F \simeq 3.56655 \times 10^{-16}$	$\delta_{F,\text{rel}} \simeq 9.71148 \times 10^{-17} < 2^{-53}$; same binary64 inputs at both precisions	PASS
Gamma rejection	$R_{\gamma,95} \simeq 1000000.0748773175 \simeq 1.000000 \times 10^6$	$R_{\gamma,95} \geq 10^6$ at 95% confidence, conditional on 2,995,731 independent zero-leakage trials	PASS
Domain 1: fatigue	$\varepsilon_{\text{tot}} = \Delta\epsilon_{\text{tot}}/2 \simeq 0.002137728$; $N_f = 52400$ cycles	Strain-life arithmetic and $N_f \geq 44820$ cycles for the 293.15–550.0 K bondline model	PASS
Domain 2: beat	$f_{\text{beat}} = 8.3280639958728$ THz; $\omega_{\text{beat}} \simeq 5.232677 \times 10^{13}$ rad/s	Wavelength-derived difference frequency and $\omega_{\text{beat}} = 2\pi f_{\text{beat}}$	PASS
Domain 2: energies	$V_{\text{bare}} \simeq 51.42714$ eV; $U_{\text{eff}} = 350.0$ eV; $V_{\text{driven}} = -298.57$ eV; $V_{\text{eff}} = 51.43$ eV	Rounded Coulomb coefficient and $V_{\text{eff}} = U_{\text{eff}} + V_{\text{driven}}$ reference convention	PASS
Domain 3: energetics	$N_{\text{dom}} = 2.53 \times 10^{14}$; $P_{\text{gross}} = 2911.4$ W	Geometric count and integral power normalization in Eqs. (141)–(142)	PASS

TABLE LXXVIII
 STATIC INVERSE-DIELECTRIC REFERENCE SAMPLES AT
 $|\mathbf{q} + \mathbf{G}| = 2.23 \text{ \AA}^{-1}$

G	G'	$\text{Re } \epsilon^{-1}$	$\text{Im } \epsilon^{-1}$	Contribution (eV)
(0, 0, 0)	(0, 0, 0)	-6.8082	0.0000	-350.00
(1, 1, 1)	(0, 0, 0)	0.1425	-0.0312	7.32
(2, 0, 0)	(0, 0, 0)	-0.0891	0.0104	-4.58
(2, 2, 0)	(1, 1, 1)	0.0314	0.0055	1.61
(3, 1, 1)	(2, 0, 0)	-0.0128	-0.0018	-0.66

TABLE LXXIX
 GAMOW ENHANCEMENT COMPARISON FOR THE ASSIGNED 350 eV SHIFT

E (eV)	$\ln \mathcal{E}_G$	Calculated ratio	Nominal ratio
10	261.641350	4.25985×10^{113}	1.83×10^{357}
100	52.482084	6.20411×10^{22}	3.63×10^{71}

APPENDIX B

CONSTITUTIVE PARAMETERS AND COMPARISON CALCULATIONS

The dielectric entries, optical constants, transport coefficients, and fixed-shift Gamow comparison retain their own constitutive definitions. They are distinct from the coupling, composition-screening, interface-flux, and Hankel-spectrum operating points.

TABLE LXXX
OPTION B OPTICAL CONSTANTS AND ISOTROPIC RELATIVE-PERMITTIVITY ENTRIES

Layer	n_{785}	k_{785}	$\epsilon_{785} = (n + ik)^2$	$n_{802.5}$	$k_{802.5}$	$\epsilon_{802.5}$
Silica prism	1.4536	0	2.112953	1.4531	0	2.111500
Pd-Ir	1.8210	5.2140	$-23.869755 + 18.989388i$	1.8840	5.3420	$-24.987508 + 20.128656i$
Ti	2.7210	3.7840	$-6.914815 + 20.592528i$	2.7650	3.8420	$-7.115739 + 21.246260i$
CVD diamond	2.3980	0	5.750404	2.3965	0	5.743212

TABLE LXXXI
MCNABB-FOSTER TRANSPORT PARAMETERS AT 550 K

Layer	D_{eff} (m ² /s)	N_j (m ⁻³)	$E_{t,j}$ (eV)	k_j (m ³ /s)	p_j (s ⁻¹)	V_H^* (m ³ /mol)	Q^* (kJ/mol)
Pd _{0.95} Ir _{0.05}	3.42×10^{-9}	1.20×10^{26}	0.22	1.45×10^{-20}	1.82×10^3	1.70×10^{-6}	18.4
Ti	8.15×10^{-12}	4.50×10^{27}	0.65	3.10×10^{-21}	2.14×10^{-1}	1.40×10^{-6}	24.1
ϵ -Cu ₃ Sn	1.12×10^{-13}	8.80×10^{26}	0.48	8.40×10^{-22}	3.65×10^1	0.95×10^{-6}	12.8

APPENDIX C

COMPLETE SUPPLEMENTARY V2.0 SOURCE RECORD

This historical appendix identifies the previously integrated 46-page v2.0 source. If its PDF is present, the facsimile is included without altering its original claims; Table LXXII governs current selection. **Source unavailable in this workspace.** The historical PDF SHBT Reactor supplementary.pdf was not supplied with this revision. Existing typed historical content and identifiers are retained; no facsimile is fabricated.

APPENDIX D

COMPLETE SUPPLEMENTARY v3.0 SOURCE RECORD

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APPENDIX E

COMPLETE SUPPLIED v4.1 MASTER RESOLUTION SOURCE

This appendix identifies the 28-page source `SHBT_Reactor_Master_Resolution.pdf`. When available, the complete PDF is included with its original tables, equations, engineering claims, and references. Original clipping, arithmetic inconsistencies, and assertions of validation remain source content and are not endorsed by inclusion.

Table [LXXII](#) governs parameter selection and release qualifications. The accompanying text-source identifier is `supplementary-4.2.txt`.

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